

Fixed-height shifted rectangles

An unconditional proof of the A181199 asymptotic,
all-order expansions, and an algebraicity dichotomy

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For every fixed number of rows m , let $T_m(n)$ count the $m \times n$ arrays increasing in rows, columns, diagonals, and downward antidiagonals. The principal result is

$$T_m(n) \sim \frac{\sqrt{m} \prod_{j=0}^{m-1} j!}{2^{m(m-1)} (2\pi)^{(m-1)/2}} \frac{m^{mn}}{n^{(m^2-1)/2}}.$$

The proof supplies every inverse-power correction, proves the asymptotic conjecture in OEIS A181199, and implies that $\sum_{n \geq 0} T_m(n) z^n$ is algebraic if and only if $m \leq 2$.

Abstract. We split the order-polytope model of a shifted rectangle at a fixed numerical threshold. Except for an explicitly bounded exponentially small boundary contribution, its enumeration is an expectation under independent binomial random variables, weighted by a squared Vandermonde and a rational pair interaction. The shifted hook formula identifies the weight exactly. Standardized binomial moments are polynomials in $1/n$, so the expectation has an effective all-order expansion with rational coefficients. Its leading Gaussian integral gives the constant above; the first two corrections are proved as explicit polynomials in m . We obtain the conjectured constant $9\sqrt{5}/(2^{17}\pi^2)$ for A181199 and five correction terms, a universal limiting ratio to ordinary rectangular tableaux, nonalgebraicity at every height at least three, and Gaussian-ensemble laws for random intermediate row lengths. Complete proofs, an exact-arithmetic coefficient engine, independent enumerators, and a source/status audit are included.

Status. AI-assisted, unrefereed research manuscript. The theorems have proofs below, but have not been independently peer reviewed or proof-assistant verified. Historical priority is not certified. No conjectured recurrence is used as a premise, and none is claimed proved here.

[write] Added 5 October 2026, batch 98: two Parts. Since 5 October 2026 this report has two Parts. Part **I** (Sections 1–13 and Appendices A–B) is the batch-85 manuscript described in the first note of Section 1.3, printed as before: its sections, equations, theorems, research questions and labels keep their numbers, and that note’s “a single manuscript, printed in full” now describes Part **I**. Part **II** (Sections 14–25 and Appendices C–D) is manuscript 02 of batch 98, *Rational diagonals and rare boundary phases in shifted rectangles* (4 October 2026, pinned at db20eb379), printed in full with the label prefix `shr:bp:`. It proves that F_m is the diagonal of a rational power series, hence D-finite, for every fixed m (Theorem 14.1), which answers the first half of Research question 7; the second half, the growth of the minimal order, stays open. It also proves a sharp all-order expansion for the number $R_m(n)$ of arrays whose first row is completed before the last row starts (Theorem 14.2). That count is a small part of the boundary error of Theorem 4.1, not its asymptotic, so Research question 4 is not answered by it (Remark 14.4); the leading term of the boundary error is the write’s Proposition 18.3. Dated notes at the end of Section 4.1, after Remark 8.1 and at Research questions 4 and 7 say so.

[write] Added 5 October 2026, batch 101: five Parts. Since 5 October 2026 this report has five Parts; Parts **I** and **II** keep every number and label, and change only by dated notes. Part **III** (Sections 26–33) is Research Report 231 of the session bundle placed in batch 101: it proves the order-two A181198 recurrence and Conjecture 18 of Kauers and Koutschan [9] for every $n \geq 1$ (Theorems 26.1, 30.1 and 30.2). Part **IV** (Sections 34–42 and Appendix E) is Research Report 233: it proves Conjecture 19 and the order-three, degree-24 A181199 recurrence (Theorems 34.1 and 34.2). These are the recurrences that Parts **I** and **II**, and the OEIS entries, call conjectural. Part **V** (Sections 43–45) prints Research Reports 85 and 91, dated 1 and 2 October 2026, before the manuscripts of Parts **I** and **II**: second routes, earlier by date line, to Theorems 2.1 and 2.3 and to Theorem 14.1. The write adds Proposition 33.2 (the A181198 operator has minimal order) and Proposition 42.2 (A181199 obeys no first-order recurrence). Dated notes at the end of Section 1.3, after Remark 8.1, at Research questions 1, 6, 7, 8 and 11, after the notes of Sections 11.3 and 23.3, after Remark 16.2 and in Section 24.1 say so. The intake’s independent check of the write (5 October 2026, end of Section 45.2) added Propositions 33.3 and 42.3: recurrences of lower degree and higher order for both sequences.

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Part I

An unconditional proof of the A181199 asymptotic, all-order expansions, and an algebraicity dichotomy

1 The problem, the open target, and the source audit

For positive integers m, n , consider arrays $(a_{i,j})$ containing each integer in $\{1, \dots, mn\}$ exactly once, subject to

$$a_{i,j} < a_{i,j+1}, \quad a_{i,j} < a_{i+1,j}, \quad (1.1)$$

$$a_{i,j} < a_{i+1,j+1}, \quad a_{i,j+1} < a_{i+1,j}, \quad (1.2)$$

whenever the indicated cells exist. Write $T_m(n)$ for their number, and set $T_m(0) = 1$. The diagonal inequality is redundant once row and column increase are imposed. The downward-antidiagonal inequality is not.

These numbers form the array A181196. Its fixed-height rows include A181197 for $m = 3$, A181198 for $m = 4$, and A181199 for $m = 5$ [1, 2, 3, 4]. Throughout this article, m is fixed and the row length n tends to infinity. Interchanging these two parameters changes the problem: already $T_2(n) = C_{n-1}$ for $n \geq 1$, whereas $T_m(2) = 1$ for every m .

1.1 The conjecture resolved here

The A181199 entry inspected on October 3, 2026, labels the following formula as a conjecture of Vaclav Kotesovec, dated February 27, 2023:

$$T_5(n) \sim \frac{9 \cdot 5^{5n+1/2}}{2^{17} \pi^2 n^{12}}. \quad (1.3)$$

The entry says it was based on Christoph Koutschan's conjectured recurrence. Kauers and Koutschan discuss the relevant guessed recurrences and finite-sum conjectures for heights four and five in their 2023 article [9, 4]. We do not assume those recurrences. We prove (1.3) directly from the combinatorial definition and extend it to every fixed height.

1.2 What is established background, and what is contributed

The shifted hook product and the order-statistics interpretation are classical tools, not new inventions of this report. Sun's treatment of shifted strips gives both tools and proves enumeration results in several small cases [11]. The known height-three leading asymptotic in A181197 is recovered, not newly conjectured. Chan's transfer-matrix theory concerns repeating strips with a fixed number of cells in each row; in the notation here that is the regime with n fixed and m growing [12]. Its constant-coefficient recurrences do not supply a fixed-height asymptotic in n .

The contributions developed in this manuscript are the exact random-threshold reduction with a boundary bound, the resulting all-height all-order theorem, universal first and second corrections, an algebraicity classification, and intermediate-shape limit laws. These statements are proved below. The source review did not locate these statements in the works inspected, but that observation is not an exhaustive priority search. In particular, a formula's continued "conjecture" label in OEIS is not by itself proof that no published proof exists.

1.3 Relationship to ProveIt

The repository was inspected at revision `6bf7f30d0352f7596e70928b3d4f304914075907`. Identifier searches for A181198 and A181199 returned no matching repository report. This is a limited search result, not a claim to have audited every repository file. The existing path-forest reports distinguish unconditional expansions, conjectured recurrences, and inverse-index approximations [18]. We retain that separation. No theorem in that report is a mathematical dependency of the present proof, and the standard inverse-index method in Section 10 is not presented as new.

Scope of the resolution. Equation (1.3) is proved here. The order-two recurrence for A181198 and the order-three recurrence for A181199 are *not* proved here. The asymptotic series is an all-order Poincaré expansion, with a fixed truncation order; no Borel summability, Stokes data, or complete exponentially improved transseries is asserted.

[write] Added 3 October 2026, batch 85. This report is a single manuscript, printed in full: manuscript 09 of batch 85 of ProveIt’s incoming-reports intake, delivered as the archive `Shifted_Rectangle_Asymptotics.zip` (343,338 bytes, wrapper directory `Shifted_Rectangle_Asymptotics/`) in commit `9d6968c8a`, and placed in commit `ddf8df5d5` in the research-report collection, among the OEIS-sequence asymptotics reports, as the report `a181199-shifted-rectangles`. The title page reads “Prepared for Vladimir Reshetnikov”; the PDF metadata names “Research report prepared for Vladimir Reshetnikov”. No human author and no tool are named. The delivered text is printed unchanged except that every label carries the prefix `shr:`, the title page no longer creates a duplicate page anchor, and dated notes such as this one are added.

Pin and repository claims. The revision quoted above, `6bf7f30d0` (3 October 2026), is the ProveIt commit the manuscript was written against. At placement, and again at this writing, the intake searched the whole tracked tree for A181196–A181199, shifted rectangles, shifted tableaux, shifted hook products and Sun’s article; apart from the arrival archive, the only hits are the batch-85 sibling named below, which points to this report. This report continues no repository material, and the sentence “Identifier searches for A181198 and A181199 returned no matching repository report” remains true. “The existing path-forest reports” is one report, `oeis-sequence-asymptotics/a189281-path-forest-expansions`, in three Parts; as the manuscript says, none of its theorems is used here. No statement of this report is formalized in Lean or Rocq, no formal development in ProveIt treats shifted tableaux, and the report’s place in the collection confers no formal status.

Neighbours. The companion reports of the same batch are `a047874-long-increasing-subsequences` (manuscript 08), whose fixed-sector problem also uses the hook count of the rectangle with m rows of length n as a normalizer, and `a182220-source-boundary` (manuscript 07, unrelated mathematics). Other conjectures of the 2023 article of Kauers and Koutschan [9] are proved in the collection reports `enumerative-combinatorics/a181280-binary-matrix-formula` (its Conjecture 20, Section 6.5, a C-finite formula), `enumerative-combinatorics/a195806-hexagonal-lattice` (Conjecture 11) and Part IV of `a215561-fixed-composition-excursions` (Conjecture 15). The present report concerns that article’s Section 6.4, and leaves the guessed recurrences of that section unproved. Two notes below relate Corollary 2.4 and Section 10 to the transseries volumes in `Analysis/Transseries/docs/series-and-transseries/`.

Notation. The manuscript reuses several letters with local meanings. They are kept as delivered; the table fixes each reading. No symbol was renamed.

[write] Notation table of the note in Section 1.3 (added 3 October 2026)

Symbol	Meanings in this report
$T_m(n)$	the shifted-rectangle count (A181196 family), <i>not</i> the number $f^{(n^m)}$ of standard Young tableaux of the $m \times n$ rectangle; the transseries volume writes $T_d(n)$ for the latter, with d rows.
d	$\binom{m}{2}$, the number of row pairs ((2.1)); in the transseries volume d is the number of rows, m here.
$K_m; K; K_j(X)$	the leading constant (2.2); $K = K_m$ in Section 10, but the number of samples below p in the proof of Theorem 9.2; Krawtchouk polynomials in Section 6.1.
$L; L_5(n)$	a truncation order (Theorem 2.1); $\log(y/K)$ in Section 10; the leading term in Section 7.3.
$h_r(q); h_j$	complete homogeneous polynomials (5.3); squared norms $j! n^j/n^j$ in Section 6.1.
λ	a state or strict partition (Section 3, Appendix A); $m \log m$ in Section 10.
$R_n; R_m; R(z); \mathcal{R}_{L,n}$	the pair weight (4.2); the radius m^{-m} in Section 8; the rational function of Appendix A; the Taylor remainder (5.12).
$p; p_r$	the threshold in $(0, 1)$; the power sums $\sum_i Y_i^r$ (Section 6).
$q_{ij}; q$	$(y_i + y_j)^2$; a derivative order in Section 8.
$Z_m(n); Z_i$	the binomial Vandermonde normalizer (6.1); standardized counts in Section 9.
$A, B; \mathcal{A}$	$\binom{m}{3}$ and $m(m-1)^2(m-2)/12$ (Section 6.1); the event that all comparisons hold.
$C(z); C_{n-1}; C_{m,L}; C_n$	the Catalan generating function; Catalan numbers; a constant in (5.12); the central event.
$E_{m,n}(p); \mathbb{E}; \mathcal{E}_z$	the boundary error (4.3); expectation; the moment functional (5.8).
$H(\lambda); H_j; H_{20}$	the hook expression of Appendix A; Hermite polynomials; the constant (6.11).
β_j	recurrence coefficients (Section 6.2); not the exponent $\beta = -\alpha_m$ of the transseries volume.

[write] **Added 5 October 2026, batch 101: the recurrences.** The note above says that this report “leaves the guessed recurrences of that section unproved”, and the box “Scope of the resolution” says the same of the manuscript. Both remain true of Part I, but no longer of the report: Part III proves the A181198 recurrence and Conjecture 18 of [9] (Theorems 30.1 and 30.2), and Part IV the A181199 recurrence and Conjecture 19 (Theorems 34.1 and 34.2), each from the array definition. Neither proof uses the asymptotic theorems of Part I; both credit them.

2 Main results

Set

$$d = \binom{m}{2}, \quad J_m = \prod_{j=0}^{m-1} j!, \quad M_m(n) = \frac{(mn)!}{(n!)^m}, \quad \alpha_m = \frac{m^2 - 1}{2}, \quad (2.1)$$

and define

$$K_m = \frac{\sqrt{m} J_m}{4^d (2\pi)^{(m-1)/2}}. \quad (2.2)$$

The multinomial coefficient $M_m(n)$ counts all row-increasing arrays, before imposing any comparisons between different rows.

Theorem 2.1 (All orders at fixed height). *For every fixed integer $m \geq 1$ there is a uniquely determined sequence $c_{m,j} \in \mathbb{Q}$, with $c_{m,0} = 1$, such that, for every fixed $L \geq 0$,*

$$T_m(n) = M_m(n) \frac{J_m}{(4n)^d} \left(\sum_{j=0}^L \frac{c_{m,j}}{n^j} + O_{m,L}(n^{-L-1}) \right). \quad (2.3)$$

Every coefficient is given by the finite moment formula (5.9). In particular,

$$c_{m,1} = \frac{m(m^2 - 1)}{12}, \quad c_{m,2} = \frac{m^2(m^2 - 1)(m^2 + 2)}{288}. \quad (2.4)$$

Equivalently, there are rational $b_{m,j}$, $b_{m,0} = 1$, such that

$$T_m(n) = K_m m^{mn} n^{-\alpha_m} \left(\sum_{j=0}^L \frac{b_{m,j}}{n^j} + O_{m,L}(n^{-L-1}) \right), \quad (2.5)$$

where

$$b_{m,1} = \frac{(m^2 - 1)^2}{12m}, \quad (2.6)$$

$$b_{m,2} = \frac{(m^2 - 1)(m^6 + 3m^2 - 1)}{288m^2}. \quad (2.7)$$

Corollary 2.2 (A181199, with five corrections). With $a(n) = T_5(n)$,

$$a(n) = \frac{9 \cdot 5^{5n+1/2}}{2^{17} \pi^2 n^{12}} \left(1 + \frac{48}{5n} + \frac{5233}{100n^2} + \frac{1028391}{5000n^3} + \frac{60248377}{100000n^4} + \frac{273939939}{250000n^5} + O(n^{-6}) \right). \quad (2.8)$$

Theorem 2.3 (Algebraicity dichotomy). The ordinary generating function

$$F_m(z) = \sum_{n \geq 0} T_m(n) z^n$$

is algebraic over $\mathbb{Q}(z)$ if and only if $m = 1$ or $m = 2$. The same classification holds for algebraicity over $\mathbb{C}(z)$.

Corollary 2.4 (A universal comparison with ordinary rectangles). Let $f^{(n^m)}$ count ordinary standard Young tableaux of rectangular shape with m rows of length n . For fixed m ,

$$\frac{T_m(n)}{f^{(n^m)}} = 2^{-m(m-1)} \left(1 + \frac{m(m^2 - 1)}{4n} + \frac{m^2(m^2 - 1)(m^2 - 2)}{32n^2} + O_m(n^{-3}) \right). \quad (2.9)$$

Thus the extra antidiagonal comparisons retain a positive limiting fraction $2^{-m(m-1)}$ of ordinary rectangular tableaux.

The constant is not obtained by treating the comparisons as independent. It comes from two shifted-hook denominators, one for each side of a threshold cut. Their product is the essential feature of the proof. The distributional counterpart is given in Section 9.

[write] Added 3 October 2026, batch 85: the comparison constant. The normalizer of Corollary 2.4 is the classical hook count of the rectangle, recorded in ProveIt as equation (3.19), label `t2:eq:tableaux`, of the volume *Combinatorial Transseries and Their Inverses* (Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/, Section 3.5), which writes it $T_d(n)$ with d rows; the hook product displayed at the end of Section 6.4 here is the same formula with $d = m$. That volume's constants (3.20) (`t2:eq:tableaux-constants`) give $f^{(n^m)} \sim C m^{mn} n^\beta$ with $a = m \log m$, $\beta = -(m^2 - 1)/2 = -\alpha_m$ and $C = \sqrt{m} (2\pi)^{(1-m)/2} \prod_{j=0}^{m-1} j!$. Comparing with (2.2), $K_m = 2^{-m(m-1)} C$ exactly, since $4^d = 2^{m(m-1)}$: the two families share the rate m^m and the exponent $-\alpha_m$, and the leading term of (2.9) is the quotient of the two constants. The intake checked this identity symbolically for $m = 2, \dots, 7$, and checked that (2.4) and the rising-factorial expansion of $\prod_{j < m} (n + j)!/n!$ give the two printed corrections of (2.9) for the same m . Neither the formula nor the constant of the ordinary rectangle is new here; the new statement is the shifted count and its ratio.

3 Prefix states and the shifted hook product

3.1 The exact state space

Fill an array by inserting $1, 2, \dots, mn$ in order. At any intermediate stage let x_i be the number of filled cells in row i . Row increase means that the filled cells in each row are a prefix. The legal states satisfy

$$n \geq x_1 \geq x_2 \geq \dots \geq x_m \geq 0, \quad x_i = x_{i+1} \implies x_i \in \{0, n\}. \quad (3.1)$$

Indeed, placing a filled cell in row $i + 1$, column $j < n$, forces the cell in row i , column $j + 1$, to have already been filled. When the lower row has length n , column increase forces the upper row to be full as well. Conversely, these conditions make the filled set an order ideal of the array poset. They therefore characterize the prefix states exactly.

A transition increases x_i by one. For $i > 1$ it is allowed precisely when

$$x_{i-1} \geq x_i + 2 \quad \text{or} \quad (x_i = n - 1 \text{ and } x_{i-1} = n), \quad (3.2)$$

with $x_i < n$; the first row has only the latter upper bound. Empty rows can tie, and completed rows can tie. Forgetting either exception would change the sequence.

For a legal state $\lambda = (\lambda_1, \dots, \lambda_m)$ define its rotated complement by

$$\lambda^c = (n - \lambda_m, \dots, n - \lambda_1). \quad (3.3)$$

Rotation through 180 degrees, together with reversal of the numerical labels, preserves the comparisons. Thus suffix enumeration at λ is prefix enumeration at λ^c .

3.2 Interior states

Call a state *interior* when

$$n > \lambda_1 > \dots > \lambda_m > 0. \quad (3.4)$$

Every path ending there has avoided the completed-row boundary. After removing trailing zero parts, it is a chain of strict partitions obtained by adding one box at a time. Such chains are standard shifted tableaux. If g^λ denotes their number and $k = |\lambda|$, the shifted hook product is

$$g^\lambda = \frac{k!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (3.5)$$

This standard formula appears, for example, in Sun's account [11]. Appendix A proves it directly from the strict-partition recursion; no shifted-tableau enumeration result is needed without proof.

For an interior state both λ and λ^c are strict positive partitions. Consequently, the prefix and suffix counts are simultaneously available in product form. This is why cutting *in the middle*, rather than tracking the first completed row, is effective for asymptotics.

4 An exact random-threshold reduction

4.1 Order statistics and the cut

Take m independent rows, each consisting of n independent uniform random numbers in $(0, 1)$, sorted increasingly. Let $\mathcal{A}$ be the event that the comparisons between rows also hold. Every interleaving of these

sorted rows has the same probability, so

$$\mathbb{P}(\mathcal{A}) = \frac{T_m(n)}{M_m(n)}. \quad (4.1)$$

This is the order-statistics form of the order-polytope interpretation [11]; it also follows immediately by sorting all mn samples at once.

Fix $p \in (0, 1)$ and let X_i count the entries in row i below p . Before conditioning on $\mathcal{A}$, the variables X_i are independent $\text{Bin}(n, p)$. Conditional on these counts, the samples below p and above p are independent sorted uniforms in their respective intervals. When the count vector is an interior state λ , the conditional probability of satisfying all comparisons below the cut is

$$\frac{g^\lambda}{k! / \prod_i \lambda_i!}.$$

The analogous suffix probability uses λ^c . All comparisons across the cut are automatic because the prefix is an order ideal. Multiplying and using (3.5) gives

$$\mathbb{P}(\mathcal{A} \mid X = \lambda) = R_n(\lambda), \quad R_n(x) = \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)}. \quad (4.2)$$

The weight is symmetric in the coordinates. Equal interior coordinates give zero, and all distinct coordinates have exactly one decreasing order.

Theorem 4.1 (Interior sum and explicit boundary error). *Let $X_1, \dots, X_m$ be independent $\text{Bin}(n, p)$ variables and $\mathcal{I} = \{1 \leq X_i \leq n - 1 \text{ for all } i\}$. Then*

$$\frac{T_m(n)}{M_m(n)} = \frac{1}{m!} \mathbb{E} [\mathbf{1}_{\mathcal{I}} R_n(X)] + E_{m,n}(p), \quad (4.3)$$

where the error is nonnegative and satisfies

$$0 \leq E_{m,n}(p) \leq m(p^n + (1 - p)^n). \quad (4.4)$$

For $p = 1/2$ the expectation is the finite rational sum

$$I_m(n) = 2^{-mn} \sum_{n > x_1 > \dots > x_m > 0} \left(\prod_{i=1}^m \binom{n}{x_i} \right) \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)}, \quad (4.5)$$

and

$$0 \leq T_m(n) - M_m(n) I_m(n) \leq 2m 2^{-n} M_m(n). \quad (4.6)$$

Proof. The contribution to $\mathbb{P}(\mathcal{A})$ from interior count vectors is the sum of their binomial probabilities times (4.2). Symmetry converts this decreasing-vector sum into $1/m!$ times the unrestricted expectation on $\mathcal{I}$. The remaining event is $\mathcal{A} \cap \mathcal{I}^c$. It has nonnegative probability, at most the probability that some row has all its samples on one side of the cut. A union bound gives (4.4). The specialization is immediate. $\square$

Remark 4.2 (A useful pointwise bound). For $0 \leq x, y \leq n$, $|x - y| \leq x + y$ and $|x - y| \leq 2n - x - y$. Thus every factor of R_n lies in $[0, 1]$ whenever its denominator is nonzero. In particular, $0 \leq R_n \leq 1$ on $\mathcal{I}$. No uncontrolled singularity is hidden near the endpoints in (4.3).

The boundary estimate is an exact, nonasymptotic statement. It is deliberately simple and is not claimed sharp. Since the main probability is of order n^{-d} , this exponentially small additive error is negligible relative to every term of the inverse-power expansion.

[write] Added 5 October 2026, batch 98. The rate 2^{-n} of the bound at $p = 1/2$ is the true one, but the bound is not sharp in its constant, nor, for $m \geq 3$, in its power of n : the write's Proposition 18.3 in Part II shows $E_{m,n}(1/2) \sim 2^{1-n} 3^{1-m} J_{m-1}(4n)^{-\binom{m-1}{2}}$, against the bound $2m 2^{-n}$.

5 Gaussian reduction and an effective all-order theorem

5.1 The scaled integrand

Use $p = 1/2$ and put

$$Y_i = \frac{2X_i - n}{\sqrt{n}}, \quad \Delta(y) = \prod_{i < j} (y_i - y_j), \quad q_{ij}(y) = (y_i + y_j)^2. \quad (5.1)$$

The pair weight becomes exactly

$$R_n(X) = (4n)^{-d} \Delta(Y)^2 \prod_{i < j} \left(1 - \frac{q_{ij}(Y)}{4n}\right)^{-1} \quad \text{on } \mathcal{I}. \quad (5.2)$$

Let $h_r(q)$ be the complete homogeneous polynomial of degree r in the d variables q_{ij} , defined by

$$\prod_{i < j} (1 - u q_{ij})^{-1} = \sum_{r \geq 0} h_r(q) u^r. \quad (5.3)$$

In particular $h_0 = 1$. For $m = 1$, the empty product is one and $h_r = 0$ for $r > 0$.

5.2 Polynomial standardized moments

A variable Y_i has the same law as $n^{-1/2}(\varepsilon_1 + \cdots + \varepsilon_n)$, where the ε_j are independent symmetric signs. Its moment generating function is

$$\mathbb{E} e^{tY_i} = (\cosh(t/\sqrt{n}))^n. \quad (5.4)$$

For every fixed nonnegative integer k , its k th moment is exactly a polynomial $\mu_k(z)$ at $z = 1/n$. One convenient formal definition is

$$\sum_{k \geq 0} \mu_k(z) \frac{t^k}{k!} = \exp \left(\frac{1}{z} \log \cosh(t\sqrt{z}) \right). \quad (5.5)$$

The apparent singularity at $z = 0$ is removable in the formal expansion. Odd moments vanish, and

$$\begin{aligned} \mu_0 &= 1, & \mu_2 &= 1, & \mu_4 &= 3 - 2z, & (5.6) \\ \mu_6 &= 15 - 30z + 16z^2, & \mu_8 &= 105 - 420z + 588z^2 - 272z^3. & (5.7) \end{aligned}$$

For $k = 2s > 0$, the degree is at most $s - 1$. This follows either by grouping terms according to equal sign indices, or from the moment–cumulant formula: a partition into even blocks contributes $z^{s-\#\text{blocks}}$ times a product of sign cumulants.

Define a linear functional $\mathcal{E}_z : \mathbb{Q}[y_1, \dots, y_m] \rightarrow \mathbb{Q}[z]$ by

$$\mathcal{E}_z(y_1^{k_1} \cdots y_m^{k_m}) = \prod_{i=1}^m \mu_{k_i}(z). \quad (5.8)$$

Then $\mathcal{E}_{1/n}$ is the exact expectation for the independent variables Y_i . At $z = 0$, it is expectation under independent standard normal variables.

5.3 The coefficient formula

Proposition 5.1 (Finite formula for each coefficient). *For $j \geq 0$, the coefficient in (2.3) is*

$$c_{m,j} = \frac{1}{m!J_m} \sum_{r=0}^j \frac{1}{4^r} [z^{j-r}] \mathcal{E}_z \left(\Delta(y)^2 h_r(q(y)) \right). \quad (5.9)$$

The formula uses finite polynomials and rational arithmetic. It does not require previously computed sequence terms or a guessed recurrence.

5.4 Uniform error control

We supply the estimates that justify the formal calculation. Let

$$C_n = \{|Y_i| \leq \sqrt{n}/2 \text{ for all } i\}.$$

Since $\cosh t \leq e^{t^2/2}$, Chernoff's inequality gives

$$\mathbb{P}(C_n^c) \leq 2me^{-n/8}. \quad (5.10)$$

The same moment generating function bound implies that the absolute moments of every fixed degree of the Y_i are bounded uniformly in n .

On C_n , each $q_{ij}(Y)/(4n)$ lies in $[0, 1/4]$. Taylor's theorem applied to the finite product in (5.2) therefore gives, for every fixed L ,

$$\prod_{i < j} \left(1 - \frac{q_{ij}(Y)}{4n} \right)^{-1} = \sum_{r=0}^L \frac{h_r(q(Y))}{(4n)^r} + \mathcal{R}_{L,n}(Y), \quad (5.11)$$

$$|\mathcal{R}_{L,n}(Y)| \leq C_{m,L} n^{-L-1} (1 + \|Y\|)^{2L+2}. \quad (5.12)$$

To see the second bound, differentiate the product $L + 1$ times with respect to $1/n$. Every differentiated term has total numerator degree $2L + 2$; all denominators are bounded below by fixed powers of $3/4$. After multiplication by $\Delta(Y)^2$, the expected remainder is still $O_{m,L}(n^{-L-1})$ by the uniform moment bounds.

We may replace polynomial expectations on C_n by unrestricted ones with exponentially small error. For example, Cauchy–Schwarz, (5.10), and a uniform moment bound control the difference for every fixed polynomial. For the original rational weight, use $R_n \leq 1$ from Remark 4.2. After restoring the factor $(4n)^d$, its discarded tail is only a fixed polynomial in n times an exponential. The endpoint error (4.4) behaves similarly. We have consequently proved

$$\frac{T_m(n)}{M_m(n)} = \frac{(4n)^{-d}}{m!} \left(\sum_{r=0}^L \frac{\mathcal{E}_{1/n}(\Delta^2 h_r)}{(4n)^r} + O_{m,L}(n^{-L-1}) \right). \quad (5.13)$$

This establishes (5.9) after collecting powers. It also proves that parity effects cannot occur in the algebraic sector: the moments are exact polynomials in $1/n$, on both parity subsequences.

5.5 The leading Gaussian integral

For independent standard normals $G_1, \dots, G_m$,

$$\mathbb{E}\Delta(G)^2 = m! \prod_{j=0}^{m-1} j! = m! J_m. \quad (5.14)$$

Here is an elementary verification. Replace the monomials in the Vandermonde determinant by the monic Hermite polynomials H_j , whose squared norms for standard normal measure are $j!$. The determinant is unchanged because this basis change is unit triangular. Expanding the square and using orthogonality leaves the $m!$ matching permutations, each contributing $\prod j!$.

Thus $c_{m,0} = 1$. Finally, Stirling's expansion gives

$$M_m(n) \sim \sqrt{m} (2\pi n)^{(1-m)/2} m^{mn}, \quad (5.15)$$

which proves the leading constant and exponent in (2.5). More explicitly, the complete conversion between the two normalizations is

$$\sum_{j \geq 0} b_{m,j} z^j = \left(\sum_{j \geq 0} c_{m,j} z^j \right) \exp \left(\sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)} (m^{1-2r} - m) z^{2r-1} \right), \quad (5.16)$$

where B_{2r} are Bernoulli numbers. All uses of this identity are formal, or truncated to a fixed finite order with Stirling's remainder estimate [14]. This completes the proof of Theorem 2.1, apart from the explicit evaluations in the next section.

6 Universal first and second corrections

The finite formula is already an all-order algorithm. To obtain short formulas valid at every height, it is better to keep the Vandermonde weight intact rather than expand it into individual monomials.

6.1 An exact binomial Vandermonde normalization

Define the normalized binomial ensemble by weighting the independent Y_i with $\Delta(Y)^2$. Its normalizing factor is

$$Z_m(n) = \mathbb{E}\Delta(Y)^2 = m! J_m \prod_{j=0}^{m-1} \frac{n^j}{n^j}. \quad (6.1)$$

This identity is valid for $n \geq m-1$ (and interpreted as zero when the support cannot accommodate m distinct points).

For completeness, let $X \sim \text{Bin}(n, 1/2)$ and define $K_j(X)$ by

$$(1+t)^X (1-t)^{n-X} = \sum_{j=0}^n K_j(X) t^j.$$

Multiplying two such generating functions and taking expectation gives $(1+st)^n$. Hence the K_j are orthogonal and $\mathbb{E}K_j^2 = \binom{n}{j}$. Their leading coefficient as polynomials in $Y = (2X - n)/\sqrt{n}$ is $n^{j/2}/j!$. The corresponding monic polynomials therefore have squared norms

$$h_j = j! n^{j/2}/n^j.$$

The same determinant argument as in (5.14) proves (6.1). These are the symmetric Krawtchouk polynomials; the standard orthogonal-polynomial framework is recorded in [15].

Put

$$A = \binom{m}{3}, \quad B = \frac{m(m-1)^2(m-2)}{12}. \quad (6.2)$$

Expanding the finite product in (6.1) gives

$$\frac{Z_m(n)}{m!J_m} = 1 - \frac{A}{n} + \frac{A^2 - B}{2n^2} + O_m(n^{-3}). \quad (6.3)$$

Indeed, $A = \sum_{j < m} \sum_{k < j} k$ and $B = \sum_{j < m} \sum_{k < j} k^2$.

6.2 The first pair statistic

Let $p_r = \sum_i Y_i^r$. The first complete homogeneous polynomial is

$$S_1 = h_1(q) = \sum_{i < j} (Y_i + Y_j)^2 = (m-2)p_2 + p_1^2. \quad (6.4)$$

Write $\langle \cdot \rangle_n$ for expectation in the normalized binomial Vandermonde ensemble. The monic polynomial recurrence is

$$yP_j(y) = P_{j+1}(y) + \beta_j P_{j-1}(y), \quad \beta_j = j \left(1 - \frac{j-1}{n}\right).$$

The diagonal coefficient is zero by symmetry. The norm quotient proves the displayed value of β_j .

We need the standard finite-projection identities

$$\langle p_2 \rangle_n = \beta_m + 2 \sum_{j=1}^{m-1} \beta_j = m^2 - \frac{m(m-1)(2m-1)}{3n}, \quad (6.5)$$

$$\langle p_1^2 \rangle_n = \beta_m = m - \frac{m(m-1)}{n}. \quad (6.6)$$

They can be seen directly from the determinant ensemble: multiplication by y is tridiagonal in the orthonormal basis, with off-diagonal entries $\sqrt{\beta_j}$. The mean of a linear statistic is its multiplication operator's trace on degrees $0, \dots, m-1$. Its variance is the squared norm of the part crossing that projection. For p_1 only the edge from degree $m-1$ to degree m crosses, giving β_m . These trace rules also follow by expanding $\mathbb{E} \prod_i (1 + tf(Y_i))$ as a determinant of the moment matrix and taking the first two derivatives at $t = 0$.

Consequently,

$$\langle S_1 \rangle_n = s_{10} + \frac{s_{11}}{n}, \quad s_{10} = m(m-1)^2, \quad s_{11} = -\frac{m(m-1)(2m^2 - 5m + 5)}{3}. \quad (6.7)$$

6.3 The second pair statistic in the Gaussian limit

Let $\langle \cdot \rangle_0$ denote the Gaussian Vandermonde ensemble. Since

$$h_2(q) = \frac{1}{2} \left(S_1^2 + \sum_{i < j} (Y_i + Y_j)^4 \right),$$

and

$$\sum_{i < j} (Y_i + Y_j)^4 = (m - 8)p_4 + 4p_1p_3 + 3p_2^2,$$

we need only a few Gaussian moments:

$$\langle p_2^2 \rangle_0 = m^2(m^2 + 2), \quad \langle p_2p_1^2 \rangle_0 = m(m^2 + 2), \quad (6.8)$$

$$\langle p_1^4 \rangle_0 = 3m^2, \quad \langle p_4 \rangle_0 = 2m^3 + m, \quad \langle p_1p_3 \rangle_0 = 3m^2. \quad (6.9)$$

There is no need to invoke random-matrix integration to establish these. Scaling and translation of the Gaussian integral give

$$\langle e^{tp_2 + sp_1} \rangle_0 = (1 - 2t)^{-m^2/2} \exp\left(\frac{ms^2}{2(1 - 2t)}\right). \quad (6.10)$$

This proves the first three identities by differentiation. Translation integration by parts, using $\Delta(y + c\mathbf{1}) = \Delta(y)$, gives $\langle p_1p_3 \rangle_0 = 3\langle p_2 \rangle_0 = 3m^2$. Finally, in the Hermite multiplication matrix, the degree- j diagonal entry of the fourth power is $6j^2 + 6j + 3$; summing over $0 \leq j < m$ yields $2m^3 + m$. Thus

$$H_{20} := \langle h_2(q) \rangle_0 = \frac{m(m - 1)(m^4 - 3m^3 + 10m^2 - 18m + 16)}{2}. \quad (6.11)$$

6.4 Collecting the terms

Combining (6.3), (6.7), and (6.11),

$$c_{m,1} = -A + \frac{s_{10}}{4}, \quad (6.12)$$

$$c_{m,2} = \frac{A^2 - B}{2} + \frac{-As_{10} + s_{11}}{4} + \frac{H_{20}}{16}. \quad (6.13)$$

Straight polynomial simplification gives (2.4). The first Stirling logarithm coefficient is $u_1 = (m^{-1} - m)/12$, so $b_{m,1} = c_{m,1} + u_1$ and $b_{m,2} = c_{m,2} + u_1c_{m,1} + u_1^2/2$. This proves (2.7).

The ordinary rectangular hook product is

$$f^{(n^m)} = M_m(n)J_m \prod_{j=0}^{m-1} \frac{n!}{(n+j)!}.$$

Its finite rising-factorial expansion, combined with (2.4), proves (2.9).

7 OEIS specializations and numerical checks

7.1 Low-height controls

For $m = 1$, $T_1(n) = 1$. For $m = 2$, the usual strict-ballot argument gives $T_2(n) = C_{n-1}$ for $n \geq 1$; equivalently,

$$T_2(n) = \binom{2n}{n} \frac{1}{2(2n-1)} = \binom{2n}{n} \frac{1}{4n} \sum_{j \geq 0} \frac{1}{(2n)^j}. \quad (7.1)$$

Thus $c_{2,j} = 2^{-j}$ at every order. This is a useful exact control of both the normalization and the coefficient generator.

For height three, the theorem recovers the existing A181197 leading term [2] and gives

$$T_3(n) = \frac{\sqrt{3} 27^n}{64\pi n^4} \left(1 + \frac{16}{9n} + \frac{755}{324n^2} + O(n^{-3}) \right). \quad (7.2)$$

For height four, corresponding to A181198,

$$T_4(n) = \frac{3 \, 256^n}{1024\sqrt{2} \pi^{3/2} n^{15/2}} \left(1 + \frac{75}{16n} + \frac{6905}{512n^2} + O(n^{-3}) \right). \quad (7.3)$$

The leading constants are consequences of the general theorem, not inputs fitted from these sequences.

7.2 Five corrections in multinomial normalization

Table 1 reports exact output of (5.9). Only the compact multinomial normalization is used in this table. The archive also supplies the corresponding $b_{m,j}$.

Table 1: Coefficients $c_{m,j}$ in (2.3); $c_{m,0} = 1$.

m	$c_{m,1}$	$c_{m,2}$	$c_{m,3}$	$c_{m,4}$	$c_{m,5}$
2	1/2	1/4	1/8	1/16	1/32
3	2	11/4	23/8	45/32	-57/16
4	5	15	33	789/16	93/16
5	10	225/4	1819/8	22045/32	1353
6	35/2	665/4	8939/8	92351/16	731621/32

7.3 How much the corrections improve A181199

For $m = 5$, let $L_5(n) = K_5 5^5 n^{-12}$ and $A_{5,L}(n) = L_5(n) \sum_{j=0}^L b_{5,j} n^{-j}$. Table 2 shows the signed relative error $A_{5,L}(n)/T_5(n) - 1$. Its values are numerical diagnostics, not rigorous interval enclosures. The counts at $n = 10, 20, 40$ were independently regenerated by the row-state dynamic program and compared with OEIS; $n = 80$ is an OEIS reference value, not regenerated in this run [6].

Table 2: A181199: signed relative errors of successive truncations.

n	$L = 0$	$L = 1$	$L = 2$	$L = 5$
10	$-6.37190 \cdot 10^{-1}$	$-2.88893 \cdot 10^{-1}$	$-9.90345 \cdot 10^{-2}$	$1.42184 \cdot 10^{-3}$
20	$-3.90471 \cdot 10^{-1}$	$-9.78967 \cdot 10^{-2}$	$-1.81550 \cdot 10^{-2}$	$1.97547 \cdot 10^{-5}$
40	$-2.16403 \cdot 10^{-1}$	$-2.83393 \cdot 10^{-2}$	$-2.71080 \cdot 10^{-3}$	$2.67266 \cdot 10^{-7}$
80	$-1.13941 \cdot 10^{-1}$	$-7.61418 \cdot 10^{-3}$	$-3.69269 \cdot 10^{-4}$	$3.76974 \cdot 10^{-9}$

The leading term alone converges rather slowly at height five because $b_{5,1} = 9.6$. This explains why the constant is difficult to recognize reliably from modestly sized data without correction terms. The proof, however, does not use the numerical agreement.

8 A complete algebraicity classification

We now prove Theorem 2.3. The argument uses the classical Newton–Puiseux property: an algebraic function has a convergent fractional power expansion at a finite point, with algebraic coefficients when both the defining polynomial and the expansion point are algebraic. It has no logarithmic leading singularity. This is standard algebraic-function background; see the algebraic-function treatment in [13].

Put $R_m = m^{-m}$ and rescale

$$G_m(z) = F_m(R_m z) = \sum_{n \geq 0} v_n z^n, \quad v_n = R_m^n T_m(n).$$

For $m \geq 2$, Theorem 2.1 gives

$$v_n = K_m n^{-\alpha_m} (1 + O(n^{-1})). \quad (8.1)$$

In particular the radius of convergence of G_m is exactly one.

8.1 Odd heights: a logarithmic obstruction

Suppose $m \geq 3$ is odd. Then α_m is a positive integer. Set $q = \alpha_m - 1$. The coefficient of z^k in $G_m^{(q)}(z)$ is

$$\frac{(k+q)!}{k!} v_{k+q} = \frac{K_m}{k} + O(k^{-2}) \quad (k \rightarrow \infty).$$

After a harmless adjustment of finitely many terms, subtraction of $K_m \sum_{k \geq 1} z^k/k$ leaves an absolutely summable coefficient sequence. Therefore, along the real interval $z \uparrow 1$,

$$G_m^{(q)}(z) = K_m \log \frac{1}{1-z} + O(1). \quad (8.2)$$

A nonzero logarithmic divergence is incompatible with a Puiseux expansion. If F_m were algebraic, so would be its rescaling and every derivative; (8.2) is a contradiction.

8.2 Even heights: a transcendental-amplitude obstruction

Suppose $m \geq 4$ is even. Now $q = \alpha_m - 1/2 = (m^2 - 2)/2$ is an integer. The derivative coefficients satisfy

$$[z^k] G_m^{(q)}(z) = K_m k^{-1/2} + O(k^{-3/2}).$$

Since $[z^k](1-z)^{-1/2} = 1/\sqrt{\pi k} + O(k^{-3/2})$, coefficient subtraction again leaves an absolutely summable series. Hence

$$G_m^{(q)}(z) = K_m \sqrt{\pi} (1-z)^{-1/2} + O(1), \quad z \uparrow 1. \quad (8.3)$$

If the function were algebraic over $\mathbb{Q}(z)$, the leading coefficient in this Puiseux expansion at the algebraic point $z = 1$ would be algebraic. But

$$K_m \sqrt{\pi} = \frac{\sqrt{m} J_m}{4^d 2^{(m-1)/2}} \pi^{-(m-2)/2} \quad (8.4)$$

is a nonzero algebraic multiple of a nonzero integer power of π^{-1} . It is transcendental: otherwise that power of π , and hence π itself, would be algebraic. This contradiction uses only the classical transcendence of π .

8.3 The two algebraic cases and descent of constants

The exceptional cases are explicit:

$$F_1(z) = \frac{1}{1-z}, \quad F_2(z) = 1 + zC(z), \quad C(z) = \frac{1 - \sqrt{1-4z}}{2z}.$$

Thus $m = 1, 2$ are algebraic.

Finally, a rational-coefficient power series algebraic over $\mathbb{C}(z)$ is algebraic over $\mathbb{Q}(z)$. Indeed, fix finite degree bounds for a nonzero polynomial relation. Equating all coefficients gives a homogeneous linear system with rational entries in finitely many unknown polynomial coefficients. The descending chain of solution spaces stabilizes after finitely many independent equations. A nonzero complex solution therefore implies a nonzero rational solution. This proves the asserted classification over either field.

Remark 8.1. Nonalgebraicity does not imply non-D-finiteness. Logarithmic singularities, and half-integer singularities with transcendental connection constants, are compatible with D-finite functions. The guessed recurrences for heights four and five remain separate questions.

[write] **Added 5 October 2026, batch 98.** F_m is in fact D-finite for every fixed m : it is the diagonal of a rational power series over $\mathbb{Q}$ (Part II, Theorem 14.1, through the multiple-binomial-sum theorem of Bostan, Lairez and Salvy). With Theorem 2.3, F_m is algebraic for $m \leq 2$ and transcendental but D-finite for $m \geq 3$. That proof is an existence proof; it produces no operator, and the guessed recurrences for heights four and five remain unproved.

[write] **Added 5 October 2026, batch 101.** The last sentence of the note above is out of date for heights four and five: Parts III and IV prove the guessed recurrences, which are explicit recurrences of orders two and three for the coefficients of F_4 and F_5 . The order two at height four is minimal (Proposition 33.2); at height five the minimal order is two or three (Proposition 42.2). For $m \geq 6$ no operator is known.

9 Gaussian laws for intermediate shapes

The same reduction describes typical partially filled arrays, not only their total number. We give both a randomized numerical cut and a fixed number of inserted labels; these are different probability models.

9.1 A cut at a fixed numerical level

Choose a tableau uniformly among the $T_m(n)$ arrays. Independently choose mn uniform random numbers in $(0, 1)$, sort them, and place them in the array in increasing order of its integer labels. This produces the uniform order-statistics model conditioned on $\mathcal{A}$. For a fixed $p \in (0, 1)$ let X_i be the number of row- i entries below p and set

$$Z_i = \frac{X_i - np}{\sqrt{np(1-p)}}. \quad (9.1)$$

Theorem 9.1 (Gaussian Vandermonde limit). *For fixed m and fixed $p \in (0, 1)$, the decreasing vector Z converges in distribution to the probability density*

$$\frac{e^{-\|z\|^2/2} \Delta(z)^2}{(2\pi)^{m/2} J_m}, \quad z_1 > z_2 > \cdots > z_m, \quad (9.2)$$

with respect to m -dimensional Lebesgue measure. Polynomial moments of fixed degree converge as well.

Proof. For interior decreasing x , the exact conditional mass is

$$\frac{\prod_i \mathbb{P}(\text{Bin}(n, p) = x_i) R_n(x)}{T_m(n)/M_m(n)}.$$

In bounded sets of standardized coordinates, the local central limit estimate for the binomial mass follows directly from Stirling's formula. Moreover,

$$x_i - x_j = \sqrt{np(1-p)}(z_i - z_j),$$

whereas the two denominator factors in (4.2) are asymptotic to $2np$ and $2n(1-p)$. Thus

$$R_n(x) = (4n)^{-d} \Delta(z)^2 (1 + o(1))$$

uniformly on every bounded set. Divide by $T_m(n)/M_m(n) \sim J_m(4n)^{-d}$ and use the lattice mesh $1/\sqrt{np(1-p)}$ in every coordinate. The Riemann sums converge to (9.2).

To justify passage beyond bounded sets, first restrict every x_i/n to a fixed compact interval inside $(0, 1)$ containing p . There the pair denominators are bounded below by a constant times n^2 , and binomial local bounds give a Gaussian upper bound times a fixed polynomial in the standardized coordinates. This is integrable, with every polynomial moment. Outside that macroscopic interval, Chernoff bounds and $R_n \leq 1$ give an exponentially small contribution even after division by the order- n^{-d} conditioning probability. The boundary term satisfies the same conclusion by (4.4). This proves tightness and moment convergence. Normalization of the limit is (5.14), with the factor $m!$ removed on the decreasing chamber. $\square$

The density (9.2) is the ordered eigenvalue density of the Gaussian unitary ensemble in the normalization with Gaussian factor $e^{-\sum z_i^2/2}$. This terminology is optional: the theorem and its normalization have already been specified by an explicit density.

9.2 A deterministic fraction of the labels

Let $\Lambda_i(k)$ be the number of cells in row i occupied by labels $1, \dots, k$ of a uniformly random tableau. Fix $p \in (0, 1)$ and take $k = \lfloor mnp \rfloor$.

Theorem 9.2 (Trace-zero limit at deterministic time). *The vector*

$$\left(\frac{\Lambda_i(k) - np}{\sqrt{np(1-p)}} \right)_{i=1}^m$$

converges to the density

$$\frac{e^{-\|z\|^2/2} \Delta(z)^2}{(2\pi)^{(m-1)/2} J_m} \quad (9.3)$$

on $\{z_1 > \dots > z_m, \sum_i z_i = 0\}$, with respect to Euclidean $(m-1)$ -dimensional measure on the trace-zero hyperplane.

Proof. In the preceding randomized-cut construction, the total number K of samples below p is $\text{Bin}(mn, p)$ and is independent of the integer tableau. Conditioning on $K = k$ therefore gives precisely $\Lambda(k)$. For an interior vector x of total size k , divide the mass used in the preceding proof by $\mathbb{P}(K = k)$. The latter is asymptotic to $(2\pi mnp(1-p))^{-1/2}$. The lattice $\{x \in \mathbb{Z}^m : \sum x_i = k\}$ has Euclidean covolume $\sqrt{m}$. In

standardized coordinates its cell volume is $\sqrt{m} (np(1-p))^{-(m-1)/2}$. These factors give exactly the density (9.3) under the local binomial estimate.

For domination, use the same compact macroscopic region as before. The product of local binomial upper bounds on the hyperplane, after the additional conditioning, is bounded by a fixed polynomial times a Gaussian in the standardized coordinates, times the hyperplane lattice-cell volume. Its tails are uniformly summable. Outside the macroscopic region, the extra factor $1/\mathbb{P}(K = k) = O(\sqrt{n})$ cannot overcome an exponential Chernoff bound. This proves convergence of the hyperplane Riemann sums.

Finally, Δ is invariant under common translation. Orthogonally splitting $\mathbb{R}^m$ into the trace-zero hyperplane and the unit trace direction shows that the full Gaussian Vandermonde integral is $\sqrt{2\pi}$ times the trace-zero integral. Equation (5.14) then verifies the normalization in (9.3). $\square$

This theorem concerns one fixed macroscopic time. It is not a claim of process convergence to a noncolliding Brownian bridge; proving tightness in a path space and identifying the joint time laws require additional work.

10 Ratios, eventual log-convexity, and inversion

10.1 Consecutive ratios

Write $\alpha = \alpha_m$ and $b_1 = b_{m,1}$. For $m \geq 2$, the all-order expansion gives

$$\frac{T_m(n+1)}{T_m(n)} = m^m \left(1 - \frac{\alpha}{n} + \frac{\alpha(\alpha+1)/2 - b_1}{n^2} + O_m(n^{-3}) \right). \quad (10.1)$$

Hence $T_m(n)$ is eventually strictly increasing. Expanding $\log T_m(n)$ at $n-1, n, n+1$ with enough terms also gives

$$\log T_m(n+1) - 2 \log T_m(n) + \log T_m(n-1) = \frac{\alpha}{n^2} + O_m(n^{-3}) > 0 \quad (10.2)$$

for sufficiently large n . Thus it is eventually strictly log-convex. These statements do not identify the smallest valid n .

10.2 Inverting the asymptotic profile

Put $\lambda = m \log m$, $K = K_m$, and $L = \log(y/K)$. The large solution of

$$\lambda x - \alpha \log x = L$$

is

$$x_0 = -\frac{\alpha}{\lambda} W_{-1} \left(-\frac{\lambda}{\alpha} e^{-L/\alpha} \right), \quad (10.3)$$

where W_{-1} is the real branch taking values at most -1 [16]. The branch choice matters: the principal branch describes the small, irrelevant solution. In elementary logarithmic form,

$$x_0 = \frac{L + \alpha \log L - \alpha \log \lambda}{\lambda} + O_m \left(\frac{\log L}{L} \right). \quad (10.4)$$

Let $\ell_2 = b_{m,2} - b_{m,1}^2/2$. Formal inversion of the continuous asymptotic profile gives

$$x = x_0 - \frac{b_{m,1}}{\lambda x_0} - \frac{\alpha b_{m,1}/\lambda^2 + \ell_2/\lambda}{x_0^2} + O_m(x_0^{-3}). \quad (10.5)$$

To verify it, insert $x = x_0 + \delta$ into $\lambda x - \alpha \log x + b_{m,1}/x + \ell_2/x^2 = L$ and compare the coefficients of x_0^{-1} and x_0^{-2} . Along the exact values $y = T_m(n)$, the approximation in (10.5) returns $n + O_m(n^{-3})$. The higher inverse terms are obtained by the same recursive substitution. This inversion technique is standard and also used elsewhere in ProveIt; its application here is a consequence of the new expansion, not a new inversion method.

For an arbitrary target y , the least integer index attaining it is a staircase function. An asymptotic estimate alone must not be rounded and advertised as an exact threshold certificate. Near an integer transition, explicit upper and lower bounds for the neighboring sequence values are required. The finite interior sum and its boundary error (4.6) supply one possible, though not always computationally efficient, route to such bounds.

[write] Added 3 October 2026, batch 85: the inversion is an instance. The inversion (10.3) is a case of Theorem J.23 (p0:thm:lambert-core, “Exact solution of the dominant block”) of the canonical transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`, which solves $aX + b \log X = L$ by $X = (b/a) W_k((a/b)e^{L/b})$. Here $a = \lambda = m \log m > 0$ and $b = -\alpha_m < 0$ for $m \geq 2$, and that theorem’s branch rule for $b < 0$ gives exactly (10.3): the large root is the W_{-1} root, and it exists precisely when $L > L_c = \alpha_m(1 - \log(\alpha_m/\lambda))$ (for $m = 5$, $K_5 e^{L_c} \approx 0.021$, so every $y \geq 1$ qualifies). The text’s own sentence that the technique “is standard and also used elsewhere in ProveIt” is accurate; nothing in this section is a new inversion result. The real substitution, branch rules, threshold and slope of Theorem J.23 are formalized in `Analysis/FabiusFunction/Lean/FabiusFunction/LinLogCoreInversion.lean` (the volume’s register rates the theorem “Partial”: the asymptotic and general complex-branch clauses are not formalized); that formalization concerns the general equation, and nothing specific to shifted rectangles is formalized. The caution about integer thresholds in the preceding paragraph is the separation condition of the volume’s staircase theorem, Theorem J.43 (p0:thm:staircase). The path-forest report cited in Section 1.3 inverts its sequences by the same apparatus (its `spf:thm:inverse-first` and `spf:thm:inverse-all`).

11 Reproducibility and computational scope

11.1 Exact coefficient generation

The program code `/coefficients.py` implements (5.9) using Python’s exact rational arithmetic. It avoids expanding the squared Vandermonde in full. For any symmetric polynomial P and any product measure with identical one-variable moments,

$$\mathcal{E}_z(\Delta^2 P) = m! \sum_{\sigma \in S_m} \text{sgn}(\sigma) \mathcal{E}_z \left(P \prod_{i=0}^{m-1} y_{i+1}^{i+\sigma(i)} \right). \quad (11.1)$$

Expand the two Vandermonde determinants, permute the integration variables in one expansion, and use symmetry of P to prove this identity. Only $m!$ permutations remain. Equal sorted exponent profiles are grouped before evaluating products of moment polynomials.

The h_r are generated from (5.3); the standardized moments are generated from sign cumulants. The Stirling conversion (5.16) is also done with rational numbers. The method is practical for modest fixed m and moderate order. Its permutation factor makes it unsuitable as presently implemented for large m ; no claim of polynomial complexity in m is made.

11.2 Independent enumeration and exact finite identities

The program code `/tableaux.py` supplies two enumerators: a row-state dynamic program using (3.2), and an independent bitmask linear-extension enumerator using the cell-level predecessor relation. The latter is intentionally restricted to small arrays. Neither uses a conjectured recurrence.

The verification run recorded in the archive passed 471 checks in the following suites:

Suite	Checks
Bitmask-poset versus row-state enumeration	27
Interior prefix counts versus the shifted hook product	238
Threshold split, rational interior sum, and boundary inequality	120
Exact Catalan control, $n = 1, \dots, 30$	30
Selected A181198 values, independently regenerated	12
Selected A181199 values, independently regenerated	12
Universal corrections versus the coefficient engine, $m = 1, \dots, 6$	30
Standardized binomial-moment controls	2

The independently regenerated OEIS comparisons use $n = 1, \dots, 10, 20, 40$ for each of heights four and five [5, 6]. The threshold identity is checked for $1 \leq m \leq 5$ and $1 \leq n \leq 8$. These are finite corroborations of the proofs, not substitutes for them.

11.3 Commands and data provenance

From the archive’s top-level directory:

```
python3 code/verify.py
python3 code/coefficients.py --height 5 --order 5
python3 code/numerics.py
bash build.sh
```

The first two commands use the Python standard library only. The numerical diagnostics additionally use `mpmath`; they were run with 90 decimal digits. The PDF build requires a conventional LaTeX installation with the packages listed in the preamble.

The reference integers in `data/oeis_selected.json` were transcribed from the entry pages and b-files inspected during the source audit. They are not outputs of the coefficient engine. Counts regenerated independently are saved separately in `data/exact_regenerated.json`. Values at $n = 80$ appear only as reference inputs to numerical diagnostics. Full third-party articles are not redistributed in this archive.

[write] Added 3 October 2026, batch 85: the shipped layout. “The archive” in this section is the delivered package. In Provel it’s files keep the subdirectories `code/` and `data/`, except four, which moved as follows:

Delivered as	Shipped as
<code>build.sh</code>	<code>code/build.sh</code>
<code>requirements-numerics.txt</code>	<code>data/requirements-numerics.txt</code>
<code>notes/SOURCES_AND_STATUS.md</code>	<code>notes-SOURCES_AND_STATUS.md</code>
<code>notes/PROPOSED_OEIS_ADDITIONS.txt</code>	<code>data/notes-PROPOSED_OEIS_ADDITIONS.txt</code>

`code/build.sh` changes to its own directory, so in its shipped place it no longer finds `article.tex`. The delivered PDF and checksum list are not shipped; they survive in the arrival commit’s archive. `code/verify.py` and `code/numerics.py` write their outputs into the report’s own `data/` directory, overwriting the shipped records, so they must be rerun on a copy; the README of the report gives the commands. At intake (3 October 2026, Python 3.14.4 on Windows) `code/verify.py` passed all 471 checks on a copy, and every regenerated data file agreed with the shipped one after removal of carriage returns, except the timing field of `data/verification.json`; `code/coefficients.py` and `code/numerics.py` (`mpmath` 1.3.0) reproduced `data/coefficients_m5.json` and `data/numerics.csv`.

`data/oeis_selected.json` holds terms of A181198 and A181199 copied from the OEIS entries and b-files (Koutschan, with earlier terms by Heinz); OEIS content is published by The OEIS Foundation Inc. under the Creative

Commons Attribution-ShareAlike 4.0 licence, so this file is third-party data under that licence, not MIT-0. The draft OEIS text in `data/notes-PROPOSED_OEIS_ADDITIONS.txt` is marked “DRAFT ONLY — NOT SUBMITTED” and stays so: nothing was submitted to OEIS by the package’s author or by the intake, and its own instruction not to mark either conjectured recurrence as proved stands.

[write] Added 5 October 2026, batch 101: the OEIS draft. The draft’s instruction “Do not mark either conjectured sequence recurrence as proved on the strength of this article” was right for this manuscript, which proves no recurrence, and the shipped file stays byte-identical. Since 5 October 2026 Parts [III](#) and [IV](#) prove both recurrences. The draft has not been updated, nothing has been submitted to OEIS, and any change to the OEIS entries is for a human editor after review.

12 Further research questions and proposed approaches

The following questions are not presented as solved. They are chosen to continue the present line of work rather than to restate the resolved leading-term conjecture.

Research question 1 (Prove the specific height-four and height-five recurrences). Can the exact enumeration be transformed into a finite hypergeometric sum or a Pfaffian identity that admits a compact creative-telelescoping certificate for the recurrences in A181198 and A181199?

The random-threshold expression is exact only after adding its boundary part. For a recurrence proof that part cannot be discarded. A useful first step is an exact decomposition according to the number of full and empty rows at the cut, followed by shifted-skew-tableau formulas for those boundary states. A proof should compare the resulting operator with the published guessed operator, not merely generate more confirming terms.

[write] Added 5 October 2026, batch 101: Research question 1 answered. Answered at both heights, through a Pfaffian identity rather than the boundary decomposition suggested above. Part [III](#) (Report 231) writes the order-polytope volume as an integral of a determinant of the Volterra transfer V^{n-1} , evaluates it by a four-variable de Bruijn Pfaffian as two kernel traces, reduces both traces to one weighted triangle period Z_n , and proves a first-order recurrence for Z_n with a six-coefficient polynomial divergence certificate (Theorem [29.1](#)); elimination gives the published A181198 operator (Theorem [30.1](#)) and variation of constants Conjecture 18 of [\[9\]](#) (Theorem [30.2](#)). Part [IV](#) (Report 233) does the same at height five with an odd augmented Pfaffian, an open-chain integral and a second period X_n , closed by polynomial contiguity and a finite moment closure; it proves the nested sum of Conjecture 19 and the published A181199 operator (Theorems [34.1](#) and [34.2](#)). Both keep every boundary contribution, compare their operators with the published ones coefficient by coefficient, and settle the initial indices: the operators hold for $n \geq 1$ and fail at $n = 0$ under $a_0 = 1$. Neither uses a creative-telelescoping certificate; a compact one at height five is Research question 17. Nothing is known for $m \geq 6$ (item (F7) and Research question 16).

Research question 2 (Polynomial dependence of every correction on height). For each fixed j , is $c_{m,j}$ a polynomial in m of degree at most $3j$? Can its factorization and denominator structure be described uniformly?

The formulas for $j = 1, 2$ support this degree scale. The question beyond those orders is not proved here. A promising approach is to reorganize [\(5.9\)](#) as Gaussian matrix moments or as connected cumulant diagrams, in which the powers of m count index loops. This could replace factorial-in- m computation with symbolic fixed-order formulas.

Research question 3 (A growing-height transition). For what growth rates $m = m(n)$ does the leading fixed-height formula remain relatively accurate? Is the natural first transition governed by m^3/n ?

The coefficient $c_{m,1} \sim m^3/12$ suggests this parameter in multinomial normalization. It does not establish a uniform regime. A proof would need uniform Vandermonde moment bounds and a growing-dimension tail analysis. For m^3/n of order one, connected logarithmic corrections may require resummation rather than ordinary fixed-order truncation.

Research question 4 (Exponentially small sectors and sharp boundary asymptotics). What is the true leading asymptotic of the discarded boundary term $E_{m,n}(1/2)$, and how does it combine with large deviations of the interior expectation?

The bound $2m2^{-n}$ is sufficient for every inverse power but is not an asymptotic formula. Configurations with a row empty or full give a natural finite hierarchy of lower-dimensional boundary strata. Computing their constants could produce a genuine transseries beyond the algebraic sector, and could clarify the signs and optimal truncation of the coefficients.

[write] **Added 5 October 2026, batch 98: Question 4 re-scoped.** Part II does not answer this question, although its introduction counts the non-sharp boundary bound among “the two gaps addressed here”. Its rare count $R_m(n)$ is the part of the boundary event in which the first row is completed before the last row starts: it satisfies $R_m(n) \leq M_m(n)E_{m,n}(1/2)$, and the ratio grows like 2^n times a power of n (Remark 14.4). The leading asymptotic of $E_{m,n}(1/2)$ itself is the write’s Proposition 18.3: exactly, $E_{m,n}(1/2)$ is twice the stratum with one boundary row (the last row empty at the cut, or, by symmetry, the first row full) plus $O(4^{-n})$, and

$$E_{m,n}(1/2) \sim 2^{1-n} 3^{1-m} J_{m-1}(4n)^{-\binom{m-1}{2}}.$$

That proposition is unrefereed and checked numerically only on finite ranges. Still open: an all-order expansion of the boundary term (the one-row stratum is an interior sum with $m-1$ rows and an extra factor $x_i/(2n-x_i)$ per row, so the method of Section 5 should apply, but this has not been carried out); the constants of the deeper strata, with two or more rows empty or full at the cut, and of the rare sectors inside them (compare Research question 10 of Part II); their combination with large deviations of the interior expectation; and any transseries reading.

Research question 5 (A full path-process limit). Do the centered row-length processes of a uniform tableau converge to a trace-zero noncolliding Gaussian bridge on compact time intervals inside $(0, 1)$? What are the entrance and exit scalings near the endpoints?

Theorem 9.2 establishes one-time marginals only. A proof of process convergence would require multi-cut enumeration, control of increments, and tightness. The complement symmetry provides a useful constraint on any proposed entrance/exit description.

Research question 6 (Certified asymptotic enclosures and integer thresholds). Can the moment expansion be packaged into efficient explicit upper and lower bounds with practical constants, avoiding the $\binom{n-1}{m}$ -term interior sum?

On a central region the pair interaction has a positive geometric expansion. Combining its finite partial sums with concentration bounds and exact moment determinants could give certified intervals. These would turn the inverse approximations into exact index decisions away from a quantified transition zone.

[write] **Added 5 October 2026, batch 101: Research question 6 advanced at heights four and five.** At $m=4$ and $m=5$, Parts III and IV turn the proved recurrences into effective statements: rational error constants for every fixed truncation (Theorems 31.1 and 40.1), explicit logarithmic remainders, finite Lambert- W_{-1} models with effective inverse errors, and two-ceiling envelopes for the integer threshold (Theorems 32.2 and 41.2), which leave at most two adjacent candidates near an integer. At height five the receipt evaluates constants at $K=0$ and $K=6$, for example $|b_n - q_6(n)| \leq 18n^{-17}$ for $n \geq 64$ (40.8). These bounds need no interior sum, but they rest on the recurrences and do not reach $m \geq 6$. Still open: practical constants at general m , and the constants that Parts III and IV construct without evaluating (items (F6) and (F10)).

Research question 7 (D-finiteness versus nonalgebraicity at arbitrary height). Is $F_m(z)$ D-finite for every fixed m , and how does the minimal annihilating order grow with m ?

Theorem 2.3 settles the algebraic/nonalgebraic distinction, not the D-finite/non-D-finite one. An exact finite-dimensional differential system in the fixed-height regime would be a stronger structural result. The opposite-orientation constant-width transfer matrix does not answer this question.

[write] Added 5 October 2026, batch 98: Question 7 answered in part. The first half is answered: for every fixed m , F_m is the diagonal of a rational power series over $\mathbb{Q}$, hence D-finite, and $(T_m(n))$ is P-recursive (Part II, Theorem 14.1: the exact sum of Theorem 16.1 over the $2m$ row births and completions, a finite sum of products of reflection determinants, is a multiple binomial sum in the sense of Bostan, Lairez and Salvy). The second half, how the minimal annihilating order (and degree) grows with m , remains open; it is Research question 11 of Part II. The existence proof supplies no operator, so the guessed recurrences of heights four and five (Research question 1) remain unproved.

[write] Added 5 October 2026, batch 101. The last sentence of the note above is out of date: Parts III and IV prove the guessed recurrences of heights four and five (note at Research question 1). On the second half of this question, the minimal order of a recurrence is two at $m = 4$ (Proposition 33.2) and two or three at $m = 5$ (Proposition 42.2).

13 Conclusion

A shifted rectangle admits a particularly effective asymptotic cut: both sides of an interior threshold state are ordinary strict shifted shapes. The two shifted hook products turn the count into a binomial Vandermonde expectation, with an explicitly controlled boundary error. This gives a proof of the A181199 leading constant independent of all guessed recurrences, an effective all-order expansion for every fixed height, and universal correction polynomials through second order.

The consequences go beyond adding coefficients to an OEIS entry. The same constant yields a complete algebraicity dichotomy for the fixed-height generating functions, while the same exact weight identifies Gaussian laws for typical intermediate shapes. The remaining recurrence, uniformity, and exponentially small questions are separate, concrete targets with clear starting points.

Part II

Rational diagonals and rare boundary phases in shifted rectangles

D-finiteness at every fixed height and a sharp exponential refinement for OEIS A181199

Prepared for Vladimir Reshetnikov, October 4, 2026

For every fixed height m , the generating function of the shifted-rectangle counts $T_m(n)$ is a diagonal of a rational function, hence D-finite. For $m \geq 2$, let $R_m(n)$ count arrays in which row 1 finishes before row m starts. Then

$$R_m(n) \sim \sqrt{\frac{m}{2}} (2\pi)^{-(m-2)/2} \frac{\prod_{j=0}^{m-3} j!}{4^{\binom{m-2}{2}} 9^{m-2}} \left(\frac{m^m}{4}\right)^n n^{-(m-2)^2/2}.$$

Both statements hold for all fixed heights, not only the tabulated cases.

Abstract. The fixed-height shifted-rectangle family combines strict inequalities between active rows with exceptional ties at empty and completed rows. We separate these boundary events from the intervening walks. The resulting finite sum of reflection determinants is a multiple binomial sum, proving rational diagonality and D-finiteness at every fixed height and resolving an explicit question in the preceding ProveIt report. We also count the rare failure of a common active phase. A first-completion cut gives a shifted-hook sum with an exponentially smaller remainder. A beta-binomial mixture then yields its sharp exponential rate, leading constant, effective all-order expansion, and a universal first correction. We compute three corrections at height five, describe the conditional Gaussian shape, and invert the rare-event profile. Exact verification programs and a source and status audit accompany the manuscript.

Status. AI-assisted, unrefereed research manuscript. The stated results have mathematical proofs below, but have not been independently peer reviewed or proof-assistant verified. Historical priority is not certified. The particular conjectured recurrences and finite-sum identities for A181198 and A181199 are *not* proved here.

[write] Added 5 October 2026, batch 98: provenance of Part II. Part II is a single manuscript, printed in full: manuscript 02 of batch 98 of ProveIt’s incoming-reports intake, *Rational diagonals and rare boundary phases in shifted rectangles*, subtitled “D-finiteness at every fixed height and a sharp exponential refinement for OEIS A181199” and dated October 4, 2026 (a 1,656-line source with 97 labels and a 24-page PDF). Its title page reads “Prepared for Vladimir Reshetnikov”; its PDF metadata name “Research report prepared for Vladimir Reshetnikov”. No human author and no tool are named. It arrived as the archive `Shifted_Rectangle_Boundary_Research.zip` (390,943 bytes, 22 files under a doubled wrapper directory of the same name) in commit 2172df76a and was placed in commit 0fba5167f (batch 98B) as an addition to this report. It pins the repository at commit db20eb37982abc0f163c6d308d392f5ac4c9bc62 (4 October 2026; Section 14). This report was not changed between the pin and the placement, so the manuscript read exactly the text printed as Part I, and its citations of that text (Theorems 2.1 and 2.3, Research question 7) carry the right numbers. Its code, data and source notes are shipped with the prefix `02-boundary-phases-`; its $\text{\LaTeX}$ source, README, PDF and checksum list are not shipped and survive in the archive of the arrival commit. The manuscript is unrefereed and AI-assisted; its proofs are ordinary mathematical proofs, nothing in it is formalized in Lean or Rocq, and its place in this report confers no formal status.

What it answers. Theorem 14.1 answers the first half of Part I’s Research question 7 (D-finiteness at every fixed height), not its second half (Remark 14.3). Theorem 14.2 is a sharp result about a sub-sector of the boundary event of Theorem 4.1; it does not answer Research question 4, which asks for the asymptotic of the whole boundary error (Remark 14.4; the write’s Proposition 18.3 gives its leading term). Dated notes at both questions, at Remark 8.1 and at the end of Section 4.1 of Part I say so.

Overlap with Part I, credited. The two texts share 1.98 % of their word 8-grams (214 of 10,822). The state space (15.1) is (3.1), re-derived; the shifted hook product (18.2) is imported (Appendix A proves it); (19.9) is (5.14) with r variables; (20.7) extends (6.1) to a general success probability and is then mixed over the beta law; (20.11) are the Gaussian limits of (6.6); the Stirling factor (20.3) is the analogue of (5.16); Lemma 19.1 is a random-cut variant of Theorem 4.1, and Theorem 21.3 the analogue, for the rare event, of Theorem 9.1. The inversion of Section 22 uses the apparatus of Section 10 with a different rate and power, and Corollary 21.2 imports Theorem 2.1 and Corollary 2.2 ($b_{5,1} = 48/5$). Everything else is new to the repository. “The preceding (ProveIt, repository, asymptotic) report” in Part II means Part I.

Editorial changes. Section k of the manuscript is Section $k + 13$ here, its Appendices A and B are Appendices C and D, and its Research questions 1–8 are Research questions 8–15 (one counter runs through both Parts); theorem and equation numbers follow their sections. Every label carries the prefix `shr:bp:`, and one label was added (Remark 16.2). The manuscript’s $d = \binom{m-2}{2}$ is renamed d_R throughout, because Part I’s d is $\binom{m}{2}$; no normalization changes. Its three citations of its predecessor (the bibliography entry `prior`, whose pin and path this note records) now point to Part I. Its title, subtitle, result box, abstract and status paragraph, from its title page, open this Part. Falling factorials are typeset as in Part I ($n^{\bar{k}}$, the manuscript’s $(n)_k$); rising factorials keep the manuscript’s $(n)^{\bar{k}}$. The bibliographies are merged: the OEIS, Kauers–Koutschan and Sun entries are Part I’s (the manuscript inspected the OEIS entries on 4 October 2026 and arXiv v2 of Kauers–Koutschan, its Section 6.4 and Conjectures 18–19), and Bostan–Lairez–Salvy [10] is added. The [write] notes, Remarks 14.3 and 14.4, Section 18.3 with Proposition 18.3, and Section 24.1 are new. No other statement, proof or number of the manuscript was changed.

[write] Added 5 October 2026, batch 98: notation of Part II. Part II keeps the manuscript’s symbols except d_R , and several of them clash with Part I’s. Within Part II every symbol has the meaning below; the right column gives Part I’s meaning, the tempting false reading. Shared without change: $T_m(n)$, $F_m(z)$, $M_m(n)$, α_m , Δ , the hook count g^λ , $J_k = \prod_{j < k} j!$ (Part II uses it at $k = r$), W_{-1} , and $\mathbb{E}$.

symbol	meaning in Part II	not: meaning in Part I
d_R	$\binom{r}{2} = \binom{m-2}{2}$, the manuscript’s d	$d = \binom{m}{2}$
d_s, b_s, r_s	completions, births and active rows among the first s event letters (Section 16)	—
r	the size of an active block in Sections 15–17; $m - 2$ from Section 18 on	the degree of h_r , the index of p_r
$R_m(n)$	the rare count $\#\{D_1 < B_m\}$ (14.3)	$R_n(x)$ the pair weight; $R_m = m^{-m}$; $R(z)$
$H_m(n); H(a)$	the core count (18.1); the indicator $\binom{a}{a}$ of $a \geq 0$ (16.4)	$H(\lambda)$, Hermite H_j , H_{20}
$\mathcal{K}_m; K_r(u, v); K$	the rare constant (14.6); the chamber kernel; $\sum_i k_i$ in (20.1)	K_m (2.2); Krawtchouk K_j
$b_{m,j}^R; b_{5,1}^{\text{total}}$	rare coefficients; Part I’s $b_{5,1} = 48/5$	$b_{m,j}$ of (2.5)
$B_i, D_i; B, D; \mathbf{B}, \mathbf{D}$	birth and completion ranks $a_{i,1}, a_{i,n}$; event letters; the birth and completion maps	$B = m(m-1)^2(m-2)/12$; Bernoulli B_{2r} (also in (20.3))
$C_r(n)$	the strict chamber (15.2)	C_n the central event
$\mathcal{I}; X_i$	$\{1 \leq X_i \leq n-2\}$; $\text{Bin}(n, U)$ counts mixed over $U \sim \text{Beta}(n, n+1)$ (Section 19)	$\{1 \leq X_i \leq n-1\}$; independent $\text{Bin}(n, p)$
$W_n(x)$	the rare weight (19.3)	$R_n(x)$ plays this role
$Y_i; Z_{i,n}, V_n, V$	$(2X_i - n)/\sqrt{n}$ for the mixed X_i ; the components of (19.5)	Y_i the same formula for independent X_i ; Z_i standardized counts (Section 9)
$\varepsilon; \mathcal{E}_\varepsilon$	$n^{-1/2}$; the formal moment functional (20.2)	ε_j random signs; $\mathcal{E}_z$ (5.8)
$\Psi_r; Q_r(z); Q_m(n)$	the interaction factor (19.8); its normalized expectation (written $Q_r(0) = 1$ once in (20.2)); the common-phase count $T_{m,B^m D^m}(n)$ (21.2)	—
$A_m(z); \mathcal{A}_{5,L}, \mathcal{L}_5$	the Stirling factor (20.3); truncations and leading term of (21.1)	$A = \binom{m}{3}$, the event $\mathcal{A}$; $A_{5,L}, L_5$ for T_5

symbol	meaning in Part II	not: meaning in Part I
$S_1, S_2; S_h(t); S$	$\sum_i y_i$ and $\sum_i y_i^2$ (Section 20); bounded-height Dyck series; the event that row m starts after row 1 ends	$S_1 = h_1(q) = \sum_{i < j} (Y_i + Y_j)^2$
$p_i, q_i; p$ $P_j; P_1(y)$	rank marks (17.5); a fixed success probability in (20.7) monic orthogonal polynomials of $\text{Bin}(n, p)$ in x ; the first coefficient of Ψ_r (20.10)	p the threshold, p_r power sums, q_{ij} monic orthogonal $P_j(y)$ of the binomial Vandermonde ensemble (Section 6.2)
λ_m, β_m, a	$m \log m - \log 4, (m-2)^2/2, \log(y/K_m)$ (22.1)	$\lambda = m \log m, \beta_j$ recurrence coefficients, $L = \log(y/K)$
L	a truncation order; $ v - u $ in Lemma 15.1	a truncation order; $\log(y/K)$
$h(w)$	the maximal Dyck height of a word	$h_r(q), h_j$
$C_a(n, p), c_k, N_k$	centred binomial moments and series coefficients (Appendix D)	$c_{m,j}, C(z), C_{n-1}, C_{m,L}$
$\mathfrak{M}, \mathfrak{S}_r, \text{Diag}, \ell_s$	signed multinomial (15.3), the symmetric group, the diagonal, event labels (17.4)	—

14 The question and the scope of the answer

14.1 The OEIS family

For $m, n \geq 1$, let $T_m(n)$ be the number of arrays $A = (a_{i,j})_{1 \leq i \leq m, 1 \leq j \leq n}$ containing each of $1, \dots, mn$ once, with increasing rows, columns, diagonals, and downward antidiagonals. Equivalently, the following comparisons suffice whenever the cells exist:

$$a_{i,j} < a_{i,j+1}, \quad a_{i,j} < a_{i+1,j}, \quad a_{i,j+1} < a_{i+1,j}. \quad (14.1)$$

The diagonal comparison follows from row and column increase. We set $T_m(0) = 1$. This is the family A181196; the height-three, height-four, and height-five rows are A181197, A181198, and A181199 [1, 2, 3, 4]. The direction of the limit matters: m is fixed and n grows. A fixed-width strip theorem with a growing number of rows is a different statement.

Kauers and Koutschan discuss the height-four and height-five problems in Section 6.4 of their 2023 paper [9]. The OEIS entries inspected for this report still label their displayed recurrences as conjectural: order two and degree nine for A181198, and order three and degree twenty-four for A181199. The same paper gives related finite-sum conjectures. These specific formulas are not used anywhere in our proofs.

14.2 What the preceding repository report already established

The report *Fixed-height shifted rectangles* in ProveIt (Part I of the present report) obtains an all-order expansion for $T_m(n)$, including

$$T_m(n) \sim K_m m^{mn} n^{-\alpha_m}, \quad \alpha_m = \frac{m^2 - 1}{2}, \quad K_m = \frac{\sqrt{m} \prod_{j=0}^{m-1} j!}{4^{\binom{m}{2}} (2\pi)^{(m-1)/2}}. \quad (14.2)$$

It proves the leading asymptotic conjecture for A181199 without assuming its recurrence, and proves nonalgebraicity at every height $m \geq 3$. Its Question 7 asks whether the generating function is D-finite for every fixed height. Its coarse exponentially small boundary bound is explicitly not claimed sharp. These are the two gaps addressed here.

The repository snapshot used for the comparison is

db20eb37982abc0f163c6d308d392f5ac4c9bc62.

The source/status file in the package records the exact report path, inspected literature, and limited identifier searches. We do not infer historical novelty merely from an OEIS conjecture label or a repository question.

14.3 New statements developed in this manuscript

Our exact-enumeration result is stronger than existence of an unspecified recurrence.

Theorem 14.1 (Rational diagonality at every fixed height). *For every fixed integer $m \geq 1$,*

$$F_m(z) = \sum_{n \geq 0} T_m(n) z^n$$

is the diagonal of a rational power series over $\mathbb{Q}$. In particular it is D-finite, and $(T_m(n))_{n \geq 0}$ is P-recursive. More generally, each fixed ordering of row births and row completions has a rational-diagonal generating function, after assigning arbitrary rational values at $n = 0, 1$.

The proof, given in Sections 15–16, is a finite construction from chamber kernels. It provides at most $m(m-1)$ free integer summation indices for each event ordering. This count concerns summation variables, not the number of variables in a minimal rational function or the order of a minimal differential equation.

For $m \geq 2$, write

$$B_i = a_{i,1}, \quad D_i = a_{i,n}, \quad R_m(n) = \#\{A : D_1 < B_m\}, \quad n \geq 1, \quad (14.3)$$

and set $R_m(0) = 0$. Thus $R_m(n)$ counts arrays for which the first row finishes before the last row starts. When $n \geq 2$, its complement has all rows active simultaneously at some intermediate stage. Put

$$r = m - 2, \quad d_R = \binom{r}{2}, \quad J_r = \prod_{j=0}^{r-1} j!, \quad J_0 = 1. \quad (14.4)$$

Empty products are one, throughout. In the chamber formulas of Sections 15–17, r may instead denote a general active-block size; from Section 18 onward it is $m - 2$.

Theorem 14.2 (Sharp rare-boundary expansion). *For each fixed $m \geq 2$ and each fixed $L \geq 0$,*

$$R_m(n) = \mathcal{K}_m \left(\frac{m^m}{4} \right)^n n^{-r^2/2} \left(\sum_{j=0}^L \frac{b_{m,j}^R}{n^j} + O_{m,L}(n^{-L-1}) \right), \quad (14.5)$$

where

$$\mathcal{K}_m = \sqrt{\frac{m}{2}} (2\pi)^{-r/2} 4^{-d_R} 9^{-r} J_r, \quad b_{m,0}^R = 1, \quad (14.6)$$

and all $b_{m,j}^R$ are effectively computable rational numbers. The first correction is

$$b_{m,1}^R = \frac{r(6r^3 - 52r^2 - 268r - 283)}{72(r+2)}. \quad (14.7)$$

At $m = 2$, the assertion reduces to the exact identity $R_2(n) = 1$.

The proof occupies Sections 18–20. It is independent of (14.2). The latter is used only when we subsequently divide $R_m(n)$ by $T_m(n)$.

What has and has not been resolved. The fixed-height D-finiteness question is answered affirmatively by an exact formula, and a sharp rare-event sector is obtained at every height. The published candidate low-order recurrences and finite sums remain unproved in this manuscript. Our all-order series are Poincaré expansions with fixed height and fixed truncation order. No Borel summability, Stokes constants, or complete exponentially improved transseries is claimed.

Remark 14.3 ([write] Research question 7 of Part I is answered in part). Added 5 October 2026, batch 98. The abstract printed at the head of this Part says that Theorem 14.1 resolves “an explicit question in the preceding ProveIt report”, and the delivered README says “This resolves Question 7”. That overstates it. Research question 7 of Part I asks two things: whether F_m is D-finite for every fixed m , and “how does the minimal annihilating order grow with m ?”. Theorem 14.1 answers the first affirmatively. It says nothing about the second: the construction gives at most $m(m-1)$ summation indices, which the text above states is not a bound on the order of a minimal differential equation, Remark 16.2 says that no minimal operator is produced, and the manuscript’s own fourth research question (here Research question 11) asks for exactly this growth. The result box above (“The fixed-height D-finiteness question is answered affirmatively”) is accurate. The second half of Research question 7 stays open; a dated note at that question says so.

Remark 14.4 ([write] Research question 4 of Part I is not settled by R_m). Added 5 October 2026, batch 98. Section 14 lists the non-sharp boundary bound of Part I among “the two gaps addressed here”. The sharp question there, Research question 4, asks for the asymptotic of the boundary error $E_{m,n}(1/2)$ of Theorem 4.1. Part II does not determine it. Its rare count is a different and much smaller sector. In the order-statistics model of Section 4.1, the rare arrays have every row-1 sample below every row- m sample, so at the cut $p = 1/2$ either $X_1 = n$ or $X_m = 0$; hence the rare event lies inside the boundary event and

$$R_m(n) \leq M_m(n) E_{m,n}(1/2)$$

(Proposition 18.3(c), which also gives the growth of the ratio). Exact values, computed at intake from the definitions (row-state dynamic program for $T_m(n)$ and $R_m(n)$, the rational sum (4.5) for $I_m(n)$, and $M_m(n)E_{m,n}(1/2) = T_m(n) - M_m(n)I_m(n)$):

m	n	$M_m(n)E_{m,n}(1/2)/R_m(n)$	$2^n E_{m,n}(1/2)$	$E_{m,n}(1/2)/(2^{1-n}3^{1-m}J_{m-1}(4n)^{-\binom{m-1}{2}})$
3	8	4.892	$7.165 \cdot 10^{-3}$	1.0317
3	12	37.20	$4.835 \cdot 10^{-3}$	1.0444
3	16	357.5	$3.594 \cdot 10^{-3}$	1.0351
3	20	3907	$2.855 \cdot 10^{-3}$	1.0276
3	30	$2.053 \cdot 10^6$	$1.885 \cdot 10^{-3}$	1.0178
3	40	$1.328 \cdot 10^9$	$1.407 \cdot 10^{-3}$	1.0131
2	20	$8.962 \cdot 10^4$	0.6817	1.0226
2	80	$5.104 \cdot 10^{22}$	0.6704	1.0056
4	12	7.183	$1.381 \cdot 10^{-6}$	1.0311
4	20	374.2	$2.995 \cdot 10^{-7}$	1.0350

At $m = 3$ the boundary error is 4.9, 37, 357 and 3907 times the rare count at $n = 8, 12, 16, 20$, and the factor keeps growing roughly like 2^n . As a probability the boundary error is 2^{-n} times a power of n (second column), while $R_m(n)/M_m(n) \leq \binom{2n}{n}^{-1}$ is at most of order 4^{-n} by (18.5). So Theorem 14.2 is a sharp theorem about a sub-sector of the boundary event, not the sharp boundary asymptotic; Research question 4 stays open, re-scoped in a dated note there. The write’s Proposition 18.3 supplies its leading term (last column). The program used for the table is the intake’s and is not shipped; the same numbers follow from the shipped enumerator count of code/02-boundary-phases-tableaux.py and the sum (4.5).

15 Boundary events and the exact walk model

15.1 Prefix states, including both exceptional boundaries

Insert the labels $1, 2, \dots, mn$ in order, and let x_i be the number of filled cells in row i . Row increase makes the filled cells in each row a prefix. The state space is exactly

$$n \geq x_1 \geq x_2 \geq \dots \geq x_m \geq 0, \quad x_i = x_{i+1} \implies x_i \in \{0, n\}. \quad (15.1)$$

Indeed, if a lower row has $j < n$ filled cells, its last filled cell forces cell $j + 1$ in the row above to be filled. If the lower row is full, column increase forces the upper row to be full. Conversely these conditions make the filled set an order ideal of (14.1). Consequently paths from $(0, \dots, 0)$ to $(n, \dots, n)$ through (15.1), using unit steps $+e_i$, are in bijection with the arrays.

A row is *unborn*, *active*, or *completed* according as its coordinate is 0, belongs to $\{1, \dots, n - 1\}$, or is n . Completed rows form a prefix and unborn rows a suffix. Thus the active rows always form one consecutive block, and their coordinates are strictly decreasing. The ties at zero and at n cannot be discarded: forbidding either kind changes the enumeration.

For $n \geq 2$, birth and completion are different unit steps. Births occur in row order $1, \dots, m$, and completions also occur in that order. Record a letter B for a birth and D for a completion. Every array gives a word w of length $2m$ with m letters of each kind and with at least as many B 's as D 's in every prefix. Such a word is a Dyck word, with B interpreted as an up-step. Conversely the i th B means the birth of row i , and the i th D means the completion of row i ; no further row-labeling data are needed.

15.2 The strict chamber between two events

For $r \geq 0$ define

$$C_r(n) = \{(u_1, \dots, u_r) : n - 1 \geq u_1 > \dots > u_r \geq 1\}, \quad C_0(n) = \{()\}. \quad (15.2)$$

During an interval with no birth or completion, its r active coordinates move by $+e_i$ in this chamber. If $u, v \in C_r(n)$, write $K_r(u, v)$ for the number of such walks from u to v , including a zero-length walk when $u = v$.

For an integer vector $a = (a_1, \dots, a_r)$ define the everywhere-defined integer multinomial

$$\mathfrak{M}(a) = \begin{cases} (a_1 + \dots + a_r)! / (a_1! \dots a_r!), & a_i \geq 0 \text{ for all } i, \\ 0, & \text{otherwise,} \end{cases} \quad \mathfrak{M}() = 1. \quad (15.3)$$

Lemma 15.1 (Reflection kernel). *For strict endpoints $u, v \in C_r(n)$,*

$$K_r(u, v) = \sum_{\sigma \in \mathfrak{S}_r} \text{sgn}(\sigma) \mathfrak{M}(v_1 - u_{\sigma(1)}, \dots, v_r - u_{\sigma(r)}). \quad (15.4)$$

Equivalently, when $L = |v| - |u| \geq 0$,

$$K_r(u, v) = L! \det_{1 \leq i, j \leq r} \frac{1}{(v_i - u_j)!}, \quad (15.5)$$

where $1/k! = 0$ for negative integers k . For $r = 0$, the value is one.

Proof. Consider the signed family of all unrestricted $+e_i$ walks from a permutation of u to v . The number starting at $(u_{\sigma(1)}, \dots, u_{\sigma(r)})$ is the multinomial in (15.4). Cancel every walk that has a collision of two coordinates: choose its earliest collision time, breaking any tie by a fixed ordering of coordinate pairs, and exchange the histories of the colliding coordinates up to that time. The starting permutation is changed by a transposition, the endpoint and step set are unchanged, and the operation is an involution. Before the chosen time, all equality relations are preserved under this exchange, so the same collision is selected on the second application.

A walk remaining after cancellation has no collisions. Its coordinate order cannot change, since a unit step cannot cross an integer coordinate without meeting it. Its strictly decreasing endpoint forces its starting permutation to be the identity, and it stays in the decreasing chamber. All coordinates are between 1 and $n - 1$: along an uncanceled walk each coordinate is nondecreasing and lies between its own endpoints. Thus no extra wall reflection is needed for the finite upper bound. The surviving walks are exactly those counted by $K_r(u, v)$. $\square$

The proof is the classical reflection argument, included to specify precisely which walls are present. Formula (15.4), rather than an unqualified factorial quotient, is the appropriate form for subsequent binomial-sum manipulations: it is defined even when an individual displacement is negative.

15.3 Birth and completion maps

If $v \in C_r(n)$, a birth changes the active vector by

$$B(v) = (v_1, \dots, v_r, 1), \quad \text{allowed if } r = 0 \text{ or } v_r \geq 2. \quad (15.6)$$

A completion changes it by

$$D(v) = (v_2, \dots, v_r), \quad \text{allowed if } r \geq 1 \text{ and } v_1 = n - 1. \quad (15.7)$$

The completion step increases the old first active coordinate from $n - 1$ to n before removing it. All previously completed coordinates are already n . These two maps account exactly for the exceptional boundary ties.

16 Finite binomial sums and rational diagonals

16.1 The event-skeleton formula

Fix a Dyck word $w = w_1 \cdots w_{2m}$. Let b_s, d_s be the numbers of births and completions in its first s letters, and set $r_s = b_s - d_s$, for $0 \leq s \leq 2m$. Let u_s be the active vector immediately after event s , and v_s the active vector immediately before event $s + 1$. The initial vector is $u_0 = ()$. A skeleton consists of vectors $v_s \in C_{r_s}(n)$ with

$$u_{s+1} = \begin{cases} B(v_s), & w_{s+1} = B, \\ D(v_s), & w_{s+1} = D, \end{cases} \quad 0 \leq s < 2m, \quad (16.1)$$

subject to the corresponding legality conditions. All u_s are thereby affine expressions in earlier endpoint coordinates and the constants $1, n$. Define $T_{m,w}(n)$ to count arrays with this event word.

Theorem 16.1 (Exact finite-phase enumeration). *For $n \geq 2$,*

$$T_{m,w}(n) = \sum_{\text{legal skeletons for } w} \prod_{s=0}^{2m-1} K_{r_s}(u_s, v_s), \quad T_m(n) = \sum_{w \in \mathcal{D}_m} T_{m,w}(n), \quad (16.2)$$

where $\mathcal{D}_m$ denotes the Dyck words with m births. Each first sum can be written with at most $m(m-1)$ free integer indices, each ranging between 0 and n , and with only affine equality and inequality constraints.

Proof. Cut a valid prefix path immediately before and after every boundary event. The intervening pieces are chamber walks with the stated endpoints. Conversely, choosing a legal skeleton and one chamber walk in each interval reconstructs a unique full prefix path. The events each contribute a single unit step, and the intervals concatenate in a prescribed order; there is no additional interleaving factor. This proves the product and sum.

There are initially $\sum_{s=0}^{2m-1} r_s$ endpoint slots. This sum is at most m^2 : at each position s the height is at most $\min(s, 2m-s)$, with equality throughout for $B^m D^m$. Each of the m completion events fixes one different endpoint coordinate to $n-1$. Eliminating these coordinates leaves at most $m^2 - m$ free slots. The chamber bounds, strict gaps, birth conditions, and all endpoint substitutions are affine in n and the remaining coordinates. $\square$

As a consistency check, every skeleton with nonzero weight satisfies

$$\sum_{s=0}^{2m-1} (|v_s| - |u_s|) = m(n-2). \quad (16.3)$$

Each of the m rows has precisely $n-2$ non-event steps. Equivalently, telescoping the active sums adds one at a birth and removes $n-1$ at a completion, which gives the same identity.

16.2 An explicit binomial encoding of every constraint

For integers a, b , use $\binom{a}{b} = [t^b](1+t)^a$ when $b \geq 0$, and zero when $b < 0$. In particular,

$$H(a) = \binom{a}{a} = \mathbf{1}_{\{a \geq 0\}}, \quad \delta(a) = H(a)H(-a) = \mathbf{1}_{\{a=0\}}. \quad (16.4)$$

Every affine inequality or equality in (16.2) is therefore a product of such factors. For example, strictness $v_i > v_{i+1}$ is $H(v_i - v_{i+1} - 1)$, and a completion uses $\delta(v_1 - n + 1)$. Moreover, for $r \geq 1$,

$$\mathfrak{M}(a_1, \dots, a_r) = \left(\prod_{i=1}^r H(a_i) \right) \prod_{i=1}^{r-1} \binom{a_i + \dots + a_r}{a_i}. \quad (16.5)$$

If a displacement is negative the leading indicator makes the term zero; all other factors remain defined. If all displacements are nonnegative, the binomial product telescopes to the multinomial.

Expand the fixed-size permutation sums in (15.4), replace each multinomial by (16.5), sum each free coordinate from 0 to n , and insert the legality factors. The result is a finite linear combination of bounded multiple sums of products of binomial coefficients with affine arguments. There are no reciprocal binomial coefficients, no factorial poles, and no hidden variable number of summation indices.

16.3 The imported diagonal theorem and its application

We use the following result of Bostan, Lairez, and Salvy [10, Theorem 3.5 and Corollary 3.6]: a univariate multiple binomial sum has a generating function that is the diagonal of a rational power series; such sequences are P-recursive. Their class is closed under affine index substitutions, products, sums, and finite partial summation.

For clarity, consider a rational function regular at the origin, with expansion

$$Q(x_1, \dots, x_k) = \sum q_{n_1, \dots, n_k} x_1^{n_1} \cdots x_k^{n_k}.$$

Its diagonal is

$$\text{Diag } Q(z) = \sum_{n \geq 0} q_{n, \dots, n} z^n.$$

A power series is D-finite if it satisfies a nonzero linear differential equation with polynomial coefficients. Equivalently its coefficient sequence satisfies a linear recurrence with polynomial coefficients.

Proof of Theorem 14.1. The preceding encoding exhibits $T_{m,w}(n)$, for $n \geq 2$, as a multiple binomial sum. To define a sequence on all integers, multiply by $H(n-2)$ and assign zero below two. The sums from 0 to n can first be interpreted as the extended partial sums of the binomial-sum algebra; the multiplier removes their values outside the intended range. Adding any finitely supported correction preserves that algebra, since shifted Kronecker deltas are allowed. The imported theorem gives a rational diagonal for each event-word sequence. Summing the finitely many words and adding the terms at $n = 0, 1$ gives F_m .

This argument proves rational diagonality, not merely a heuristic closure claim about hypergeometric expressions. D-finiteness and P-recursiveness then follow from the same theorem. At $m = 1$, one may also observe directly that $F_1(z) = 1/(1-z)$. $\square$

Remark 16.2 (Constructive does not mean a compact telescope is supplied). The skeleton and binomial formulas are explicit finite input to the rational-integral and diagonal constructions of [10]. This is an effective existence proof. We have not carried out that elimination to produce a small rational function or a minimal differential operator for general m . In particular, it does not certify that either operator currently displayed in OEIS is the correct one.

[write] **Added 5 October 2026, batch 101.** The last sentence of the remark concerns this existence proof only. Parts III and IV certify both displayed operators by another method (Theorems 30.1 and 34.2), from Pfaffian formulas for the transfer integral rather than from the event sum. Report 91, printed in Part V, proves Theorem 14.1 by the same route as this section (Theorem 44.1).

Combined with the previous report's algebraicity classification (Part I, Theorem 2.3), the result places the entire family in a precise hierarchy: F_m is algebraic for $m \leq 2$, and transcendental but D-finite for $m \geq 3$. The nonalgebraicity part of that statement is inherited from the previous report, not reproved or claimed as new here.

17 Which event words occur, and which statistics remain D-finite?

17.1 A complete realizability criterion

Let $h(w)$ be the maximum Dyck height of w .

Theorem 17.1 (Event-word realizability). *For $m \geq 1$, $n \geq 2$, and $w \in \mathcal{D}_m$,*

$$T_{m,w}(n) > 0 \iff h(w) \leq n - 1. \quad (17.1)$$

Consequently every Dyck word of semilength m occurs as soon as $n \geq m + 1$.

Proof. At height r there must be r distinct active coordinates in $\{1, \dots, n-1\}$, so $r \leq n-1$ is necessary.

For sufficiency, follow the prescribed word. Before a birth at active height r , the height bound implies $r \leq n-2$. Move the existing coordinates, in top-to-bottom order, to $(n-1, n-2, \dots, n-r)$. Each target dominates the present coordinate, since any decreasing vector in $C_r(n)$ has $u_i \leq n-i$. The moves are legal because the coordinate above has already reached its target. The last target is at least two, so a new coordinate one can be appended. Before a completion, simply move the top active coordinate to $n-1$ and complete that row. Neither operation creates any additional boundary event. Continuing to the end constructs a valid path with exactly w . $\square$

The number of event words available at width n is thus a classical bounded-height Dyck number. If $S_h(t)$ is its generating function by semilength with height at most h , the first-return decomposition gives

$$S_0(t) = 1, \quad S_h(t) = \frac{1}{1 - tS_{h-1}(t)} \quad (h \geq 1). \quad (17.2)$$

This counts possible event types, not arrays: the sector weights are generally far from equal.

At $n = 2$ there is only the alternating word $(BD)^m$, and $T_m(2) = 1$. At $n = 3$, height is at most two. A one-row active phase has only one coordinate to move, while a two-row active phase has the unique vector $(2, 1)$. Thus each permitted word has weight one. Formula (17.2) gives $T_m(3) = 2^{m-1}$ for $m \geq 1$, recovering a known small-width formula [11], not a new one.

17.2 Small worked examples

For two rows and $n \geq 2$, the alternating word has weight one. For the other word,

$$\begin{aligned} T_{2,BBDD}(n) &= \sum_{a=2}^{n-1} \sum_{b=1}^{n-2} K_2((a, 1), (n-1, b)) \\ &= \sum_{a=2}^{n-1} \sum_{b=1}^{n-2} (\mathfrak{M}(n-1-a, b-1) - \mathfrak{M}(n-2, b-a)). \end{aligned} \quad (17.3)$$

This is an explicit binomial sum without exceptional denominator values. Adding one recovers $T_2(n) = C_{n-1}$, the usual Catalan control.

At $m = 3, n = 4$, the sector weights are

Event word	<i>BBBDDD</i>	<i>BBDBDD</i>	<i>BBDDBD</i>	<i>BDBBDD</i>	<i>BDBDBD</i>
Number of arrays	4	16	4	4	1

Their sum is 29, and the rare subcount is 25. This also illustrates that having a full-height phase is possible but not yet typical at small width.

17.3 Birth and completion ranks

In a skeleton, immediately before event $s+1$ there are d_s completed rows and $|v_s|$ cells in active rows. Hence its label is the affine form

$$\ell_{s+1} = d_s n + |v_s| + 1. \quad (17.4)$$

Let $p_1, \dots, p_m, q_1, \dots, q_m$ be independent indeterminates and define

$$T_m(n; \mathbf{p}, \mathbf{q}) = \sum_A \prod_{i=1}^m p_i^{B_i} q_i^{D_i}. \quad (17.5)$$

The birth and completion ranks are labels, not column positions.

Corollary 17.2 (Marked enumeration and moment totals). *For fixed m , the generating function in n of $T_m(n; \mathbf{p}, \mathbf{q})$ is a rational diagonal over $\mathbb{Q}(\mathbf{p}, \mathbf{q})$. For every fixed polynomial P with rational coefficients in the $2m$ event ranks,*

$$n \mapsto \sum_A P(B_1, \dots, B_m, D_1, \dots, D_m)$$

is a multiple binomial sum and is P -recursive. The same statements hold within each fixed event sector.

Proof. Insert the weights from (17.4) into (16.2). They are geometric sequences with affine exponents over the indicated coefficient field. Polynomial weights are also in the binomial-sum algebra, since an integer index is $\binom{k}{1}$. The same closure proof applies. When $n = 1$, use the appropriate finitely supported marked term; birth and completion coincide within a row only in this exceptional case. $\square$

This proves the assertion for *unnormalized moment totals*. An expectation divides by $T_m(n)$; P -recursiveness of such quotients is not asserted. Rotation of the array by 180 degrees followed by replacement of each label a by $mn + 1 - a$ gives the useful exact symmetry

$$B_i \longleftrightarrow mn + 1 - D_{m+1-i}, \quad T_{m,w}(n) = T_{m,\overline{w}^{\text{rev}}}(n), \quad (17.6)$$

where the bar interchanges B and D .

18 A first-completion cut for the rare sector

18.1 The dominant rare configuration

For $m \geq 2$ and $n \geq 2$, define the core subcount

$$H_m(n) = \#\{A : B_{m-1} < D_1 < B_m\}. \quad (18.1)$$

For $m = 2$ this equals $R_2(n) = 1$. For $m \geq 3$, the remainder has $D_1 < B_{m-1}$: the first row finishes while at least the last two rows are still unborn. This extra separation will give an exponentially smaller rate.

The classical shifted hook product for a strict positive partition $\lambda = (\lambda_1 > \dots > \lambda_k > 0)$ is

$$g^\lambda = \frac{|\lambda|!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (18.2)$$

It counts chains of strict partitions obtained by adding one cell at a time, equivalently standard shifted tableaux; see [11]. This is an imported classical enumeration formula, not a new hook theorem.

Proposition 18.1 (Exact core sum). *With $r = m - 2$, for all $m, n \geq 2$,*

$$H_m(n) = \sum_{n-2 \geq x_1 > \dots > x_r \geq 1} g^{(n-1, x_1, \dots, x_r)} g^{(n, n-x_r, \dots, n-x_1)}. \quad (18.3)$$

For $r = 0$ the sum has one term, equal to one.

Proof. Immediately before the first completion, row 1 has length $n - 1$ and the last row is empty. In the core sector every intervening row is active, so their lengths are $n - 2 \geq x_1 > \cdots > x_r \geq 1$. Before this point no row has completed. Removing zero rows, the prefix path is a chain of strict partitions ending at $(n - 1, x_1, \dots, x_r)$, so it has the first hook count in (18.3).

The next step completes row 1. Reverse all remaining steps and rotate rows; the complementary lengths are $(n, n - x_r, \dots, n - x_1)$, a strict positive partition. Every reversed prefix stays in the strict-partition region until the endpoint. In particular, a coordinate equal to n at the endpoint causes no repeated completed-row boundary, since all other parts are strictly smaller. Thus the suffix has the second hook count. Their label sets are fixed by the size of the prefix, and the intervening completion step is unique. The two chains concatenate bijectively to the core arrays. $\square$

18.2 Two useful nonasymptotic error bounds

Write

$$M_m(n) = \frac{(mn)!}{(n!)^m}. \quad (18.4)$$

This counts all interleavings of m individually sorted rows, before imposing cross-row comparisons.

Proposition 18.2 (Separation bounds). *For $m \geq 2$ and $n \geq 1$,*

$$R_m(n) \leq \frac{M_m(n)}{\binom{2n}{n}}. \quad (18.5)$$

For $m \geq 3$ and $n \geq 2$,

$$0 \leq R_m(n) - H_m(n) \leq \frac{M_m(n)}{\binom{3n}{n}}. \quad (18.6)$$

For $m = 2$ the difference is zero.

Proof. In a uniform row interleaving, restrict attention to rows 1 and m . Of their $\binom{2n}{n}$ possible relative words, exactly one places all of row 1 before all of row m . Dropping other comparisons bounds the rare sector and proves (18.5).

For the remainder, rows $m - 1$ and m must both start after row 1 has finished. Among the relative interleavings of these three rows, the probability that the first n letters all come from row 1 is

$$\frac{(2n)!/(n!)^2}{(3n)!/(n!)^3} = \frac{1}{\binom{3n}{n}}.$$

Again dropping all other constraints gives an upper bound. $\square$

These estimates have explicit constants and hold at finite n . Once the core asymptotic is known, (18.6) is smaller than the core by a polynomial factor times $(16/27)^n$. It is therefore negligible at every inverse-power order on the rare scale. We do not claim that (18.6) gives the sharp next exponential sector.

18.3 [write] The boundary error of Part I's central cut

[write] **Added 5 October 2026, batch 98.** This subsection is the write's, not the manuscript's. It gives the leading asymptotic of the boundary error asked for in Research question 4 of Part I, and places the rare sector of this Part inside that error (Remark 14.4). It uses only Part I's model and estimates and the hook argument of Proposition 18.1. Like the

rest of the report it is unrefereed; the exact identity (a) and the asymptotic (b) were checked against exact values for $m = 2, 3, 4$ (table of Remark 14.4).

Dated note (5 October 2026). An independent adversarial check of the results added at this write, made by the intake after the write, found all of them valid: Proposition 18.3, parts (a), (b) and (c); the table of Remark 14.4, all ten rows to every printed digit; and the completed arguments for (19.6) and (19.11) in the notes after them. The check was a careful reading with independent exact computations: $T_m(n)$ by a new prefix-state recursion, cross-checked against a direct linear-extension count of the poset for $mn \leq 22$; $E_{m,n}(1/2)$ in exact rationals from conditional path counts, without the hook formula; $R_m(n)$ recomputed; 116 exact rows ($m = 2, \dots, 5$, n up to 80 at $m = 2$), with floating extensions to $n = 1600$ at $m = 3$; and the bound (19.11) tested at every atom of the beta-binomial law for $n \leq 400$. In those rows $D_{m,n}$ lies far inside its bound. It is not a formal verification or an external review. No fix was needed; the observed rate in (b) is recorded after the proof.

Proposition 18.3 ([write] Leading boundary error at the central cut). *Let $m \geq 1$, $n \geq 2$, and let $E_{m,n}(1/2)$ be the boundary error of Theorem 4.1 at $p = 1/2$. Let $X'_1, \dots, X'_{m-1}$ be independent $\text{Bin}(n, 1/2)$ variables, $I' = \{1 \leq X'_i \leq n - 1 \text{ for all } i\}$, and*

$$W'_n(x) = \prod_{1 \leq i < j \leq m-1} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)} \prod_{i=1}^{m-1} \frac{x_i}{2n - x_i}.$$

(a) *Exactly,*

$$E_{m,n}(1/2) = \frac{2^{1-n}}{(m-1)!} \mathbb{E}[\mathbf{1}_{I'} W'_n(X')] + D_{m,n}, \quad 0 \leq D_{m,n} \leq 2m(m-1)4^{-n}. \quad (18.7)$$

(b) *As $n \rightarrow \infty$,*

$$E_{m,n}(1/2) \sim 2^{1-n} 3^{1-m} J_{m-1} (4n)^{-\binom{m-1}{2}}, \quad J_{m-1} = \prod_{j=0}^{m-2} j!. \quad (18.8)$$

(c) *Consequently $M_m(n)E_{m,n}(1/2)/T_m(n) \sim 2^{1-n} 3^{1-m} (4n)^{m-1}/(m-1)!$, and, for $m \geq 2$, $R_m(n) \leq M_m(n)E_{m,n}(1/2)$ for every $n \geq 2$, with*

$$\frac{M_m(n)E_{m,n}(1/2)}{R_m(n)} \sim \frac{2 \cdot 3^{m-3} (m-2)!}{\sqrt{\pi} 4^{m-2}} 2^n n^{-(2m-3)/2}.$$

Proof. Use the order-statistics model of Section 4.1 at $p = 1/2$: X_i counts the row- i samples below $1/2$, the X_i are independent $\text{Bin}(n, 1/2)$, and the proof of Theorem 4.1 identifies $E_{m,n}(1/2)$ with $\mathbb{P}(\mathcal{A} \cap \mathcal{I}^c)$.

(a) On $\mathcal{A}$ the cells with values below $1/2$ form an order ideal, so X is a prefix state (3.1): $X_i = 0$ forces $X_{i+1} = \dots = X_m = 0$, and $X_i = n$ forces $X_1 = \dots = X_i = n$. Hence on $\mathcal{A} \cap \mathcal{I}^c$ either exactly one coordinate lies in $\{0, n\}$, and then it is $X_m = 0$ or $X_1 = n$ with every other coordinate in $[1, n-1]$, or at least two coordinates lie in $\{0, n\}$. The second event, whose probability on $\mathcal{A}$ we call $D_{m,n}$, has probability at most $\binom{m}{2} (2 \cdot 2^{-n})^2 = 2m(m-1)4^{-n}$.

Let $x = (x_1, \dots, x_{m-1}, 0)$ with $n-1 \geq x_1 > \dots > x_{m-1} \geq 1$. As in Section 4.1, $\mathbb{P}(\mathcal{A} \mid X = x)$ is the product of a prefix and a suffix probability. Every prefix path to x has last coordinate 0 and first coordinate below n , so its states are strict partitions and the prefix count is $g^{(x_1, \dots, x_{m-1})}$. The rotated complement (3.3) of x is $(n, n-x_{m-1}, \dots, n-x_1)$, a strict positive partition in which only the first part can reach n ; as in the proof of Proposition 18.1, the suffix count is $g^{(n, n-x_{m-1}, \dots, n-x_1)}$. Dividing both counts by the multinomial numbers of interleavings and using (3.5), the pairs among $x_1, \dots, x_{m-1}$ contribute $\frac{x_i - x_j}{x_i + x_j} \cdot \frac{x_i - x_j}{2n - x_i - x_j}$ and the pairs of the part n with the parts $n - x_i$ contribute $\frac{x_i}{2n - x_i}$, so $\mathbb{P}(\mathcal{A} \mid X = x) = W'_n(x_1, \dots, x_{m-1})$. The weight W'_n is symmetric and vanishes at ties; summing over decreasing vectors and multiplying by $\mathbb{P}(X_m = 0) = 2^{-n}$, the

stratum $\{X_m = 0\}$ contributes $2^{-n} \mathbb{E}[\mathbf{1}_{\mathcal{I}'} W'_n(X')]/(m-1)!$. Reflecting every sample $u \mapsto 1-u$ and reversing the rows preserves the model and $\mathcal{A}$ and maps X to $(n-X_m, \dots, n-X_1)$, so the stratum $\{X_1 = n\}$ contributes the same. This proves (18.7). (For $m = 1$ there are no pairs, $W'_n = 1$, $D_{1,n} = 0$, and $E_{1,n}(1/2) = 2^{1-n}$.)

(b) Put $c = \binom{m-1}{2}$, $Y_i = (2X'_i - n)/\sqrt{n}$ and $C'_n = \{|Y_i| \leq \sqrt{n}/2 \text{ for all } i\}$; then $C'_n \subset \mathcal{I}'$. On C'_n , exactly as in (5.2),

$$(4n)^c W'_n(X') = \Delta(Y)^2 \prod_{i < j} \left(1 - \frac{(Y_i + Y_j)^2}{4n}\right)^{-1} \prod_i \frac{1 + Y_i/\sqrt{n}}{3 - Y_i/\sqrt{n}},$$

where every pair factor lies in $[1, 4/3]$ and every single factor in $[1/7, 3/5]$. The right side is at most $(4/3)^c \Delta(Y)^2$, a uniformly integrable family because the moments of each fixed degree of the Y_i are bounded uniformly in n (Section 5.4), and it converges to $3^{1-m} \Delta(y)^2$ uniformly on compact sets. Since Y converges in distribution to a vector G of independent standard normals, $(4n)^c \mathbb{E}[\mathbf{1}_{C'_n} W'_n(X')] \rightarrow 3^{1-m} \mathbb{E} \Delta(G)^2 = 3^{1-m} (m-1)! J_{m-1}$ by (5.14) in $m-1$ variables. Off C'_n , $0 \leq W'_n \leq 1$ (Remark 4.2 and $x_i \leq n$), and $\mathbb{P}(C_n^c) \leq 2(m-1)e^{-n/8}$ by (5.10), which is $o(n^{-c})$. Inserting this into (18.7), where $D_{m,n} = O(4^{-n})$ is $o(2^{-n} n^{-c})$, gives (18.8).

(c) The first relation is (18.8) divided by $T_m(n)/M_m(n) \sim J_m(4n)^{-\binom{m}{2}}$ (2.3), since $J_m = (m-1)! J_{m-1}$ and $\binom{m}{2} - \binom{m-1}{2} = m-1$. In the same model the rare event $\{D_1 < B_m\}$ is the event that every row-1 sample is below every row- m sample. If the largest row-1 sample is below $1/2$, then $X_1 = n$; otherwise the smallest row- m sample exceeds $1/2$ and $X_m = 0$. Either way $X \notin \mathcal{I}$, so $R_m(n)/M_m(n) = \mathbb{P}(\mathcal{A} \cap \{D_1 < B_m\}) \leq \mathbb{P}(\mathcal{A} \cap \mathcal{I}^c) = E_{m,n}(1/2)$. The ratio follows from (18.8), Theorem 14.2 and $M_m(n) \sim \sqrt{m}(2\pi n)^{(1-m)/2} m^{mn}$ (Section 5.5): the powers of m^m cancel, $\sqrt{m}/\sqrt{m/2} = \sqrt{2}$, $(2\pi)^{(1-m)/2 + (m-2)/2} = (2\pi)^{-1/2}$, $3^{1-m} 9^{m-2} = 3^{m-3}$, $J_{m-1}/J_{m-2} = (m-2)!$, $4^{-\binom{m-1}{2} + \binom{m-2}{2}} = 4^{2-m}$, and the power of n is $-\binom{m-1}{2} + \frac{1-m}{2} + \frac{(m-2)^2}{2} = -\frac{2m-3}{2}$. $\square$

At $m = 2$, (c) reads $\binom{2n}{n} E_{2,n}(1/2) \sim \frac{2}{3\sqrt{\pi}} 2^n n^{-1/2}$, since $R_2(n) = 1$.

The rate in (b): an observation (dated 5 October 2026). Part (b) states no rate. The independent check recorded at the head of this subsection observed numerically that the ratio of $E_{m,n}(1/2)$ to the right side of (18.8) behaves like $1 + c_m/n + O(n^{-2})$, with no $n^{-1/2}$ term (consistent with the symmetry $Y \mapsto -Y$): n times the excess tends to about 0.500 at $m = 3$ (to $n = 1600$), to about 0.667 at $m = 4$ (to $n = 400$), and is still settling near 1.42 at $m = 5$ (to $n = 100$). For $m = 2$ the constant is $c_2 = 4/9$, by a short calculation. There are no pairs and $J_1 = 1$, so by (18.7) the ratio is $3\mathbb{E}[\mathbf{1}_{\mathcal{I}'} X'_1/(2n - X'_1)] + O(2^{-n})$. The excluded atom $X'_1 = 0$ has weight 0 and $X'_1 = n$ has probability 2^{-n} . With $t = (2X'_1 - n)/n \in [-1, 1]$,

$$\frac{X'_1}{2n - X'_1} = \frac{1+t}{3-t} = \frac{1}{3} + \frac{4t}{9} + \frac{4t^2}{27} + \frac{4t^3}{81} + O(t^4)$$

uniformly, and $\mathbb{E}t = \mathbb{E}t^3 = 0$, $\mathbb{E}t^2 = 1/n$, $\mathbb{E}t^4 = O(n^{-2})$, so the ratio is $1 + \frac{4}{9n} + O(n^{-2})$. The values $c_3 \approx \frac{1}{2}$ and $c_4 \approx \frac{2}{3}$ are numerical observations only; the expansion to all orders is item (F2) of Section 24.1.

Part (a) is also the first step of an all-order expansion: the expectation in (18.7) is an interior sum of Part I's type with $m-1$ rows and one extra rational factor per row, so the method of Section 5 should expand it to every order in $1/n$. That expansion is not carried out here (Section 24.1).

19 A beta-binomial representation and the leading constant

19.1 A random cut at the first completion

Take m independent rows of n independent uniform random variables on $(0, 1)$ and sort within each row. Their global ranks are uniform among the $M_m(n)$ row interleavings. Let U be the maximum of the top row, and let $\mathcal{S}$ be the event that every entry in the bottom row exceeds U . Its probability is $\binom{2n}{n}^{-1}$. The joint density of U and $\mathcal{S}$ is $nu^{n-1}(1-u)^n$. Consequently

$$U \mid \mathcal{S} \sim \text{Beta}(n, n+1). \quad (19.1)$$

Conditional on $U = u$ and $\mathcal{S}$, the numbers of middle-row samples below u are independent variables

$$X_i \mid U = u \sim \text{Bin}(n, u), \quad 1 \leq i \leq r. \quad (19.2)$$

The X_i are exchangeable but *not* independent after integrating out U .

Set $\mathcal{I} = \{1 \leq X_i \leq n-2 \text{ for all } i\}$ and define

$$W_n(x) = \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)} \prod_{i=1}^r \frac{(n-1-x_i)x_i}{(n-1+x_i)(2n-x_i)}. \quad (19.3)$$

Denominators are positive on $\mathcal{I}$; equal coordinates give zero. Every pair factor and every single-coordinate factor belongs to $[0, 1]$, so $0 \leq W_n \leq 1$ there.

Lemma 19.1 (Exact mixture identity). *For all $m, n \geq 2$,*

$$\frac{H_m(n)}{M_m(n)} = \frac{1}{r! \binom{2n}{n}} \mathbb{E}[\mathbf{1}_{\mathcal{I}} W_n(X)], \quad (19.4)$$

with the beta-binomial law (19.1)–(19.2).

Proof. Given the counts x , the samples below U in the top and middle rows are independent sorted uniforms. Relative to their unconstrained multinomial interleavings, the first hook count contributes

$$\prod_i \frac{n-1-x_i}{n-1+x_i} \prod_{i < j} \frac{x_i - x_j}{x_i + x_j}$$

when $x_1 > \cdots > x_r$. Above U , the complementary hook count similarly contributes

$$\prod_i \frac{x_i}{2n-x_i} \prod_{i < j} \frac{x_i - x_j}{2n-x_i-x_j}.$$

All cross-cut comparisons are automatic for the legal prefix state. The product is $W_n(x)$. Distinct middle counts have exactly one decreasing ordering; the mixture law and the weight are symmetric in those counts. Thus summing over decreasing counts is $1/r!$ times the unrestricted expectation on $\mathcal{I}$. Multiplying by $\mathbb{P}(\mathcal{S})$ proves the identity. This is also a probabilistic derivation of (18.3). $\square$

19.2 Scaling and the common Gaussian mode

Let

$$\varepsilon = n^{-1/2}, \quad Y_i = \frac{2X_i - n}{\sqrt{n}}, \quad V_n = 2\sqrt{n} \left(U - \frac{1}{2}\right), \quad Z_{i,n} = \frac{2(X_i - nU)}{\sqrt{n}}. \quad (19.5)$$

Then $Y_i = Z_{i,n} + V_n$. A direct Laplace expansion of the beta density gives $V_n \Rightarrow V \sim N(0, 1/2)$. Conditional binomial characteristic functions, uniformly for U near $1/2$, give joint convergence

$$(Z_{1,n}, \dots, Z_{r,n}, V_n) \Rightarrow (Z_1, \dots, Z_r, V), \quad (19.6)$$

where the Z_i are independent $N(0, 1)$ variables and are independent of V . For example, expanding $(1 - U + Ue^{2it/\sqrt{n}})^n e^{-2itnU/\sqrt{n}}$ gives the limiting factor $e^{-t^2/2}$ on $U = 1/2 + o(1)$; the beta mass outside that neighborhood tends to zero. The same argument applies to a fixed vector of t_i .

[write] Added 5 October 2026, batch 98: the convergence (19.6) in detail. The two sentences above are a sketch; here is the argument with its uniformity made explicit. The density of U given $\mathcal{S}$ is $f(u) = n \binom{2n}{n} u^{n-1} (1-u)^n$, so for $|v| < \sqrt{n}$ the density of V_n is

$$\sqrt{n} \binom{2n}{n} 4^{-n} \left(1 - \frac{v}{\sqrt{n}}\right) \left(1 - \frac{v^2}{n}\right)^{n-1} \longrightarrow \frac{e^{-v^2}}{\sqrt{\pi}},$$

the $N(0, 1/2)$ density, and Scheffé's lemma gives $V_n \Rightarrow V$. Conditionally on $U = u$ the $Z_{i,n}$ are independent with characteristic function $\varphi_n(t; u) = (\mathbb{E} e^{ih(\xi - u)})^n$, where $h = 2t/\sqrt{n}$ and ξ is Bernoulli(u). The cumulant function $\kappa(h) = \log \mathbb{E} e^{ih(\xi - u)}$ has $\kappa(0) = \kappa'(0) = 0$, $\kappa''(0) = -u(1-u)$ and, since $|\mathbb{E} e^{ih(\xi - u)}| \geq 1 - h^2/8$, a third derivative bounded uniformly in $u \in [0, 1]$ for $|h| \leq 1/2$. Hence $\log \varphi_n(t; u) = -2u(1-u)t^2 + O(|t|^3/\sqrt{n})$ uniformly in u , for $|t| \leq \sqrt{n}/4$. As $|\varphi_n| \leq 1$,

$$\mathbb{E} \exp\left(isV_n + i \sum_i t_i Z_{i,n}\right) = \mathbb{E} \left[e^{isV_n} e^{-2U(1-U) \sum_i t_i^2} \right] + o(1) \longrightarrow \mathbb{E} e^{isV} e^{-\sum_i t_i^2/2},$$

because $2U(1-U) = \frac{1}{2} - V_n^2/(2n)$ and $V_n \Rightarrow V$. This is (19.6).

Let $\Delta(y) = \prod_{i < j} (y_i - y_j)$. Algebraic substitution into (19.3) gives the exact factorization

$$W_n(X) = (4n)^{-d_R} 9^{-r} \Delta(Y)^2 \Psi_r(\varepsilon, Y) \quad \text{on } \mathcal{I}, \quad (19.7)$$

where

$$\begin{aligned} \Psi_r(\varepsilon, y) &= \prod_{i < j} \left(1 - \frac{\varepsilon^2 (y_i + y_j)^2}{4}\right)^{-1} \\ &\quad \times \prod_{i=1}^r \frac{9(1 - 2\varepsilon^2 - \varepsilon y_i)(1 + \varepsilon y_i)}{(3 - 2\varepsilon^2 + \varepsilon y_i)(3 - \varepsilon y_i)}. \end{aligned} \quad (19.8)$$

The factor 9^{-r} has a simple origin: at $x_i \sim n/2$, the interaction with the completed top row and the unborn bottom row tends to $(1/3)(1/3) = 1/9$ for each middle row.

19.3 Gaussian normalization

For independent standard normal Z_i ,

$$\mathbb{E} \Delta(Z)^2 = r! J_r. \quad (19.9)$$

One elementary proof writes the Vandermonde determinant in the basis of monic Hermite polynomials. Their squared norms for the standard normal measure are $j!$; determinant expansion and orthogonality give $r! \prod_{j=0}^{r-1} j!$. The Hermite norms themselves follow from

$$\mathbb{E} \exp(tZ - t^2/2) \exp(sZ - s^2/2) = e^{ts}.$$

Since adding a common scalar does not change a Vandermonde,

$$\Delta(Z_1 + V, \dots, Z_r + V) = \Delta(Z). \quad (19.10)$$

Thus the extra beta fluctuation contributes a common Gaussian shift, but does not alter the leading Vandermonde normalization.

For completeness, moment convergence is justified by uniform tails. The beta density is proportional to $[u(1-u)]^{n-1}(1-u)$, and $u(1-u) = 1/4 - (u - 1/2)^2$. It follows, by comparing the normalizing integral with its central part, that $\mathbb{P}(|V_n| > t) \leq C e^{-c t^2}$ in the relevant range, with harmless changes of constants near the endpoints. Conditional binomial exponential bounds give the same form for $Z_{i,n}$. In particular, all fixed moments of the Y_i are uniformly bounded, and for any fixed $0 < \eta < 1/2$,

$$\mathbb{P}\{\max_i |Y_i| > n^\eta\} \leq C_r e^{-c_r n^{2\eta}}. \quad (19.11)$$

These facts also follow from the exact moment formula in the next section.

[write] Added 5 October 2026, batch 98: explicit constants in (19.11). The phrase “with harmless changes of constants near the endpoints” can be replaced by explicit constants. With $\delta = u - \frac{1}{2}$, $1 - u \leq 1$ and the standard bound $\binom{2n}{n} \leq 4^n / \sqrt{\pi n}$,

$$f(u) \leq n \binom{2n}{n} \left(\frac{1}{4} - \delta^2\right)^{n-1} \leq 4\sqrt{n/\pi} e^{-4(n-1)\delta^2}.$$

Integrating over $|\delta| > s$ and using $(s+v)^2 \geq s^2 + v^2$ gives $\mathbb{P}(|U - \frac{1}{2}| > s) \leq 2\sqrt{n/(n-1)} e^{-4(n-1)s^2}$; with $s = t/(2\sqrt{n})$ and $n \geq 2$, $\mathbb{P}(|V_n| > t) \leq 2\sqrt{2} e^{-t^2/2}$ for all $t \geq 0$. Given $U = u$, Hoeffding’s inequality for the n Bernoulli variables of row i gives $\mathbb{P}(|Z_{i,n}| > t \mid U = u) \leq 2e^{-t^2/2}$, uniformly in u . Since $Y_i = Z_{i,n} + V_n$, it follows that $\mathbb{P}(|Y_i| > t) \leq 5e^{-t^2/8}$ for all $n \geq 2$ and $t \geq 0$. This gives the uniform moment bounds, and (19.11) with $C_r = 5r$ and $c_r = 1/8$, with no exception near the endpoints.

On the central event in (19.11), $\Psi_r \rightarrow 1$ with denominators uniformly separated from zero. Outside it, $W_n \leq 1$ makes its contribution to (19.4) smaller than any power of n^{-1} , even after multiplication by n^{d_R} . Therefore

$$\frac{H_m(n)}{M_m(n)} \sim \frac{J_r}{\binom{2n}{n} (4n)^{d_R} 9^r}. \quad (19.12)$$

Using Stirling’s formula for $M_m(n)$ and $\binom{2n}{n}$ gives (14.6) and the leading term of Theorem 14.2. Finally (18.6) transfers it to $R_m(n)$. The power is

$$\frac{m-1}{2} - \frac{1}{2} + d_R = \frac{r}{2} + \binom{r}{2} = \frac{r^2}{2},$$

which explains the drop from the total-count exponent.

20 All-order coefficients and the universal first correction

20.1 Exact factorial moments of the mixture

For nonnegative integers $k_1, \dots, k_r$, let $K = k_1 + \dots + k_r$. Conditional binomial factorial moments and the beta integral give

$$\mathbb{E} \prod_{i=1}^r X_i^{k_i} = \left(\prod_{i=1}^r n^{\underline{k_i}} \right) \frac{(n)^{\overline{K}}}{(2n+1)^{\overline{K}}}. \quad (20.1)$$

Here falling and rising factorials are distinguished typographically. Every joint ordinary moment of X is a finite rational function of n , by the Stirling-number identity $x^a = \sum_k \left\{ \begin{smallmatrix} a \\ k \end{smallmatrix} \right\} x^{\underline{k}}$. Therefore every joint moment of Y has an effective Laurent expansion in $\varepsilon = n^{-1/2}$. For a monomial of total degree A , only powers of ε of parity A occur. The bounded-moment argument above ensures that no negative powers survive.

Let $\mathcal{E}_\varepsilon$ be the linear functional on polynomials in $y_1, \dots, y_r$ obtained by replacing each monomial by this formal moment expansion. Set

$$Q_r(z) = \frac{1}{r! J_r} \mathcal{E}_\varepsilon \left[\Delta(y)^2 \Psi_r(\varepsilon, y) \right] \Big|_{\varepsilon^2=z}, \quad Q_r(0) = 1. \quad (20.2)$$

This is interpreted coefficient by coefficient, not as convergence of a formal infinite series. The symmetry $\Psi_r(-\varepsilon, -y) = \Psi_r(\varepsilon, y)$ shows that its coefficient of ε^k has polynomial parity k . Since Δ^2 has even total degree, the moment parity rule shows that the right-hand side of (20.2) contains only even powers of ε . Thus Q_r is a well-defined series in z with rational coefficients.

The remaining Stirling factor is

$$A_m(z) = \exp \left\{ \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} \left(m^{1-2k} - m + 2 - 2^{1-2k} \right) z^{2k-1} \right\}, \quad (20.3)$$

where B_{2k} are Bernoulli numbers. We obtain the explicit all-order rule

$$b_{m,j}^R = [z^j] A_m(z) Q_{m-2}(z). \quad (20.4)$$

For a prescribed j , all operations are finite: expand Ψ through ε^{2j} , take finitely many moments from (20.1), and truncate (20.3) at z^j .

20.2 Why the formal calculation is an asymptotic theorem

Completion of the all-order part of Theorem 14.2. Fix L . On $\max |Y_i| \leq n^\eta$, choose any $0 < \eta < 1/2$. Each rational factor of (19.8) has denominator bounded away from zero for large n . Taylor expansion through order $2L+1$ has a remainder bounded by

$$C_{r,L} \varepsilon^{2L+2} (1 + \|Y\|)^{N_{r,L}}$$

there, for some fixed exponent $N_{r,L}$. This follows either by finite geometric-series remainders in the displayed denominators, or by Taylor's formula on the line segment from 0 to ε ; throughout that segment $|\varepsilon Y_i|$ tends uniformly to zero. After multiplication by Δ^2 , uniform bounded moments bound the expected remainder by $O_{r,L}(n^{-L-1})$.

The complement of the central event contributes exponentially little to the original weighted expectation, by $0 \leq W_n \leq 1$ and (19.11). It also contributes exponentially little to each polynomial Taylor term, by the

same tail bound and higher bounded moments. We may consequently replace each truncated polynomial expectation by its full exact expectation. Formula (20.1) expands these rational functions to the desired order with an ordinary rational remainder estimate. The parity argument removes half-integral orders. This proves the Poincaré expansion specified by Q_r .

Stirling's formula with its fixed-order remainder yields A_m and (20.4). Finally

$$\frac{M_m(n)/\binom{3n}{n}}{M_m(n)/\{\binom{2n}{n}n^{d_R}\}} = n^{d_R} \frac{\binom{2n}{n}}{\binom{3n}{n}} = O_r\left(n^{d_R}(16/27)^n\right).$$

Thus the difference between R_m and H_m affects no coefficient of the rare-scale inverse-power expansion. All coefficients are rational by construction, and the case $r = 0$ is the exact identity already noted. $\square$

The theorem asserts effective coefficients and existence of fixed-order error constants. The package does not supply optimized numerical values for these constants or a rigorously certified onset of the asymptotic regime.

20.3 An exact Vandermonde identity

It is useful to derive the first correction without a large polynomial expansion. For $X \sim \text{Bin}(n, p)$ consider

$$F_X(t) = (1 + (1 - p)t)^x (1 - pt)^{n-x}. \quad (20.5)$$

The polynomial $P_j(x) = j![t^j]F_X(t)$ is monic of degree j . Direct binomial expectation gives

$$\mathbb{E}F_X(t)F_X(s) = (1 + p(1 - p)ts)^n. \quad (20.6)$$

Hence the P_j are orthogonal and have squared norms $j!n^j[p(1 - p)]^j$. Replacing Vandermonde columns by these monic polynomials and expanding the determinant gives, for independent binomial variables conditional on $U = p$,

$$\mathbb{E}[\Delta(Y)^2 \mid U = p] = r!J_r \left(\prod_{j=0}^{r-1} \frac{n^j}{n^j} \right) [4p(1 - p)]^{d_R}. \quad (20.7)$$

Integrating the beta law,

$$\mathbb{E}[U^{d_R}(1 - U)^{d_R}] = \frac{(n)^{\overline{d_R}}(n + 1)^{\overline{d_R}}}{(2n + 1)^{2\overline{d_R}}}. \quad (20.8)$$

It follows by expanding these finite products that

$$\frac{\mathbb{E}\Delta(Y)^2}{r!J_r} = 1 - \frac{r(r - 1)(2r - 1)}{12n} + O_r(n^{-2}). \quad (20.9)$$

This identity isolates the correction caused by finite binomial and beta moments before inserting the rational interaction factors.

20.4 Evaluation of the interaction correction

Write $S_1 = \sum_i y_i$ and $S_2 = \sum_i y_i^2$. Expanding (19.8) begins with

$$\Psi_r(\varepsilon, y) = 1 + \varepsilon^2 P_1(y) + O(\varepsilon^3), \quad P_1(y) = \frac{9r - 50}{36} S_2 + \frac{1}{4} S_1^2 - \frac{4r}{3}. \quad (20.10)$$

The $O(\varepsilon^3)$ notation here denotes a formal polynomial expansion; the previous proof supplies its expectation-level meaning.

Under the Gaussian law tilted by $\Delta(Z)^2$, two elementary identities are

$$\mathbb{E}_* \sum_i Z_i^2 = r^2, \quad \mathbb{E}_* \left(\sum_i Z_i \right)^2 = r. \quad (20.11)$$

For the first, scale the Gaussian integral: the degree of Δ^2 is $r(r-1)$ and the volume dimension is r , so differentiation in the Gaussian variance gives r^2 . For the second, separate the center of mass from its orthogonal complement. The Vandermonde is independent of the center of mass, whose sum coordinate is $N(0, r)$. Since $Y_i = Z_i + V$ and $\text{Var } V = 1/2$, with the tilt still independent of V ,

$$\mathbb{E}_* S_2 = r^2 + r/2, \quad \mathbb{E}_* S_1^2 = r + r^2/2. \quad (20.12)$$

Substitution into (20.10) yields

$$\mathbb{E}_* P_1 = \frac{r(9r^2 - 41r - 64)}{36}. \quad (20.13)$$

Combining with (20.9), the first coefficient of Q_r is

$$[z]Q_r(z) = \frac{r(3r^2 - 32r - 67)}{36}. \quad (20.14)$$

The first coefficient of A_m is

$$[z]A_m(z) = \frac{1}{8} + \frac{1/m - m}{12}. \quad (20.15)$$

Adding (20.14) and (20.15), with $m = r + 2$, gives (14.7). This proves the universal first correction independently of the higher-order implementation.

21 Height five, conditional shapes, and exponential separation

21.1 Three corrections for the A181199 rare subcount

Applying (20.4) at $m = 5$ gives

Corollary 21.1 (Height-five rare expansion). *For $R_5(n) = \#\{A : a_{1,n} < a_{5,1}\}$,*

$$\begin{aligned} R_5(n) = & \frac{\sqrt{5}}{93312\pi^{3/2}} \left(\frac{3125}{4} \right)^n n^{-9/2} \\ & \times \left(1 - \frac{1393}{120n} + \frac{656449}{28800n^2} + \frac{9833844689}{155520000n^3} + O(n^{-4}) \right). \end{aligned} \quad (21.1)$$

The rational coefficients are finite evaluations of the proven formula, not a fit to sequence values. Appendix D describes the implementation-independent arithmetic used to reproduce them. For comparison, the first three corrections at adjacent heights are

m	$b_{m,1}^R$	$b_{m,2}^R$	$b_{m,3}^R$
2	0	0	0
3	$-199/72$	$-6479/10368$	$-491815/2239488$
4	$-979/144$	$74083/13824$	$25693405/1990656$
5	$-1393/120$	$656449/28800$	$9833844689/155520000$

Here R_5 is a newly considered refinement of the A181199 counting problem; it is not assigned a new OEIS identifier in this report.

21.2 How rare is the absence of a common active phase?

Let $Q_m(n) = T_{m,B^m D^m}(n)$ for $n \geq 2$. Then exactly

$$T_m(n) = Q_m(n) + R_m(n). \quad (21.2)$$

Both Q_m and R_m separately have rational-diagonal generating functions: each is a fixed finite sum of the event sectors in Theorem 14.1, up to the harmless terms at $n = 0, 1$. Using (14.2) from the preceding report gives

Corollary 21.2 (Rare-event probability). *For fixed $m \geq 2$, a uniformly chosen shifted rectangle satisfies*

$$\mathbb{P}(D_1 < B_m) = \frac{R_m(n)}{T_m(n)} \sim \frac{\sqrt{\pi} 4^{2m-3}}{9^{m-2}(m-2)!(m-1)!} n^{2m-5/2} 4^{-n}. \quad (21.3)$$

At height five, the first correction is

$$\frac{R_5(n)}{T_5(n)} = \frac{1024\sqrt{\pi}}{6561} n^{15/2} 4^{-n} \left(1 - \frac{509}{24n} + O(n^{-2}) \right). \quad (21.4)$$

Proof. Divide the leading constants and powers in (14.5) and (14.2). The factorial quotient is $J_{m-2}/J_m = 1/((m-2)!(m-1)!)$, and the powers of 4 give 4^{2m-3} . For the correction at height five, the preceding report gives $b_{5,1}^{\text{total}} = 48/5$, so subtraction gives $-1393/120 - 48/5 = -509/24$. $\square$

Thus Q_m has the same dominant-scale inverse-power expansion as T_m to every fixed order, yet their difference has the explicitly identified exponential scale $(m^m/4)^n$. This is a combinatorially defined, beyond-all-orders refinement. It does *not* by itself identify a Stokes multiplier of a chosen analytic interpolation or summation of the dominant formal series. There may be other exponentially small structures inside Q_m , and the next boundary sector remains to be analyzed.

21.3 The middle shape conditioned on the rare event

The same argument gives a probabilistic limit theorem. Use the auxiliary uniform-sample model of Section 19. Condition on a legal array and on $D_1 < B_m$. Let X_i count middle-row entries below the top-row maximum U , and use the scaling (19.5).

Theorem 21.3 (Rare conditional shape). *For fixed $m \geq 3$, conditional on the rare event, the probability that all middle rows are active at the first completion tends to one. On that core event, the limiting joint law is*

$$(Y_1, \dots, Y_r, V_n) \Rightarrow (\Lambda_1 + V, \dots, \Lambda_r + V, V), \quad (21.5)$$

where $V \sim N(0, 1/2)$ is independent of $\Lambda_1 > \dots > \Lambda_r$, whose density on the decreasing chamber is

$$\frac{1}{J_r} \Delta(\lambda)^2 \prod_{i=1}^r \frac{e^{-\lambda_i^2/2}}{\sqrt{2\pi}}. \quad (21.6)$$

All fixed polynomial moments converge as well.

Proof. The first statement is (18.6) divided by the asymptotic for R_m . In the mixture representation, conditioning on legality and the core multiplies the joint law by $W_n(X)$ and restricts the counts to the decreasing chamber. After normalization, (19.7), (19.6), and the tail estimates show that the limiting weight is $\Delta(Z + V)^2 = \Delta(Z)^2$. It is independent of V , so the common Gaussian mode is not tilted. Restricting the Z coordinates to decreasing order changes the normalization from $r!J_r$ to J_r . The tail estimates with additional polynomial factors give moment convergence. The omitted noncore configurations have exponentially small conditional probability; their scaled coordinates are bounded by $O(\sqrt{n})$, so they do not affect any fixed moment. $\square$

The auxiliary U is a uniform-order-statistic realization of the combinatorial cut. The theorem is not a claim that a tableau itself contains real-valued sample times. In particular, its statement about Y_i , the scaled integer row lengths, is intrinsic; the joint statement with U uses the specified coupling.

22 Inverting the rare-count asymptotic profile

For $m \geq 3$ put

$$\lambda_m = m \log m - \log 4 > 0, \quad \beta_m = \frac{(m-2)^2}{2} > 0, \quad a = \log(y/\mathcal{K}_m). \quad (22.1)$$

The leading continuous profile is $y = \mathcal{K}_m e^{\lambda_m x} x^{-\beta_m}$. Its large solution is

$$x_0 = -\frac{\beta_m}{\lambda_m} W_{-1} \left(-\frac{\lambda_m}{\beta_m} e^{-a/\beta_m} \right), \quad (22.2)$$

where W_{-1} is the real lower branch. The argument belongs to $(-1/e, 0)$ for sufficiently large y . The branch is forced by the requirement $x_0 \rightarrow \infty$; the principal branch gives the small solution instead. Lambert-core inversion and inverse-power reversion are standard tools, also used for the total sequence in Part I (Section 10). Their application here uses the different rare-event rate and power.

Proposition 22.1 (Two corrections to the inverse profile). *Write $b_1 = b_{m,1}^R$, $b_2 = b_{m,2}^R$, $\lambda = \lambda_m$, and $\beta = \beta_m$. The formal large inverse has*

$$x = x_0 - \frac{b_1}{\lambda x_0} - \frac{1}{x_0^2} \left(\frac{b_2 - b_1^2/2}{\lambda} + \frac{\beta b_1}{\lambda^2} \right) + O(x_0^{-3}). \quad (22.3)$$

Every subsequent coefficient is obtained recursively from (20.4). For any fixed sufficiently long truncation of the real asymptotic profile, its genuine large real inverse has the corresponding finite expansion.

Proof. Take logarithms:

$$\lambda x - \beta \log x + \log \left(1 + b_1/x + b_2/x^2 + \cdots \right) = a.$$

Substitute $x = x_0 + c_1/x_0 + c_2/x_0^2 + \cdots$ and use $\lambda x_0 - \beta \log x_0 = a$. At orders x_0^{-1} and x_0^{-2} ,

$$\lambda c_1 + b_1 = 0, \quad \lambda c_2 - \beta c_1 + b_2 - b_1^2/2 = 0.$$

This gives (22.3). At every later order the next unknown coefficient occurs linearly with nonzero coefficient λ . For a fixed real profile truncation, its logarithmic derivative tends to $\lambda > 0$. Taylor's theorem or the real implicit function theorem then turns the recursive formal truncation into the stated inverse asymptotic. $\square$

For height five, $b_2 - b_1^2/2 = -535/12$. Thus, with $\lambda = \log(3125/4)$ and $\beta = 9/2$,

$$x = x_0 + \frac{1393}{120\lambda x_0} + \frac{1}{x_0^2} \left(\frac{535}{12\lambda} + \frac{4179}{80\lambda^2} \right) + O(x_0^{-3}). \quad (22.4)$$

The leading logarithmic scale can also be read without W :

$$x_0 = \frac{1}{\lambda} \left(a + \beta \log a - \beta \log \lambda + O\left(\frac{\log a}{a}\right) \right).$$

22.1 A discrete inverse is a staircase

The sequence $R_m(n)$ is eventually strictly increasing for $m \geq 3$, since its successive ratio tends to $e^{\lambda_m} > 1$. Define $N_m(y) = \min\{n \geq 1 : R_m(n) \geq y\}$ for large y . Let $\xi_L(y)$ be the large inverse of the truncated continuous profile

$$\mathcal{K}_m e^{\lambda_m x} x^{-\beta_m} \sum_{j=0}^L b_{m,j}^R x^{-j}.$$

The fixed-order remainder and a logarithmic derivative bounded away from zero imply that, for some $C_{m,L}$ and all sufficiently large y ,

$$\left[\xi_L(y) - C_{m,L} \xi_L(y)^{-L-1} \right] \leq N_m(y) \leq \left[\xi_L(y) + C_{m,L} \xi_L(y)^{-L-1} \right]. \quad (22.5)$$

Indeed, the asymptotic relative error is $O(n^{-L-1})$, so the two smooth comparison profiles cross the target within that horizontal distance. When the two displayed ceilings coincide, their common value is the threshold. No such agreement can be asserted uniformly near integer crossings without explicit error constants and distance information. The package's numerical inverse checks are therefore profile diagnostics, not integer-threshold certificates.

[write] **Added 5 October 2026, batch 98: the inversion is an instance.** As for Part I's (10.3) (note at the end of Section 10), (22.2) is a case of Theorem J.23 (p0:thm:lambert-core) of the transseries volume Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/, with $a = \lambda_m > 0$ and $b = -\beta_m < 0$, whose branch rule for $b < 0$ selects W_{-1} ; the caution about integer thresholds and (22.5) are the separation condition of that volume's staircase theorem J.43 (p0:thm:staircase). The manuscript itself calls these tools standard. Two steps are argued in one sentence each and are recorded in Section 24.1: the last sentence of Proposition 22.1 (that the genuine real inverse has the formal expansion) and the staircase (22.5), whose constant $C_{m,L}$ and range of validity are existential.

23 Exact checks, numerical diagnostics, and reproducibility

23.1 Independent computations

The package includes three genuinely different finite enumerations: unit-step prefix-state dynamic programming; an independent bitmask order-ideal enumerator built directly from the cell comparisons; and an event-skeleton evaluator using the signed multinomial kernel. The hook core sum and the symbolic beta-moment expansion are separate again. No candidate OEIS recurrence is used to generate a test value.

The verification script completed **1,954 exact checks**. The categories and counts are recorded in data/verification.json:

Check	Number
Independent poset enumeration versus prefix states	28
Reflection kernel versus independent chamber walks	887
Event decomposition versus prefix states	25
Event-word realizability and rotation symmetry	640
Rare event-word sum and the worked example	21
OEIS reference values (heights four and five)	20
Core lower bound, separation bound, and core error bound	228
Classical Catalan and small-width controls	64
Universal first correction and saved higher coefficients	7
Common-mode Gaussian moments and exact beta moments	34
Total	1954

The reference terms are small data excerpts from A181198 and A181199, with source URLs and access date in the audit. The tests regenerate these values rather than assuming their recurrence. Finite checks corroborate the arguments; they are not proofs of an infinite identity or a substitute for peer review.

23.2 Accuracy on the rare exponential scale

Let $\mathcal{L}_5(n)$ denote the leading term in (21.1), and put $\mathcal{A}_{5,L}(n) = \mathcal{L}_5(n) \sum_{j=0}^L b_{5,j}^R n^{-j}$. The following entries are relative errors $\mathcal{A}_{5,L}(n)/R_5(n) - 1$ computed against exact integer rare counts.

n	$L = 0$	$L = 1$	$L = 2$	$L = 3$
10	4.46198	-1.87847	-0.633499	-0.288127
20	1.05994	-0.135683	-0.0183006	-0.00201881
40	0.379221	-0.0210405	-0.00139232	-0.0000296524
60	0.229776	-0.00815163	-0.000365317	-0.00000531141

These values illustrate both the usefulness of the corrections and the slow onset of the leading asymptotic at height five. The negative first-order approximation at $n = 10$ is not a contradiction: fixed-order asymptotic expansions are not guaranteed positive or accurate at small n .

The core already captures the rare count particularly well at moderate width:

n	$R_5(n)/\mathcal{L}_5(n)$	$H_5(n)/R_5(n)$
10	0.183083833013	0.862684900894
20	0.485450790529	0.999319561936
40	0.725047021487	0.999999971834
60	0.813156328776	0.99999999998899

The underlying integer values and 70-digit calculations are provided in `data/numerics.csv`. Displayed decimals are diagnostics, not interval enclosures.

23.3 Reproducing the package

From the package root, a Python environment with SymPy and mpmath can run

```
python code/verify.py
```

```
python code/coefficients.py --height 5 --order 3
python code/numerics.py
python code/inverse_profile.py --index 60 --height 5
```

The first command reruns the exact tests and writes their log. The second regenerates the three height-five rational corrections. The third regenerates the decimal comparison tables from exact rare counts. The fourth checks the inverse profile at a target obtained from an exact rare count; it is not a threshold-certification procedure. The tested environment is Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. The standard-library enumerators themselves do not require either symbolic or numerical package. No Maple diagonal-elimination computation, Lean proof, or Rocq proof is claimed.

[write] Added 5 October 2026, batch 98: the shipped layout. “The package” in this section is the delivered archive. In ProveIt its files carry the prefix 02-boundary-phases-:

Delivered as	Shipped as
<code>code/name.py</code> (five scripts)	<code>code/02-boundary-phases-name.py</code>
<code>build.sh</code>	<code>code/02-boundary-phases-build.sh</code>
<code>data/name</code> (nine records)	<code>data/02-boundary-phases-name</code>
<code>requirements.txt</code>	<code>data/02-boundary-phases-requirements.txt</code>
<code>notes/PROPOSED_OEIS_NOTE.md</code>	<code>data/02-boundary-phases-notes-PROPOSED_OEIS_NOTE.md</code>
<code>notes/SOURCES_AND_STATUS.md</code>	<code>02-boundary-phases-notes-SOURCES_AND_STATUS.md</code>

The scripts import one another and read and write `data/` under their delivered names (`verify.py` imports `tableaux`, and every script sets its root to the parent of `code/`), and `build.sh` changes to its own directory; so they run only on a copy with the delivered names restored, never in place, and the README of the report gives the commands. At intake (5 October 2026, on such a copy; Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0) `verify.py` passed all 1,954 checks in 66 s (the package recorded 23 s), `numerics.py`, `coefficients.py -height 5 -order 3` and `inverse_profile.py -index 60 -height 5` reproduced the shipped records after removal of carriage returns, and the only differences were the timing fields of `verification.json`, `verification.txt` and `numerics.json`. The check table above merges some of the script’s sixteen categories (its $640 = 320 + 320$, $21 = 20 + 1$, $228 = 3 \cdot 76$, $7 = 4 + 3$ and $34 = 6 + 28$); the total agrees.

`code/02-boundary-phases-verify.py` embeds the first ten terms of A181198 and A181199 as reference values. They are OEIS content, published by The OEIS Foundation Inc. under the Creative Commons Attribution-ShareAlike 4.0 licence, so those values are third-party data under that licence, not MIT-0. The draft `data/02-boundary-phases-notes-PROPOSED_OEIS_NOTE.md` is headed “NOT SUBMITTED” and stays so: nothing was submitted to OEIS by the package’s author or by the intake, and its instruction not to remove the conjectural status of the displayed recurrences stands.

[write] Added 5 October 2026, batch 101: the OEIS draft. The draft’s instruction “Do not remove the conjectural status of the displayed specific recurrences or finite-sum formulas on the basis of this result alone” was right for this manuscript, which proves no recurrence, and the shipped file stays byte-identical. Since 5 October 2026 Parts III and IV prove those recurrences and finite sums. The draft has not been updated, nothing has been submitted to OEIS, and any change to the OEIS entries is for a human editor after review.

24 Further research questions and concrete next steps

Research question 8 (Certify the particular low-order recurrences). Convert the event sum at heights four and five to a reduced rational integral, compute a telescoper, and compare it with the conjectured OEIS operators. A successful certificate must also control singular initial indices. This would settle a stronger statement than the existence theorem proved here and could resolve Conjectures 18 and 19 of [9].

[write] Added 5 October 2026, batch 101: Research question 8 answered. Conjectures 18 and 19 of [9] and both OEIS operators are proved in Parts [III](#) and [IV](#), with the singular initial indices controlled: the operators hold for $n \geq 1$, and their residuals at $n = 0$ under $a_0 = 1$ are -2160 and 5621993879040000 . The route is not the one proposed here: no reduced rational integral of the event sum and no telescoper, but Pfaffian identities for the transfer integral, one or two polynomial periods, and explicit rational certificates (a divergence certificate at height four, contiguity and a moment closure at height five). A compact certificate from the event sum or the order polytope remains of interest (Research question 17).

Research question 9 (Reduce the dimension of the exact formula). The bound $m(m - 1)$ is deliberately structural rather than optimal. Can adjacent chamber kernels be convolved before summation, or can a Pfaffian/determinantal identity absorb the ordered births and deaths? A substantially smaller representation would make the constructive D-finiteness theorem computationally practical at larger heights.

Research question 10 (Determine deeper boundary sectors). The error in (18.6) contains arrays for which two or more bottom rows are unborn at the first completion. Determine its sharp rate, constant, and all-order expansion, then stratify by the number of unborn rows and early completions. The crude multinomial bound supplies an upper rate, not the next asymptotic term.

Research question 11 (Find minimal differential complexity). How do the minimal recurrence order, polynomial degree, and minimal rational-diagonal dimension grow with m ? Does the event decomposition impose a useful uniform bound much smaller than a general elimination bound? The conjectured small orders at heights four and five suggest substantial cancellation that the existence proof does not explain.

[write] Added 5 October 2026, batch 101: Research question 11 advanced. At $m = 4$ the minimal order of a recurrence is two, and among recurrences of order two the minimal degree is nine (Proposition 33.2); at $m = 5$ the minimal order is two or three (Proposition 42.2). The small orders are explained by the Pfaffian reductions of Parts [III](#) and [IV](#), which leave one period at height four and two at height five. Open: the degrees over all orders, order two against three at $m = 5$ (items (F5) and (F9)), and the growth with m . (Added 5 October 2026, after the intake's independent check: at $m = 4$ the smallest degree is eight at order three and seven at orders four to eight, Proposition 33.3; at $m = 5$ there are recurrences of order four and degree 18 and of order five and degree 16, Proposition 42.3. The degrees at higher orders stay open.)

Research question 12 (Describe event-rank limit laws). Corollary 17.2 gives exact D-finite moment totals. Use these or the beta representation to derive joint laws for B_i, D_i , both unconditionally and under $D_1 < B_m$. Determine whether the one-time conditional shape theorem extends to a process of interacting paths near the first completion.

Research question 13 (Make height growth uniform). The proofs fix m . Determine ranges $m = m(n)$ in which the Gaussian reduction and rare-sector asymptotics remain uniform. The dimensions of the Vandermonde and the interaction product grow quadratically, so fixed height error estimates cannot simply be reused. Identify the crossover between a probable common active phase and a boundary-dominated regime.

Research question 14 (Obtain certified finite-width bounds and inversion). Replace the existential remainders in (14.5) by computable, useful bounds with explicit onset. Combine them with (22.5) to certify integer thresholds, including inputs very close to an integer crossing. This requires more than high-precision floating-point evaluation of the asymptotic series.

Research question 15 (Generalize and formalize the finite-event method). Investigate unequal row lengths and other fixed-offset antidiagonal constraints whose active states may still be described by a finite collection of reflection chambers. Separately, formalize the prefix-state bijection, reflection involution, and binomial-sum encoding in a proof assistant. Each is a finite combinatorial component with a clear interface; a formal proof of the analytic rare-event expansion is a larger second stage.

[write] Added 5 October 2026, batch 98: the questions of Part II and Part I. The eight questions above are the manuscript’s Research questions 1–8, numbered 8–15 here. Question 8 refines Part I’s Research question 1 (a telescoper route from the event sum, and Conjectures 18–19 of [9]); Question 11 is the open second half of Part I’s Research question 7; Questions 12, 13 and 14 continue Research questions 5, 3 and 6 of Part I.

24.1 Further questions and research recorded at the write

Added 5 October 2026, batch 98, under Vladimir Reshetnikov’s standing rule that unproved claims are recorded as open questions rather than dropped. Each item names its source, the argument it came with, and what is missing.

- (F1) *Other exponentially small structures* (the manuscript, after Corollary 21.2). The text says there “may be other exponentially small structures inside Q_m , and the next boundary sector remains to be analyzed”. No argument is offered either way; Proposition 18.3 shows that the boundary error of Part I’s central cut has rate 2^{-n} , far above the rare rate, but that error is an artefact of the cut, not a sector of T_m . Missing: a classification of the event sectors of Theorem 16.1 by exponential rate.
- (F2) *All orders for the boundary error* (the write; Research question 4 of Part I, re-scoped). Proposition 18.3 gives the leading term of $E_{m,n}(1/2)$. Sketch: by (18.7) the leading stratum is an interior sum with $m - 1$ independent binomial rows and the extra factor $\prod_i (1 + \varepsilon Y_i)/(3 - \varepsilon Y_i)$, whose odd powers of ε cancel in expectation by the symmetry $Y \mapsto -Y$; Section 5’s moment polynomials and Taylor control should give an expansion in powers of $1/n$. Missing: writing that argument, computing its coefficients, and the analogous analysis of the $O(4^{-n})$ strata $D_{m,n}$ and of how the interior sum and the boundary terms combine.
- (F3) *Sketch-level steps of Sections 19, 21 and 22*. The joint convergence (19.6) and the tail bound (19.11) were sketches; the write’s notes after them supply complete arguments. Still argued only in outline: the moment convergence in Theorem 21.3 (“the tail estimates with additional polynomial factors give moment convergence”; with the explicit tails it reduces to uniform integrability of $|Y|^k W_n(X)/\mathbb{E}W_n(X)$, which needs $W_n \leq Cn^{-d_R} \Delta(Y)^2$ on the central event and the exponential tails off it); the last sentence of Proposition 22.1 (the genuine real inverse has the formal expansion: a real implicit-function argument, as in Theorem J.23 of the transseries volume); and the staircase (22.5), whose constant $C_{m,L}$ and threshold for “sufficiently large y ” are existential (Research question 14 asks for explicit ones).
- (F4) *Priority of Theorem 14.1* (the manuscript and its source audit). The audit searched the OEIS entries, Kauers–Koutschan, Bostan–Lairez–Salvy, Sun and the web for the sequence identifiers with “proved”, “D-finite”, “shifted rectangles” and “diagonal”, and found no earlier proof; it did not search the general literature on D-finiteness of linear-extension counts of fixed-height posets or of walks in Weyl chambers, where the theorem might be a known special case. Missing: that search.

[write] Added 5 October 2026, batch 101: item (F4). Report 91, dated 2 October 2026 and printed in Part V, proves Theorem 14.1 by the same route (Theorem 44.1): Dyck words of births and saturations, reflection determinants, everywhere-defined binomial products, and Theorem 3.5 and Corollary 3.6 of Bostan, Lairez and Salvy. Neither manuscript knew the other. Within this report the earlier proof by date line is therefore Report 91’s (Section 45); the search of the general literature that (F4) asks for is still missing.

25 Conclusion

The obstruction in the shifted-rectangle model is not the strict interior walk but the coexistence of births, completions, and two kinds of boundary ties. Recording those events turns the exact enumeration into a finite binomial construction. That proves rational diagonality and D-finiteness for every fixed height without a

guessed recurrence.

The same boundaries also carry information invisible to the dominant inverse-power expansion. The absence of a common active phase has rate 4^{-n} relative to the main sequence, an explicit polynomial multiplier, and its own all-order expansion. A random first-completion cut exposes both the factor 9^{-r} and a Gaussian common mode independent of the Vandermonde repulsion. These statements are precise additions to the preceding asymptotic report. They leave clear next targets: compact operators, the specific OEIS recurrence certificates, and deeper exponential boundary strata.

Part III

A proof of the height four shifted rectangle conjecture

Exact enumeration for A181198 and Kauers and Koutschan Conjecture 18

Research report 231, prepared for Vladimir Reshetnikov, October 5, 2026

Main conclusion. The full height-four array count satisfies the specific second-order recurrence recorded in OEIS A181198 and the finite sum printed as Conjecture 18 of Kauers and Koutschan (2023). The proof starts with the literal array inequalities, reduces their order-polytope volume to a weighted triangle integral, and supplies an explicit polynomial divergence certificate with all boundary terms.

Abstract. Column interlacing gives an exterior Volterra transfer. An elementary four-variable Pfaffian calculation reduces the entire count to two kernel traces. A rank-two perturbation and integrations by parts express those traces in one weighted period. A six-coefficient polynomial vector field then proves a first-order inhomogeneous recurrence. Exact elimination gives the prescribed OEIS operator; variation of constants gives the actual printed finite sum, including its initial value. We also derive rational all-fixed-order asymptotic coefficients directly from the now-proved recurrence, with effective contraction remainder bounds. Finite Lambert- W_{-1} models yield rigorously qualified inverse-index envelopes. The leading asymptotic and existence of all-order expansions were already established in the earlier fixed-height report and are credited as such.

Status and limits. AI-assisted research manuscript, not a peer-reviewed or proof-assistant formalization. Exact verification code is supplied. The proof concerns height four only; Conjecture 19 for height five remains open here. Historical priority is not certified. No canonical exponentially small transseries is asserted.

[write] Added 5 October 2026, batch 101: provenance of Part III. Part III is a single manuscript, printed in full: Research Report 231 of the session bundle that arrived in commit `60f54ea06` and was placed in commit `f7c612c72` (batch 101 of ProveIt’s incoming-reports intake) as an addition to this report. Its title is *A proof of the height four shifted rectangle conjecture*, subtitled “Exact enumeration for A181198 and Kauers and Koutschan Conjecture 18”, dated October 5, 2026 (a 25-page PDF; a main file and eight section files, 1,518 lines with 108 labels). It arrived as the archive `Report231.zip` (514,097 bytes, 20 files under the doubled wrapper directory `Report231/Report231/`). Its title page reads “Prepared for Vladimir Reshetnikov” and its PDF metadata name “Research report prepared for Vladimir Reshetnikov”; no human author and no tool are named. The “Status and limits” paragraph above (“Conjecture 19 for height five remains open here”) describes the manuscript; Part IV proves Conjecture 19.

Pin. The manuscript names no ProveIt commit. It pins this report’s source by its Git blob `2b40ded9daa6549f786a52b880d5477de3b5adfc`, which is `article.tex` as committed in `6fef5383b` (3 October 2026, the batch-85 write) and unchanged until `b71545625` added Part II on 5 October 2026. So it read Part I alone, with the numbers printed here. It also inspected a 24-page “user-held copy” of the manuscript that is now Part II, which it calls “this private companion” and cites for overlap disclosure only, not as a mathematical dependency; that manuscript is public as Part II.

Files. Its four programs, its receipt, its code README and its source record are shipped with the prefix `03-height-four-`, in `code/`, in `data/` and at the report root; the note at the end of Section 33.3 lists them. Its $\text{\LaTeX}$ source, delivered README, PDF and `MANIFEST.sha256` (19 entries, all verified at placement) are not shipped and survive in the arrival commit’s archive. The manuscript is unrefereed and AI-assisted; its proofs are ordinary mathematical proofs, nothing in it is formalized in Lean or Rocq, and its place in this report confers no formal status.

What it answers. Theorems 26.1, 30.1 and 30.2 prove, from the array definition and for every $n \geq 1$ (the finite sum for $n > 1$), the order-two, degree-nine recurrence displayed as conjectural in OEIS A181198 and the finite sum printed as Conjecture 18 of Kauers and Koutschan [9]. This answers the height-four half of Research question 1 of Part I and of Research question 8 of Part II; Part IV answers the height-five half. Dated notes at both questions, after Remark 8.1, at Research questions 6, 7 and 11, and after Remark 16.2 say so. The manuscript also shows that the operator does not extend to $n = 0$ under $a_0 = 1$ (residual -2160). The write adds Section 33.5: the operator has minimal order, and every order-two recurrence for A181198 is a polynomial multiple of it.

Overlap, credited. Section 31 recovers, from the recurrence, the case $m = 4$ of Part I's Theorem 2.1, as the manuscript says: its c_j are $\frac{3}{1024}c_{4,j}$ ($J_4/4^6 = 3/1024$), its d_j are $b_{4,j}$, and $d_1, \dots, d_5$ equal the entries of `data/coefficients_m1_to_m6.json`. The sixth coefficient $d_6 = -79147528275/268435456$ is new to the repository; at placement, Part I's engine code/`coefficients.py` run at order six on a copy gave the same $b_{4,6}$, so two independent routes agree. Its C_* , β_* and λ are K_4 , α_4 and $4 \log 4$. The effective error constants of Sections 31 and 32 are new; the finite Lambert model and the two-ceiling envelope use the apparatus of Section 10, Section 22 and Theorems J.23 and J.43 of the transseries volume (see the note at the end of Section 10.2). The transfer, the rank-two identity and the scalar integrals of Sections 27–28 are re-derived independently in Part IV, and so is the period recurrence, Theorem 29.1 (Remark 42.1). The text shares 0.0 % and 0.6 % of its word 8-grams with Parts I and II, and 5.7 % with Part IV.

Source checks. The manuscript transcribes Conjecture 18 from arXiv:2303.02793v2 (24 April 2023, Section 6.4, printed page 33) and the operator from the A181198 internal record, retrieved 5 October 2026 (revision 39 of 1 January 2024, still labelled conjectural); its bounded priority search, which included arXiv:2607.24832, found no matching proof (Section 26). At placement the intake did not re-inspect the printed page of [9]; it checked that p_0, p_1, p_2 equal the polynomials of the live A181198 entry, checked the certificate (29.4), the edge identities and (29.8) in SymPy (residual zero), and checked Theorems 26.1, 30.1 and 30.2, (28.21) and (29.1) for $n \leq 30$ against an independent row-state count. Its verifier, rerun at the write on a copy (Section 33.3), printed its receipt byte for byte.

Editorial changes. Section k of the manuscript is Section $k + 25$ here; theorem and equation numbers follow their sections. Every label carries the prefix `shr:h4:`. Its title, subtitle and title-page paragraphs open this Part; its table of contents is dropped. Its two citations of its predecessor now point to Part I and its citation of the companion to Part II. The bibliographies are merged: Kauers–Koutschan, Sun and DLMF §5.11 are Part I's entries (the manuscript retrieved DLMF 5.11.1 and §5.11(ii) on 5 October 2026 for the positive-real Stirling remainder bound), and its OEIS entry is [7]. Its bibliography said of Part I, pinned by the blob above, that “its all-height asymptotics are prior results relative to the present report; its scope leaves the particular recurrences unproved”, and of Part II that “the inspected copy proves rational diagonality and D-finiteness and records the particular recurrence problem as open”. Remarks take this report's theorem style, and the four command-line options of Section 33.3 are typeset `--option`, because this report's typewriter font would join their double hyphens into a dash. The **[write]** notes, Section 33.5 and Section 33.6 are new. No other statement, proof or number of the manuscript was changed.

[write] Added 5 October 2026, batch 101: notation of Part III. Part III keeps the manuscript's symbols. Many clash with those of the other Parts; within Part III each has the meaning below, and the right column gives the tempting false reading from elsewhere in this report. The count is $a_n = T_4(n)$ and the array entries are $x_{i,j}$. Shared without change: $T_m(n)$, Δ , the Lambert branch W_{-1} and the Bernoulli numbers B_{2j} .

symbol	meaning in Part III	not: meaning elsewhere
$a_n; M_n; b_n$	$T_4(n); M_4(n) = (4n)!/(n!)^4; a_n/M_n$ (26.2)	array entries $a_{i,j}$; the coefficients $b_{m,j}$ of (2.5)
A_n, B_n, D_n	coefficients of (26.3)	$A = \binom{m}{3}, B = m(m-1)^2(m-2)/12$ (Section 6.1); Part II's ranks B_i, D_i
V, K, W	the Volterra kernel, $K = V^{n-1}$ (27.5), $W = V^n(V^*)^n$ (28.2)	K_m (2.2); Part II's $K_r(u, v)$ and $W_n(x)$
$E; E_K, E'$	the sign kernel (27.6); error constants (Sections 31–32)	the boundary error $E_{m,n}(p); \mathbb{E}$
$J; J_0; J_n$	$\mathbf{1} \otimes \mathbf{1}$; a scalar (28.8); $D_n M_n$ in Section 30.2	$J_m = \prod_{j < m} j!$
$\mathcal{A}; B$	the kernels KEK^* and $E\mathcal{A}$ (27.8)	the event $\mathcal{A}$ of Section 4
$\alpha(t), \beta(t); \beta_*$	$t^n/n!$ and $t^{n-1}/(n-1)!$ (28.1); $15/2 = \alpha_4$	$\alpha_m = (m^2 - 1)/2$; recurrence coefficients β_j
$h, e, w; c, H, U, I,$ K_0, L	functions and scalars of Section 28	$h_r(q), h_j; H(\lambda); L$ a truncation order

symbol	meaning in Part III	<i>not</i> : meaning elsewhere
N	$n!$ (Sections 28–29); an integer onset (Section 31); the threshold $N(y)$ (32.10)	—
Z_n	the weighted triangle period (28.20), (28.23), the same as Part IV's Z_n	$Z_m(n)$ (6.1); Z_i (Section 9)
$C_m; \mathcal{D}, f; I_j$	the ordered simplex; the triangle and $uv(u+v-1)$ (28.22); triangle integrals (Section 29)	the central event C_n ; Part II's chamber $C_r(n)$ and Dyck words $\mathcal{D}_m$
$P, Q; p_{ij}; \gamma_n$	the certificate field (29.3); its coefficients (29.2)	Part I's P_j ; Part II's $Q_m(n), p_i$; Part IV's $P(n), Q(n), \gamma_n$
g_n, r_n, m_n, q_n, c_n	Section 30	$c_{m,j}$; $r = m - 2$ (Part II)
p_0, p_1, p_2	the A181198 operator (30.6)	power sums p_r
F_n, s_n, t_k	variation of constants (30.7), (30.8)	the generating function $F_m(z)$; Part IV's $F_n = a_n/M_n$
$(x)_j$	the rising factorial (Section 30.4)	Part I's falling factorial $n^{\bar{k}}$; Part II and Part IV write rising factorials $(x)^{\bar{k}}$
$R(n), S(n)$	the coefficients (31.2), (31.3)	$R_n, R_m, R(z), R_m(n)$; Part IV's ratios $R(n), S(n)$
$C(t), c_j; d_j$	the formal series of $n^6 b_n$, with $c_j = \frac{3}{1024} c_{4,j}$; $d_j = b_{4,j}$	the Catalan series $C(z)$; $c_{m,j}$; $d = \binom{m}{2}$
$C_*, \lambda, \ell_j, s_j$	K_4 ; $\log 256$; logarithmic and inverse coefficients	$\lambda = m \log m$ (Section 10)
$\Psi_K, \xi_K, w(y), \Delta_K,$ $G_K, H_K, W_K, X_K,$ C_K	the finite inverse model of Section 32	the Vandermonde Δ ; W_{-1} ; Part II's $C_{m,L}$

26 The theorem and its scope

For integers $m, n \geq 1$, let $T_m(n)$ count the $m \times n$ arrays $(x_{i,j})$ containing the integers $1, \dots, mn$ exactly once and satisfying

$$\begin{aligned} x_{i,j} &< x_{i,j+1}, & x_{i,j} &< x_{i+1,j}, \\ x_{i,j} &< x_{i+1,j+1}, & x_{i,j+1} &< x_{i+1,j}, \end{aligned} \quad (26.1)$$

whenever the cells exist. Rows increase from left to right and columns from top to bottom. The third inequality follows from the first two. The fourth, for a downward antidiagonal, is essential. Throughout this report *height is four and width is n* . Write

$$a_n = T_4(n), \quad M_n = \frac{(4n)!}{(n!)^4}, \quad b_n = \frac{a_n}{M_n}. \quad (26.2)$$

The initial values are

$$a_1 = a_2 = 1, \quad a_3 = 8, \quad a_4 = 169, \quad a_5 = 6392.$$

Width two gives a single interlaced chain, explaining $a_2 = 1$ without recourse to a recurrence.

Theorem 26.1 (The exact inhomogeneous identity). *For every integer $n \geq 1$,*

$$A_n a_{n+1} + B_n a_n = D_n M_n, \quad (26.3)$$

where

$$\begin{aligned} A_n &= 3(n-1)(n+1)(3n+1)(3n+2), \\ B_n &= 8n(2n+1)(4n-1)(4n+1), \\ D_n &= \frac{6(4n+1)(7n^2-1)}{(n+1)^2(2n-1)^2(2n+1)}. \end{aligned} \quad (26.4)$$

For $n \geq 2$ this identity and $a_2 = 1$ determine the whole sequence. It implies the exact order-two operator in Theorem 30.1 and the printed finite sum in Theorem 30.2.

The first-order coefficient A_1 is zero, so the identity at $n = 1$ fixes a_1 rather than a_2 . We keep this degeneracy explicit. The resulting second-order recurrence is valid for $n \geq 1$. It does *not* hold at $n = 0$ if one sets the empty-array value $a_0 = 1$.

26.1 What is resolved and what is reused

Kauers and Koutschan [9] give the height-four formula as Conjecture 18 in Section 6.4, printed page 33 of arXiv version 2. Their statement is a finite sum, rather than merely an unnamed D-finiteness assertion. The corresponding order-two recurrence is printed in OEIS A181198 [7]. Sections 27–30 prove both exact statements from the full array definition.

The earlier report *Fixed-height shifted rectangles* (Part I) already proves the leading asymptotic and all-fixed-order expansions for every fixed height. Its scope expressly leaves the specific height-four and height-five recurrences unproved. A later companion report on rational diagonals and rare boundary phases (Part II) establishes D-finiteness by a different route and likewise leaves these particular recurrences unresolved. We do not reclassify those earlier asymptotic or D-finiteness results as new. The additional exact result here is the counting-to-operator certificate for height four, together with the simpler coefficient and remainder machinery that this recurrence permits.

The proof uses classical order-polytope, determinant-integration, and Pfaffian ideas. The finite-dimensional integral identities needed here are proved directly, so no unstated Fredholm determinant theorem or symbolic-integration oracle is a mathematical dependency. Sun’s work on shifted strips [11] supplies relevant background; the orientation with fixed width and growing height is a different enumeration problem.

26.2 Source status and limitations

The source check used the primary 2023 article, the A181198 entry retrieved on October 5, 2026, and the existing report source. The retrieved OEIS internal snapshot still calls the recurrence and formula conjectural; its revision record is revision 39, January 1, 2024. That label is evidence of the inspected entry’s status, not proof of global novelty. Targeted identifier and exact-coefficient searches, including inspection of the 2026 paper arXiv:2607.24832, did not locate a matching proof. These are bounded search results. The report makes no first-ever or exhaustive priority claim.

The height-five operator and Conjecture 19 are outside the theorem. Splitting the count into asymptotic event sectors does not separately annihilate those sectors with the whole-sequence operator; the proof below avoids that issue by retaining the entire order polytope. The expansions in Section 31 are Poincaré expansions at every *fixed* order. They are not assertions about convergence, optimal truncation, Borel summation, or a complete exponentially improved transseries.

26.3 Outline of the proof

The exact argument has four stages:

1. Compose the interlacing indicator along consecutive columns, with ordered-simplex normalization retained exactly.
2. Evaluate the resulting four-variable determinant integral in terms of two traces, and reduce those traces to one scalar period Z_n .

3. Prove a first-order recurrence for Z_n by a polynomial divergence identity on a triangle, evaluating every edge.
4. Substitute back into the full count, eliminate the hypergeometric forcing term, and solve by variation of constants.

Every identity is uniform in the integer parameter. Finite counting tests are separate transcription and implementation checks, never replacements for these arguments.

27 Column transfer and the Pfaffian identity

27.1 The order polytope and interlacing determinant

Let

$$C_m = \{(x_1, \dots, x_m) : 0 < x_1 < \dots < x_m < 1\}.$$

Give each array cell a real coordinate in $(0, 1)$ and impose the same strict comparisons as in (26.1). Coordinates that coincide form a null set. Every admissible total order of the mn coordinates contributes an open simplex of volume $1/(mn)!$. Hence the volume of this order polytope is $T_m(n)/(mn)!$.

For two consecutive columns $x, y \in C_m$, the comparisons are exactly

$$x_1 < y_1 < x_2 < y_2 < \dots < x_m < y_m. \quad (27.1)$$

Rows give $x_i < y_i$ and antidiagonals give $y_i < x_{i+1}$. These comparisons also imply all column and downward-diagonal comparisons involving these columns. For $n \geq 2$ the adjacent-column conditions therefore include all constraints, with no restriction on whether a row has begun or finished in a label-prefix description.

Define the Volterra kernel and its operator by

$$V(t, s) = \mathbf{1}_{\{s < t\}}, \quad (Vg)(t) = \int_0^t g(s) ds.$$

For ordered distinct x, y , we have

$$\det[V(y_i, x_j)]_{i,j=1}^m = \mathbf{1}_{\{x_1 < y_1 < \dots < x_m < y_m\}}. \quad (27.2)$$

Indeed, let k_i be the number of x_j below y_i . Row i consists of k_i ones followed by zeros, with $0 \leq k_1 \leq \dots \leq k_m \leq m$. A nonzero determinant requires distinct nonzero rows, hence $k_i = i$. In that case the matrix is unit lower triangular. This proves both the indicator and its positive sign.

Lemma 27.1 (Composition on the ordered simplex). *For bounded kernels F, G on $[0, 1]^2$,*

$$\int_{C_m} \det[F(y_i, z_j)] \det[G(z_i, x_j)] dz = \det \left[\int_0^1 F(y_i, t) G(t, x_j) dt \right]. \quad (27.3)$$

Proof. The product of determinants is symmetric in the z variables, because permuting them changes the sign of each factor. Its cube integral is thus $m!$ times its ordered-simplex integral. Expand both determinants in the cube integral and integrate each product by Fubini. For every relative permutation contributing to the determinant on the right there are $m!$ choices of the first permutation, all with the same resulting sign. That factor cancels the ordered-simplex factor. No factorial remains. $\square$

Composing the $n - 1$ adjacent-column indicators now gives, for $n \geq 2$,

$$\frac{T_m(n)}{(mn)!} = \int_{C_m} \int_{C_m} \det[K(y_i, x_j)] dx dy, \quad K = V^{n-1}, \quad (27.4)$$

where

$$K(t, s) = \frac{(t - s)_+^{n-2}}{(n - 2)!}. \quad (27.5)$$

For $n = 2$ the exponent-zero notation denotes the indicator $s < t$; its value on equality is immaterial. Formula (27.5) follows by integrating the simplex $s < t_1 < \cdots < t_{n-2} < t$. The case $n = 1$ is handled by its direct count and never requires an identity distribution as an integral kernel.

27.2 An elementary four-variable de Bruijn calculation

Set

$$E(t, s) = \operatorname{sgn}(s - t), \quad J = \mathbf{1} \otimes \mathbf{1}, \quad E = J - 2V = 2V^* - J. \quad (27.6)$$

Values on the diagonal may be assigned arbitrarily. For an antisymmetric 4×4 matrix C , our Pfaffian convention is

$$\operatorname{Pf} C = C_{12}C_{34} - C_{13}C_{24} + C_{14}C_{23}.$$

In particular, for $x \in C_4$, every upper-triangular entry of $[E(x_i, x_j)]$ is 1, so its Pfaffian is $1 - 1 + 1 = 1$.

Let f_i be bounded functions and put

$$A_{ij} = \int_0^1 \int_0^1 f_i(s) E(s, t) f_j(t) ds dt.$$

Then

$$\int_{C_4} \det[f_i(x_j)] dx = \operatorname{Pf} A. \quad (27.7)$$

Here is a direct derivation. If $D(x) = \det[f_i(x_j)]$, alternation gives

$$\operatorname{Pf} A = \frac{1}{8} \int_{[0,1]^4} D(x) E(x_1, x_2) E(x_3, x_4) dx.$$

This follows by expanding the standard $1/(2^2 2!)$ permutation formula for the Pfaffian. Relabeling variables shows that the integral of $D(x)(-E_{13}E_{24})$ and that of $D(x)E_{14}E_{23}$ each equal the integral of $D(x)E_{12}E_{34}$. Thus the right side is

$$\frac{1}{24} \int_{[0,1]^4} D(x) \operatorname{Pf}[E(x_i, x_j)] dx.$$

The integrand is symmetric, and on the ordered simplex its Pfaffian factor is 1. This is exactly the left side of (27.7). The orientation can also be checked in size two: the corresponding entry is

$$\int_{s < t} (f_i(s)f_j(t) - f_i(t)f_j(s)) ds dt.$$

Apply (27.7) to $f_i(s) = K(y_i, s)$ and define

$$\mathcal{A} = KEK^*, \quad B = E\mathcal{A} = EKEK^*. \quad (27.8)$$

The kernel $\mathcal{A}$ is antisymmetric. The inner integral in (27.4) is $\operatorname{Pf}[\mathcal{A}(y_i, y_j)]$.

27.3 The two traces and their exact constants

The product $\text{Pf}[\mathcal{A}(y_i, y_j)] \text{Pf}[E(y_i, y_j)]$ is symmetric, so

$$\frac{a_n}{(4n)!} = \frac{1}{24} \int_{[0,1]^4} \text{Pf}[\mathcal{A}_{ij}] \text{Pf}[E_{ij}] dy. \quad (27.9)$$

In this subsection, a trace denotes its finite kernel integral. Thus

$$\text{tr } B = \int_0^1 B(t, t) dt, \quad \text{tr}(B^2) = \int_0^1 \int_0^1 B(t, s) B(s, t) ds dt.$$

Expanding the two three-term Pfaffians gives nine products. The three matching pairings each integrate to $(\text{tr } B)^2$. For example, $\int E_{12} \mathcal{A}_{12} = -\text{tr } B$, and its square has the positive sign. The six signed crossed pairings each integrate to $-\text{tr}(B^2)$. One representative follows from

$$\text{tr}(B^2) = \int E_{12} \mathcal{A}_{23} E_{34} \mathcal{A}_{41} dy = - \int E_{12} E_{34} \mathcal{A}_{14} \mathcal{A}_{23} dy;$$

the others follow by relabeling the four variables and retaining the Pfaffian signs. Consequently

$$\boxed{\frac{a_n}{(4n)!} = \frac{(\text{tr } B)^2}{8} - \frac{\text{tr}(B^2)}{4}, \quad n \geq 2.} \quad (27.10)$$

The constants are exactly $3/24$ and $6/24$. All the kernels used here are bounded, piecewise polynomial functions. Products are absolutely integrable on compact cubes, so Fubini and the relabelings are justified. This is a finite integral calculation, not a formal infinite-dimensional Fredholm series argument.

28 Rank reduction to one weighted period

Throughout this section $n \geq 2$ and $N = n!$. To keep the count a_n distinct from scalar functions, use

$$\alpha(t) = \frac{t^n}{N}, \quad \beta(t) = \frac{t^{n-1}}{(n-1)!}, \quad \alpha' = \beta. \quad (28.1)$$

Write $p \otimes q$ for the rank-one kernel $p(t)q(s)$, and let $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$. Define

$$\begin{aligned} W &= V^n (V^*)^n, \\ W(t, s) &= \frac{1}{((n-1)!)^2} \int_0^{\min(t,s)} (t-r)^{n-1} (s-r)^{n-1} dr, \\ h(t) &= W(t, 1), \quad e(t) = 2h(t) - \frac{\beta(t)}{N}. \end{aligned} \quad (28.2)$$

The kernel W is symmetric and continuous, with polynomial expressions on the two triangles of the square.

28.1 An exact rank-two perturbation

Use both expressions for E in (27.6), preserving factor order:

$$\begin{aligned} B &= (JV^{n-1} - 2V^n)(2(V^*)^n - J(V^*)^{n-1}) \\ &= -4W + \mathbf{1} \otimes e + 2\alpha \otimes \beta. \end{aligned}$$

Thus

$$B = -4W + \mathbf{1} \otimes e + 2\alpha \otimes \beta. \quad (28.3)$$

For clarity, the four product reductions are

$$\begin{aligned} JV^{n-1}(V^*)^n &= \mathbf{1} \otimes h, & JV^{n-1}J(V^*)^{n-1} &= \frac{\mathbf{1} \otimes \beta}{N}, \\ V^n(V^*)^n &= W, & V^nJ(V^*)^{n-1} &= \alpha \otimes \beta. \end{aligned}$$

For the first, integrate the first argument of V^{n-1} to obtain $(1-r)^{n-1}/(n-1)!$, leaving exactly h . For the second, the intervening scalar integral is $1/N$. The remaining two follow directly from $V^n\mathbf{1} = \alpha$ and $V^{n-1}\mathbf{1} = \beta$.

The kernel identities needed below are

$$W(t, 0) = 0, \quad (\partial_t + \partial_s)W(t, s) = \beta(t)\beta(s), \quad (28.4)$$

$$W(rt, rs) = r^{2n-1}W(t, s), \quad (28.5)$$

and the scalar integrals are

$$\begin{aligned} \int_0^1 h &= \frac{1}{2N^2}, & \int_0^1 \beta &= \frac{1}{N}, & c := \langle \alpha, \beta \rangle &= \frac{1}{2N^2}, \\ \int_0^1 e &= 0, & \text{tr } W &= \frac{1}{2n(2n-1)((n-1)!)^2}. \end{aligned} \quad (28.6)$$

To prove the differential identity, the sum of the two outer derivatives of the integrand of (28.2) is minus its derivative in r . The lower endpoint produces $\beta(t)\beta(s)$. The upper endpoint vanishes for $n \geq 2$. The identities hold almost everywhere, which is sufficient for the following integrations. When the square is split at $t = s$, its inner boundary contributions cancel because the kernels are continuous. Homogeneity follows by the substitution $r \mapsto r\rho$ with $\rho > 0$. The integral of h can be checked explicitly by Fubini:

$$\int_0^1 h(t) dt = \frac{1}{n((n-1)!)^2} \int_0^1 (1-r)^{2n-1} dr = \frac{1}{2N^2}.$$

The remaining integrals are monomial integrals.

Taking the trace in (28.3) now gives

$$\text{tr } B = -4 \text{tr } W + \int e + 2c = -\frac{1}{(2n-1)N^2}. \quad (28.7)$$

28.2 Every integration by parts for the square trace

Introduce

$$\begin{aligned} H &= \langle \alpha, h \rangle, & U &= \text{tr}(W^2), & w &= W\mathbf{1}, \\ J_0 &= \langle h, w \rangle, & K_0 &= \langle \beta, w \rangle, & I &= \langle \beta, W\alpha \rangle. \end{aligned} \quad (28.8)$$

Integrate the sum of the two derivatives of $\alpha(t)\alpha(s)W(t, s)$ over $[0, 1]^2$. The lower edges vanish, because $\alpha(0) = 0$ and $W(t, 0) = 0$. Each upper edge contributes H/N . The differentiated integrand integrates to $2I + c^2$, using symmetry of W and (28.4). Hence

$$2I + c^2 = \frac{2H}{N}, \quad I = \frac{H}{N} - \frac{c^2}{2}. \quad (28.9)$$

Next apply the same operation to $\alpha(t)W(t, s)$. Its two upper edges are $(\int h)/N$ and H , and its two lower edges vanish. Its interior integral is $K_0 + c/N$. By (28.6),

$$K_0 + \frac{c}{N} = H + \frac{1}{N} \int h, \quad K_0 = H. \quad (28.10)$$

For a rank-one kernel, $\text{tr}(p \otimes q) = \langle q, p \rangle$ and $(p \otimes q)(r \otimes s) = \langle q, r \rangle p \otimes s$. Squaring (28.3) therefore gives exactly

$$\text{tr}(B^2) = 16U - 8\langle e, w \rangle - 16I + 4c^2 + \frac{4}{N}\langle e, \alpha \rangle. \quad (28.11)$$

The term $(\int e)^2$ has vanished; the cross term of the two rank-one kernels supplies the last term, including its factor 4. Since

$$\langle e, w \rangle = 2J_0 - \frac{H}{N}, \quad \langle e, \alpha \rangle = 2H - \frac{c}{N},$$

substitution of (28.9) reduces (28.11) to

$$\text{tr}(B^2) = 16U - 16J_0 + \frac{1}{N^4}. \quad (28.12)$$

All occurrences of H/N in this substitution cancel.

Integrating (28.4) with respect to s gives

$$w'(t) + h(t) = \frac{\beta(t)}{N}, \quad w(0) = 0, \quad w(1) = \frac{1}{2N^2}.$$

Multiply by w and integrate. The only integration-by-parts boundary is $[w(t)^2/2]_0^1 = 1/(8N^4)$. By (28.10),

$$J_0 = \frac{K_0}{N} - \frac{1}{8N^4} = \frac{H}{N} - \frac{1}{8N^4}. \quad (28.13)$$

Thus

$$\text{tr}(B^2) = 16U - \frac{16H}{N} + \frac{3}{N^4}. \quad (28.14)$$

This calculation has retained the full trace, including its boundary terms. No asymptotic replacement has been made.

28.3 Reducing the remaining square integral

Symmetry and (28.5) give, by $t = rs$ on $0 < t < s < 1$,

$$\begin{aligned} U &= 2 \int_0^1 \int_0^s W(t, s)^2 dt ds \\ &= 2 \int_0^1 \int_0^1 s^{4n-1} h(r)^2 dr ds = \frac{1}{2n} \int_0^1 h(t)^2 dt. \end{aligned} \quad (28.15)$$

The Jacobian is s , which is responsible for the exponent $4n - 1$. The defining integral for h obeys

$$(1-t)h'(t) + (2n-1)h(t) = \frac{t^{n-1}}{((n-1)!)^2}. \quad (28.16)$$

For $n \geq 2$, integrate minus the r derivative of $(t-r)^{n-1}(1-r)^n$. The negative derivative is

$$(n-1)(1-t)(t-r)^{n-2}(1-r)^{n-1} + (2n-1)(t-r)^{n-1}(1-r)^{n-1}.$$

The lower endpoint contributes t^{n-1} and the upper endpoint vanishes. This proves (28.16). At $n = 1$, the same equation is checked directly from $h(t) = t$.

Let $L = \int_0^1 t^{n-1} h(t) dt$. Multiplying (28.16) by $2h$ and integrating yields

$$(4n-1) \int_0^1 h^2 = \frac{2L}{((n-1)!)^2}. \quad (28.17)$$

Indeed, the first term integrates to $[(1-t)h(t)^2]_0^1 + \int h^2 = \int h^2$; both endpoints are zero, by $h(0) = 0$ and the factor $1-t$. Multiplication instead by t^n gives

$$-nL + 3n \int_0^1 t^n h(t) dt = \frac{1}{2n((n-1)!)^2}.$$

Here $[t^n(1-t)h(t)]_0^1 = 0$, and differentiating $t^n(1-t)$ produces $nt^{n-1} - (n+1)t^n$. Since $\int t^n h = NH$, this says

$$L = 3NH - \frac{1}{2N^2}. \quad (28.18)$$

Combining (28.15)–(28.18) gives

$$U = \frac{3nH}{(4n-1)N} - \frac{n}{2(4n-1)N^4}. \quad (28.19)$$

28.4 The exact full-count formula

Substitute (28.7), (28.14), and (28.19) into (27.10). Define

$$Z_n = N^3 H = N^2 \int_0^1 t^n h(t) dt. \quad (28.20)$$

After collecting rational terms, the result is

$$\frac{a_n}{M_n} = \frac{4(n-1)}{4n-1} Z_n - \frac{32n^3 - 56n^2 + 28n - 5}{8(2n-1)^2(4n-1)}. \quad (28.21)$$

For example, before the final simplification the right side is

$$\frac{1}{8(2n-1)^2} - 4N^4 U + 4N^3 H - \frac{3}{4},$$

which provides an explicit check on the normalization.

In the double integral defining (28.20), set $u = t$ and $v = 1 - r$. The domain $0 < r < t < 1$ becomes the triangle

$$\mathcal{D} = \{(u, v) : 0 < u < 1, 0 < v < 1, u + v > 1\}, \quad f(u, v) = uv(u + v - 1). \quad (28.22)$$

The Jacobian has absolute value 1, and the factorial factor is $N^2/((n-1)!)^2 = n^2$. Thus

$$Z_n = n^2 \int_{\mathcal{D}} u f^{n-1} du dv = \frac{n^2}{2} \int_{\mathcal{D}} (u + v) f^{n-1} du dv. \quad (28.23)$$

The second equality uses the symmetry exchanging u and v . Formula (28.21) was derived for $n \geq 2$. It also holds at $n = 1$ by direct checking: its coefficient of Z_1 is zero, and its remaining value is $1/24 = a_1/M_1$. Independently, (28.23) gives $Z_1 = 1/3$. This extension uses no singular kernel calculation.

29 A polynomial certificate on the triangle

Set

$$I_j = \int_{\mathcal{D}} (u+v)f(u,v)^j du dv, \quad j \geq 0,$$

so $Z_n = n^2 I_{n-1}/2$. We now prove the exact period recurrence.

Theorem 29.1 (The weighted triangle recurrence). *For every integer $n \geq 1$,*

$$3(3n+1)(3n+2)Z_{n+1} + (n+1)^2 Z_n = \frac{28n^3 + 50n^2 + 29n + 5}{2(2n+1)}. \quad (29.1)$$

29.1 The six coefficients

For a fixed integer $n \geq 1$, define

$$\begin{aligned} \gamma_n &= \frac{3(3n+1)(3n+2)}{n^2}, \\ p_{21} &= \frac{(n+3)(3n+1)(3n+2)}{n^2(n+1)^2}, & p_{20} &= -\frac{3n+1}{n(n+1)}, \\ p_{12} &= \frac{3(3n+1)(3n+2)}{n^2(n+1)}, & p_{11} &= -\frac{(3n+1)(7n^2+14n+6)}{n^2(n+1)^2}, \\ p_{10} &= \frac{n-1}{n(n+1)}, & p_{00} &= \frac{1}{n}. \end{aligned} \quad (29.2)$$

Use the polynomial vector field

$$\begin{aligned} P(u,v) &= uv(p_{21}u^2v + p_{20}u^2 + p_{12}uv^2 + p_{11}uv + p_{10}u + p_{00}), \\ Q(u,v) &= P(v,u). \end{aligned} \quad (29.3)$$

Its certificate identity is

$$f(P_u + Q_v) + (n-1)(Pf_u + Qf_v) = f(\gamma_n f + 1)(u+v). \quad (29.4)$$

This is a polynomial identity in u, v , over rational functions of n . It can be verified simply by substituting (29.2) and comparing coefficients. More concretely, multiply it by $n^2(n+1)^2$ to obtain an identity in $\mathbb{Q}[n, u, v]$ with no rational-function arithmetic left. The packaged verifier performs that coefficient comparison with integer and rational arithmetic, and the residual polynomial is identically zero. This displayed certificate, rather than the procedure that discovered it, is the proof object.

For $n \geq 2$, multiplication of (29.4) by f^{n-2} gives

$$\operatorname{div}(f^{n-1}P, f^{n-1}Q) = f^{n-1}(\gamma_n f + 1)(u+v). \quad (29.5)$$

For $n = 1$, cancel the common polynomial factor f in (29.4) first. This gives $P_u + Q_v = (\gamma_1 f + 1)(u+v)$, which is precisely (29.5). In particular there is no division by a vanishing boundary function. For all the stated indices, both sides of (29.5) are polynomials on the closed triangle.

29.2 The slanted edge including the exceptional exponent

The lower slanted edge is $v = 1 - u$, with outward normal parallel to $(-1, -1)$. Its normal flux, in the parameter u , is $-f^{n-1}(P + Q) du$. The stronger polynomial identity

$$P + Q = f \left((p_{21} + p_{12})uv + p_{20}(u + v) - \frac{2}{n} \right) \quad (29.6)$$

holds by substitution. Therefore $P(u, 1 - u) + Q(u, 1 - u) = 0$ identically. The flux vanishes even at $n = 1$, when $f^{n-1} = 1$ and vanishing of that power alone would not settle the boundary. The vertices require no separate contribution in the planar divergence theorem.

29.3 Both outer edges and their exact integral

At the vertical edge $u = 1$, the outward flux is $f(1, v)^{n-1} P(1, v) dv$, where

$$\begin{aligned} f(1, v) &= v^2, \\ P(1, v) &= p_{12}v^3 - \frac{(3n+1)(4n+3)}{n(n+1)^2}v^2 - \frac{v}{n}. \end{aligned} \quad (29.7)$$

At the horizontal edge $v = 1$, symmetry gives the same integral. Consequently their sum is

$$\begin{aligned} 2 \int_0^1 v^{2n-2} P(1, v) dv &= 2 \left(\frac{p_{12}}{2n+2} - \frac{(3n+1)(4n+3)}{n(n+1)^2(2n+1)} - \frac{1}{2n^2} \right) \\ &= \frac{28n^3 + 50n^2 + 29n + 5}{n^2(n+1)^2(2n+1)}. \end{aligned} \quad (29.8)$$

Each exponent being integrated is nonnegative for $n \geq 1$. All integrals are ordinary proper integrals of polynomials; no convergence or limiting boundary prescription is needed.

Proof of Theorem 29.1. Integrate (29.5) and apply the divergence theorem on the closed triangle. Equations (29.6) and (29.8) account for all three edges. Thus

$$\gamma_n I_n + I_{n-1} = \frac{28n^3 + 50n^2 + 29n + 5}{n^2(n+1)^2(2n+1)}.$$

Multiplying by $n^2(n+1)^2/2$ and using $Z_j = j^2 I_{j-1}/2$ proves (29.1) for every $n \geq 1$. $\square$

Direct period integration gives

$$Z_1 = \frac{1}{3}, \quad Z_2 = \frac{13}{45}, \quad Z_3 = \frac{461}{1680}.$$

These values provide useful checks on orientation and normalization; the uniform recurrence follows from the certificate independently of them.

30 The exact recurrence and printed finite sum

30.1 Returning from the period to the array count

Write the full-count formula (28.21) as

$$b_n = g_n Z_n + r_n, \quad g_n = \frac{4(n-1)}{4n-1}, \quad r_n = -\frac{32n^3 - 56n^2 + 28n - 5}{8(2n-1)^2(4n-1)}.$$

The multinomial normalizer has ratio

$$m_n := \frac{M_{n+1}}{M_n} = \frac{8(2n+1)(4n+1)(4n+3)}{(n+1)^3}. \quad (30.1)$$

For $n \geq 2$, solve (29.1) for Z_{n+1} and substitute in

$$A_n m_n (g_{n+1} Z_{n+1} + r_{n+1}) + B_n (g_n Z_n + r_n).$$

The coefficient of Z_n cancels because

$$\frac{A_n m_n g_{n+1} (n+1)^2}{3(3n+1)(3n+2)} = B_n g_n.$$

The remaining rational expression is

$$A_n m_n \left(g_{n+1} \frac{28n^3 + 50n^2 + 29n + 5}{6(2n+1)(3n+1)(3n+2)} + r_{n+1} \right) + B_n r_n = D_n. \quad (30.2)$$

This equality follows by a common-denominator expansion and is one of the exact polynomial checks supplied with the report. Thus $A_n m_n b_{n+1} + B_n b_n = D_n$, proving (26.3) for $n \geq 2$.

For $n = 1$, $A_1 = 0$, $B_1 = 360$, and $D_1 M_1 = 360$, so the identity holds by $a_1 = 1$. The width-two count is independently $a_2 = 1$. All factors of A_n are nonzero for $n \geq 2$. This proves Theorem 26.1, including its precise initial-index statement.

30.2 Eliminating the hypergeometric forcing term

Let $J_n = D_n M_n$. Direct cancellation gives

$$q_n := \frac{J_{n+1}}{J_n} = \frac{8(2n-1)^2(4n+3)(4n+5)(7n^2+14n+6)}{(n+1)(n+2)^2(2n+3)(7n^2-1)}. \quad (30.3)$$

Subtract q_n times (26.3) at n from that identity at $n+1$. Multiply the result by

$$c_n = \frac{(n+1)(n+2)^2(2n+3)(7n^2-1)}{n}. \quad (30.4)$$

The denominator is nonzero for the stated range $n \geq 1$.

Theorem 30.1 (The A181198 operator). *For every integer $n \geq 1$,*

$$p_2(n)a_{n+2} + p_1(n)a_{n+1} + p_0(n)a_n = 0, \quad (30.5)$$

where

$$\begin{aligned}
 p_2(n) &= 3(n+1)(2n+3)(3n+4)(3n+5)(7n^2-1)(n+2)^3, \\
 p_1(n) &= -8(n+1)(2n+1)(4n+3)(4n+5) \\
 &\quad \cdot (364n^5 + 84n^4 - 1025n^3 - 534n^2 + 157n + 54), \\
 p_0(n) &= -64(2n-1)^2(2n+1)(4n-1)(4n+1)(4n+3)(4n+5) \\
 &\quad \cdot (7n^2 + 14n + 6).
 \end{aligned} \tag{30.6}$$

These are the coefficient polynomials recorded for OEIS A181198.

Proof. The three coefficient identities are

$$p_2 = c_n A_{n+1}, \quad p_1 = c_n (B_{n+1} - q_n A_n), \quad p_0 = -c_n q_n B_n.$$

They follow by direct factorization, or by clearing denominators and comparing polynomial coefficients. The elimination just performed thus proves (30.5). Its leading coefficient is nonzero for every positive integer n , so $a_1 = a_2 = 1$ uniquely specifies its solution. $\square$

The natural empty-array convention would give $a_0 = 1$. Substitution of $a_0 = a_1 = a_2 = 1$ in the polynomial left side at $n = 0$ gives -2160 , not zero. The pole in (30.4) is therefore substantive. Neither the proof nor the theorem extends this operator to $n = 0$.

30.3 Solving the first-order recurrence

For $n \geq 2$, put

$$F_n = \frac{(-1)^{n+1} n(n-1)(4n-2)!}{n!(3n)!}, \tag{30.7}$$

and for $k \geq 2$ put

$$t_k = \frac{3(-1)^k (7k^2 - 1)(3k)!}{(k!)^3 (k-1)k(k+1)^2 (2k-1)^2 (2k+1)^2}. \tag{30.8}$$

Factorial cancellation proves the two exact relations

$$\frac{F_{n+1}}{F_n} = -\frac{B_n}{A_n}, \quad A_n F_{n+1} t_n = D_n M_n \quad (n \geq 2). \tag{30.9}$$

Writing $a_n = F_n s_n$ therefore turns (26.3) into $s_{n+1} - s_n = t_n$. Since $F_2 = -1$ and $a_2 = 1$, $s_2 = -1$, and

$$a_n = F_n \left(-1 + \sum_{k=2}^{n-1} t_k \right), \quad n \geq 2. \tag{30.10}$$

The sum is empty for $n = 2$, giving its correct value without a separate limit convention.

30.4 Literal correspondence with Conjecture 18

For a nonnegative integer j , write $(x)_j = x(x+1) \cdots (x+j-1)$ for the rising factorial, with $(x)_0 = 1$. Generalized binomial coefficients have the usual polynomial definition.

Theorem 30.2 (Kauers and Koutschan Conjecture 18). *For each integer $n > 1$,*

$$a_n = \frac{(-64)^n (n-1) \left(-\frac{1}{2}\right)_{2n} \left(\frac{1}{2}\right)_n}{4(3n)!} \left(-1 + 3 \sum_{k=2}^{n-1} \frac{(-4)^k (7k^2 - 1) \binom{3k}{2k} \binom{k+1/2}{k}}{(k-1)k(k+1)^2(2k-1)^2(2k+1)^3} \right). \quad (30.11)$$

Proof. The elementary products

$$\left(-\frac{1}{2}\right)_{2n} = -\frac{(4n-2)!}{2^{4n-1}(2n-1)!}, \quad \left(\frac{1}{2}\right)_n = \frac{(2n)!}{4^n n!} \quad (30.12)$$

reduce the prefactor in (30.11) exactly to F_n in (30.7). Also

$$\binom{k+1/2}{k} = \frac{(2k+1)!}{4^k (k!)^2}.$$

Insert this and $\binom{3k}{2k} = (3k)! / ((2k)!k!)$ into three times the summand in (30.11). The factors 4^k cancel, and one factor $2k+1$ cancels from its denominator, leaving exactly t_k . Thus (30.11) is (30.10), already proved. $\square$

The rising factorials, summation range, factor 3, and prefactor have all been retained. This establishes the statement actually printed in the primary paper, rather than only a recurrence empirically agreeing with a few of its values.

31 Rational asymptotics with effective fixed-order errors

The asymptotic conclusions of this section overlap the earlier all-height theorem (Part I). The recurrence gives a short independent coefficient algorithm and convenient explicit error bounds. In particular, the leading constant is a recovered result, not a new priority claim.

31.1 The normalized recurrence is a contraction

Divide (26.3) by $A_n M_{n+1}$. For $n \geq 2$,

$$b_{n+1} = R(n)b_n + S(n), \quad (31.1)$$

where

$$R(n) = -\frac{n(n+1)^2(4n-1)}{3(n-1)(3n+1)(3n+2)(4n+3)}, \quad (31.2)$$

$$S(n) = \frac{7n^2 - 1}{4(n-1)(3n+1)(3n+2)(2n-1)^2(2n+1)^2(4n+3)}. \quad (31.3)$$

Both rational functions are regular for $n \geq 2$, and

$$R(n) \longrightarrow -\frac{1}{27}, \quad S(n) = \frac{7}{2304}n^{-6} + O(n^{-7}). \quad (31.4)$$

A convenient global bound is

$$|R(n)| \leq \frac{1}{6} \quad (n \geq 2). \quad (31.5)$$

Indeed, $n + 1 \leq 3n/2$, $4n - 1 \leq 4n$, $n - 1 \geq n/2$, and each of $3n + 1, 3n + 2, 4n + 3$ is at least its leading monomial. The numerator is at most $9n^4$ and the denominator is at least $54n^4$.

Let $t = 1/n$, $r(t) = R(1/t)$ and $s(t) = t^{-6}S(1/t)$. These functions are rational and analytic at zero. Seek

$$b_n \sim n^{-6}C(1/n), \quad C(t) = \sum_{j \geq 0} c_j t^j.$$

The formal equation is

$$(1+t)^{-6}C\left(\frac{t}{1+t}\right) = r(t)C(t) + s(t). \quad (31.6)$$

In the coefficient of t^j , the new unknown c_j occurs with coefficient $1 - r(0) = 28/27$. Therefore this equation determines one and only one formal series, with every $c_j \in \mathbb{Q}$. Its first coefficient is

$$c_0 = \frac{7/2304}{1 + 1/27} = \frac{3}{1024}.$$

Only finitely many rational operations are required for any chosen order.

Theorem 31.1 (Every fixed order with a computable error). *For every fixed integer $K \geq 0$, there are explicitly computable rational constants $E_K > 0$ and integer $N_K \geq 2$ such that*

$$\left| b_n - n^{-6} \sum_{j=0}^K c_j n^{-j} \right| \leq E_K n^{-K-7} \quad (n \geq N_K). \quad (31.7)$$

The coefficients are exactly those of (31.6). In particular,

$$\begin{aligned} b_n = & \frac{3}{1024}n^{-6} + \frac{15}{1024}n^{-7} + \frac{45}{1024}n^{-8} + \frac{99}{1024}n^{-9} \\ & + \frac{2367}{16384}n^{-10} + \frac{279}{16384}n^{-11} - \frac{56751}{65536}n^{-12} + O(n^{-13}). \end{aligned} \quad (31.8)$$

Proof. Fix K , write $p = K + 7$ and $q_K(n) = n^{-6} \sum_{j=0}^K c_j n^{-j}$. Its exact residual is

$$d_K(n) = q_K(n+1) - R(n)q_K(n) - S(n).$$

The formal construction says $d_K(n) = O(n^{-p})$. It also supplies an effective rational bound as follows. Express

$$t^{-p}d_K(1/t) = \frac{P(t)}{H(t)}, \quad H(0) > 0, \quad (31.9)$$

with rational polynomials P, H . Remove any common powers of t before this expression; the rational function is analytic at zero. Write $P(t) = \sum P_i t^i$, $H(t) = \sum H_i t^i$. Choose an integer $N \geq 2$ large enough that

$$h_N := H_0 - \sum_{i \geq 1} |H_i| N^{-i} > 0, \quad \rho_N := \left(\frac{N}{N+1} \right)^p > \frac{1}{6}. \quad (31.10)$$

Both conditions eventually hold. Then an explicit bound is

$$C_D = \frac{\sum_i |P_i| N^{-i}}{h_N}, \quad |d_K(n)| \leq C_D n^{-p} \quad (n \geq N). \quad (31.11)$$

The initial b_N is an exact rational number, obtained from the proved recurrence or directly from the period formula. Define

$$E_K = \max \left\{ N^P |b_N - q_K(N)|, \frac{C_D}{\rho_N - 1/6} \right\}. \quad (31.12)$$

If this maximum is zero one may replace it by any positive rational. Set $e_n = b_n - q_K(n)$. From (31.1), $e_{n+1} = R(n)e_n - d_K(n)$. If $|e_n| \leq E_K n^{-P}$, then

$$|e_{n+1}| \leq (E_K/6 + C_D)n^{-P} \leq E_K \rho_N n^{-P} \leq E_K (n+1)^{-P}.$$

The last inequality uses $(n/(n+1))^P \geq \rho_N$. The initial inequality follows from (31.12), proving the bound by induction. This is a proof for the actual counting solution, not just a formal-series calculation. $\square$

The constants in (31.9)–(31.12) can be obtained by finite polynomial arithmetic and finitely many exact recurrence steps. The bound deliberately favors transparency over numerical sharpness. It does not impose an upper limit on the fixed order K .

31.2 Restoring the multinomial factor

The logarithmic Stirling expansion gives

$$\begin{aligned} \log M_n &\sim n \log 256 - \frac{3}{2} \log n - \log(\sqrt{2} \pi^{3/2}) \\ &\quad + \sum_{j \geq 1} \frac{B_{2j}}{2j(2j-1)} (4^{-(2j-1)} - 4) n^{-(2j-1)}, \end{aligned} \quad (31.13)$$

interpreted at each fixed truncation, with B_{2j} the Bernoulli numbers. In particular, the first logarithmic correction is $-5/(16n)$. The constant follows directly from $\sqrt{8\pi n}/(2\pi n)^2 = 1/(\sqrt{2} \pi^{3/2} n^{3/2})$.

Corollary 31.2 (The count expansion). *For every fixed $K \geq 0$,*

$$a_n = \frac{3}{1024\sqrt{2} \pi^{3/2}} 256^n n^{-15/2} \left(1 + \sum_{j=1}^K d_j n^{-j} + O_K(n^{-K-1}) \right), \quad (31.14)$$

with rational coefficients. The first six are

$$\begin{aligned} d_1 &= \frac{75}{16}, & d_2 &= \frac{6905}{512}, & d_3 &= \frac{233985}{8192}, \\ d_4 &= \frac{20845307}{524288}, & d_5 &= -\frac{66392235}{8388608}, & d_6 &= -\frac{79147528275}{268435456}. \end{aligned} \quad (31.15)$$

Proof. Multiply the expansion of Theorem 31.1 by the fixed-order exponential of (31.13). Its leading coefficient is $\frac{c_0}{\sqrt{2} \pi^{3/2}}$. Every further coefficient is obtained by finite addition and multiplication of rational numbers. Stirling's remainder and (31.7) justify multiplication at each fixed order. $\square$

The leading term in (31.14) agrees with the general fixed-height constant of the prior report at $m = 4$. The factor n^{-6} in b_n accounts for the difference between the multinomial exponent $3/2$ and the count exponent $15/2$.

31.3 An explicit logarithmic remainder for inversion

The following construction makes the error term usable without concealing an unknown constant. Define

$$C_* := \frac{3}{1024\sqrt{2}\pi^{3/2}}, \quad \lambda = \log 256, \quad \beta_* := \frac{15}{2}. \quad (31.16)$$

For each fixed K , there are rational $\ell_1, \dots, \ell_K$ and computable $C_K > 0, N_K$ such that

$$\left| \log a_n - \left(\lambda n - \beta_* \log n + \log C_* + \sum_{j=1}^K \ell_j n^{-j} \right) \right| \leq C_K n^{-K-1} \quad (n \geq N_K). \quad (31.17)$$

The ℓ_j are the coefficients of $\log(1 + \sum_{j \geq 1} d_j t^j)$.

Here is one fully explicit construction of C_K . Retain an $E = E_K$ and initial lower bound from Theorem 31.1. Put

$$u(t) = \sum_{j=1}^K \frac{c_j}{c_0} t^j, \quad U = \sum_{j=1}^K \left| \frac{c_j}{c_0} \right| N^{-(j-1)}.$$

Increase the integer N until $U/N \leq 1/4$ and $(E/c_0)N^{-K-1} \leq 1/4$. Increasing N preserves the already-proved error bound. For $n \geq N$, both $1 + u(1/n)$ and $b_n n^6/c_0$ are at least $1/2$, so the mean value theorem for log gives

$$\left| \log \frac{b_n n^6}{c_0} - \log(1 + u(1/n)) \right| \leq \frac{2E}{c_0} n^{-K-1}.$$

Let

$$L_K(t) = \sum_{h=1}^K \frac{(-1)^{h+1}}{h} u(t)^h, \quad V = \sum_{j>K} |[t^j] L_K(t)| N^{-(j-K-1)}.$$

The omitted tail of the scalar logarithm series is bounded by $4U^{K+1}n^{-K-1}/(3(K+1))$. The part of the finite polynomial L_K above degree K is bounded by Vn^{-K-1} . For $K = 0$, take $u = L_K = 0$ and $U = V = 0$. Thus the b_n part contributes at most

$$C_b = \frac{2E}{c_0} + V + \frac{4U^{K+1}}{3(K+1)}. \quad (31.18)$$

Let $L = \lfloor (K+1)/2 \rfloor$ and $q = 2L+1$. The classical positive-real Stirling remainder after L Bernoulli terms has absolute value at most $|B_{2L+2}|/((2L+2)(2L+1)x^{2L+1})$; this follows, for example, from the integral remainder in the Euler–Maclaurin formula for $\log \Gamma$ [14]. Apply it separately at $x = 4n$ and at $x = n$. Since $q \geq K+1$, the multinomial logarithm contributes at most

$$C_M = \frac{|B_{2L+2}|}{(2L+2)(2L+1)} (4^{-q} + 4) N^{K+1-q} \quad (31.19)$$

times n^{-K-1} . Take $C_K = C_b + C_M$ in (31.17). This provides a finite, effective remainder certificate for every fixed order, using only the displayed polynomial and rational bounds plus the standard Stirling inequality.

31.4 Why this does not specify a full transseries

The homogeneous equation associated with (31.1) has solution proportional to $\prod R(j)$, whose successive absolute ratio tends to $1/27$. The first-order equation therefore explains stability of the algebraic expansion. It does not make a particular all-orders formal series into a canonical summed particular solution. Changing an exact particular solution by a homogeneous solution changes its exponentially small component without changing any inverse-power coefficient. Neither the finite-order bounds nor the formal branches alone select Stokes data or a canonical exponentially small transseries decomposition. No such conclusion is used here.

32 Finite Lambert W models and threshold inverses

The sequence is defined at integers. We therefore distinguish a smooth finite model, which can be inverted as a real function, from the exact integer threshold of the count. This distinction prevents an unjustified rounding claim near an integer and avoids differentiating an unspecified discrete remainder.

32.1 A monotone finite model

Fix $K \geq 0$ and use the coefficients and error bound from (31.17). Define, for real $x > 0$,

$$\Psi_K(x) = \lambda x - \beta_* \log x + \log C_* + \sum_{j=1}^K \ell_j x^{-j}. \quad (32.1)$$

Choose X_K large enough that

$$\frac{\beta_*}{X_K} + \sum_{j=1}^K j |\ell_j| X_K^{-j-1} \leq \frac{\lambda}{2}. \quad (32.2)$$

Then $\Psi'_K(x) \geq \lambda/2$ for $x \geq X_K$. Hence for every $y \geq \exp \Psi_K(X_K)$ there is a unique $\xi_K(y) \geq X_K$ with

$$\Psi_K(\xi_K(y)) = \log y. \quad (32.3)$$

This defines an actual finite smooth model and a genuine inverse, rather than an unspecified interpolation of the sequence.

For $K = 0$, inversion is explicit:

$$w(y) := \xi_0(y) = -\frac{\beta_*}{\lambda} W_{-1} \left(-\frac{\lambda}{\beta_*} \left(\frac{C_*}{y} \right)^{1/\beta_*} \right). \quad (32.4)$$

For sufficiently large y the argument is in $(-1/e, 0)$. The branch W_{-1} is selected because it tends to $-\infty$ as the argument tends to 0 from below, giving the large positive solution. Indeed, taking β_* th roots in $y = C_* e^{\lambda w} w^{-\beta_*}$ gives $w e^{-\lambda w/\beta_*} = (C_*/y)^{1/\beta_*}$, which proves (32.4). The principal branch would give the small solution on the decreasing part of this same leading model.

32.2 Arbitrary fixed corrections to the Lambert solution

The finite model admits an explicitly recursive inverse expansion. Put $t = 1/w$ and seek

$$\widehat{\xi}_K(y) = w + \Delta_K(1/w), \quad \Delta_K(t) = \sum_{j=1}^K s_j t^j. \quad (32.5)$$

The coefficients are determined by setting the coefficients of $t, t^2, \dots, t^K$ to zero in

$$G_K(t) = \lambda \Delta_K(t) - \beta_* \log(1 + t \Delta_K(t)) + \sum_{j=1}^K \ell_j t^j (1 + t \Delta_K(t))^{-j}. \quad (32.6)$$

At order j , the new s_j occurs only in $\lambda \Delta_K$, so the recursion is uniquely solvable and $s_j \in \mathbb{Q}[\lambda^{-1}]$. For example,

$$\ell_1 = \frac{75}{16}, \quad \ell_2 = \frac{5}{2}, \quad s_1 = -\frac{75}{16\lambda}, \quad s_2 = -\frac{5}{2\lambda} - \frac{1125}{32\lambda^2}. \quad (32.7)$$

Thus corrections begin at w^{-1} ; the leading Lambert model already incorporates the logarithmic displacement.

Proposition 32.1 (A finite inverse with an effective error). *For each fixed K , constants H_K, W_K can be computed from the finite expression (32.6) such that, whenever $w(y) \geq W_K$ and $y \geq \exp \Psi_K(X_K)$,*

$$|\xi_K(y) - \widehat{\xi}_K(y)| \leq \frac{2H_K}{\lambda} w(y)^{-K-1}. \quad (32.8)$$

Proof. Choose $W_K \geq 2X_K$ large enough that $|t\Delta_K(t)| \leq 1/2$ for $0 \leq t \leq 1/W_K$. This can be ensured by the finite coefficient bound $\sum_j |s_j| W_K^{-j-1} \leq 1/2$. Then $\widehat{\xi}_K = w(1 + t\Delta_K(t)) \geq w/2 \geq X_K$. Since G_K has its coefficients through degree K equal to zero, Taylor's theorem gives

$$|G_K(t)| \leq H_K t^{K+1} \quad (0 \leq t \leq 1/W_K).$$

One explicit way to obtain the constant is to differentiate $K+1$ times. Writing $z(t) = 1 + t\Delta_K(t)$, the derivative has the form

$$G_K^{(K+1)}(t) = \frac{P_K(t)}{z(t)^{2K+1}}$$

for a computable real polynomial P_K with coefficients in $\mathbb{Q}[\lambda, \lambda^{-1}]$. For $K=0$ take $P_0 = 0$. The claimed form follows by repeated differentiation of the logarithm and the finitely many negative integer powers, then bringing them to a common denominator. Thus one may take

$$H_K = \frac{2^{2K+1}}{(K+1)!} \sum_i |[t^i] P_K(t)| W_K^{-i}. \quad (32.9)$$

The identity $\Psi_0(w) = \log y$ shows exactly that $\Psi_K(\widehat{\xi}_K) - \log y = G_K(1/w)$. Both ξ_K and $\widehat{\xi}_K$ lie on the interval where the derivative is at least $\lambda/2$. The mean value theorem now gives (32.8). $\square$

All objects in this proof are finite functions. No convergence of the infinite inverse series is asserted or needed. Interval evaluations of $\lambda = \log 256$ and of the finite polynomials suffice if numerical certified constants rather than the displayed exact expressions are wanted.

32.3 The integer threshold and its rounding envelope

Define

$$N(y) = \min\{n \geq 1 : a_n \geq y\}. \quad (32.10)$$

This is defined for all positive y because $a_n \rightarrow \infty$. The sequence is eventually strictly increasing. One direct effective proof uses (31.17) at $K=0$: for large enough n ,

$$\begin{aligned} \log a_{n+1} - \log a_n &\geq \lambda - \beta_* \log(1 + 1/n) - C_0(n^{-1} + (n+1)^{-1}) \\ &\geq \lambda - \frac{\beta_* + 2C_0}{n}. \end{aligned}$$

Taking $n \geq 2(\beta_* + 2C_0)/\lambda$, as well as the onset required by (31.17), makes this at least $\lambda/2$. No derivative of an error sequence has been taken.

Fix K , let $p = K+1$, and choose an integer N_* beyond this monotonicity onset, beyond the onset of (31.17), and beyond X_K . Choose y large enough that

$$\begin{aligned} y &> \max_{1 \leq n < N_*} a_n, \quad \xi := \xi_K(y) \geq \max(2N_* + 2, 4), \\ \delta &:= \frac{2^{p+1} C_K}{\lambda} \xi^{-p} \leq 1. \end{aligned} \quad (32.11)$$

Every condition is verifiable from the finite bounds and initial values.

Theorem 32.2 (Qualified threshold envelope). *Under (32.11),*

$$\lceil \xi_K(y) - \delta \rceil \leq N(y) \leq \lceil \xi_K(y) + \delta \rceil, \quad \delta = \frac{2^{K+2}C_K}{\lambda} \xi_K(y)^{-K-1}. \quad (32.12)$$

Proof. Let $m = \lceil \xi + \delta \rceil$. The derivative bound gives $\Psi_K(m) - \log y \geq \lambda\delta/2$. Since $m \geq \xi \geq N_*$, its logarithmic error is at most $C_K m^{-p} \leq C_K \xi^{-p} \leq \lambda\delta/2$. Therefore $a_m \geq y$.

Let $j = \lceil \xi - \delta \rceil - 1$. Then $j < \xi - \delta$ and $j \geq \xi - 2 \geq \xi/2 \geq N_*$. Hence $\Psi_K(j) - \log y < -\lambda\delta/2$, whereas its logarithmic error is at most $C_K j^{-p} \leq 2^p C_K \xi^{-p} = \lambda\delta/2$. Thus $a_j < y$. Eventual monotonicity and the finite-prefix condition in (32.11) rule out an earlier crossing. These two integer comparisons prove (32.12). $\square$

For sufficiently large y , the envelope has width less than one in its real argument. It gives a unique integer whenever its two ceilings coincide. If an endpoint approaches an integer, the theorem may leave two adjacent possibilities; checking their exact recurrence values resolves the threshold. An unconditional instruction to round a real asymptotic inverse would lose this necessary qualification.

The explicit Lambert correction (32.5) can replace ξ_K . For example, write $E' = 2H_K/\lambda$ and $D = 2^{p+1}C_K/\lambda$. Enlarge the onset until $|\xi_K - \widehat{\xi}_K| \leq E'w^{-p}$ and $\xi_K \geq w/2$. Then (32.12) implies the entirely explicit envelope

$$\left\lceil \widehat{\xi}_K - (E' + 2^p D)w^{-p} \right\rceil \leq N(y) \leq \left\lceil \widehat{\xi}_K + (E' + 2^p D)w^{-p} \right\rceil. \quad (32.13)$$

Here w is the Lambert expression (32.4), $\widehat{\xi}_K$ is a finite correction polynomial, and every error constant has already been specified. The increasing onset is part of the conclusion; it is not a claim of sharp small- n rounding accuracy.

33 Exact verification and reproducibility

33.1 Proof certificates and finite checks have different roles

The proof's universal algebraic objects are (29.4), its boundary identities, the rational substitution (30.2), and the three elimination coefficients. These are identities in polynomial or rational-function rings. Their verification does not depend on a finite numerical sample of n . The order-polytope, Pfaffian, integration-by-parts, contraction, and inverse arguments are the mathematical proofs in the preceding sections.

The accompanying core code uses the Python standard library only. It implements sparse polynomials over exact rational numbers and rational functions reduced by polynomial arithmetic. All proof-critical tests raise explicit exceptions on failure, rather than relying on statements removed by Python's optimized mode.

The principal files are:

- `code/exact_algebra.py`: rational sparse polynomial and rational-function operations;
- `code/formal_series.py`: fixed-order formal arithmetic, shifts, Bernoulli numbers, and the Stirling product;
- `code/verify_report231.py`: certificate identities, independent exact counting checks, and series checks;
- `code/receipt.json`: deterministic output of the verifier;
- `build.py`: frozen-inventory verification and isolated PDF/archive replay.

The package includes the editable L^AT_EX source and this PDF. It does not redistribute third-party source papers.

33.2 Three independent ways to evaluate the finite count

A legal prefix of an array is described by the number of assigned entries in each row, $(\lambda_1, \dots, \lambda_4)$. Rows increase, so each row prefix is an initial interval. Literal cell comparisons say exactly that

$$n \geq \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0, \quad \lambda_i = \lambda_{i+1} \implies \lambda_i \in \{0, n\}. \quad (33.1)$$

To see the strictness clause, an occupied lower-row cell in column $j < n$ requires its upper-row neighbor in column $j + 1$ by the antidiagonal comparison. If the lower row has length n , the upper row must also have length n . Empty adjacent rows impose no strictness. Conversely these conditions include the row, column, and antidiagonal predecessors of each occupied cell; the diagonal condition is redundant. Starting from $(0, 0, 0, 0)$, increment one legal coordinate and accumulate integer path counts until (n, n, n, n) . This dynamic program uses no candidate recurrence or triangle integral.

A second evaluator integrates (28.23) directly in positive barycentric coordinates on the triangle. Put

$$r = 1 - u, \quad s = 1 - v, \quad t = 1 - r - s = u + v - 1.$$

Then $u = s + t$, $v = r + t$, and

$$Z_n = n^2 \int_{r,s>0, r+s<1} (s+t)^n (r+t)^{n-1} t^{n-1} dr ds.$$

Expanding the two positive binomial powers and using

$$\int_{r,s>0, r+s<1} r^a s^b (1-r-s)^c dr ds = \frac{a!b!c!}{(a+b+c+2)!}$$

gives a finite positive rational sum. It is independent of both the period recurrence and the prefix-state dynamic program. The displayed Dirichlet integral follows by two beta integrals, or by integrating monomials on a simplex.

A third check uses the exact polynomial

$$h(t) = \sum_{j=0}^{n-1} \frac{(-1)^j t^{n+j}}{(n-1-j)!(n+j)!}, \quad (33.2)$$

obtained by expanding $(1-r)^{n-1}$ in the defining integral and applying a beta integral. It yields

$$Z_n = (n!)^2 \sum_{j=0}^{n-1} \frac{(-1)^j}{(n-1-j)!(n+j)!(2n+j+1)}. \quad (33.3)$$

The opposite signs make this a less stable floating-point formula, but exact rational arithmetic has no cancellation error. It is convenient for independent symbolic checking of the trace reduction.

The package verifies these count evaluations, the printed finite sum, and both recurrences for every width $1 \leq n \leq 20$ where their indices apply. The first ten values are

n	a_n	n	a_n
1	1	6	352184
2	1	7	25097600
3	8	8	2152061145
4	169	9	212012802584
5	6392	10	23263015359672

These tests help detect transcription errors. They do not establish a uniform recurrence by induction on the number of successful samples.

33.3 Replaying the public package

From the package directory, run

```
python -B code/verify_report231.py --self-test
```

for the standard-library certificate checks. It prints deterministic JSON; `--receipt` accepts a fresh destination and refuses an existing file. To verify the frozen package inventory without rebuilding, use

```
python -B build.py --verify-only
```

To rebuild, provide a new directory outside the package:

```
python -B build.py --output /path/to/new-output
```

The build repeats exact checks, creates a private TeX format/cache, compiles the source, compares the PDF with the frozen copy, and writes a deterministic archive. A TeX Live installation with the packages named in `article.tex` is needed for PDF reproduction; the exact checker itself needs only Python. The build never needs network access. The same commands may be run with `-O`; every Python child process explicitly receives `-B`, and optimized-mode guards remain active.

The manifest covers every packaged file except itself and rejects missing, changed, or extra files and unexpected directories. Output paths must be new, outside the source package, and free of symlink ancestors. Source inventory is compared before and after the build. The shipped PDF and archive are pinned by their public digest records. Byte-for-byte PDF reproduction is expected under the recorded TeX toolchain; a different font or TeX version can legitimately require a new rendering review.

[write] Added 5 October 2026, batch 101: the shipped layout. “The package” in this section is the delivered archive. In Provelt its programs and receipt carry the prefix `03-height-four-`:

Delivered as	Shipped as
<code>code/name.py</code> (three modules)	<code>code/03-height-four-name.py</code>
<code>build.py</code>	<code>code/03-height-four-build.py</code>
<code>code/receipt.json</code>	<code>data/03-height-four-receipt.json</code>
<code>code/README.md</code>	<code>03-height-four-code-README.md</code>
<code>SOURCES.md</code>	<code>03-height-four-SOURCES.md</code>

The editable source, this PDF and `MANIFEST.sha256` named above are not shipped. `verify_report231.py` imports `exact_algebra` and `formal_series` under their delivered names, and `build.py` needs the manifest and the frozen PDF, so neither runs in place; the verifier runs on a copy with the delivered names restored, and the README of the report gives the commands. At the write (5 October 2026, Python 3.14.4 on Windows, on such a copy) `python -B code/verify_report231.py --self-test` printed `data/03-height-four-receipt.json` byte for byte after removal of carriage returns, with and without `-O`, in about three seconds. `build.py` was not run. The verifier hard-codes the three coefficient polynomials as displayed in the A181198 entry and checks the elimination against them; as OEIS content they are published by The OEIS Foundation Inc. under the Creative Commons Attribution-ShareAlike 4.0 licence.

33.4 What remains outside this report

The height-five Conjecture 19 is not implied by the four-variable Pfaffian calculation. Higher Pfaffian sizes contain further trace invariants, so an analogous low-dimensional reduction would require new work.

Likewise, the stable first-order normalized recurrence controls every fixed algebraic order but does not by itself canonically resolve exponentially small sectors. Neither extension is claimed here.

The result established in full is the height-four count, its specified operator, and the primary paper's finite sum. The proof and exact certificate are available for independent checking without trusting a recurrence guess, a finite match, or a symbolic algebra package.

33.5 [write] The order-two operator is minimal

Added 5 October 2026, batch 101. This subsection is the write's, not the manuscript's, which makes no minimality claim. It shows that A181198 obeys no first-order recurrence, so the operator of Theorem 30.1 has the smallest possible order, and that every second-order recurrence of the sequence is a polynomial multiple of that operator. The proof uses only the period recurrence (29.1), the inhomogeneous identity (26.3) and the affine formula (28.21). In this subsection a *recurrence of order r* for a sequence (s_n) means polynomials $q_0, \dots, q_r \in \mathbb{Q}[n]$ with $q_r \neq 0$ and $\sum_{j=0}^r q_j(n)s_{n+j} = 0$ for all sufficiently large integers n .

Lemma 33.1 ([write] The period is not eventually rational). *There is no rational function $z \in \mathbb{C}(n)$ with $Z_n = z(n)$ for all sufficiently large integers n .*

Proof. Put $a(n) = 3(3n+1)(3n+2)$, $b(n) = (n+1)^2$ and $\phi(n) = (28n^3 + 50n^2 + 29n + 5)/(2(2n+1))$. The numerator of ϕ equals $-1/2$ at $n = -1/2$, so ϕ has a simple pole at $-1/2$ and no other pole. Suppose $Z_n = z(n)$ for $n \geq n_1$. By (29.1), the rational function $a(n)z(n+1) + b(n)z(n) - \phi(n)$ vanishes at every integer $n \geq n_1$, hence identically:

$$a(n)z(n+1) + b(n)z(n) = \phi(n) \quad \text{in } \mathbb{C}(n). \quad (33.4)$$

Suppose z has a pole, and fix a coset $O = \theta + \mathbb{Z}$ of $\mathbb{C}/\mathbb{Z}$ that contains one. The poles of z in O form a finite nonempty set; let β be its largest and γ its smallest element (they differ by a nonnegative integer).

(i) Near $n = \beta$, the term $a(n)z(n+1)$ is regular, because $\beta+1 \in O$ is not a pole. By (33.4), $b(n)z(n) = \phi(n) - a(n)z(n+1)$ is regular at β unless ϕ has a pole there. Since z has a pole at β , either $b(\beta) = 0$, that is $\beta = -1$, or $\beta = -1/2$.

(ii) Near $n = \gamma - 1$, the term $b(n)z(n)$ is regular, because $\gamma - 1 \in O$ is not a pole. By (33.4), $a(n)z(n+1) = \phi(n) - b(n)z(n)$ is regular at $\gamma - 1$ unless $\gamma - 1 = -1/2$. Since $z(n+1)$ has a pole at $n = \gamma - 1$, either $a(\gamma - 1) = 0$, that is $\gamma - 1 \in \{-1/3, -2/3\}$, or $\gamma - 1 = -1/2$. Thus $\gamma \in \{1/3, 2/3, 1/2\}$.

By (i), $O = \mathbb{Z}$ or $O = \frac{1}{2} + \mathbb{Z}$. If $O = \mathbb{Z}$, (ii) is impossible. If $O = \frac{1}{2} + \mathbb{Z}$, then (i) gives $\beta = -1/2$ and (ii) gives $\gamma = 1/2$, contradicting $\gamma \leq \beta$. Hence z has no pole and is a polynomial (possibly zero). Then the left side of (33.4) is a polynomial, while ϕ has a pole at $-1/2$. This contradiction proves the lemma. $\square$

Proposition 33.2 ([write] Minimality at height four). *The sequence $a_n = T_4(n)$ (A181198) has no recurrence of order zero or one; in particular it is not hypergeometric. If q_0, q_1, q_2 give a recurrence of order two, then $(q_0, q_1, q_2) = \rho(p_0, p_1, p_2)$ for a polynomial $\rho \in \mathbb{Q}[n]$, with p_0, p_1, p_2 as in (30.6). Hence the operator of Theorem 30.1 has minimal order, and degree nine is the smallest degree of any recurrence of order two.*

Proof. Every a_n with $n \geq 1$ is a positive integer (the column-by-column interlaced filling is one array).

Order zero. If $q_0(n)a_n = 0$ for all large n , then q_0 has infinitely many zeros, so $q_0 = 0$; this is excluded.

Order one. Suppose $q_1(n)a_{n+1} + q_0(n)a_n = 0$ for $n \geq n_0$ with $q_1 \neq 0$. Enlarging n_0 , we may assume $n_0 \geq 2$ and $q_1(n) \neq 0$ for $n \geq n_0$, so $a_{n+1} = \varrho(n)a_n$ there, with $\varrho = -q_0/q_1 \in \mathbb{Q}(n)$. Substituting into (26.3) gives

$$a_n(A_n\varrho(n) + B_n) = D_nM_n \quad (n \geq n_0).$$

For $n \geq 1$ both D_n and M_n are positive, so $A_n Q(n) + B_n \neq 0$ for every $n \geq n_0$; the rational function $A_n Q + B_n$ is therefore not zero, and

$$b_n = \frac{a_n}{M_n} = \frac{D_n}{A_n Q(n) + B_n} \quad (n \geq n_0)$$

is the value of a rational function. By (28.21), $Z_n = (b_n - r_n)/g_n$ for $n \geq 2$, where $g_n = 4(n-1)/(4n-1) \neq 0$ and r_n are the rational functions of Section 30.1. So Z_n coincides with a rational function for all $n \geq n_0$, contradicting Lemma 33.1.

Order two. Write $(La)(n) = p_2(n)a_{n+2} + p_1(n)a_{n+1} + p_0(n)a_n$ and $(L'a)(n) = q_2(n)a_{n+2} + q_1(n)a_{n+1} + q_0(n)a_n$. By Theorem 30.1 and the hypothesis, both vanish for all large n , and hence so does

$$q_2(n)(La)(n) - p_2(n)(L'a)(n) = (q_2 p_1 - p_2 q_1)(n) a_{n+1} + (q_2 p_0 - p_2 q_0)(n) a_n.$$

By the cases of order zero and one, both polynomial coefficients vanish identically. With $\rho = q_2/p_2 \in \mathbb{Q}(n)$ this says $(q_0, q_1, q_2) = \rho(p_0, p_1, p_2)$. The polynomials p_0, p_1, p_2 have no common factor: the irreducible factors of p_2 are $n+1, n+2, 2n+3, 3n+4, 3n+5$ and $7n^2-1$, and none of them divides p_0 , whose irreducible factors are $2n \pm 1, 4n-1, 4n+1, 4n+3, 4n+5$ and $7n^2+14n+6$. Write $\rho = u/v$ in lowest terms. Since $q_j = up_j/v$ is a polynomial, v divides up_j , hence p_j , for $j = 0, 1, 2$; so v is a constant and ρ is a polynomial. Each p_j has degree nine, so $\deg q_2 = 9 + \deg \rho \geq 9$. $\square$

The same proof works over $\mathbb{C}$ in place of $\mathbb{Q}$. The proposition says nothing about recurrences of order three or more: a left multiple of the operator by a rational operator can have polynomial coefficients of lower degree, and Proposition 33.3 below shows that this happens here. (Revised 5 October 2026 after the intake's independent check, recorded at the end of Section 45.2; this sentence first ended "and whether this happens here is item (F5) below".) The corresponding statement at height five is Proposition 42.2.

Proposition 33.3 was found and proved by the intake's independent check, 5 October 2026; it is neither the manuscript's nor the write's. Below, E is the forward shift, $Ef(n) = f(n+1)E$, and $L = p_2(n)E^2 + p_1(n)E + p_0(n)$ is the operator of Theorem 30.1; a product $(\sum_k Q_k E^k)L$ with rational Q_k is taken in the Ore ring $\mathbb{Q}(n)[E]$.

Proposition 33.3 ([check]) Lower degree at orders three to eight). *Let $a_n = T_4(n)$ (A181198).*

(a) *The left multiple*

$$\frac{3E + 4(1673n + 942)}{7n^2 + 14n + 6} L = \sum_{j=0}^3 q_j(n) E^j$$

has the polynomial coefficients

$$\begin{aligned} q_3 &= 9(n+2)(n+3)^3(2n+5)(3n+7)(3n+8), \\ q_2 &= 12(n+2)(2n+3)(15057n^6 + 97156n^5 + 222866n^4 + 200444n^3 \\ &\quad + 33411n^2 - 21546n + 6340), \\ q_1 &= -32(2n+1)(4n+3)(4n+5)(86996n^5 - 17552n^4 - 200235n^3 \\ &\quad - 68149n^2 + 40579n + 13581), \\ q_0 &= -256(2n-1)^2(2n+1)(4n-1)(4n+1)(4n+3)(4n+5)(1673n+942), \end{aligned}$$

and $\sum_{j=0}^3 q_j(n)a_{n+j} = 0$ for every $n \geq 1$: A181198 obeys a recurrence of order three and degree eight.

(b) *The left multiple*

$$\left(\frac{E^2}{7n^2 + 28n + 27} - \frac{4(7315n^2 + 21945n + 16406)E}{(7n^2 + 14n + 6)(7n^2 + 28n + 27)} + \frac{239136}{7n^2 + 14n + 6} \right) L = \sum_{j=0}^4 \tilde{q}_j(n) E^j$$

has the polynomial coefficients

$$\begin{aligned}
 \tilde{q}_4 &= 3(n+3)(n+4)^3(2n+7)(3n+10)(3n+11), \\
 \tilde{q}_3 &= -4(n+3)(2n+5)(29879n^5 + 358932n^4 + 1691327n^3 \\
 &\quad + 3891444n^2 + 4346534n + 1869504), \\
 \tilde{q}_2 &= 32(2n+3)(1069163n^6 + 8737560n^5 + 26637818n^4 \\
 &\quad + 34145688n^3 + 7805720n^2 - 19807932n - 12702660), \\
 \tilde{q}_1 &= -256(2n+1)(4n+3)(4n+5)(321716n^4 - 767080n^3 \\
 &\quad - 1948559n^2 - 1282655n - 449532), \\
 \tilde{q}_0 &= -15304704(2n-1)^2(2n+1)(4n-1)(4n+1)(4n+3)(4n+5),
 \end{aligned}$$

and $\sum_{j=0}^4 \tilde{q}_j(n)a_{n+j} = 0$ for every $n \geq 1$: A181198 obeys a recurrence of order four and degree seven.

- (c) Every recurrence of A181198, in the sense of this subsection, holds for every $n \geq 1$. No recurrence of order at most three has degree below eight, and no recurrence of order at most eight has degree below seven. Hence the smallest degree of a recurrence of order r is nine for $r = 2$, eight for $r = 3$, and seven for $4 \leq r \leq 8$.

Proof. (a) and (b). Since $7(n+1)^2 - 1 = 7n^2 + 14n + 6$ and $7n^2 + 28n + 27$ is $7n^2 + 14n + 6$ at $n+1$, the first quadratic divides $p_0(n)$ and $p_2(n+1)$, and the second divides $p_0(n+1)$ and $p_2(n+2)$. Expanding the products with $Ef(n) = f(n+1)E$, the coefficient of E^j is $\sum_k Q_k(n)p_{j-k}(n+k)$ ($p_i = 0$ for $i \notin \{0, 1, 2\}$); clearing the denominators and comparing coefficients gives the displayed polynomials, a finite identity over $\mathbb{Q}[n]$ checked exactly. The quadratic $7n^2 + 14n + 6$ has discriminant 28 and no rational zero, so neither quadratic vanishes at an integer, and for $n \geq 1$

$$\sum_j q_j(n)a_{n+j} = \frac{3(La)(n+1) + 4(1673n + 942)(La)(n)}{7n^2 + 14n + 6} = 0$$

by Theorem 30.1; the same argument with three terms proves (b).

(c) Let $R = \sum_{j=0}^r q_j E^j$ be a recurrence of order r , valid for $n \geq n_0$; by Proposition 33.2, $r \geq 2$. Since $p_2 \neq 0$, right division by L in $\mathbb{Q}(n)[E]$ gives unique rational functions $Q_0, \dots, Q_{r-2}, s_0, s_1$ with $R = (\sum_{k=0}^{r-2} Q_k E^k)L + s_1 E + s_0$. Comparing the coefficients of $E^r, E^{r-1}, \dots, E^2$ determines them from the top: $Q_{r-2} = q_r/p_2(n+r-2)$ and

$$Q_k = \frac{q_{k+2} - Q_{k+1}(n)p_1(n+k+1) - Q_{k+2}(n)p_0(n+k+2)}{p_2(n+k)} \quad (k = r-3, \dots, 0),$$

with $Q_k = 0$ for $k > r-2$; then $s_1 = q_1 - Q_0 p_1 - Q_1(n)p_0(n+1)$ and $s_0 = q_0 - Q_0 p_0$. By induction from the top, the denominator of Q_k divides $\prod_{i=k}^{r-2} p_2(n+i)$, and those of s_0, s_1 divide $\prod_{i=0}^{r-2} p_2(n+i)$. The real zeros of p_2 are $-2, -5/3, -3/2, -4/3, -1$ and $\pm 1/\sqrt{7}$, all below one, so no $p_2(n+i)$ with $i \geq 0$ vanishes at an integer $n \geq 1$, and every Q_k, s_0, s_1 is finite there. At such n the coefficient identity gives

$$(Ra)(n) = \sum_{k=0}^{r-2} Q_k(n)(La)(n+k) + s_1(n)a_{n+1} + s_0(n)a_n.$$

For $n \geq n_0$ the left side and every $(La)(n+k)$ vanish, so $s_1(n)a_{n+1} + s_0(n)a_n = 0$ there. Multiplied by a common denominator, this is a polynomial relation of order at most one that holds for all large n ; by the cases of order zero and one in the proof of Proposition 33.2 both of its coefficients vanish identically, so

$s_0 = s_1 = 0$. Hence $(Ra)(n) = \sum_k Q_k(n)(La)(n+k) = 0$ for every $n \geq 1$, by Theorem 30.1. This proves the first assertion.

Now let R have order at most $\bar{r}$ and degree at most d , and write $q_j = \sum_{i=0}^d x_{ji} n^i$ for $0 \leq j \leq \bar{r}$, with $x_{ji} = 0$ for j above the order of R . By the first assertion the vector $(x_{ji}) \neq 0$ solves the integer linear system $\sum_{j,i} n^i a_{n+j} x_{ji} = 0$, $n = 1, \dots, N$, whose $(\bar{r}+1)(d+1)$ columns are indexed by (j, i) and whose entries are the exact values of a_n that Theorem 30.1 determines from $a_1 = a_2 = 1$. For $(\bar{r}, d, N) = (3, 7, 72)$ the 72×32 matrix has rank 32 over $\mathbb{Q}$, and for $(\bar{r}, d, N) = (8, 6, 103)$ the 103×63 matrix has rank 63 (exact fraction-free computation; also full rank modulo $2^{61} - 1$, which by itself implies full rank over $\mathbb{Q}$; recorded in `data/check-lower-order-recurrences.json`). So no such R exists. Proposition 33.2 gives degree nine at order two, (a) gives degree eight at order three, and (b) and its shifts $\sum_{j=0}^4 \tilde{q}_j(n+k)a_{n+k+j} = 0$, $1 \leq k \leq 4$, give degree seven at orders four to eight. $\square$

Proposition 42.3 is the height-five counterpart. The coefficients of (a) and (b) and the ranks used in (c) are also in `data/check-lower-order-recurrences.json`.

33.6 Further questions and research recorded at the write

Added 5 October 2026, batch 101, under Vladimir Reshetnikov's standing rule that unproved claims are recorded as open questions rather than dropped. The manuscript states no unproved claim; the items below are what it constructs without evaluating, what it leaves outside its scope, and the minimality question that the batch-101 placement assigned to this section. The labels continue those of Section 24.1.

- (F5) *Minimality of the A181198 recurrence* (the placement; no claim in the manuscript). Settled for the order, and for the degree among recurrences of order two, by Proposition 33.2. At orders three to eight it is settled by Proposition 33.3: the smallest degree is eight at order three and seven at each order from four to eight. Still open: whether some recurrence of order nine or more has degree below seven. Missing: a degree bound for left multiples QL that holds at every order (the rank computation in the proof of Proposition 33.3 treats each range of orders separately). (Re-scoped 5 October 2026 after the intake's independent check, which found and proved Proposition 33.3; this item first read: "Still open: whether some recurrence of order three or more has polynomial coefficients of degree below nine. Missing: a degree bound for left multiples QL with Q a rational operator, or an exhaustive search over small orders and degrees.")
- (F6) *Numerical constants* (Sections 31 and 32). The manuscript constructs $E_K, N_K, C_K, H_K, W_K, X_K$ and the onset (32.11) by finite rational arithmetic, but evaluates none of them; its verifier does not compute them (its README says so). Missing: the evaluation, for example at $K = 0$ and $K = 6$, as Part IV's receipt does at height five (40.13), and a worked certified threshold.
- (F7) *Even heights six and more* (Section 33.4). "Higher Pfaffian sizes contain further trace invariants, so an analogous low-dimensional reduction would require new work." Missing: a reduction of the traces $\text{tr}(B^j)$, $j \leq m/2$, to a few polynomial periods, with their contiguity; Research question 16 asks the same at odd heights.
- (F8) *Exponentially small content* (Section 31.4, a non-claim). The homogeneous mode of (31.1) has ratio tending to $-1/27$; the manuscript selects no canonical exponentially small component and no Stokes data. Missing: the analysis that Research question 18 asks for at height five.

Part IV

A complete exact formula for height five shifted rectangles

A proof of A181199 and Kauers and Koutschan Conjecture 19

Research report 233, prepared for Vladimir Reshetnikov, October 5, 2026

Main conclusion. The full height-five array count satisfies the nested finite sum printed as Conjecture 19 of Kauers and Koutschan and the literal third-order, degree-24 recurrence recorded in OEIS A181199, for every integer $n \geq 1$. All boundary contributions are retained.

Abstract. Column interlacing turns the counting problem into an exterior Volterra transfer. Two odd augmented Pfaffians reduce its exact volume to traces and three open-chain integrals. A rank-two calculation closes the length-four open chain and expresses the full count through two polynomial periods. We derive differential equations, polynomial contiguity, and a finite moment closure with every endpoint specified. Rational identities then give a first difference, the actual printed nested sum with its rising factorials, and an explicit factorization of the prescribed OEIS operator. The proof is uniform in the width, with separate treatment of width one. Supplementary exact arithmetic verifies the symbolic certificates and independent finite counts. Recurrence-based asymptotic and inverse-index consequences are separated from the exact enumeration; the earlier fixed-height asymptotic results receive explicit credit.

Status and limits. AI-assisted research manuscript, not a peer-reviewed or proof-assistant formalization. Historical priority is not certified. The recurrence is proved on $n \geq 1$; it fails at $n = 0$ under the empty-array convention.

[write] Added 5 October 2026, batch 101: provenance of Part IV. Part IV is a single manuscript, printed in full: Research Report 233 of the session bundle that arrived in commit `60f54ea06` and was placed in commit `f7c612c72` (batch 101) as an addition to this report. Its title is *A complete exact formula for height five shifted rectangles*, subtitled “A proof of A181199 and Kauers and Koutschan Conjecture 19”, dated October 5, 2026 (a 30-page PDF; a main file and eleven section files, 1,860 lines with 135 labels). It arrived as the archive `Report233.zip` (614,266 bytes, 22 files under the doubled wrapper directory `Report233/Report233/`). Its title page reads “Prepared for Vladimir Reshetnikov” and its PDF metadata name “Research report prepared for Vladimir Reshetnikov”; no human author and no tool are named.

Pin. Like Part III, the manuscript names no commit and pins this report’s source by the blob `2b40ded9daa6549f786a52b880d5477de3b5adfc`, the text of `6fef5383b` (3 October 2026): it read Part I only. It cites neither the manuscript of Part II nor that of Part III.

Files. Its five programs (its `exact_algebra.py` is not the same file as Part III’s), its receipt and its source record are shipped with the prefix `04-height-five-`, in `code/`, in `data/` and at the report root; the note at the end of Section 42.3 lists them. Its $\text{\LaTeX}$ source, delivered `README`, `PDF` and `MANIFEST.sha256` (21 entries, all verified at placement) are not shipped and survive in the arrival commit’s archive. The manuscript is unrefereed and AI-assisted; nothing in it is formalized, and its place in this report confers no formal status.

What it answers. Theorems 34.1 and 34.2 prove, from the array definition and for every $n \geq 1$, the nested finite sum printed as Conjecture 19 of [9] and the order-three, degree-24 recurrence displayed as conjectural in OEIS A181199. With Part III this answers Research question 1 of Part I and Research question 8 of Part II; dated notes there say so. The manuscript also shows that the operator does not extend to $n = 0$ under $a_0 = 1$ (residual 5621993879040000) and that

its leading coefficient c_3 is positive for $n \geq 1$. The write adds Section 42.5: a remark that the period and its recurrence are Part III's, and a proof that A181199 obeys no first-order recurrence, so the minimal order is two or three.

Two independent derivations. Reports 231 and 233 bear the same date and neither cites the other. Part IV re-derives, at height five, the column transfer, the composition identity, the rank-two form of B and the scalar integrations by parts of Part III, and it defines the same period Z_n ; its recurrence (34.17) is Theorem 29.1, proved here by contiguity and a moment closure instead of a divergence certificate (Remark 42.1). New in Part IV: the odd augmented Pfaffian and the enumeration of its 225 products, the open-chain reduction $q_2 = 16G - 12H/N^2 + 2/N^5$, which has no analogue at height four, the second period X_n , the polynomial contiguity, the moment closure, the first difference (34.14), and the factorization (39.17). The two texts share 4.3 % of their word 8-grams (233 against 231), and Part IV shares 0.2 % and 0.5 % with Parts I and II.

Overlap with Part I, credited. Section 40 recovers, from the recurrence, the case $m = 5$ of Theorem 2.1 and Corollary 2.2, as the manuscript says: its c_j/c_0 are $c_{5,j}$ (Table 1), with $c_0 = 9/32768 = J_5/4^{10}$, and its $d_1, \dots, d_5$ are those of (2.8). The sixth corrections $-6103/16$ and $d_6 = -2709616559/3125000$ are new to the repository, “without a novelty claim”; at placement Part I's engine run at order six on a copy gave the same $b_{5,6}$. The effective constants are new. Section 41 uses the same apparatus as Section 32.

Source checks. The manuscript transcribes Conjecture 19 from arXiv:2303.02793v2 (Section 6.4, printed page 34; “the rising-factorial overbars were checked on the printed page”) and the operator from the A181199 internal record, consulted 5 October 2026. At placement the intake did not re-inspect the printed page; it checked that the four coefficients $c_0, \dots, c_3$ of Appendix E equal those of the live A181199 entry, checked the y -contiguity certificate and the bridge (39.1) in SymPy (residual zero), and checked Theorem 34.1, (37.2) and (34.14) for $n \leq 20$ against an independent row-state count; the transcribed operator annihilates those counts for $n = 1, \dots, 15$. Its verifier, rerun at the write on a copy (Section 42.3), printed its receipt byte for byte.

Editorial changes. Section k of the manuscript is Section $k + 33$ here and its Appendix A is Appendix E, printed after Appendix D; theorem and equation numbers follow their sections, and its table is Table 8. Its four Questions are Research questions 16–19 (one counter runs through the report). Every label carries the prefix `shr:h5:`, including its `asy:` and `inv:` labels. Its title, subtitle and title-page paragraphs open this Part; its table of contents is dropped. Its three citations of its predecessor now point to Part I. The bibliographies are merged: Kauers–Koutschan, Sun and DLMF §§4.13 and 5.11 are Part I's entries, and its OEIS entry is [8]. Its bibliography cites Sun's article, which its text does not cite. It said of Part I: “Its all-height asymptotic results precede the present report; its explicit scope leaves the particular recurrences unproved. The source is cited for credit, not redistributed in this package.” Remarks take this report's theorem style. The [write] notes and Section 42.5 with Remark 42.1, Proposition 42.2 and Section 42.6 are new. No other statement, proof or number of the manuscript was changed.

[write] Added 5 October 2026, batch 101: notation of Part IV. Part IV keeps the manuscript's symbols, including the reuses that it flags itself (q_j , V and V_j , E , b and b_n , J_0 and J_j). The count is $a_n = T_5(n)$, with array entries $a_{i,j}$ as in Part I. Within Part IV each symbol has the meaning below; the right column gives the tempting false reading from elsewhere in this report. The kernel functions W , h , e , w and the scalars c , H , U , I , J_0 are those of Part III at height five, with $f = t^n/N$ for Part III's α and $b = t^{n-1}/(n-1)!$ for its β .

symbol	meaning in Part IV	not: meaning elsewhere
$T_j; B$	$\text{tr}(B^j)$ (35.7); the kernel EA	the count $T_m(n)!$; Part I's B , Part II's B_i
$P(n), Q(n), D(n)$	the polynomials (34.8)	Part III's field P , Q and D_n ; Part II's $Q_m(n)$, D_i
$u(n), v(n); u, v$	(34.9), (34.10); the triangle variables of (34.15)	—
γ_n	(34.11)	Part III's certificate constant γ_n
$N; M_n; F_n$	$n!$; $M_5(n)$; a_n/M_n (also written b_n in Section 40)	$F_m(z)$; Part III's prefactor F_n ; $b_{m,j}$
$Z_n; X_n$	the period, equal to Part III's; the second period (37.1)	$Z_m(n)$; Part II's binomial counts X_i
v_n, δ_n	the period recurrence (34.17)	Part V's $v = \alpha_m$
C_m	the ordered simplex	Part V's leading constant $C_m = K_m$
$V; V_j$	the Volterra operator; moments (38.1)	—
$K, J, E, A; b$	kernels (35.4); $b = K1$	$K_m; J_m; E_{m,n}(p)$; $A = \binom{m}{3}$; $b_{m,j}$
E	the sign kernel (Section 35); the forward shift (Section 39)	—

symbol	meaning in Part IV	not: meaning elsewhere
$q_j; q(t), r(t)$	$\langle b, B^j \mathbf{1} \rangle$ (Sections 35–36); contiguity polynomials (37.14) with coefficients q_j, r_j	$q_{ij} = (y_i + y_j)^2; r = m - 2$
$G; L$	$\langle f, Wh \rangle$ (36.4); $\langle b, Ww \rangle$ (36.15)	L a truncation order
$y_n, x_n; k(n), \ell(n); c_n^*$	polynomials (37.1); (37.3); (37.10)	Part I's Y_i, X_i
λ, μ, ν	contiguity constants (37.12), $\nu = \nu_n$; later $\lambda = \log 3125$	$\lambda = m \log m$
$V_j, Y_j, U_j, J_j; [p]v$	moment families (38.1) and their linear forms	Part I's Y_i ; Part III's U
$A_5(n), \rho_n, \sigma_n, \tau_n, P_7, P_9$	the X transport (38.16)–(38.24)	Part V's cut proportion ρ
H_n, I_n	hypergeometric terms (39.2)	Part II's $H_m(n)$; Part III's I_j
$R(n), S(n)$	ratios (39.6)	Part III's $R(n), S(n)$
$c_0, \dots, c_3; c_j$	the OEIS coefficients (Appendix E); β_{10+j} in Section 40	$c_{m,j}$; Part III's c_j
$r(t), v_*, d_*, \alpha(t), \eta(t), \mathcal{Z}, \mathcal{B}, z_j, \beta_j$		$\alpha_m; \beta_j$ (Section 6.2)
$C_*, d_j, \ell_j; J$	$K_5, b_{5,j}$, logarithmic coefficients; $\lfloor (K+1)/2 \rfloor$ in Section 40.3	$d = \binom{m}{2}; J_m$
$N(y)$	the threshold (41.9)	Part III's $N(y)$
$B_{21}, C_{19}, b_j, d_j, Q_+, P_+$	integer coefficient data of Appendix E	the d_j of Section 40

34 The exact target and the scope of the result

For a positive integer n , let a_n be the number of arrays $(a_{i,j})_{1 \leq i \leq 5, 1 \leq j \leq n}$ containing $1, \dots, 5n$ exactly once and satisfying

$$\begin{aligned} a_{i,j} &< a_{i,j+1}, & a_{i,j} &< a_{i+1,j}, \\ a_{i,j} &< a_{i+1,j+1}, & a_{i,j+1} &< a_{i+1,j}, \end{aligned} \quad (34.1)$$

whenever the indicated cells exist. These are the literal row, column, downward-diagonal, and downward-antidiagonal inequalities. The downward diagonal is redundant; the antidiagonal is essential. Height is fixed at five throughout. In particular, the width-two count is one, although the height-two problem at variable width has a different answer.

The sequence is OEIS A181199 [8]. Kauers and Koutschan state a specific nested finite sum as Conjecture 19 in Section 6.4 of their article [9], printed page 34. We first give that formula in a compact notation which is also used in the proof. Set

$$P(n) = 25216n^8 + 9888n^7 - 14496n^6 + 11208n^5 + 23832n^4 \quad (34.2)$$

$$+ 7383n^3 - 1522n^2 - 939n - 90, \quad (34.3)$$

$$Q(n) = 137855872n^{11} + 860969696n^{10} + 2047036856n^9 + 2032587274n^8 \quad (34.4)$$

$$- 24192441n^7 - 1894061166n^6 - 1671661480n^5 - 524330624n^4 \quad (34.5)$$

$$+ 36004789n^3 + 62751860n^2 + 13865604n + 927360, \quad (34.6)$$

$$D(n) = 100864n^{11} + 241280n^{10} + 109376n^9 - 61744n^8 + 105456n^7 \quad (34.7)$$

$$+ 247584n^6 + 103814n^5 - 40103n^4 - 33803n^3 - 1331n^2 + 2445n + 306. \quad (34.8)$$

For $r \geq 0$, the notation $(z)^{\overline{r}} = z(z+1) \cdots (z+r-1)$ denotes a *rising factorial*, with the empty product equal

to one. Define

$$u(n) = \frac{8P(n)}{(2n-1)(4n-1)(3n+1)^3(4n+1)^4}, \quad (34.9)$$

$$v(n) = \frac{(3n+1)(3n+2)(4n+3)Q(n)}{(n+1)^2(n+2)^2(2n-1)(2n+1)(2n+3)P(n)P(n+1)}, \quad (34.10)$$

$$\gamma_n = \frac{(4n+1)D(n)}{16(n+1)^2(2n-1)(2n+1)P(n)}. \quad (34.11)$$

The overbars in the source's denominator of u are important: replacing the rising factorials by powers changes the conjecture. The other degree-eight factor printed in the denominator of v is exactly $P(n+1)$, as explicitly verified in Appendix E.

Theorem 34.1 (The printed finite sum). *For every integer $n \geq 1$,*

$$a_n = 1 - \frac{27}{4} \sum_{k=1}^{n-1} \left\{ (-1)^k u(k) \frac{(5k)!}{(3k)!(k!)^2} \sum_{i=1}^{k-1} (-1)^i v(i) \frac{(3i)!}{(i!)^3} \right\}. \quad (34.12)$$

All displayed denominators are nonzero on this range. In particular, $a_1 = a_2 = 1$ and $a_3 = 16$.

Theorem 34.2 (The literal OEIS recurrence). *Let c_0, c_1, c_2, c_3 be the polynomials displayed in Appendix E. They are precisely the four coefficients of the third-order, degree-24 recurrence in the inspected A181199 entry. Then*

$$c_0(n)a_n + c_1(n)a_{n+1} + c_2(n)a_{n+2} + c_3(n)a_{n+3} = 0 \quad (n \geq 1). \quad (34.13)$$

The coefficient $c_3(n)$ is strictly positive there, so these three initial values determine the sequence uniquely.

34.1 A smaller identity that proves both statements

Put $N = n!$ and $M_n = (5n)!/N^5$. The proof will establish

$$a_{n+1} - a_n = 2M_n u(n)(Z_n - \gamma_n), \quad (34.14)$$

where the period is the positive triangle integral

$$Z_n = n^2 \iint_{\substack{0 < u, v < 1 \\ u+v > 1}} u \{uv(u+v-1)\}^{n-1} du dv. \quad (34.15)$$

It obeys the elementary first-order affine recurrence

$$Z_{n+1} = v_n Z_n + \delta_n, \quad (34.16)$$

$$v_n = -\frac{(n+1)^2}{3(3n+1)(3n+2)}, \quad \delta_n = \frac{28n^3 + 50n^2 + 29n + 5}{6(2n+1)(3n+1)(3n+2)}. \quad (34.17)$$

The initial value is $Z_1 = 1/3$. Equations (34.14) and (34.17), together with an explicit rational identity for γ_n , will prove the actual nested sum, rather than only a compatible recurrence. A subsequent exact Ore-operator identity proves Theorem 34.2.

34.2 What the present proof adds

The earlier report *Fixed-height shifted rectangles: asymptotics and algebraicity* (Part I) already established the leading asymptotic for A181199, all-fixed-order expansions at every fixed height, and further algebraicity results. Its stated scope explicitly leaves the particular A181198 and A181199 recurrences unproved. We do not present its asymptotic conclusions as new. The new mathematical task here is exact full-count enumeration: the odd augmented Pfaffian, its open-chain reduction, polynomial period transport, and the specified finite sum and operator. No asymptotic estimate is used in their proof.

The source comparison is bounded. It covers the cited article, the inspected OEIS record, and the specified earlier report, pinned in the bibliography. Exact-identifier searches located the OEIS and original article descriptions; a conjectural label in a source is not an exhaustive priority search. No global novelty or external peer-review claim follows. The standard tools, including order-polytope volumes, determinant composition, and Pfaffian integration, are not inventions of this report. Their required special cases are proved below so that the exact result is self-contained.

34.3 Indexing and boundary discipline

The proof concerns positive widths only. At width one, the count is one and the required period formulas are checked directly. This avoids using a singular transfer kernel at that index. Moreover, if one adopts the optional convention $a_0 = 1$, the left side of (34.13) at $n = 0$ is

$$5621993879040000,$$

not zero. The recurrence must not be extended to zero without modification. A boundary term that is exponentially small for large n can still affect an exact recurrence. The integral proof below retains every such term.

35 Column transfer and the odd augmented Pfaffian

All integrals in this section are ordinary Lebesgue integrals on compact cubes or ordered simplices. Equalities among coordinates occupy a set of measure zero and have no effect on the volumes.

35.1 The transfer determinant and its normalization

Write

$$C_m = \{0 < x_1 < \cdots < x_m < 1\}, \quad V(t, s) = \mathbf{1}_{s < t}.$$

The same letter V denotes the integral operator on $[0, 1]$. For adjacent ordered columns x, y , the inequalities (34.1) are equivalent to

$$x_1 < y_1 < x_2 < y_2 < \cdots < x_5 < y_5. \quad (35.1)$$

Indeed the row and antidiagonal inequalities give the displayed chain, which implies the two within-column orders and the required downward diagonal comparisons. Thus no extra restriction has been introduced.

Lemma 35.1. *On $C_m \times C_m$, the determinant $\det[V(y_i, x_j)]_{i,j=1}^m$ is the indicator of $x_1 < y_1 < \cdots < x_m < y_m$.*

Proof. Each row consists of an initial string of ones followed by zeros, and these strings are nested. Let their lengths be $r_1 \leq \cdots \leq r_m$. If any length is zero, or two lengths coincide, the determinant vanishes. Otherwise

the lengths must be $1, \dots, m$, in which case the matrix is lower triangular with diagonal one. Those lengths are exactly the asserted interlacing conditions. $\square$

For kernels K_1, K_2 with integrable products, determinant expansion and Fubini give

$$\int_{C_m} \det[K_1(y_i, z_j)] \det[K_2(z_i, x_j)] dz = \det[(K_1 K_2)(y_i, x_j)]. \quad (35.2)$$

To check the factor, the product of determinants is symmetric in the z variables. Its cube integral is $m!$ times its ordered-simplex integral, while expansion of the cube integral gives $m!$ times the determinant on the right. The two factors cancel.

Every linear extension of the array poset occupies a simplex of volume $1/(5n)!$ in the unit cube. Applying Lemma 35.1 between successive columns and (35.2) at the intermediate columns therefore proves, for $n \geq 2$,

$$\frac{a_n}{(5n)!} = \int_{C_5 \times C_5} \det[K(y_i, x_j)] dx dy, \quad K = V^{n-1}, \quad (35.3)$$

where

$$K(t, s) = \frac{(t-s)_+^{n-2}}{(n-2)!}.$$

At $n = 2$, this exponent-zero truncated-power notation means the indicator $\mathbf{1}_{s < t}$, consistent with $K = V$; no value of 0^0 is used. There is no factorial left over from any intermediate column. Width one will be handled separately in Section 37.

35.2 The bordered identity

Let $J = \mathbf{1} \otimes \mathbf{1}$ be the constant kernel and set

$$E(t, s) = \operatorname{sgn}(s - t) = J - 2V = 2V^* - J, \quad A = KEK^*, \quad b = K\mathbf{1} = \frac{t^{n-1}}{(n-1)!}. \quad (35.4)$$

The kernel A is skew-symmetric. The odd Pfaffian integral is

$$\int_{C_5} \det[K(y_i, x_j)] dx = \operatorname{Pf} \begin{pmatrix} (A(y_i, y_j))_{i,j=1}^5 & (b(y_i))_{i=1}^5 \\ -(b(y_j))_{j=1}^5 & 0 \end{pmatrix}. \quad (35.5)$$

We spell out its normalization. For an even number $2r$ of rows, put

$$M_{ij} = \iint K(y_i, s) E(s, t) K(y_j, t) ds dt.$$

The Pfaffian definition is

$$\operatorname{Pf}(M) = \frac{1}{2^r r!} \sum_{\sigma \in S_{2r}} \operatorname{sgn}(\sigma) \prod_{j=1}^r M_{\sigma(2j-1), \sigma(2j)}.$$

Expand the pairwise integrals and combine the alternating sum into a determinant. With $D(z) = \det[K(y_i, z_j)]$, this gives

$$\operatorname{Pf}(M) = \frac{1}{2^r r!} \int_{[0,1]^{2r}} D(z) \prod_{j=1}^r E(z_{2j-1}, z_{2j}) dz.$$

Relabeling the integration variables in each term of $\text{Pf}(E(z_i, z_j))$ then gives

$$\text{Pf}(M) = \frac{1}{(2r)!} \int_{[0,1]^{2r}} D(z) \text{Pf}(E(z_i, z_j)) dz.$$

Both factors change by the same sign under permutation, and the sign Pfaffian is one on the ordered chamber by its elementary recursion. Partitioning the cube into its $(2r)!$ ordered chambers proves the even integration formula with factor one. For the odd version, adjoin a point $\star$ of mass one, ordered after $[0, 1]$. Extend K by $K(\star, \star) = 1$ and zero mixed entries, and use six rows with the last row at $\star$. In a nonzero determinant integral the atom appears exactly once. The remaining determinant is the original five-variable determinant; its pairwise skew integrals form A , and the last column is $K\mathbf{1} = b$. This proves (35.5) with factor one, including the border sign.

For ordered y , the corresponding sign Pfaffian is

$$\text{Pf} \begin{pmatrix} (E(y_i, y_j)) & (1) \\ -(1) & 0 \end{pmatrix} = 1. \quad (35.6)$$

All entries above its diagonal equal one. The Pfaffian expansion along the first row gives an alternating sum of five ones, equal to one, and the same recursion starts from the 2×2 case. Multiplying the right side of (35.5) by (35.6) allows us to replace its ordered integral by the cube integral divided by $5!$: permuting the five ordinary labels changes both Pfaffians by the same sign.

35.3 The complete enumeration of the 225 products

Put

$$B = EA, \quad T_j = \text{tr}(B^j), \quad q_j = \langle b, B^j \mathbf{1} \rangle. \quad (35.7)$$

Each six-label perfect matching has one edge incident to the border. There are $5 \cdot 3 = 15$ such matchings, hence 225 products of terms in the two bordered Pfaffians. The following argument accounts for every product and every integrated sign, rather than suppressing the odd open path.

Superpose the two matchings, remove their border vertices, and retain the border weights b and 1 at the two exposed ordinary vertices. The component containing those vertices is an alternating open path. Every other component is an alternating even cycle. The open path has one, three, or five ordinary vertices. The four exhaustive classes are

Open path and remaining components	Number	Integrated term
One vertex and two doubled edges	$5 \cdot 3 = 15$	$q_0 T_1^2$
One vertex and one four-cycle	$5 \cdot 6 = 30$	$-q_0 T_2$
Three vertices and one doubled edge	$\binom{5}{3} \cdot 6 = 60$	$-q_1 T_1$
Five vertices	$5 \cdot 4 \cdot 3! = 120$	q_2

For the three-vertex class, choose its labels and its ordered endpoints; the middle vertex and remaining doubled edge are forced. For the five-vertex class, choose the ordered endpoints and the order of the three internal labels. For the one-vertex class, fix its label; the two matchings on the other four labels either coincide in one of three ways, or differ in one of six ordered ways. These counts sum to 225.

Here are canonical signed representatives of the four classes, with $E_{ij} = E(y_i, y_j)$ and similarly for A :

$$\begin{aligned} & E_{12}E_{34}A_{12}A_{34}b_5, \\ & - E_{12}E_{34}A_{13}A_{24}b_5, \\ & - E_{12}E_{34}A_{12}A_{35}b_4, \\ & E_{12}E_{34}A_{13}A_{25}b_4. \end{aligned}$$

The missing border weight in each case is one. Integrating $E_{12}A_{12}$ gives $-T_1$, since $A_{21} = -A_{12}$. The product $E_{12}E_{34}A_{13}A_{24}$ integrates to T_2 . Further, $b_4E_{34}A_{35}$ integrates to $-q_1$. In the last representative, reversing both E_{34} and A_{13} gives precisely the oriented product defining $\langle b, EAEA1 \rangle = q_2$. These establish the table's four signs. Relabeling variables changes the two original Pfaffian signs together, so it preserves their product; therefore the canonical sign applies to every term in its class.

For complete algorithmic enumeration, pair the smallest unpaired label i with each possible j , multiply the recursive sign by $(-1)^{p-1}$ when j is in position p among the remaining labels, and continue. This generates all 15 signed matchings on labels $1, \dots, 6$. Taking the Cartesian square and following each exposed path generates exactly the table above. The accompanying verifier implements that finite procedure; the proof of the counts and signs is the preceding argument.

Division by $120 = 5!$ gives the crucial full-volume formula

$$\boxed{\frac{a_n}{(5n)!} = q_0 \left(\frac{T_1^2}{8} - \frac{T_2}{4} \right) - \frac{q_1 T_1}{2} + q_2.} \quad (35.8)$$

All kernels here are bounded and piecewise polynomial for $n \geq 2$. The notation $\text{tr}(B^j)$ means the finite cyclic kernel integral, so Fubini is sufficient. No formal infinite-dimensional determinant or unjustified trace expansion is being used.

36 The Volterra rank reduction and the open chain

The odd Pfaffian formula contains a term q_2 that has no analogue in the height-four trace-only calculation. Its exact reduction is the key extra step. Throughout this section $n \geq 2$; the kernel identities themselves will also be checked at $n = 1$ where needed later.

36.1 The kernel and its elementary identities

Write $N = n!$ and introduce the function $f(t) = t^n/N$, so $f' = b$. Define the symmetric kernel and its boundary functions by

$$W = V^n(V^*)^n, \quad W(t, s) = \frac{1}{((n-1)!)^2} \int_0^{\min(t, s)} (t-r)^{n-1} (s-r)^{n-1} dr, \quad (36.1)$$

$$h(t) = W(t, 1), \quad e(t) = 2h(t) - \frac{b(t)}{N}, \quad (36.2)$$

$$w = W\mathbf{1}, \quad z = Wf. \quad (36.3)$$

Inner products are real integrals on $[0, 1]$. Set

$$H = \langle f, h \rangle, \quad U = \text{tr}(W^2), \quad G = \langle f, Wh \rangle = \langle z, h \rangle, \quad c = \langle f, b \rangle = \frac{1}{2N^2}. \quad (36.4)$$

Preserving the order of noncommuting factors in (35.4) gives

$$B = EKEK^* = (JV^{n-1} - 2V^n)(2(V^*)^n - J(V^*)^{n-1}) \quad (36.5)$$

$$= -4W + \mathbf{1} \otimes e + 2f \otimes b. \quad (36.6)$$

Here $(r \otimes s)g = r\langle s, g \rangle$. For example, $V^n J(V^*)^{n-1} = f \otimes b$, and the first term containing J is $2\mathbf{1} \otimes h$. The remaining double- J term is $-\mathbf{1} \otimes b/N$. This supplies a direct check of the border normalization in the rank reduction.

The defining integral gives

$$W(t, 0) = 0, \quad (\partial_t + \partial_s)W(t, s) = b(t)b(s), \quad W(rt, rs) = r^{2n-1}W(t, s). \quad (36.7)$$

For the differential identity, the sum of the external derivatives of the integrand is its negative derivative in r . The endpoint $r = 0$ therefore supplies $b(t)b(s)$, while the moving upper endpoint vanishes for $n \geq 2$. At $n = 1$, $W(t, s) = \min(t, s)$ and the same identity holds on both sides of the diagonal directly. The kernel is continuous across the diagonal; integrations by parts on the two triangles have cancelling internal boundary fluxes.

Direct integration also gives

$$\int_0^1 h = \frac{1}{2N^2}, \quad \int_0^1 b = \frac{1}{N}, \quad \int_0^1 e = 0, \quad \text{tr } W = \frac{1}{2n(2n-1)((n-1)!)^2}. \quad (36.8)$$

For instance, integrating t first in $\int h$ leaves $((n-1)!)^{-2}n^{-1} \int_0^1 (1-r)^{2n-1} dr = 1/(2N^2)$. On the diagonal $W(t, t) = t^{2n-1}/((2n-1)((n-1)!)^2)$, giving the trace.

36.2 The scalar integration by parts identities

We need three scalar integrals:

$$\langle b, w \rangle = H, \quad I := \langle b, z \rangle = \frac{H}{N} - \frac{1}{8N^4}, \quad J_0 := \langle h, w \rangle = \frac{H}{N} - \frac{1}{8N^4} = I. \quad (36.9)$$

Here J_0 is a temporary scalar name, unrelated to the moment J_j introduced later. To prove the first identity, integrate the sum of the t and s derivatives of $f(t)W(t, s)$ over the square. The interior integral is $\langle b, w \rangle + c/N$. The upper boundaries give $H + (\int h)/N$; both lower boundaries vanish because $f(0) = 0$ and $W(t, 0) = 0$. Since $c = \int h$, the first assertion follows.

Next use $f(t)f(s)W(t, s)$. The interior integral is $2I + c^2$, and the upper boundaries sum to $2H/N$. Thus $2I + c^2 = 2H/N$, proving the second assertion. Finally, integrate the kernel differential identity in s :

$$w' + h = b/N, \quad w(0) = 0, \quad w(1) = 1/(2N^2). \quad (36.10)$$

Consequently

$$J_0 = \frac{\langle b, w \rangle}{N} - \int_0^1 w'w = \frac{H}{N} - \frac{w(1)^2}{2} = \frac{H}{N} - \frac{1}{8N^4}.$$

Every nonzero outer endpoint has been included in these equalities.

36.3 The two traces

For rank-one operators, $\text{tr}(r \otimes s) = \langle s, r \rangle$ and $\text{tr}((r \otimes s)(u \otimes v)) = \langle s, u \rangle \langle v, r \rangle$. Using these rules in (36.6) and (36.8) gives

$$T_1 = -4 \text{tr } W + 2c = -\frac{1}{(2n-1)N^2}. \quad (36.11)$$

The full trace-square expansion, before any simplification, is

$$T_2 = 16U - 8\langle e, w \rangle - 16I + 4c^2 + \frac{4\langle e, f \rangle}{N}. \quad (36.12)$$

The square of $\mathbf{1} \otimes e$ has trace zero since $\int e = 0$; the two cross terms between the rank-one summands produce the last term. Now

$$\langle e, w \rangle = 2J_0 - H/N, \quad \langle e, f \rangle = 2H - c/N.$$

Substitution of (36.9) yields

$$T_2 = 16U - \frac{16H}{N} + \frac{3}{N^4}. \quad (36.13)$$

36.4 The open path of length four

The first two open-chain integrals follow from $B\mathbf{1} = -4w + 2f/N$:

$$q_0 = 1/N, \quad q_1 = -4H + 1/N^3. \quad (36.14)$$

One more application of (36.6) gives

$$q_2 = 16L - \frac{8I}{N} - \frac{4\langle e, w \rangle}{N} + \frac{2\langle e, f \rangle}{N^2} + 2cq_1, \quad L = \langle b, Ww \rangle. \quad (36.15)$$

It remains to show $L = G$, without dropping any boundary contribution. Multiply (36.7) by $f(s)$ and integrate in s . The s derivative has upper flux h/N and lower flux zero, so

$$Wb = z' - cb + h/N. \quad (36.16)$$

Symmetry of W and a final integration by parts give

$$L = \langle w, Wb \rangle = \langle w, z' \rangle - cH + J_0/N \quad (36.17)$$

$$= [wz]_0^1 - \langle w', z \rangle - cH + J_0/N \quad (36.18)$$

$$= \frac{H}{2N^2} - \frac{I}{N} + G - cH + \frac{J_0}{N} = G. \quad (36.19)$$

Here $z(1) = H$, $w(1) = 1/(2N^2)$, $z(0) = w(0) = 0$, $w' = b/N - h$, $c = 1/(2N^2)$, and $I = J_0$. Thus the new open-chain term is exactly

$$q_2 = 16G - \frac{12H}{N^2} + \frac{2}{N^5}. \quad (36.20)$$

This identity is an equality of the full integrals. It is not a leading term approximation to the path contribution.

36.5 Eliminating the remaining trace square

Symmetry of W lets us integrate its square over twice the lower triangle. Put $s = rt$, or equivalently parametrize the larger coordinate by r and the smaller by rt . Homogeneity gives

$$U = 2 \int_0^1 r^{4n-1} dr \int_0^1 h(t)^2 dt = \frac{1}{2n} \int_0^1 h(t)^2 dt. \quad (36.21)$$

The factor two includes both triangles; their diagonal has measure zero. The boundary function satisfies

$$(1-t)h' + (2n-1)h = \frac{t^{n-1}}{((n-1)!)^2}. \quad (36.22)$$

A direct derivation, including the width-one case, is given in the next section. Multiply (36.22) first by $2h$ and then by t^n , and integrate by parts. In both fluxes the upper endpoint is killed by $1-t$ and the lower endpoint by $h(0) = 0$. One obtains

$$(4n-1) \int h^2 = \frac{2}{((n-1)!)^2} \int t^{n-1} h, \quad (36.23)$$

$$\int t^{n-1} h = 3NH - \frac{1}{2N^2}. \quad (36.24)$$

Combining these with (36.21) proves

$$U = \frac{3nH}{(4n-1)N} - \frac{n}{2(4n-1)N^4}. \quad (36.25)$$

Equations (36.11), (36.13), (36.14), (36.20), and (36.25) now express every term in the Pfaffian volume using H, G and rational functions of n, N .

37 Polynomial periods and their contiguous functions

Define the normalized polynomial functions and their periods by

$$y_n = N^2 h, \quad x_n = N^3 z, \quad Z_n = \int_0^1 t^n y_n(t) dt = N^3 H, \quad X_n = \int_0^1 x_n(t) y_n(t) dt = N^5 G. \quad (37.1)$$

These definitions make sense for every $n \geq 1$, directly through $W = V^n(V^*)^n$, regardless of the transfer formula's narrower domain.

Proposition 37.1 (The exact full-count representation). *For every integer $n \geq 1$,*

$$F_n := \frac{a_n}{M_n} = 16X_n + k(n)Z_n + \ell(n), \quad (37.2)$$

where

$$k(n) = -\frac{2(44n^2 - 26n + 3)}{(2n-1)(4n-1)}, \quad \ell(n) = \frac{224n^3 - 232n^2 + 76n - 7}{8(2n-1)^2(4n-1)}. \quad (37.3)$$

Proof. For $n \geq 2$, substitute all the reductions of Section 36 into (35.8) and multiply by N^5 . Before eliminating U the result is

$$16N^5G - 4N^4U - \left(8 + \frac{2}{2n-1}\right)N^3H + \frac{5}{4} + \frac{1}{2(2n-1)} + \frac{1}{8(2n-1)^2}.$$

Using (36.25) gives precisely (37.2). At $n = 1$, direct integration of $W(t, s) = \min(t, s)$ gives

$$y_1(t) = t, \quad x_1(t) = t/2 - t^3/6, \quad Z_1 = 1/3, \quad X_1 = 2/15. \quad (37.4)$$

The right side of (37.2) is then $1/120 = a_1/M_1$, since $a_1 = 1$ and $M_1 = 120$. This proves the stated range. $\square$

The integral for Z_n follows by inserting the kernel definition:

$$Z_n = n^2 \int_0^1 \int_0^t t^n (t-r)^{n-1} (1-r)^{n-1} dr dt.$$

The change $u = t$, $v = 1 - r$ gives exactly (34.15), including its factor u and the triangular domain.

37.1 Differential identities with all boundary data

Lemma 37.2. *For every $n \geq 1$ the functions x_n, y_n are polynomials and satisfy*

$$(1-t)y'_n + (2n-1)y_n = n^2 t^{n-1}, \quad (37.5)$$

$$tx'_n - 3nx_n = -y_n, \quad (37.6)$$

$$x_n(0) = y_n(0) = 0, \quad x_n(1) = Z_n. \quad (37.7)$$

Also $y_n(1) = n^2/(2n-1)$.

Proof. The kernel formula gives

$$y_n(t) = n^2 \int_0^t (t-r)^{n-1} (1-r)^{n-1} dr.$$

For $n \geq 2$, put $f_0 = (t-r)^{n-1}(1-r)^{n-1}$ and $g_0 = (t-r)^{n-1}(1-r)^n$. Direct differentiation yields

$$(1-t)\partial_t f_0 + (2n-1)f_0 = -\partial_r g_0.$$

The moving-endpoint contribution $f_0(t, t)$ vanishes, and integrating the right side gives $g_0(t, 0) - g_0(t, t) = t^{n-1}$. This proves (37.5) for $n \geq 2$ and also proves (36.22). At $n = 1$, the moving-endpoint contribution does *not* vanish: $f_0 = 1$, $g_0 = 1 - r$. It contributes $1 - t$ to $(1-t)y'_1$, while the integrated $-g'_0$ contributes t . Their sum is one. Equivalently, $y_1 = t$ in (37.4) verifies the identity directly. No width-one endpoint is discarded.

Euler homogeneity in (36.7) says $tW_t + sW_s = (2n-1)W$. Since $x_n = N^2 \int_0^1 s^n W(t, s) ds$, integration by parts gives

$$\begin{aligned} tx'_n &= (2n-1)x_n - N^2 \int_0^1 s^{n+1} W_s(t, s) ds \\ &= 3nx_n - N^2 W(t, 1) = 3nx_n - y_n. \end{aligned}$$

The upper endpoint is exactly $-y_n$; the lower one is zero. Splitting at $s = t$ is legitimate even at $n = 1$, and the two internal boundary terms cancel by continuity of W . The endpoint identities follow from $W(t, 0) = 0$, symmetry, and the definitions; the value of $y_n(1)$ is a direct integral. Polynomiality also follows from the next explicit form. $\square$

37.2 The homogeneous coefficient is fixed by the kernel

For $s \geq t$, beta integration after expanding $(s - r)^{n-1}$ gives

$$W(t, s) = \sum_{j=0}^{n-1} \frac{(-1)^j t^{n+j} s^{n-1-j}}{(n-1-j)!(n+j)!}. \quad (37.8)$$

In particular,

$$y_n(t) = N^2 \sum_{j=0}^{n-1} \frac{(-1)^j t^{n+j}}{(n-1-j)!(n+j)!}. \quad (37.9)$$

The first-order equation (37.6) alone allows an arbitrary multiple of t^{3n} . The defining kernel fixes it. Indeed $z = V^n(V^*)^n f$ implies $z^{(2n)} = (-1)^n f$, by differentiating the two iterated integral operators. Its leading coefficient is $(-1)^n/(3n)!$. Consequently, writing

$$c_n^* = \frac{(-1)^n (n!)^3}{(3n)!}, \quad (37.10)$$

we have the exact polynomial

$$x_n(t) = \sum_{j=n}^{2n-1} \frac{[t^j]y_n(t)}{3n-j} t^j + c_n^* t^{3n}. \quad (37.11)$$

Here $[t^j]$ means coefficient extraction. Alternatively, (37.8) on the two halves of the square gives x_n by ordinary polynomial integration and obtains the same highest coefficient. Thus (37.11) is not an imposed choice of homogeneous solution.

37.3 Explicit polynomial contiguity

Set

$$\lambda = -\frac{(n+1)^2}{2n(2n+1)}, \quad \mu = \frac{(n+1)^2}{2(2n+1)}, \quad \nu = -\frac{(n+1)^2}{3(3n+1)(3n+2)}. \quad (37.12)$$

In this subsection coefficients are evaluated at the current n . Define $q(t) = q_0 + q_1 t + q_2 t^2$ and $r(t) = r_0 + r_1 t + r_2 t^2$ by

$$q_0 = -\frac{n+1}{6n(2n+1)}, \quad q_1 = \frac{(n+1)(5n+3)}{6n(2n+1)(3n+2)}, \quad q_2 = -\frac{n+1}{3(3n+1)(3n+2)}, \quad (37.13)$$

$$r_0 = \frac{n+1}{6(2n+1)}, \quad r_1 = \frac{7n^2+9n+3}{6(2n+1)(3n+2)}, \quad r_2 = -\frac{n^2(n+1)}{3(2n+1)(3n+1)(3n+2)}. \quad (37.14)$$

These q_j are coefficients of a polynomial, no longer the open-chain integrals of Section 35.

Proposition 37.3. For every $n \geq 1$, identically in t ,

$$y_{n+1} = \lambda(1-t)^2 y_n + \mu t^n (1+t), \quad (37.15)$$

$$x_{n+1} = \nu t^3 x_n + (1-t)q(t)y_n + t^n r(t). \quad (37.16)$$

Proof. Substitute the right side of (37.15) into the new y equation. After dividing its residual by t^{n-1} , the coefficient condition is the elementary polynomial identity

$$\begin{aligned} \lambda n^2(1-t)^2 + \mu\{n(1-t)(1+t) + t(1-t) \\ + (2n+1)t(1+t)\} - (n+1)^2 t = 0. \end{aligned} \quad (37.17)$$

Its coefficients vanish for (37.12). Both the actual and candidate y_{n+1} vanish at zero. Their difference satisfies $(1-t)d' + (2n+1)d = 0$, hence is a constant multiple of $(1-t)^{2n+1}$ on the interior; the value at zero forces that constant to vanish. Polynomial continuity covers $t = 1$.

For the second identity put $Q_0(t) = (1-t)q(t)$. Substitution into the new x equation reduces, using the two current ODEs, to

$$-vt^3 + tQ'_0 - 3(n+1)Q_0 - (2n-1)tq + \lambda(1-t)^2 = 0, \quad (37.18)$$

$$tr' - (2n+3)r + n^2q + \mu(1+t) = 0. \quad (37.19)$$

These are degree-three and degree-two polynomial identities in t , verified by inserting (37.14); they constitute a finite rational certificate for the substitution. The difference between the candidate and actual x_{n+1} solves $td' - 3(n+1)d = 0$, so it is a multiple of t^{3n+3} . Its highest coefficient vanishes because $c_{n+1}^*/c_n^* = v$. All other terms on the proposed right side have degree at most $2n+2 < 3n+3$, including when $n = 1$. Thus the homogeneous multiple is zero. $\square$

The next section provides a second check of this normalization at $t = 1$: it derives $Z_{n+1} = vZ_n + \delta_n$ from y alone and finds $r(1) = \delta_n$. Hence the candidate also has the required endpoint $x_{n+1}(1) = Z_{n+1}$ independently of its leading coefficient.

38 Finite moment closure and the full first difference

The contiguous functions contain only fixed-degree polynomial factors. Consequently, their product can be integrated through a finite family of moments. This section derives each reduction with its flux, so that no endpoint convention is hidden in the symbolic calculation.

38.1 Four families and four flux identities

At a fixed positive n , abbreviate $x = x_n$, $y = y_n$, $X = X_n$, $Z = Z_n$, and define

$$V_j = \int_0^1 t^{n+j} y \, dt, \quad Y_j = \int_0^1 t^j y^2 \, dt, \quad U_j = \int_0^1 t^{n+j} x \, dt, \quad J_j = \int_0^1 t^j xy \, dt. \quad (38.1)$$

We need nonnegative j , together with V_{-1} ; its weight has exponent $n-1 \geq 0$, even at $n = 1$. The use of V_j for scalar moments is separate from the Volterra operator V .

The differential equations of Lemma 37.2 give the following polynomial flux identities:

$$[t^{n+j}(1-t)y]' = (n+j)t^{n+j-1}y - (3n+j)t^{n+j}y + n^2t^{2n+j-1}, \quad (38.2)$$

$$[t^j(1-t)y^2]' = jt^{j-1}y^2 - (4n+j-1)t^jy^2 + 2n^2t^{n+j-1}y, \quad (38.3)$$

$$[t^{n+j+1}x]' = (4n+j+1)t^{n+j}x - t^{n+j}y, \quad (38.4)$$

$$[t^{j+1}(1-t)xy]' = (3n+j+1)t^jxy - (5n+j+1)t^{j+1}xy \quad (38.5)$$

$$- (1-t)t^jy^2 + n^2t^{n+j}x. \quad (38.6)$$

In (38.3) the $j = 0$ first term is absent; it is not an undefined product at zero. The product identity underlying the last line is worth recording separately:

$$t(1-t)(xy)' = [3n - (5n-1)t]xy + n^2t^n x - (1-t)y^2. \quad (38.7)$$

The final factor $1 - t$ cannot be omitted.

Here are every upper and lower flux in the four integrations:

- For (38.2), $t^{n+j}(1-t)y$ is zero at one by $1-t$, and at zero by $y(0) = 0$ and $n+j \geq 1$.
- For (38.3), $t^j(1-t)y^2$ is zero at one by $1-t$, and at zero by $y(0) = 0$, including $j = 0$.
- For (38.4), the upper flux is $x(1) = Z$. The lower flux is zero because $t^{n+j+1}x(t)$ vanishes there.
- For (38.6), both fluxes are zero because $t^{j+1}(1-t)xy$ contains both endpoint factors and x, y are bounded.

These are polynomial equalities on the closed interval, also for $n = 1$. Dividing the ODEs by t or $1-t$ on the interior has not suppressed any nonzero boundary value.

Integration therefore yields the finite closure

$$V_0 = Z, \quad V_{-1} = 3Z - \frac{1}{2}, \quad V_j = \frac{(n+j)V_{j-1} + n^2/(2n+j)}{3n+j} \quad (j \geq 1), \quad (38.8)$$

$$Y_0 = \frac{2n^2V_{-1}}{4n-1}, \quad Y_j = \frac{jY_{j-1} + 2n^2V_{j-1}}{4n+j-1} \quad (j \geq 1), \quad (38.9)$$

$$U_j = \frac{Z + V_j}{4n+j+1} \quad (j \geq 0), \quad J_0 = X, \quad (38.10)$$

$$J_{j+1} = \frac{(3n+j+1)J_j + n^2U_j - Y_j + Y_{j+1}}{5n+j+1} \quad (j \geq 0). \quad (38.11)$$

The value $V_{-1} = 3Z - 1/2$ is simply (38.2) with $j = 0$, integrated and divided by n . This division explains again why positive width is the natural domain.

38.2 A short certificate for both shifted periods

For a polynomial $p(t) = \sum p_j t^j$, write $[p]_V = \sum p_j V_j$ and similarly for Y, U, J . Multiply (37.15) by (37.16) and integrate. Every term is covered by the four moment families:

$$X_{n+1} = [\lambda \nu t^3(1-t)^2]_J + [\lambda(1-t)^3 q]_Y \quad (38.12)$$

$$+ [\lambda(1-t)^2 r + \mu(1+t)(1-t)q]_V \quad (38.13)$$

$$+ [\mu \nu t^3(1+t)]_U + \sum_j \frac{[t^j] \{ \mu(1+t)r \}}{2n+j+1}. \quad (38.14)$$

The largest J and Y indices are five; the largest V and U indices are four. Thus finitely many applications of (38.8)–(38.11) evaluate the whole expression as an affine form in the independent symbols X, Z .

The simpler calculation from y alone is

$$Z_{n+1} = \lambda(V_1 - 2V_2 + V_3) + \mu \left(\frac{1}{2n+2} + \frac{1}{2n+3} \right). \quad (38.15)$$

Its Z coefficient and constant term simplify to exactly ν_n and δ_n in (34.17). Also $r(1) = \delta_n$, so this establishes the independent endpoint check promised after Proposition 37.3.

For explicit reproducibility, the full X transport is

$$X_{n+1} = \rho_n X_n + \sigma_n Z_n + \tau_n, \quad (38.16)$$

where, with $A_5(n) = \prod_{j=1}^4 (5n + j)$,

$$\rho_n = \frac{(n+1)^4}{5A_5(n)} = \frac{M_n}{M_{n+1}}, \quad (38.17)$$

$$\sigma_n = -\frac{(n+1)^2 P_7(n)}{30(2n+1)(3n+1)(3n+2)(4n+1)(4n+3)A_5(n)}, \quad (38.18)$$

$$\tau_n = \frac{P_9(n)}{120(2n+1)^2(3n+1)(3n+2)(4n+3)A_5(n)}, \quad (38.19)$$

with

$$P_7(n) = 138688n^7 + 507872n^6 + 772828n^5 + 630758n^4 \quad (38.20)$$

$$+ 296881n^3 + 80204n^2 + 11463n + 666, \quad (38.21)$$

$$P_9(n) = 744128n^9 + 4138176n^8 + 10020608n^7 + 13828392n^6 \quad (38.22)$$

$$+ 11948992n^5 + 6682909n^4 + 2410801n^3 \quad (38.23)$$

$$+ 538853n^2 + 67431n + 3582. \quad (38.24)$$

Equations (38.14) and (38.15), with the explicit moment recursions, are small rational certificates for these somewhat larger coefficients. Checking them means expanding fixed-degree polynomials and testing zero numerators in $\mathbb{Q}(n)$. No evaluation at finitely many widths, guessed recurrence, or polynomial interpolation is involved. The supplied standard-library verifier performs exactly those operations.

38.3 Eliminating the second period from the first difference

Substitute (38.16) and (34.17) into the full-count representation (37.2) at $n+1$. After subtracting $\rho_n F_n$, the X_n coefficient vanishes because it is $16\rho_n - 16\rho_n$. The other two coefficients are

$$[Z_n](F_{n+1} - \rho_n F_n) = \frac{2(n+1)^2 P(n)}{15(2n-1)(2n+1)(3n+1)(3n+2)} \quad (38.25)$$

$$\cdot \frac{1}{(4n-1)(4n+1)(4n+3)A_5(n)}, \quad (38.26)$$

$$[1](F_{n+1} - \rho_n F_n) = -\frac{D(n)}{120(2n-1)^2(2n+1)^2(3n+1)(3n+2)} \quad (38.27)$$

$$\cdot \frac{1}{(4n-1)(4n+3)A_5(n)}. \quad (38.28)$$

Here $[1]$ denotes the coefficient independent of X_n, Z_n , rather than substitution of their actual values. Direct cancellation with (34.9), (34.11), and (38.17) identifies these as $2\rho_n u(n)$ and $-2\rho_n u(n)\gamma_n$. We have proved

$$F_{n+1} - \rho_n F_n = 2\rho_n u(n)(Z_n - \gamma_n). \quad (38.29)$$

Multiplying by M_{n+1} proves (34.14), including at $n=1$. This completes the derivation from the literal arrays to the smaller exact period system announced in Section 34.

As a width-one check independent of this simplification, direct polynomials give

$$y_2 = 2t^2 - 2t^3/3, \quad x_2 = t^2/2 - 2t^3/9 + t^6/90,$$

$$Z_2 = 13/45, \quad X_2 = 1322/14175, \quad F_2 = 1/113400.$$

Since $M_2 = 113400$, this is $a_2 = 1$, as the interlacing total order requires. These values are checks of endpoints and normalization, not premises of the all-width certificate.

39 The printed nested sum and the exact OEIS operator

The previous sections proved the full-count identities directly. We now perform the finite algebra that identifies them with the specified source statements, including their summation constants and the orientation of the shift operator.

39.1 Variation of constants gives the actual finite sum

Recall $c_n^* = (-1)^n (n!)^3 / (3n)!$ and $c_{n+1}^* = v_n c_n^*$. The following rational identity links the period's forcing to the printed summand:

$$\delta_n + v_n \gamma_n - \gamma_{n+1} = -\frac{27}{8} v_n v(n) = \frac{9(n+1)^2 v(n)}{8(3n+1)(3n+2)}. \quad (39.1)$$

Every quantity is explicit in (34.8)–(34.11) and (34.17). Multiplication by the displayed denominators reduces (39.1) to a zero polynomial. This is the last rational certificate required for the finite sum.

Put

$$H_n = (-1)^n u(n) \frac{(5n)!}{(3n)!(n!)^2} = M_n u(n) c_n^*, \quad I_n = (-1)^n v(n) \frac{(3n)!}{(n!)^3} = \frac{v(n)}{c_n^*}. \quad (39.2)$$

From (34.17) and (39.1),

$$\frac{Z_{n+1} - \gamma_{n+1}}{c_{n+1}^*} - \frac{Z_n - \gamma_n}{c_n^*} = -\frac{27}{8} I_n. \quad (39.3)$$

The constant of summation is determined, rather than chosen: direct substitution gives $\gamma_1 = Z_1 = 1/3$. Hence

$$\frac{Z_k - \gamma_k}{c_k^*} = -\frac{27}{8} \sum_{i=1}^{k-1} I_i.$$

Inserting this in the proved first difference (34.14) yields

$$a_{k+1} - a_k = -\frac{27}{4} H_k \sum_{i=1}^{k-1} I_i. \quad (39.4)$$

Summation over $k = 1, \dots, n-1$ with $a_1 = 1$ gives (34.12), proving Theorem 34.1. This argument matches both nested summation limits in the printed source; it does not merely construct another solution of a recurrence.

The empty sums give $a_1 = a_2 = 1$. Exact substitution gives

$$u(1) = 4/5, \quad v(1) = 5/8208, \quad H_1 = -16, \quad H_2 = 608, \quad I_1 = -5/1368. \quad (39.5)$$

Therefore $a_3 = 1 - (27/4) 608 (-5/1368) = 16$.

39.2 The rational factorization and shift order

Let E denote the forward shift. Coefficients and shifts obey $E f(n) = f(n+1)E$. Define

$$R(n) = \frac{H_{n+1}}{H_n}, \quad S(n) = \frac{H_{n+2} I_{n+1}}{H_{n+1} I_n}. \quad (39.6)$$

We use S rather than U for the second ratio to distinguish it from the kernel trace and the moment family. Explicit cancellation of the factorials gives

$$R(n) = -\frac{5(2n-1)(4n-1)(4n+1)A_5(n)P(n+1)}{3(n+2)^2(2n+3)(3n+4)(3n+5)(4n+5)(4n+7)P(n)}, \quad (39.7)$$

$$S(n) = \frac{5(2n-1)(2n+1)(3n+4)(3n+5)(4n+5)(4n+7)}{(n+3)^4(2n+5)^2(3n+7)(3n+8)(4n+9)(4n+11)} \quad (39.8)$$

$$\cdot \frac{\prod_{j=6}^9 (5n+j)P(n)Q(n+1)}{P(n+1)Q(n)}. \quad (39.9)$$

The rightmost operator acts first. Equation (39.4) therefore implies

$$(E - R(n))(E - 1)a_n = -\frac{27}{4}H_{n+1}I_n. \quad (39.10)$$

The ratio of the forcing at $n+1$ to that at n is $S(n)$. Thus

$$(E - S(n))(E - R(n))(E - 1)a_n = 0. \quad (39.11)$$

For a direct check of Ore multiplication, the coefficients in increasing powers of E are

$$-SR, \quad R(n+1) + S(1+R), \quad -(1+R(n+1)+S), \quad 1. \quad (39.12)$$

Replacing $R(n+1)$ by $R(n)$ would be an incorrect commutative multiplication; the shift in this formula is essential.

Define

$$c_3(n) = 3(n+2)^2(2n+3)(2n+5)^2(3n+7)(3n+8) \times (4n+9)(4n+11)Q(n)(n+3)^4. \quad (39.13)$$

The exact polynomial identities are

$$c_0 = -c_3SR, \quad (39.14)$$

$$c_1 = c_3\{R(n+1) + S(1+R)\}, \quad (39.15)$$

$$c_2 = -c_3\{1 + R(n+1) + S\}, \quad (39.16)$$

with the literal coefficient polynomials in Appendix E. They are verified coefficientwise over $\mathbb{Q}[n]$ after clearing explicit denominators. That appendix gives the full independent source transcription, not coefficients defined circularly by these identities. It follows that

$$L_{\text{OEIS}} = c_3(n)(E - S(n))(E - R(n))(E - 1). \quad (39.17)$$

Together with (39.11) this proves Theorem 34.2.

There is also a useful explicit forcing version of (39.10):

$$a_{n+2} - (1 + R(n))a_{n+1} + R(n)a_n = M_n\Phi(n), \quad (39.18)$$

where

$$\Phi(n) = \frac{15A_5(n)Q(n)}{4(n+1)^4(n+2)^4(2n-1)(2n+1)^2(2n+3)^2} \quad (39.19)$$

$$\cdot \frac{1}{(3n+4)(3n+5)(4n+5)(4n+7)P(n)}. \quad (39.20)$$

This is obtained directly by cancelling $-(27/4)H_{n+1}I_n/M_n$. Thus the forcing is fixed by the summands and its ratio is fixed by (39.6).

39.3 Positivity and uniqueness on the correct range

All affine denominators in the rank, differential, and moment formulas are positive for $n \geq 1$. The only further possible zeros occur in $P(n)$ or $Q(n)$. For $m = n - 1 \geq 0$, the coefficients of $P(m + 1)$, in ascending order, are

$$(60480, 353502, 971955, 1683047, 1973632, 1543976, 760768, 211616, 25216).$$

Those of $Q(m + 1)$ are

$$(1077753600, 23068181160, 128320688700, 354332674386, \\ 588363536007, 638861237719, 471293180347, 238738633847, \\ 81945774178, 18238806776, 2377384288, 137855872).$$

Every coefficient is strictly positive, so both polynomials are positive for every real $n \geq 1$. In particular, $u(n)$, $v(n)$, the ratios, and all eliminations are defined there, while $c_3(n) > 0$. The factorial c_n^* never vanishes. This proves the denominator assertion in Theorem 34.1 and the uniqueness assertion in Theorem 34.2.

The first terms, with index starting at one, are

$$1, 1, 16, 985, 141696, 36372976, 14083834704, 7372392431849.$$

These are useful checks of orientation and indexing. The proof is the uniform integral and rational derivation, not agreement of a finite prefix. Finally, literal substitution at $n = 0$ with the empty-array values $(a_0, a_1, a_2, a_3) = (1, 1, 1, 16)$ gives the nonzero residual stated in Section 34; no uniqueness claim from that starting index is made.

40 Recurrence-based asymptotics with effective errors

The leading constant and the existence of all fixed inverse-power orders were already proved in the earlier all-height report (Part I). This section recovers those results from the exact recurrence and supplies finite rational residual certificates. It does not claim priority for the asymptotic expansion.

40.1 A triangular stable system

Write $b_n = F_n = a_n/M_n$, where $M_n = (5n)!/(n!)^5$. The proved identities give

$$Z_{n+1} = v_n Z_n + \delta_n, \\ b_{n+1} = \rho_n b_n + 2\rho_n u(n)(Z_n - \gamma_n), \quad \rho_n = \frac{(n+1)^4}{5(5n+1)(5n+2)(5n+3)(5n+4)}, \quad (40.1)$$

with $Z_1 = 1/3$ and $b_1 = 1/120$. Here b_n is a scalar normalized count, distinct from any function $b(t)$ used in the kernel calculation. For $n \geq 1$,

$$|v_n| \leq \frac{4}{27} < \frac{1}{6}, \quad 0 < \rho_n \leq \frac{16}{3125} < \frac{1}{100}. \quad (40.2)$$

These bounds follow by replacing $n + 1$ by $2n$ in the numerators and each positive denominator factor by its leading monomial. The coupling $2\rho_n u(n)$ is $O(n^{-1})$, which is important for the error orders.

Put $t = 1/n$, $f(t) = t/(1 + t)$, and define rational functions

$$r(t) = \rho_{1/t}, \quad v_*(t) = v_{1/t}, \quad d_*(t) = \delta_{1/t}, \quad \alpha(t) = 2\rho_{1/t}u(1/t), \quad \eta(t) = -\alpha(t)\gamma_{1/t}.$$

All extend analytically to zero; $\alpha(0) = \eta(0) = 0$, $r(0) = 1/3125$, and $v_*(0) = -1/27$. The two formal series $\mathcal{Z}(t) = \sum_{j \geq 0} z_j t^j$ and $\mathcal{B}(t) = \sum_{j \geq 0} \beta_j t^j$ are specified by

$$\begin{aligned}\mathcal{Z}(f(t)) &= v_*(t)\mathcal{Z}(t) + d_*(t), \\ \mathcal{B}(f(t)) &= r(t)\mathcal{B}(t) + \alpha(t)\mathcal{Z}(t) + \eta(t).\end{aligned}\tag{40.3}$$

In the first equation the new coefficient z_j has multiplier $28/27$; in the second the new coefficient β_j has multiplier $3124/3125$. Moreover, $\alpha(0) = 0$ means the second equation at order j needs only $z_0, \dots, z_{j-1}$. Thus both series are uniquely determined over $\mathbb{Q}$ by finite coefficient arithmetic, to any prescribed degree.

For completeness, direct expansion gives

$$\mathcal{Z}(t) = \frac{1}{4} + \frac{t}{16} + \frac{t^2}{32} + \frac{t^3}{128} - \frac{t^4}{128} - \frac{9t^5}{512} - \frac{21t^6}{1024} - \frac{41t^7}{4096} + \frac{29t^8}{1024} + \frac{925t^9}{8192} + O(t^{10}).$$

Substitution of the displayed rational γ_n shows

$$\mathcal{Z}(t) - \gamma_{1/t} = \frac{189783}{1613824}t^9 + O(t^{10}), \quad \alpha(t) = \frac{197}{84375}t + O(t^2).$$

Consequently $\beta_0 = \dots = \beta_9 = 0$ and $\beta_{10} = 9/32768$. These are finite rational identities, not fitted values. The companion standard-library script checks them coefficientwise. Set $c_j = \beta_{10+j}$ and $c_0 = 9/32768$.

Theorem 40.1 (Every fixed order with a rational certificate). *For every integer $K \geq 0$, one can compute an integer $N_K \geq 2$ and positive rational constants $E_{Z,K}, E_{b,K}$ such that*

$$\begin{aligned}\left| Z_n - \sum_{j=0}^{K+9} z_j n^{-j} \right| &\leq E_{Z,K} n^{-K-10}, \\ \left| b_n - n^{-10} \sum_{j=0}^K c_j n^{-j} \right| &\leq E_{b,K} n^{-K-11} \quad (n \geq N_K).\end{aligned}\tag{40.4}$$

In particular,

$$\begin{aligned}b_n &= \frac{9}{32768} n^{-10} \left(1 + \frac{10}{n} + \frac{225}{4n^2} + \frac{1819}{8n^3} + \frac{22045}{32n^4} + \frac{1353}{n^5} \right. \\ &\quad \left. - \frac{6103}{16n^6} + O(n^{-7}) \right).\end{aligned}\tag{40.5}$$

Proof. Fix K , set $L = K + 9$, $p_Z = K + 10$, $p_b = K + 11$, and let

$$q_Z(n) = \sum_{j=0}^L z_j n^{-j}, \quad q_b(n) = n^{-10} \sum_{j=0}^K c_j n^{-j}.$$

Their exact rational residuals are

$$\begin{aligned}D_Z(n) &= q_Z(n+1) - v_n q_Z(n) - \delta_n, \\ D_b(n) &= q_b(n+1) - \rho_n q_b(n) - 2\rho_n u(n)(q_Z(n) - \gamma_n).\end{aligned}$$

Equation (40.3) gives $D_Z = O(n^{-p_Z})$ and $D_b = O(n^{-p_b})$. The latter order holds although q_Z stops one degree before q_b , because the coupling vanishes to first order at zero.

Here is an explicit finite procedure to turn these formal cancellations into bounds. For each of

$$t^{-p_Z} D_Z(1/t), \quad t^{-p_b} D_b(1/t), \quad \alpha(t)/t,$$

form an exact ratio $P(t)/H(t)$ of rational polynomials with $H(0) > 0$. Remove any common power of t first. For an integer $N \geq 2$, put

$$h_N = H_0 - \sum_{i \geq 1} |H_i| N^{-i}, \quad C_N(P, H) = \frac{\sum_i |P_i| N^{-i}}{h_N}. \quad (40.6)$$

Increase N , for example by doubling, until all three h_N are positive and

$$\theta_Z = \left(\frac{N}{N+1} \right)^{p_Z} > \frac{1}{6}, \quad \theta_b = \left(\frac{N}{N+1} \right)^{p_b} > \frac{1}{100}.$$

This search terminates because $H(0) > 0$ and both powers tend to one. Let C_Z, C_b, A be the three respective values of C_N . Then for every $n \geq N$,

$$|D_Z(n)| \leq C_Z n^{-p_Z}, \quad |D_b(n)| \leq C_b n^{-p_b}, \quad |2\rho_n u(n)| \leq A/n.$$

Compute Z_N, b_N exactly from (40.1) and define

$$\begin{aligned} E_Z &= \max \left\{ N^{p_Z} |Z_N - q_Z(N)|, \frac{C_Z}{\theta_Z - 1/6} \right\}, \\ E_b &= \max \left\{ N^{p_b} |b_N - q_b(N)|, \frac{C_b + A E_Z}{\theta_b - 1/100} \right\}. \end{aligned} \quad (40.7)$$

If $E_Z = 0$, replace it by any positive rational before defining E_b ; if $E_b = 0$, replace it by any positive rational as well.

Writing $e_Z = Z - q_Z$ and $e_b = b - q_b$, the exact error equations give

$$\begin{aligned} |e_Z(n+1)| &\leq |e_Z(n)|/6 + C_Z n^{-p_Z}, \\ |e_b(n+1)| &\leq |e_b(n)|/100 + (A/n)|e_Z(n)| + C_b n^{-p_b}. \end{aligned}$$

The first estimate propagates $|e_Z(n)| \leq E_Z n^{-p_Z}$ because $E_Z/6 + C_Z \leq E_Z \theta_Z$. The second propagates $|e_b(n)| \leq E_b n^{-p_b}$ because $p_Z + 1 = p_b$ and $E_b/100 + A E_Z + C_b \leq E_b \theta_b$. Finally $(n/(n+1))^p \geq (N/(N+1))^p$ for $n \geq N$. The initial inequalities in (40.7) start both inductions. This proves the claim for the actual counting solution. $\square$

The rational certificate at $K = 6$ permits the particularly simple inequality

$$|b_n - q_6(n)| \leq 18n^{-17} \quad (n \geq 64), \quad (40.8)$$

where $q_6(n)$ is the displayed truncation through n^{-16} in (40.5). This onset and constant are deliberately conservative.

40.2 Restoring the multinomial factor

Stirling's expansion and its positive-real remainder bound [14] give

$$\begin{aligned} \log M_n &\sim n \log 3125 - 2 \log n + \log \frac{\sqrt{5}}{4\pi^2} \\ &+ \sum_{h \geq 1} \frac{B_{2h}}{2h(2h-1)} (5^{-(2h-1)} - 5) n^{-(2h-1)}. \end{aligned} \quad (40.9)$$

Here and below infinite expansions are interpreted at each fixed order. The first logarithmic correction is $-2/(5n)$. Multiplication with Theorem 40.1 yields the previously established leading term and all its rational relative corrections:

$$a_n = C_* 3125^n n^{-12} \left(1 + \sum_{j=1}^K d_j n^{-j} + O_K(n^{-K-1}) \right), \quad C_* := \frac{9\sqrt{5}}{2^{17}\pi^2}. \quad (40.10)$$

For example,

$$\begin{aligned} d_1 &= \frac{48}{5}, & d_2 &= \frac{5233}{100}, & d_3 &= \frac{1028391}{5000}, \\ d_4 &= \frac{60248377}{100000}, & d_5 &= \frac{273939939}{250000}, & d_6 &= -\frac{2709616559}{3125000}. \end{aligned}$$

All coefficients follow by exponentiating finite rational series. The first five displayed normalized and count corrections already appear in the prior report (Part I); the sixth displayed correction is obtained here from the current recurrence certificate, without a novelty claim.

40.3 A finite logarithmic remainder

For inversion we need a bound, not an unspecified O term. Let $\lambda = \log 3125$, and let ℓ_j be the rational coefficients in $\log(1 + \sum_{j \geq 1} d_j t^j)$. For example,

$$\ell_1 = \frac{48}{5}, \quad \ell_2 = \frac{25}{4}, \quad \ell_3 = -\frac{8889}{5000}, \quad \ell_4 = -\frac{335}{8}. \quad (40.11)$$

For each fixed K there are a computable rational $C_K > 0$ and integer $N_K^{\log}$ such that

$$\left| \log a_n - \left(\lambda n - 12 \log n + \log C_* + \sum_{j=1}^K \ell_j n^{-j} \right) \right| \leq C_K n^{-K-1} \quad (n \geq N_K^{\log}). \quad (40.12)$$

Here is an explicit construction. Retain $E = E_{b,K}$ from (40.4), put

$$u_K(t) = \sum_{j=1}^K \frac{c_j}{c_0} t^j, \quad U = \sum_{j=1}^K \left| \frac{c_j}{c_0} \right| N^{-(j-1)},$$

and increase N beyond N_K until $U/N \leq 1/4$ and $(E/c_0)N^{-K-1} \leq 1/4$. Then the mean value theorem for $\log$ on $[1/2, \infty)$ gives

$$\left| \log \frac{n^{10} b_n}{c_0} - \log(1 + u_K(1/n)) \right| \leq \frac{2E}{c_0} n^{-K-1}.$$

Define the finite polynomial and coefficient bound

$$L_K(t) = \sum_{h=1}^K \frac{(-1)^{h+1}}{h} u_K(t)^h, \quad V = \sum_{j>K} |[t^j]L_K(t)|N^{-(j-K-1)}.$$

The scalar logarithm tail is at most $4U^{K+1}n^{-K-1}/(3(K+1))$, while the terms of L_K of degree above K contribute at most Vn^{-K-1} . Thus the normalized-count contribution is bounded by $C^{(b)}n^{-K-1}$, where

$$C^{(b)} = 2E/c_0 + V + \frac{4U^{K+1}}{3(K+1)}.$$

For $K = 0$, use $u_K = L_K = 0$ and $U = V = 0$.

Set $J = \lfloor (K+1)/2 \rfloor$ and $q = 2J+1$. The Stirling remainder following J Bernoulli terms in $\log(n!)$ is at most $|B_{2J+2}|/((2J+2)(2J+1)n^q)$. Applying this separately at $5n$ and at n gives the explicit additional constant

$$C^{(M)} = \frac{|B_{2J+2}|}{(2J+2)(2J+1)}(5^{-q} + 5)N^{K+1-q}.$$

Since $q \geq K+1$, this proves (40.12) with $C_K = C^{(b)} + C^{(M)}$ and $N_K^{\log} = N$. The exact companion certificates, with harmless upward rounding, give

$$(K, N_K^{\log}, C_K) = (0, 512, 202) \quad \text{and} \quad (6, 64, 6100000). \quad (40.13)$$

Every bound in this construction is rational; the constants π and $\sqrt{5}$ enter only the displayed leading factor, which is retained exactly.

Remark 40.2 (Scope of the expansions). The contraction proof fixes all algebraic coefficients and controls every fixed truncation. It does not assert convergence of the infinite formal series or select a canonical decomposition into exponentially small sectors. The two homogeneous contraction ratios tend to $1/3125$ and $-1/27$, but those ratios alone are not Stokes data. None of the finite error or inverse conclusions requires such a decomposition.

41 Finite Lambert models and certified integer thresholds

The count exists at integer widths. We first invert an explicitly defined finite real function and then compare it with the exact integer threshold. This avoids differentiating an unspecified discrete error or rounding an asymptotic approximation without a margin.

41.1 A finite monotone model

Fix $K \geq 0$ and use (40.12). For real $x > 0$ set

$$\Psi_K(x) = \lambda x - 12 \log x + \log C_* + \sum_{j=1}^K \ell_j x^{-j}. \quad (41.1)$$

Choose $X_K > 0$ large enough that

$$\frac{12}{X_K} + \sum_{j=1}^K j|\ell_j|X_K^{-j-1} \leq \frac{\lambda}{2}. \quad (41.2)$$

Then $\Psi'_K(x) \geq \lambda/2$ for $x \geq X_K$. For each $y \geq \exp \Psi_K(X_K)$, there is exactly one $\xi_K(y) \geq X_K$ satisfying $\Psi_K(\xi_K(y)) = \log y$. This is the inverse of a particular finite function, rather than a choice of interpolation of the discrete count.

For the leading model the increasing-branch inverse is

$$w(y) := \xi_0(y) = -\frac{12}{\lambda} W_{-1} \left(-\frac{\lambda}{12} \left(\frac{C_*}{y} \right)^{1/12} \right). \quad (41.3)$$

For large y , the argument lies in $(-1/e, 0)$. Indeed, $y = C_* e^{\lambda w} w^{-12}$ is equivalent to $w e^{-\lambda w/12} = (C_*/y)^{1/12}$. The real branch W_{-1} [16] gives $w \rightarrow \infty$ as $y \rightarrow \infty$; the principal branch describes the small solution of the same leading model and is not the large-width inverse.

41.2 Arbitrary fixed inverse corrections

Let $t = 1/w$ and seek the finite approximation

$$\widehat{\xi}_K(y) = w + \Delta_K(1/w), \quad \Delta_K(t) = \sum_{j=1}^K s_j t^j. \quad (41.4)$$

Choose $s_1, \dots, s_K$ to cancel the coefficients of degrees 1 through K in

$$G_K(t) = \lambda \Delta_K(t) - 12 \log(1 + t \Delta_K(t)) + \sum_{j=1}^K \ell_j t^j (1 + t \Delta_K(t))^{-j}. \quad (41.5)$$

At degree j , the new coefficient s_j occurs only as λs_j . The recursion therefore has a unique solution in $\mathbb{Q}[\lambda^{-1}]$. The first two coefficients are

$$s_1 = -\frac{48}{5\lambda}, \quad s_2 = -\frac{25}{4\lambda} - \frac{576}{5\lambda^2}. \quad (41.6)$$

The logarithmic displacement is already incorporated in w , so these additional corrections start at order w^{-1} .

Proposition 41.1 (Effective finite inverse error). *For each fixed K , one can compute $H_K \geq 0$ and $W_K > 0$ such that*

$$|\xi_K(y) - \widehat{\xi}_K(y)| \leq \frac{2H_K}{\lambda} w(y)^{-K-1} \quad (41.7)$$

whenever $w(y) \geq W_K$ and $y \geq \exp \Psi_K(X_K)$.

Proof. Choose $W_K \geq 2X_K$ so that $\sum_{j=1}^K |s_j| W_K^{-j-1} \leq 1/2$. Then $z(t) := 1 + t \Delta_K(t) \geq 1/2$ for $0 \leq t \leq 1/W_K$ and $\widehat{\xi}_K = w z(1/w) \geq w/2 \geq X_K$. The Taylor coefficients of G_K through degree K vanish. For $K \geq 1$, repeated differentiation of (41.5) gives

$$G_K^{(K+1)}(t) = \frac{P_K(t)}{z(t)^{2K+1}},$$

where P_K is a finite, computable polynomial with coefficients in $\mathbb{Q}[\lambda, \lambda^{-1}]$. The largest denominator power comes from $K+1$ derivatives of z^{-K} ; derivatives of the logarithm have lower powers and can be put over the same denominator. Taylor's theorem therefore gives $|G_K(t)| \leq H_K t^{K+1}$ with

$$H_K = \frac{2^{2K+1}}{(K+1)!} \sum_i |[t^i] P_K(t)| W_K^{-i}. \quad (41.8)$$

For $K = 0$, take $\Delta_0 = 0$ and $H_0 = 0$. The identity $\Psi_0(w) = \log y$ gives exactly $\Psi_K(\widehat{\xi}_K) - \log y = G_K(1/w)$. Both $\widehat{\xi}_K$ and ξ_K belong to the interval where $\Psi'_K \geq \lambda/2$, so the mean value theorem proves the bound. $\square$

This proof uses only finite functions. It makes no convergence assertion about the infinite inverse series. Certified intervals for λ suffice to turn the finite formulas for W_K, H_K into numerical bounds.

41.3 The exact integer threshold

For $y > 0$, define

$$\mathcal{N}(y) = \min\{n \geq 1 : a_n \geq y\}. \quad (41.9)$$

It exists since (40.10) implies $a_n \rightarrow \infty$. Eventual strict increase follows without differentiating an error term: from (40.12) at $K = 0$,

$$\begin{aligned} \log a_{n+1} - \log a_n &\geq \lambda - 12 \log(1 + 1/n) - C_0(n^{-1} + (n+1)^{-1}) \\ &\geq \lambda - \frac{12 + 2C_0}{n}. \end{aligned}$$

Thus any integer beyond $N_0^{\log}$ and $2(12 + 2C_0)/\lambda$ is an effective monotonicity onset.

Fix K , put $p = K + 1$, and choose an integer N_* beyond this onset, $N_K^{\log}$, and X_K . Suppose y is large enough that

$$\begin{aligned} y &> \max_{1 \leq n < N_*} a_n, \quad \xi := \xi_K(y) \geq \max(2N_* + 2, 4), \\ \delta &:= \frac{2^{p+1}C_K}{\lambda} \xi^{-p} \leq 1. \end{aligned} \quad (41.10)$$

Every condition is finite and checkable. The prefix condition is necessary unless one has separately established monotonicity on the entire prefix.

Theorem 41.2 (A qualified ceiling enclosure). *Under (41.10),*

$$\lceil \xi_K(y) - \delta \rceil \leq \mathcal{N}(y) \leq \lceil \xi_K(y) + \delta \rceil, \quad \delta = \frac{2^{K+2}C_K}{\lambda} \xi_K(y)^{-K-1}. \quad (41.11)$$

Proof. Put $m = \lceil \xi + \delta \rceil$. The derivative bound gives $\Psi_K(m) - \log y \geq \lambda\delta/2$. Its logarithmic error is at most $C_K m^{-p} \leq C_K \xi^{-p} \leq \lambda\delta/2$, so $a_m \geq y$. Next put $j = \lceil \xi - \delta \rceil - 1$. We have $j < \xi - \delta$ and $j \geq \xi - 2 \geq \xi/2 \geq N_*$. Hence $\Psi_K(j) - \log y < -\lambda\delta/2$, whereas $C_K j^{-p} \leq 2^p C_K \xi^{-p} = \lambda\delta/2$. Thus $a_j < y$. Eventual monotonicity and the prefix condition rule out all earlier crossings. These comparisons prove both inequalities. $\square$

When the two ceilings agree, the integer threshold is certified. For sufficiently large y one has $2\delta < 1$, so at most two adjacent integers remain. Near an integer boundary, an unconditional rule to round the asymptotic inverse is not justified. Instead evaluate the remaining exact recurrence values and compare them with y ; for an integer or rational target these are exact terminating rational comparisons, including equality. For a general real target, the needed exact comparison must be supplied or certified by the representation of that target. Numerical intervals alone need not resolve equality with an integer count.

The finite Lambert approximation also yields an explicit enclosure. Write $E' = 2H_K/\lambda$ and $D = 2^{p+1}C_K/\lambda$, and enlarge the onset so that $|\xi_K - \widehat{\xi}_K| \leq E'w^{-p}$ and $\xi_K \geq w/2$. Then (41.11) implies

$$\left\lceil \widehat{\xi}_K - (E' + 2^p D)w^{-p} \right\rceil \leq \mathcal{N}(y) \leq \left\lceil \widehat{\xi}_K + (E' + 2^p D)w^{-p} \right\rceil. \quad (41.12)$$

Here w is (41.3), $\widehat{\xi}_K$ is a finite correction polynomial, and every constant has a specified finite construction. Interval evaluation must preserve the outward direction of both endpoints. If an endpoint cannot be separated from an integer, retain both candidates and use the exact comparison rather than assuming a rounding direction.

42 Exact verification and questions left open

The accompanying code is an auditable supplement to the mathematical proof. Its principal checks take place in exact polynomial and rational rings. Finite values are reported in a separate part of the receipt so that their logical role is not confused with an all-width theorem.

42.1 What the uniform verifier establishes

The standard-library program `code/verify_certificate.py` uses dense polynomials over $\mathbb{Q}$ for rational functions of n and sparse polynomials for the finitely many additional formal variables. Rational equality is tested by exact zero numerators, not by evaluating at sample widths. Its certificate covers:

1. the coefficient identities for both contiguous functions, including the homogeneous coefficient ratio and the endpoint at $t = 1$;
2. the general moment flux identities, all moment reductions used in the product, and each coefficient of the resulting affine period shift;
3. the trace/open-chain scalar reductions, the full-count expression, and each independent coefficient in the first-difference identity;
4. the γ bridge and initial value, the rising-factorial source polynomials, and positive coefficients after shifting P, Q by one;
5. all four literal OEIS coefficients, the explicit inhomogeneous forcing, and a separate direct elimination of the three-state period system against the literal operator.

The receipt records each reduced moment as a three-component row in the state $(X, Z, 1)$, with polynomial numerators and denominators in ascending powers of n . These rows are one concrete machine-readable form of Section 38's certificate. The algebra library has its own small consistency checks; every essential test uses explicit exceptions rather than assertions that disappear under Python optimization.

The program also generates all 15 signed six-label matchings and all 225 products. Its path/cycle classification reproduces the four exact counts and signs in Section 35. This is a complete enumeration of a fixed finite Pfaffian calculation. The reduction of arbitrary kernels to its four classes remains the sign and symmetry argument in the text.

42.2 Independent finite checks and their limits

Three separate exact constructions are compared for $1 \leq n \leq 20$:

- A literal-cell-graph dynamic program uses the four local inequalities in (34.1) and counts its linear extensions. It does not use the proposed recurrence or the Pfaffian formula.
- Direct polynomial integration of the kernel on both halves of the square constructs x_n, y_n and evaluates the full-count formula. It does not obtain the periods by their contiguous recurrence.
- The printed nested sum is evaluated with exact factorials and fractions, including the rising-factorial denominator.

These agree, and the operator's finite residuals vanish wherever their terms are available. Separately, three arbitrary integer-matrix examples compare the discrete odd Pfaffian identities against literal sums of minors.

Such tests are useful for catching transcription, sign, normalization, and indexing mistakes. They neither replace the integral argument nor establish a rational identity from finitely many values. The distinct zero-index failure is tested explicitly.

42.3 Rebuilding the manuscript and certificates

The package contains the editable TeX, the frozen PDF, exact-arithmetic programs and receipts, a source-attribution record, and a frozen SHA-256 inventory. The commands are described in `README.md`; the verifier alone requires only Python's standard library. A complete build additionally requires the stated TeX Live packages and fonts. There is no network access or downloaded input during reproduction.

The build program verifies every source member, reruns the certificates, compiles the TeX in a private temporary directory, and checks byte equality against the frozen receipt and PDF. It creates a deterministic ZIP with fixed member order, timestamps, and permissions. The output path must be a new directory outside the source package; existing files, symlink components, and unsafe paths are rejected. Each Python subprocess has bytecode writing disabled and preserves the requested optimization level. The small integer-to-string limit used in the recorded builds is stated in the README. Source hashes are checked again after reproduction.

Reproducing a fixed PDF byte-for-byte depends on the recorded TeX and font versions. A different toolchain can produce different bytes from identical mathematics; it requires a fresh rendering inspection rather than a claim that the discrepancy is harmless. The inventory guarantees consistency with the supplied package, not independent authentication or peer review. No third-party source PDF or private review material is redistributed.

[write] Added 5 October 2026, batch 101: the shipped layout. “The package” in this section is the delivered archive. In ProveIt its programs and receipt carry the prefix `04-height-five-`: `code/name.py` (four modules) is shipped as `code/04-height-five-name.py`, `build.py` as `code/04-height-five-build.py`, `code/receipt.json` as `data/04-height-five-receipt.json`, and `SOURCES.md` as `04-height-five-SOURCES.md`. The editable source, the frozen PDF, the delivered README and the SHA-256 inventory named above are not shipped. `verify_certificate.py` imports `exact_algebra`, `printed_coefficients` and `asymptotic_certificate` under their delivered names, and `build.py` needs the inventory and the frozen PDF, so neither runs in place; the verifier runs on a copy with the delivered names restored, and the README of the report gives the commands. At the write (5 October 2026, Python 3.14.4 on Windows, on such a copy) `python -B code/verify_certificate.py` printed `data/04-height-five-receipt.json` byte for byte after removal of carriage returns, in about 19 seconds; the receipt reports status PASS and 58 named uniform certificates. `build.py` was not run. `code/04-height-five-printed_coefficients.py` is a literal transcription of the A181199 operator (its docstring cites the OEIS internal entry); as OEIS content that operator is published by The OEIS Foundation Inc. under the Creative Commons Attribution-ShareAlike 4.0 licence. `code/04-height-five-asymptotic_certificate.py` lists the first eight terms of A181199 as expected values of its own recurrence run.

42.4 Further questions

Research question 16. For heights seven, nine, and higher, can the open paths in the odd augmented-Pfaffian expansion be reduced to a comparably small family of polynomial periods?

The present height-five calculation shows why traces alone are insufficient at odd height: the length-four open path must be retained. A systematic relation between path length, period dimension, and the minimal recurrence order would give a useful general theory.

Research question 17. Can one derive the stated order-three operator directly by a compact creative-telescoping certificate on the original order polytope, with a size comparable to the two-period contiguity certificate?

The factorization in (39.17) is a factorization over $\mathbb{Q}(n)$; it is not a proof that order three is minimal among homogeneous rational-coefficient recurrences for this particular solution. Minimality is a separate question, not needed for either source statement.

Research question 18. What is the precise exponentially small information carried by the homogeneous modes of the triangular period system?

Their limiting multipliers are $1/3125$ and $-1/27$ for the normalized system. The fixed-order contraction estimates establish Poincaré expansions and effective finite bounds. They do not identify a canonical exponentially improved expansion, Stokes data, or Borel sum. Those stronger questions require additional normalization and analysis.

Research question 19. Can the finite inverse-index bounds be sharpened enough to certify most threshold ceilings with little or no exact recurrence refinement?

The effective constants here are deliberately conservative. Refining them is an algorithmic problem distinct from proving the recurrence, and it must retain the near-integer caveat in any ceiling rule.

[write] **Added 5 October 2026, batch 101.** The four questions above are the manuscript’s Questions 1–4, numbered 16–19 here. Question 17 continues Research question 1 of Part I and Research question 8 of Part II, now for a compact certificate rather than a proof; Question 18 is the height-five form of Research question 4 and of item (F8); Question 19 continues Research question 6.

42.5 [write] One period, and no first-order recurrence

Added 5 October 2026, batch 101. This subsection is the write’s, not the manuscript’s.

Remark 42.1 ([write] The period of Parts III and IV). The period (34.15) is the period (28.23) of Part III: both are $n^2 \int_{\mathcal{D}} u f^{n-1}$ over the same triangle, and both arise as $N^3 H$ with $H = \langle t^n/N, h \rangle$ ((28.20), (37.1)). The recurrence (34.17) is Theorem 29.1 divided by $3(3n+1)(3n+2)$: indeed $3(3n+1)(3n+2)v_n = -(n+1)^2$ and $3(3n+1)(3n+2)\delta_n = (28n^3 + 50n^2 + 29n + 5)/(2(2n+1))$. Part III proves it by the divergence certificate (29.4); Part IV derives it from the y -contiguity (37.15) and the moment closure (38.15), without citing Part III. These are two independent proofs of one identity. In the same way (36.11), (36.13), (36.25) and the scalar identities (36.9) are (28.7), (28.14), (28.19), (28.9), (28.10) and (28.13) with the same proofs, and (35.3), (35.2) and Lemma 35.1 are (27.4), Lemma 27.1 and (27.2) with $m = 5$. The kernels K , W , h and their scalars depend on n only, not on the height, so these identities are the same statements after renaming Part III’s α, β to Part IV’s f, b . Where the printed forms differ they agree by $c = 1/(2N^2)$ and $\int e = 0$: thus $c^2/2 = 1/(8N^4)$ in (28.9), and the term $\int e$ of (28.7) is absent from (36.11). Part IV’s $J_0 = I$ in (36.9) is Part III’s display (28.13) combined with (28.9). The proof of Lemma 33.1 uses only the recurrence, which Part IV proves for its own Z_n ; so Proposition 42.2 below does not depend on this identification of the integrals. (Reworded 5 October 2026 after the intake’s independent check, recorded at the end of Section 45.2. The list above first cited only (28.9) and (28.10) for the scalar identities, and the last sentence first read: “The kernels K , W , h and their scalars depend on n only, not on the height, so these identities are literally the same statements.”)

Proposition 42.2 ([write] A181199 is not hypergeometric). *The sequence $a_n = T_5(n)$ has no recurrence of order zero or one, in the sense of Section 33.5. Hence the smallest order of a recurrence for A181199 is two or three, and the operator of Theorem 34.2 has minimal order unless A181199 obeys a recurrence of order two.*

Proof. Every a_n with $n \geq 1$ is a positive integer, so a recurrence of order zero is impossible, as in Proposition 33.2. Suppose $q_1(n)a_{n+1} + q_0(n)a_n = 0$ for $n \geq n_0$ with $q_1 \neq 0$; enlarging n_0 , $a_{n+1} = \varrho(n)a_n$

for $n \geq n_0$ with $\varrho \in \mathbb{Q}(n)$. By Section 39, $P(n) > 0$ and $Q(n) > 0$ for $n \geq 1$, so $u(n) > 0$ and $v(n) > 0$ there. Put $Y_n = Z_n - \gamma_n$. By (34.17) and (39.1),

$$Y_{n+1} - v_n Y_n = \omega(n) := \frac{9(n+1)^2 v(n)}{8(3n+1)(3n+2)} > 0 \quad (n \geq 1), \quad (42.1)$$

and by (34.14), $(\varrho(n) - 1)a_n = 2M_n u(n)Y_n$ for $n \geq n_0$. If $\varrho(n) = 1$ for infinitely many n , then $\varrho = 1$ identically, so $Y_n = 0$ for all $n \geq n_0$, contradicting (42.1). Enlarging n_0 , we therefore have $\varrho(n) \neq 1$, and hence $Y_n \neq 0$, for $n \geq n_0$. Then

$$\sigma(n) := \frac{Y_{n+1}}{Y_n} = \frac{(\varrho(n+1) - 1)\varrho(n)u(n)}{(\varrho(n) - 1)u(n+1)} \cdot \frac{M_n}{M_{n+1}}$$

is the value of a rational function, since $M_n/M_{n+1} = \rho_n$ (38.17). By (42.1), $Y_n(\sigma(n) - v_n) = \omega(n) \neq 0$ for $n \geq n_0$, so $\sigma - v_n$ is a nonzero rational function and

$$Z_n = \gamma_n + \frac{\omega(n)}{\sigma(n) - v_n} \quad (n \geq n_0)$$

coincides with a rational function for all large n . By Remark 42.1 this Z_n is the period of Part III, so Lemma 33.1 gives a contradiction. Theorem 34.2 supplies a recurrence of order three. $\square$

Whether A181199 obeys a recurrence of order two is item (F9) below. At height four, excluding order one settled the question; at height five it leaves orders two and three.

As at height four, the degree drops at higher order. Proposition 42.3 was found and proved by the intake's independent check, 5 October 2026; it is neither the manuscript's nor the write's. Notation as before Proposition 33.3, with $L_{\text{OEIS}} = \sum_{j=0}^3 c_j(n)E^j$ the operator of Theorem 34.2 and Q the polynomial of (34.8).

Proposition 42.3 ([check] Orders four and five at height five). *Let $a_n = T_5(n)$ (A181199).*

(a) *There are polynomials $e_0, \dots, e_4$ of degree 18 and $U_0, U_1 \in \mathbb{Q}[n]$ of degree five with*

$$\sum_{j=0}^4 e_j(n)E^j = \frac{U_1(n)E + U_0(n)}{Q(n+1)} L_{\text{OEIS}},$$

and $\sum_{j=0}^4 e_j(n)a_{n+j} = 0$ for every $n \geq 1$: A181199 obeys a recurrence of order four and degree 18.

(b) *There are polynomials $g_0, \dots, g_5$ of degree 16 and $V_0, V_1, V_2 \in \mathbb{Q}[n]$ with*

$$\sum_{j=0}^5 g_j(n)E^j = \left(\frac{V_2(n)E^2}{Q(n+2)} + \frac{V_1(n)E}{Q(n+1)Q(n+2)} + \frac{V_0(n)}{Q(n+1)} \right) L_{\text{OEIS}},$$

and $\sum_{j=0}^5 g_j(n)a_{n+j} = 0$ for every $n \geq 1$: A181199 obeys a recurrence of order five and degree 16.

(c) *If A181199 has no recurrence of order two, then every recurrence of A181199 holds for every $n \geq 1$, and the smallest degree of a recurrence is 24 at order three, 18 at order four and 16 at order five.*

The polynomials e_j , g_j , U_k and V_k are given in data/check-lower-order-recurrences. json.

Proof. (a) and (b). Expanding the right sides with $Ef(n) = f(n+1)E$, clearing the denominators and comparing coefficients with the left sides is a finite identity over $\mathbb{Q}[n]$, checked exactly from the data file and

the coefficients of Appendix E. By Section 39, $Q(x) > 0$ for every real $x \geq 1$, so for $n \geq 1$ every coefficient of the rational left factors is finite, and in (a)

$$\sum_{j=0}^4 e_j(n) a_{n+j} = \frac{U_1(n) (L_{\text{OEIS}} a)(n+1) + U_0(n) (L_{\text{OEIS}} a)(n)}{Q(n+1)} = 0$$

by Theorem 34.2; (b) is the same with three terms.

(c) Let R be a recurrence of order r , valid for $n \geq n_0$; by Proposition 42.2 and the hypothesis, $r \geq 3$. As in the proof of Proposition 33.3, right division by L_{OEIS} gives $R = (\sum_{k=0}^{r-3} W_k E^k) L_{\text{OEIS}} + s_2 E^2 + s_1 E + s_0$, where the denominators of the W_k and s_j divide $\prod_{i=0}^{r-3} c_3(n+i)$. The real zeros of c_3 (39.13) are $-3, -11/4, -8/3, -5/2, -7/3, -9/4, -2, -3/2$ and those of Q , which lie below one since $Q(x) > 0$ for $x \geq 1$; so no $c_3(n+i)$ with $i \geq 0$ vanishes at an integer $n \geq 1$. For $n \geq n_0$ it follows that $s_2(n) a_{n+2} + s_1(n) a_{n+1} + s_0(n) a_n = 0$; after clearing denominators this is a polynomial relation of order at most two, which the hypothesis and Proposition 42.2 force to be trivial. So $s_0 = s_1 = s_2 = 0$, and $(Ra)(n) = \sum_k W_k(n) (L_{\text{OEIS}} a)(n+k) = 0$ for every $n \geq 1$. The linear-algebra step of the proof of Proposition 33.3 then applies, with the exact values that Theorem 34.2 determines from $a_1 = a_2 = 1, a_3 = 16$: for $(\bar{r}, d, N) = (3, 23, 126), (4, 17, 120)$ and $(5, 15, 126)$ the matrices have full column rank 96, 90 and 96 over $\mathbb{Q}$. So no recurrence of order at most three has degree below 24, none of order at most four has degree below 18, and none of order at most five has degree below 16; Theorem 34.2, (a) and (b) attain these degrees. $\square$

Without the hypothesis of (c), the same ranks show only that no recurrence of those orders and degrees holds for every n in the range of rows.

42.6 Further questions and research recorded at the write

Added 5 October 2026, batch 101, under Vladimir Reshetnikov's standing rule. The manuscript's own open questions are Research questions 16–19 above; its text after Question 17 states the non-claim that the factorization over $\mathbb{Q}(n)$ “is not a proof that order three is minimal”.

(F9) *Minimality of the A181199 recurrence* (the manuscript's non-claim after Research question 17; the placement). Proposition 42.2 excludes order one, and Proposition 42.3 gives recurrences of order four and degree 18 and of order five and degree 16, and the smallest degrees 24, 18 and 16 at orders three, four and five if order two is excluded. Open: whether some recurrence of order two exists; the smallest degrees at orders three to five without that hypothesis, and at orders six and more. Evidence, not proof: no recurrence of order at most two and degree at most 60 holds for every $5 \leq n \leq 214$, since the 210×183 integer matrix $(n^i a_{n+j})$ ($0 \leq j \leq 2, 0 \leq i \leq 60$) with those rows has rank 183 over $\mathbb{Q}$ (exact computation, recorded in `data/check-lower-order-recurrences.json`). This is not a proof, because a recurrence need only hold for large n , and for a recurrence of order two nothing like Proposition 42.3(c) shows that it holds from $n = 5$. (Re-scoped 5 October 2026 after the intake's independent check, which found and proved Proposition 42.3; this item first read: “Open: whether some recurrence of order two exists, and the smallest degree at orders three and more.”) Sketch: if $a_{n+2} = r_1(n) a_{n+1} + r_0(n) a_n$ with rational r_0, r_1 , then $\Delta_n = a_{n+1} - a_n = -\frac{27}{4} H_n \sum_{i=1}^{n-1} I_i$ (39.4) satisfies $\Delta_{n+1} = (r_0 + r_1 - 1) a_n + (r_1 - 1) \Delta_n$. The coefficient $r_0 + r_1 - 1$ cannot vanish identically, since Δ_n would then be hypergeometric and the proof above applies to $Y_n = \Delta_n / (2M_n u(n))$; so a_n lies in the $\mathbb{Q}(n)$ -span of Δ_n and Δ_{n+1} . Conversely such a representation gives order two. Deciding it is a rational-solution problem for a first-order 2×2 difference system whose coefficients come from R and S (39.6). Missing: that computation, or an argument like Lemma 33.1 for the sum $\sum_{i < n} I_i$.

- (F10) *Numerical constants* (Sections 40 and 41). The receipt evaluates $E_{Z,K}$, $E_{b,K}$, C_K and the onsets at $K = 0$ and $K = 6$ ((40.8), (40.13)); the inverse constants H_K , W_K , X_K and the onset (41.10) of the threshold envelope are constructed but not evaluated. Missing: those values and a worked certified threshold (compare Research question 19).

Part V

Two earlier independent routes: all-order asymptotics, rational diagonals and transcendence

Research reports 85 and 91, prepared for Vladimir Reshetnikov, October 1 and 2, 2026

[write] Added 5 October 2026, batch 101: provenance of Part V. Part V prints two manuscripts in full, one section each:

- Section 43 is Research Report 85 of the session bundle, *Fixed-Height Shifted Strips: All-Orders Asymptotics and Inversion*, dated 1 October 2026 (a 7-page PDF; one 330-line source with 23 labels and four bibliography entries). It arrived as the archive `ProveIt_Fixed_Height_Shifted_Strips_Asymptotics.zip` (357,992 bytes, 21 files, no wrapper directory). Its code and records are shipped with the prefix `05-strip-asymptotics-`.
- Section 44 is Research Report 91, *Rational Diagonals and Transcendence for Fixed Height Shifted Strips*, dated 2 October 2026 (a 9-page PDF; one 448-line source with 27 labels and eight bibliography entries). It arrived as the archive `ProveIt_Shifted_Strips_Rational_Diagonals_and_Transcendence.zip` (371,265 bytes, 12 files, no wrapper directory). Its code and records are shipped with the prefix `06-strip-diagonals-`.

Both arrived in commit `60f54ea06` and were placed in commit `f7c612c72` (batch 101) as additions to this report. Both title lines and PDF metadata read “Research report prepared for Vladimir Reshetnikov with OpenAI”: unlike the sources of Parts I–IV, they name a tool. Neither names a ProveIt commit or read the repository; Report 91 cites Report 85 as an “unpublished project report supplied with this research task”. Their $\LaTeX$ sources, delivered READMEs, PDFs, SHA-256 manifests (20 and 11 entries, verified at placement) and two generic helper files of Report 85 (a build script and an integrity checker that are byte copies of files shipped elsewhere in the collection) and the integrity checker of Report 91 are not shipped; they survive in the arrival commit’s archives. Both manuscripts are unrefereed and AI-assisted; nothing in them is formalized, and their place in this report confers no formal status.

Chronology. The two manuscripts are dated before the manuscripts of Part I (3 October) and Part II (4 October), neither of which knew them; they reached ProveIt on 5 October 2026. They are printed after Parts I–IV only because they were placed later, and no number of those Parts moved. Report 85 proves Part I’s Theorem 2.1 (with (2.7), the A181199 constant and two corrections) by another route; Report 91 proves Part II’s Theorem 14.1 by the same route, and Part I’s Theorem 2.3 with elementary radial lemmas in place of the Puiseux argument. Section 45 sets out the correspondence; by their date lines these are the earliest proofs of those statements in this report. The dates are the manuscripts’ own and certify nothing outside ProveIt.

Editorial changes. Each manuscript’s sections are subsections of its section here, and its subsections are subsubsections. Equation (k) of either manuscript is (43.k) or (44.k) here. Theorem-like statements are numbered through the section: Report 85’s Theorem 1.1 is Theorem 43.1; Report 91’s Theorems 1.1–1.2, Lemmas 2.1, 3.1 and 3.2, Propositions 4.1 and 6.1, Lemmas 7.1–7.3 and its closing Remark are 44.1–44.11 in that order. Labels carry the prefixes `shr:sa:` and `shr:sd:`. Their titles, author lines, dates and abstracts open the two sections. In Report 85, `\tr` is typeset as `Tr`, as delivered. Report 91’s citation of Report 85 now points to Section 43. The bibliographies are merged: the OEIS entries (Report 85 accessed them on 1 October, Report 91 on 2 October 2026), Kauers–Koutschan, Sun and Bostan–Lairez–Salvy are Part I’s and Part II’s entries, and Lindemann’s article [17] is added. Remarks take this report’s theorem style. The [write] notes and Section 45 are new. No other statement, proof or number of either manuscript was changed.

[write] Added 5 October 2026, batch 101: notation of Part V. Both manuscripts write $g(m, n)$ for $T_m(n)$ and use the same letters for the same objects in their shared asymptotic argument. Within Part V each symbol has the meaning below; the last column gives the tempting false reading from elsewhere in this report. The worst trap is Report 85’s $c_j(m)$: it is Part I’s $b_{m,j}$, the coefficient in the normalization $K_m m^{mn} n^{-\alpha_m}$, not Part I’s $c_{m,j}$.

symbol	Report 85 (Section 43)	Report 91 (Section 44)	not: meaning elsewhere
$g(m, n)$	$T_m(n)$	$T_m(n)$, with $g(m, 0) = 1$	—
$c_j(m)$	$b_{m,j}$ of (2.5)	—	$c_{m,j}$ of (2.3)
C_m	K_m (2.2) (as $2^{-m(m-1)} = 4^{-d}$)	K_m	Part IV's ordered simplex C_m
$v; d$	$\alpha_m; \binom{m}{2}$	the same	Part IV's v_n ; Parts III–IV's d_j
$\alpha_m; L_m$	—	$2^{-m(m-1)-(m-1)/2} \sqrt{m} \prod_{j < m} j!$; $\alpha_m \pi^{-(m-2)/2}$	$\alpha_m = (m^2 - 1)/2$
$G_m(z); A_m(t)$	—	$F_m(z); G_m(m^{-m}t)$	Part II's Stirling factor $A_m(z)$
$R_n; R_m$	$\frac{(mn)!}{(n!)^m f(n^m)}$	the same R_n ; a rational function with $G_m = \text{diag } R_m$	the pair weight $R_n(x)$; $R_m = m^{-m}$; $R_m(n)$
$f^\lambda, g^\lambda, \delta, \Delta$	ordinary and shifted hook counts; $(m-1, \dots, 0)$; the Vandermonde	the same	$f^{(n^m)}$ as in Corollary 2.4; Part II's g^λ
$\mu_n, \eta_n, \rho, H(x)$	the cut law, the hypergeometric law, the cut proportion, binary entropy	the same, with $\rho = 1/m$	$H(\lambda)$; Part IV's ρ_n
$y_i; \mathbb{E}_m; S_r$	$(\lambda_i - \rho n)/\sqrt{n}$; the trace-zero Gaussian–Vandermonde expectation; power sums	y_i as in Report 85	Part I's $Y_i, \mathbb{E}, p_r$; Part II's S_1, S_2
$\beta_k; \mathcal{L}_m, \mathcal{B}, L_k,$ H_j	$B_{2k}/(2k(2k-1))$; the formal logarithm and its coefficients	—	recurrence coefficients β_j
L	$m \log m$ (Section 43.6)	a path length; L_j gap lengths	a truncation order; Part I's $\lambda = m \log m$
$d_1, d_2; w, y$	inverse constants; Ln , $\log(x/(C_m L^y))$ (Section 43.6)	$d_i = b_i - a_{\sigma(i)}$	Parts III–IV's $d_j = b_{m,j}$
$N(x); J; K$	the threshold; a truncation order; a gap	—	Part III's $N(y)$
$B, S; h_j$	—	birth and saturation letters; heights of the Dyck word	Part II's letters B, D ; Part I's h_j
$K_k(a, b); H(t),$ $\delta(t)$	—	the chamber count; the Heaviside and Kronecker sequences	Part II's $K_r(u, v), H(a)$
$a^j, b^j, \mathcal{D}_m,$ $\mathcal{B}_w(n), D_w,$ $E_m(n)$	—	event vectors, Dyck words, admissible vectors, dimension, the binomial sum	Part II's $u_s, v_s; E_{m,n}$
$(x)_j$	the rising factorial	the same	Part I's falling factorial

43 Report 85: fixed-height shifted strips, all-orders asymptotics and inversion

Research report prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026

Abstract. Let $g(m, n)$ count the $m \times n$ arrays filled by $1, \dots, mn$ whose rows, columns, diagonals and downward antidiagonals increase. For every fixed height m , we prove a complete expansion in powers of $1/n$, with leading constant $2^{-m(m-1)}$ times that for ordinary rectangular standard Young tableaux. At height five this proves the equivalent currently conjectured in OEIS A181199, and gives relative corrections $48/(5n)$ and $5233/(100n^2)$. The proof splits a path at an intermediate shape, applies two shifted hook products, and uses a symmetric lattice expansion. We give a finite exact rule for every coefficient and an inverse expansion, retaining the necessary qualification when rounding to integer indices. No conjectured recurrence is used.

[write] Added 5 October 2026, batch 101. Every statement of this section was already proved in Part I, two days later by date line and by another route; Section 45 lists the correspondence. The abstract's “currently conjectured” refers to the A181199 entry as accessed on 1 October 2026. Three steps of Subsection 43.4 are argued in outline (item (F11)); Part I's proof of the same theorem is complete.

43.1 The result and its scope

The array counts form the table A181196. Its height-four and height-five rows are A181198 and A181199. The latter entry states the conjecture

$$g(5, n) \sim \frac{9\sqrt{5}}{2^{17}\pi^2} \frac{3125^n}{n^{12}}.$$

Kauers and Koutschan's recurrence and closed-form conjectures for these two rows remain separate questions [3, 4, 9]. Sun proves formulas for $g(3, n)$ and for $g(n, 4)$, $g(n, 5)$ [11]; the last two vary the other dimension.

Put

$$d = \binom{m}{2}, \quad \nu = \frac{m^2 - 1}{2}, \quad C_m = 2^{-m(m-1)} \sqrt{m} (2\pi)^{-(m-1)/2} \prod_{j=0}^{m-1} j!.$$

Theorem 43.1. Fix an integer $m \geq 2$. There are effectively computable rational numbers $c_j(m)$ such that, for every fixed $J \geq 0$,

$$g(m, n) = C_m m^{mn} n^{-\nu} \left(1 + \sum_{j=1}^J \frac{c_j(m)}{n^j} + O_{m,J}(n^{-J-1}) \right). \quad (43.1)$$

The first two coefficients are

$$c_1(m) = \frac{(m^2 - 1)^2}{12m}, \quad c_2(m) = \frac{(m^2 - 1)(m^6 + 3m^2 - 1)}{288m^2}. \quad (43.2)$$

A single formal coefficient procedure applies to every fixed height.

Thus the explicit height-five expansion is

$$g(5, n) = \frac{9\sqrt{5}}{2^{17}\pi^2} \frac{3125^n}{n^{12}} \left(1 + \frac{48}{5n} + \frac{5233}{100n^2} + O(n^{-3}) \right). \quad (43.3)$$

For comparison, the height-four row has

$$g(4, n) = \frac{3}{1024\sqrt{2}\pi^{3/2}} \frac{256^n}{n^{15/2}} \left(1 + \frac{75}{16n} + \frac{6905}{512n^2} + O(n^{-3}) \right).$$

The theorem is an all-orders Poincaré expansion of the dominant exponential sector. It does not classify every exponentially smaller sector or prove the unused recurrence conjectures. Height one is the immediate identity $g(1, n) = 1$.

43.2 Exact paths and the two hook products

Fill the array in increasing label order, and let λ_i be the number of filled cells in row i . Its row-choice word determines the filling uniquely. The state conditions are exactly

$$n \geq \lambda_1 \geq \cdots \geq \lambda_m \geq 0, \quad \text{with repetitions permitted only at 0 and } n. \quad (43.4)$$

Indeed, filling cell $(i + 1, j)$ for $j < n$ requires $(i, j + 1)$ to be filled already. Filling $(i + 1, n)$ requires the entire preceding row. These are the downward-antidiagonal and column conditions; the southeast diagonal conditions follow from row and column increase.

If an endpoint satisfies $\lambda_1 < n$, its admissible prefixes are chains of strict partitions, with zero parts omitted. For an interior strict endpoint, the classical shifted hook product is

$$g^\lambda = \frac{|\lambda|!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (43.5)$$

This formula and the shifted-strip interpretation appear in [11]. Reversing the path and its row order gives the complementary partition

$$\bar{\lambda} = (n - \lambda_m, \dots, n - \lambda_1),$$

which obeys the same state condition. Consequently the number of admissible complete paths through an interior strict state is exactly

$$g^\lambda g^{\bar{\lambda}}. \quad (43.6)$$

We do not apply (43.5) to a saturated boundary state with repeated parts n .

Let f^λ be the ordinary standard-tableau count, and set $\delta = (m - 1, \dots, 0)$. The ordinary hook product and rectangular specialization are

$$f^\lambda = \frac{|\lambda|! \Delta(\lambda + \delta)}{\prod_i (\lambda_i + m - i)!}, \quad f^{(n^m)} = \frac{(mn)! \prod_{j=0}^{m-1} j!}{\prod_{j=0}^{m-1} (n + j)!}, \quad (43.7)$$

where $\Delta(x) = \prod_{i < j} (x_i - x_j)$. One way to obtain the first identity is to shift weakly ordered row-growth paths by δ , apply the reflection principle, and evaluate $|\lambda|! \det(1/(\lambda_i - i + j)!) as a Vandermonde determinant.$

Ordinary rectangular branching at any time k gives the probability law

$$\mu_n(\lambda) = \frac{f^\lambda f^{\bar{\lambda}}}{f^{(n^m)}}, \quad |\lambda| = k. \quad (43.8)$$

Every admissible path is an ordinary path. The ratio of the admissible through-count to the ordinary through-count therefore always belongs to $[0, 1]$, including at boundary states.

43.3 Concentration and the leading constant

We first cut at $k = n$, so $\rho = k/(mn) = 1/m$ is fixed without any parity restriction. Compare (43.8) to the labeled multivariate hypergeometric law

$$\eta_n(\ell) = \frac{\prod_i \binom{n}{\ell_i}}{\binom{mn}{n}}, \quad \sum_i \ell_i = n.$$

The ratio of unrestricted row words to ordinary tableaux is

$$R_n = \frac{(mn)!/(n!)^m}{f^{(n^m)}} = \frac{\prod_{j=0}^{m-1} (n + j)!}{(n!)^m \prod_{j=0}^{m-1} j!} = O_m(n^d).$$

Bounding each prefix and suffix by its unrestricted multinomial count yields

$$\mu_n(\lambda) \leq R_n \eta_n(\lambda). \quad (43.9)$$

Stirling bounds and strict concavity of binary entropy imply that a fixed linear deviation from $\lambda_i = \rho n$ has exponentially small probability. The polynomial factor R_n does not affect this conclusion.

A refined bound also excludes bounded row gaps. On a fixed central region $\epsilon n \leq \lambda_i \leq (1 - \epsilon)n$, put $y_i = (\lambda_i - \rho n)/\sqrt{n}$, so $\sum_i y_i = 0$. From (43.7),

$$\mu_n(\lambda) = R_n \eta_n(\lambda) \frac{\Delta(\lambda + \delta)^2}{\prod_i (\lambda_i + 1)_{m-i} (n - \lambda_i + 1)_{i-1}}, \quad (43.10)$$

where $(x)_j = x(x+1) \cdots (x+j-1)$. Reversing the complementary partition makes its Vandermonde equal to the first one. The denominator is at least a positive constant times n^{2d} , and the numerator is at most $C_m n^d (1 + \|y\|)^{2d}$.

Uniform factorial estimates give

$$\eta_n(\lambda) \leq C n^{-(m-1)/2} e^{-c\|y\|^2}.$$

To see the uniformity, write each binomial logarithm as $nH(\lambda_i/n) - \frac{1}{2} \log n + O(1)$. The denominator has the corresponding $n^{-1/2}$ prefactor, the linear entropy terms cancel because the sum is fixed, and H'' is bounded above by a negative constant. Thus

$$\mu_n(\lambda) \leq C n^{-(m-1)/2} (1 + \|y\|)^{2d} e^{-c\|y\|^2}. \quad (43.11)$$

Gaussian lattice-sum bounds give tightness of y . In a fixed box $\|y\| \leq M$, a specified gap $\lambda_i - \lambda_j \leq K$ occupies at most $O_M((K+1)n^{(m-2)/2})$ lattice points. Its probability is therefore $O_M((K+1)/\sqrt{n})$. Letting first n and then M tend to infinity proves

$$\lambda_i/n \longrightarrow \rho, \quad \min_{i < j} (\lambda_i - \lambda_j) \longrightarrow \infty \quad \text{in probability.} \quad (43.12)$$

In particular, interior strict states have probability tending to one.

For such a state, the two hook products give

$$\frac{g^\lambda}{f^\lambda} = \frac{\prod_i (\lambda_i + m - i)! / \lambda_i!}{\prod_{i < j} (\lambda_i + \lambda_j)} \frac{\Delta(\lambda)}{\Delta(\lambda + \delta)}.$$

By (43.12), its first factor tends to 2^{-d} and its second to one. The complementary ratio has the same limit. The exact statewise ratios are bounded by one, so bounded convergence in probability applied to (43.8) proves

$$\frac{g(m, n)}{f(n^m)} \longrightarrow 2^{-2d}. \quad (43.13)$$

Finally, Stirling's formula in (43.7) gives

$$f(n^m) \sim \sqrt{m} (2\pi)^{-(m-1)/2} \left(\prod_{j=0}^{m-1} j! \right) m^{mn} n^{-\nu}.$$

This proves the leading assertion and the constant in Theorem 43.1. All the preceding bounds are uniform when ρ ranges over a fixed compact subinterval of $(0, 1)$ and $k = \rho mn$ is an integer.

43.4 A symmetric lattice expansion to every order

At a general integer cut $k = \rho mn$, the exact interior through-count (43.6) becomes

$$\frac{k!(mn - k)! \Delta(\lambda)^2}{\prod_i \lambda_i! (n - \lambda_i)! \prod_{i < j} (\lambda_i + \lambda_j) (2n - \lambda_i - \lambda_j)}. \quad (43.14)$$

Restrict all coordinates to a fixed central region as above. Its complement contributes an exponentially small amount relative to $f^{(n^m)}$, hence also to $g(m, n)$ by (43.13). The summand (43.14) is symmetric. Coincidence terms vanish because of $\Delta(\lambda)^2$, while its denominators stay positive. Therefore the ordered strict sum is exactly $1/m!$ times the sum over all labeled central integer vectors of sum k . There are no chamber-wall correction terms.

Write $\lambda_i = \rho n + \sqrt{n} y_i$. The leading weight on the hyperplane $\sum y_i = 0$ is

$$\Delta(y)^2 \exp\left(-\frac{\sum_i y_i^2}{2\rho(1-\rho)}\right). \quad (43.15)$$

The affine lattice has covolume $\sqrt{m} n^{-(m-1)/2}$ in this hyperplane.

Here are the fixed-order error estimates. For a chosen J , restrict first to $\|y\| \leq n^\varepsilon$, with $\varepsilon > 0$ sufficiently small depending on J . Uniform Stirling expansion and Taylor expansion of the positive pair-sum factors yield polynomial coefficients at every half-order. Retain the half-orders through $n^{-J-1/2}$ before integration. Taylor's theorem, with a sufficiently long finite expansion, bounds the remaining normalized summand by

$$C_J n^{-J-1} (1 + \|y\|)^{M_J} e^{-c\|y\|^2}.$$

Choosing ε small permits exponentiation of the remainder without losing an integrable Gaussian majorant. Outside this growing window, (43.11) and (43.9) give a bound of the form a fixed polynomial times $e^{-cn^{2\varepsilon}}$, which is negligible to every algebraic order. These estimates are uniform for ρ in a compact subset of $(0, 1)$.

For each polynomial-Gaussian coefficient, Poisson summation replaces the lattice sum by its integral with an error smaller than every power of $1/n$, uniformly in the affine lattice translation. Its Fourier transform is another polynomial times a Gaussian, and every nonzero dual frequency has length at least $c\sqrt{n}$. The growing-window cutoff can first be removed at a superalgebraically small cost. The coefficient of half-order j has parity j under $y \mapsto -y$; odd coefficients integrate to zero. We obtain

$$g(m, n) = C_m m^{mn} n^{-\nu} \left(\sum_{j=0}^J A_j(m, \rho) n^{-j} + O_{m,J,I}(n^{-J-1}) \right), \quad A_0 = 1. \quad (43.16)$$

The coefficients are analytic in ρ on $(0, 1)$.

They are independent of the cut. Choose two rational cut proportions and take n through common multiples for which both cuts are integers. The left side of (43.16) is identical. Successive uniqueness of asymptotic coefficients gives equality at both rational values, at every order. Continuity extends the equality to all cut proportions. Thus coefficients may be calculated at the balanced cut $\rho = 1/2$. They hold for all integers n by applying (43.16) at the parity-free cut $k = n$, $\rho = 1/m$. This proves the all-orders assertion without extrapolating from an even subsequence.

43.5 An exact coefficient rule

Let $\mathbb{E}_m$ denote expectation on $\sum y_i = 0$ with normalized density proportional to $\Delta(y)^2 e^{-2\sum y_i^2}$. Put $\beta_k = B_{2k}/(2k(2k-1))$, with B_j the Bernoulli numbers. The balanced-cut calculation gives

$$c_j(m) = [t^j] \mathbb{E}_m \exp \mathcal{L}_m(t, y), \quad (43.17)$$

where the formal logarithm is

$$\begin{aligned}\mathcal{L}_m(t, y) = & \sum_{k \geq 1} \beta_k (2^{2k} m^{1-2k} - m) t^{2k-1} + \sum_i \mathcal{B}(t, y_i) \\ & + \sum_{i < j} \sum_{r \geq 1} \frac{(y_i + y_j)^{2r}}{r} t^r,\end{aligned}\tag{43.18}$$

with

$$\begin{aligned}\mathcal{B}(t, y) = & - \sum_{r \geq 2} \frac{2^{2r} y^{2r}}{2r(2r-1)} t^{r-1} + \sum_{r \geq 1} \frac{2^{2r-1} y^{2r}}{r} t^r \\ & + \sum_{k \geq 1} \beta_k t^{2k-1} \left\{ 1 - 2^{2k-1} \left[(1 + 2\sqrt{t} y)^{-(2k-1)} + (1 - 2\sqrt{t} y)^{-(2k-1)} \right] \right\}.\end{aligned}$$

The braces contain only integer powers of t . The first line of (43.18) accounts for the factorial prefactor and central binomials; its last line expands the paired denominators. Every requested coefficient uses finitely many terms.

For an exact rational evaluation, write $S_r = \sum_i y_i^r$, with $S_0 = m$ and $S_1 = 0$. Integration by parts on the trace-zero hyperplane gives, for $k \geq 1$ and any polynomial P in the power sums,

$$\begin{aligned}4\mathbb{E}_m[S_{k+1}P] = & \mathbb{E}_m \left[\left(\sum_{a=0}^{k-1} S_a S_{k-1-a} - \frac{k}{m} S_{k-1} \right) P \right] \\ & + \mathbb{E}_m \left[\sum_{j \geq 1} j \left(S_{k+j-1} - \frac{S_k S_{j-1}}{m} \right) \frac{\partial P}{\partial S_j} \right].\end{aligned}\tag{43.19}$$

Use the tangent vector field $(y_i^k - S_k/m)P$. The Gaussian derivative contributes $-4S_{k+1}$; the Vandermonde derivative gives the difference-quotient sum, and tangent projection gives the $1/m$ terms. Gaussian decay removes boundary terms, and the squared Vandermonde removes the apparent singularity at a coincidence. Each use of (43.19) decreases total polynomial degree by two. Starting from $\mathbb{E}_m 1 = 1$, it therefore computes every moment by finite rational arithmetic. This also proves the effective and rational coefficient assertion in Theorem 43.1.

For implementing (43.18), the useful identity

$$\sum_{i < j} (y_i + y_j)^{2r} = \frac{1}{2} \left(\sum_{a=0}^{2r} \binom{2r}{a} S_a S_{2r-a} - 2^{2r} S_{2r} \right)$$

expresses every term in power sums. If $\mathcal{L} = \sum_{k \geq 1} L_k t^k$ and $e^{\mathcal{L}} = \sum_{j \geq 0} H_j t^j$, then $H_0 = 1$ and $H_j = j^{-1} \sum_{k=1}^j k L_k H_{j-k}$. This is the finite procedure used by the bundled coefficient generator.

43.5.1 The first two coefficients

The first two logarithmic terms are

$$L_1 = -\frac{m^2 - 1}{3m} + mS_2 - \frac{4}{3}S_4, \quad L_2 = -\frac{4}{3}S_2 + \frac{m}{2}S_4 - \frac{32}{15}S_6 + \frac{3}{2}S_2^2.$$

The moments required by $c_1 = \mathbb{E}_m L_1$ and $c_2 = \mathbb{E}_m L_2 + \frac{1}{2}\mathbb{E}_m L_1^2$ are

$$\begin{aligned}\mathbb{E}_m S_2 &= \frac{m^2 - 1}{4}, & \mathbb{E}_m S_4 &= \frac{(m^2 - 1)(2m^2 - 3)}{16m}, \\ \mathbb{E}_m S_6 &= \frac{5(m^2 - 1)(m^4 - 3m^2 + 3)}{64m^2}, & \mathbb{E}_m S_2^2 &= \frac{(m^2 - 1)(m^2 + 1)}{16}, \\ \mathbb{E}_m (S_2 S_4) &= \frac{(m^2 - 1)(m^2 + 3)(2m^2 - 3)}{64m}, \\ \mathbb{E}_m S_4^2 &= \frac{(m^2 - 1)(4m^6 + 20m^4 - 99m^2 + 135)}{256m^2}.\end{aligned}$$

Substitution gives (43.2); in particular,

$$c_2 - \frac{c_1^2}{2} = \frac{m^2(m^2 - 1)}{96}. \quad (43.20)$$

A second calculation uses a traceless Hermitian Gaussian matrix with covariance

$$\mathbb{E}[X_{ij}X_{k\ell}] = \frac{1}{4} \left(\delta_{i\ell}\delta_{jk} - \frac{\delta_{ij}\delta_{k\ell}}{m} \right).$$

Its eigenvalue density is the one defining $\mathbb{E}_m$. Replacing S_r by $\text{Tr}(X^r)$ and summing Wick pairings gives the same symbolic moment formulas. A separate direct Gaussian integration check verifies the two corrections at heights two, three and four.

There are also exact low-height checks. Since $g(2, n)$ is the $(n - 1)$ st Catalan number, its corrections are $3/8$ and $25/128$. Sun's proved recurrence simplifies to

$$g(3, n + 1) + g(3, n) = \frac{(3n)!}{(n!)^3} \frac{7n + 1}{2(n + 1)^2(4n^2 - 1)}.$$

Substituting (43.2) agrees with its expansion through the second correction.

43.6 Inversion and integer thresholds

Set $L = m \log m > 0$ and define

$$y = \log \frac{x}{C_m L^\nu}, \quad d_1 = L c_1(m), \quad d_2 = L^2 \left(c_2(m) - \frac{c_1(m)^2}{2} \right).$$

Writing $w = Ln$, the logarithmic forward expansion is $y = w - \nu \log w + d_1/w + d_2/w^2 + \dots$. Formal substitution gives the smooth inverse

$$\begin{aligned}w &= y + \nu \log y + \frac{\nu^2 \log y - d_1}{y} \\ &\quad + \frac{-\frac{1}{2}\nu^3(\log y)^2 + (\nu^3 + \nu d_1) \log y - \nu d_1 - d_2}{y^2} + O\left(\frac{(\log y)^3}{y^3}\right).\end{aligned} \quad (43.21)$$

Every further coefficient is obtained recursively and is a polynomial in $\log y$. The derivative of the logarithmic forward approximation tends to L , so the forward remainder transfers to the inverse by the mean-value theorem.

For precision about the discrete sequence, (43.1) implies $g(m, n+1)/g(m, n) \rightarrow m^m > 1$, hence eventual strict increase. Let $N(x) = \min\{n : g(m, n) \geq x\}$, and let $n_J(x)$ solve the positive J -term forward approximation in (43.1) for sufficiently large real n . Comparing the two adjacent threshold integers gives

$$N(x) = \lceil n_J(x) + \varepsilon_J(x) \rceil, \quad |\varepsilon_J(x)| = O_{m,J}((\log x)^{-J-1}). \quad (43.22)$$

Thus taking the ceiling is certified away from an error-sized neighborhood of an integer. An unqualified rounding rule at every x is not asserted. In particular,

$$N(x) = \frac{\log x + \nu \log \log x - \log C_m - \nu \log L}{L} + O(1).$$

The displayed logarithmic formula (43.21) has its own stated truncation error; the sharper error in (43.22) refers to solving the corresponding finite forward approximation.

43.7 Verification and relation to prior results

The source package contains an endpoint-path enumerator checking sixty small rectangles, thirty published initial terms across A181198 and A181199, and 121 interior hook identities. These are checks of the exact combinatorial correspondence, not a substitute for the uniform proof. Two symbolic moment methods agree on (43.2); a third, direct Gaussian-integral calculation checks the indicated small heights. A separate formal substitution checks (43.21). The generator implements the finite procedure (43.17)–(43.19) at a requested order.

The hook products, Gaussian expansion and Poisson summation are classical tools. The argument proves the fixed-height comparison and its complete dominant-sector expansion directly for the array counts. The live A181199 entry and the cited primary sources were checked on 1 October 2026; the displayed equivalent remains conjectural there. This is a bounded literature-positioning statement, not an exhaustive priority claim. The order-two and order-three recurrence conjectures in [9] are not consequences asserted by this report. No proof-assistant certification is claimed.

43.8 Further questions

The conjectured exact recurrences and closed forms for heights four and five remain natural targets. The present proof determines their asymptotic behavior without establishing those identities.

A second question is to identify the exponentially smaller sectors omitted by the Poincaré expansion, including the contributions of saturated boundary shapes. This would turn the dominant-sector expansion into a fuller transseries.

It is also useful to let the height grow with the width. Since the first relative correction is asymptotic to $m^3/(12n)$, the regime m^3/n of constant order is a concrete candidate for a new uniform approximation. This observation is a proposed scale, not a growing-height theorem. Finally, effective error constants would turn the qualified inverse bounds into numerical certificates for specific large thresholds.

[write] Added 5 October 2026, batch 101: the shipped layout of Report 85. “The source package” of Subsection 43.7 is the delivered archive. Its `verify.py` and six programs `code/name.py` are shipped as `code/05-strip-asymptotics-verify.py` and `code/05-strip-asymptotics-name.py`; the six records `code/name.json`, `CHECKS.json` and `verification.json` as `data/05-strip-asymptotics-name.verify.py` runs each program with working directory `code/`, and each program writes its JSON beside itself, so they run only on a copy with the delivered layout restored, never in place; the README of the report gives the commands. At placement the whole suite passed on such a copy (SymPy 1.14.0; six and a half minutes on a loaded machine), and every refreshed record equalled the

shipped one after removal of carriage returns, except the timing fields of `verification.json`. The suite runs its programs with `-0`, which is harmless: they test with `raise`, not `assert`. `code/05-strip-asymptotics-check_midpoint.py` embeds fifteen terms each of A181198 and A181199 copied from OEIS; as OEIS content they are published by The OEIS Foundation Inc. under the Creative Commons Attribution-ShareAlike 4.0 licence.

44 Report 91: rational diagonals and transcendence for fixed-height shifted strips

Research report prepared for Vladimir Reshetnikov with OpenAI, 2 October 2026

Abstract. For every fixed positive height m , the number $g(m, n)$ of increasing shifted-strip arrays is a multiple binomial sum in the width n . Its ordinary generating function is therefore a rational diagonal and is D-finite. The proof records each row's birth and saturation, leaving a fixed number of strict-chamber walks. Their reflection determinants become everywhere-defined binomial products, and the resulting sum has at most m^2 integer coordinates. We also prove that the generating function is algebraic exactly for $m = 1, 2$. For larger heights, the leading asymptotic gives either logarithmic radial growth after differentiation or an impossible inverse-square-root coefficient. A concise proof of the leading equivalent is included. Exact checks use cell-poset counts, event sums and independent chamber walks. The argument proves existence of polynomial-coefficient recurrences, without asserting either of the particular recurrences conjectured for OEIS A181198 and A181199.

[write] **Added 5 October 2026, batch 101.** Theorem 44.1 is Part II's Theorem 14.1, proved by the same route two days earlier by date line, and Theorem 44.2 is Part I's Theorem 2.3 (Section 45). Its bound of m^2 coordinates and Part II's $m(m-1)$ free indices agree: both proofs count the same $\sum_j h_{j-1} \leq m^2$ slots, and Part II then removes the m coordinates that the saturation events fix at $n-1$, which Report 91 keeps and sums against the Kronecker factor $\delta(b_1^j - n + 1)$. The (3, 4) table of Subsection 44.4 is the table of Section 17.2. Since 5 October 2026 the particular recurrences that the abstract leaves aside are proved in Parts III and IV. Subsection 44.6 reproduces the argument of Subsection 43.3, as its first paragraph says, adding the explicit entropy exponent and the finite union over pairs; it is printed as delivered.

44.1 The statements and their scope

Let $g(m, n)$ count the $m \times n$ arrays filled bijectively by $1, \dots, mn$ in which rows increase left to right, columns increase downward, southeast diagonals increase downward, and antidiagonals increase from upper right to lower left. Set $g(m, 0) = 1$, the empty-array convention, and write

$$G_m(z) = \sum_{n \geq 0} g(m, n) z^n.$$

This is the array model in OEIS A181196. Its height-four and height-five sequences are A181198 and A181199 [1, 3, 4]. It agrees with the truncated shifted-strip tableau model discussed by Sun [11].

Theorem 44.1. *For every fixed integer $m \geq 1$, the sequence $n \mapsto g(m, n)$ is a multiple binomial sum over $\mathbb{Q}$. There is a rational function R_m in finitely many variables, regular at the origin and with rational coefficients, such that*

$$G_m(z) = \text{diag } R_m(z).$$

Consequently G_m is D-finite and $g(m, n)$ is P-recursive: there exist $r \geq 0$ and polynomials $p_0, \dots, p_r \in \mathbb{Q}[n]$, not all zero, with

$$\sum_{j=0}^r p_j(n) g(m, n+j) = 0$$

for all sufficiently large integers n . The binomial-sum construction uses at most m^2 summation coordinates for each of finitely many event patterns.

Here a diagonal means $[z^n] \operatorname{diag} R = [x_1^n \cdots x_d^n] R$. D-finiteness means that a nonzero linear differential operator with polynomial coefficients annihilates the series.

Theorem 44.2. *For positive integer heights, $G_m(z)$ is algebraic over $\mathbb{C}(z)$ exactly when $m = 1$ or $m = 2$. In those cases,*

$$G_1(z) = \frac{1}{1-z}, \quad G_2(z) = \frac{3 - \sqrt{1-4z}}{2}.$$

Thus every fixed height $m \geq 3$ gives a transcendental D-finite rational diagonal.

Kauers and Koutschan [9, Section 6.4] conjecture specific recurrences of order two and degree nine for height four, and order three and degree twenty-four for height five. The OEIS entries still label those formulas conjectural as checked on 2 October 2026. Theorem 44.1 proves existence of a recurrence at every height; it does not identify a minimal operator or prove either displayed conjecture. Sun's formulas for $g(n, 4)$ and $g(n, 5)$ fix the other dimension, while his formula for $g(3, n)$ fixes height three. These distinctions matter because the array is not symmetric in its dimensions.

44.2 Paths and their boundary events

Fill an array in increasing label order. Let λ_i be the number of cells already filled in row i . The row-choice word determines the array uniquely.

Lemma 44.3. *Legal fillings are in bijection with paths using positive coordinate unit steps from $(0, \dots, 0)$ to $(n, \dots, n)$ through states satisfying*

$$n \geq \lambda_1 \geq \dots \geq \lambda_m \geq 0, \quad \text{with repetitions allowed only at 0 and } n. \quad (44.1)$$

Proof. If row $i + 1$ contains j filled cells, with $1 \leq j < n$, its last cell requires the upper-right cell $(i, j + 1)$ to be filled, so $\lambda_i \geq j + 1$. If $j = n$, column increase requires the upper row to be full. This proves necessity. Conversely, when a cell is inserted along a path satisfying (44.1), its same-row predecessor, upper predecessor and upper-right predecessor, when present, have already been inserted. These are the required local inequalities. Southeast inequalities follow from row and column increase. Thus every path gives a legal array. $\square$

For the moment let $n \geq 2$. Mark each row's *birth*, the increment $0 \rightarrow 1$, and its *saturation*, the increment $n - 1 \rightarrow n$. They are distinct events even at $n = 2$. Births occur in row order, and saturations occur in row order. At every time the number of births is at least the number of saturations. The event word therefore belongs to the set $\mathcal{D}_m$ of Dyck words with m letters B and m letters S , where B raises height and S lowers it.

After r births and s saturations, the state has the form

$$\underbrace{(n, \dots, n, a_1, \dots, a_{r-s})}_s, \underbrace{(0, \dots, 0)}_{m-r}, \quad n > a_1 > \dots > a_{r-s} > 0.$$

The active rows are one contiguous strict block. Between events, only its coordinates change.

Fix $w \in \mathcal{D}_m$, and let h_j be its height after event j , with $h_0 = h_{2m} = 0$. Immediately before event j record a vector b^j of length h_{j-1} ; immediately afterward record a^j . Set $a^0 = ()$. The event rules are

$$w_j = B : a^j = (b_1^j, \dots, b_{h_{j-1}}^j, 1), \quad b_{h_{j-1}}^j \geq 2 \quad \text{if } h_{j-1} > 0; \quad (44.2)$$

$$w_j = S : a^j = (b_2^j, \dots, b_{h_{j-1}}^j), \quad b_1^j = n - 1. \quad (44.3)$$

All recorded pre-event vectors satisfy

$$n - 1 \geq b_1^j > \dots > b_{h_{j-1}}^j \geq 1. \quad (44.4)$$

Empty vectors obey the empty condition. A birth into an empty active block needs no additional guard: any preceding row is already saturated at $n \geq 2$. A Dyck word has positive height before a saturation.

44.3 The chamber transition as a binomial sum

For strictly decreasing integer vectors a, b of length k , let $K_k(a, b)$ count paths from a to b , with steps in $\{e_1, \dots, e_k\}$, staying strictly decreasing. Set $K_0((), ()) = 1$.

Lemma 44.4. *If $L = \sum_i (b_i - a_i) \geq 0$, then*

$$K_k(a, b) = L! \det_{1 \leq i, j \leq k} \left(\frac{1}{(b_i - a_j)!} \right), \quad (44.5)$$

where a reciprocal factorial with negative argument is zero. If $L < 0$, then $K_k(a, b) = 0$.

Proof. Expand the determinant as a signed sum over starting vectors obtained by permuting a . For a permutation σ , the unrestricted walks to b require $b_i - a_{\sigma(i)}$ steps of type e_i . Their number is the corresponding multinomial, or zero if one difference is negative; the total length is always L .

Cancel walks having a coordinate collision by a sign-reversing involution. Choose the first collision time and the lexicographically first colliding pair at that time. Exchange those two coordinate histories up to and including that collision. The endpoint does not change, while the starting permutation changes by a transposition. Before the chosen time the set of coordinate values is unchanged, and at the collision the whole vector is unchanged, so the chosen first collision is preserved. This is an involution. In an uncanceled walk no two coordinates collide. Since one coordinate changes by one at each step, coordinate ordering cannot change without a collision. The strictly decreasing endpoint forces the initial permutation to be the identity. The survivors are precisely the counted chamber walks. Negative total length is impossible. $\square$

We now remove every problematic factorial from this expression. For all $N, K \in \mathbb{Z}$, use the convention

$$\binom{N}{K} = [x^K](1+x)^N, \quad \binom{N}{K} = 0 \quad (K < 0). \quad (44.6)$$

For negative N , the right side is a formal power series at zero. This is the convention of Bostan, Lairez and Salvy [10, Section 1.1].

Lemma 44.5. *For all strict integer endpoints,*

$$K_k(a, b) = \sum_{\sigma \in \mathfrak{S}_k} \text{sgn}(\sigma) \prod_{i=1}^k \binom{d_i + \dots + d_k}{d_i}, \quad d_i = b_i - a_{\sigma(i)}. \quad (44.7)$$

Every binomial factor is defined for all integer values of the endpoints.

Proof. If every d_i is nonnegative, the product telescopes to $(\sum_i d_i)! / \prod_i d_i!$. If some d_i is negative, its own factor vanishes by (44.6), even when an upper argument elsewhere is negative. Thus the signed sum is (44.5) whenever $L \geq 0$. When $L < 0$, every permutation has a negative d_i , so every product vanishes. The empty product and the unique empty permutation give $K_0 = 1$. $\square$

The unrestricted multinomials in this proof already count the ordering of the interior increments. It would be incorrect to multiply the final event sum by another factor for interleaving them.

44.4 An exact event sum with bounded dimension

Let $\mathcal{B}_w(n)$ be the integer pre-event vectors satisfying (44.2)–(44.4), with a^j defined there. We may include the redundant coordinatewise inequalities $b_i^j \geq a_i^{j-1}$: if they fail, the transition count is already zero.

Proposition 44.6. For $n \geq 2$,

$$g(m, n) = \sum_{w \in \mathcal{D}_m} \sum_{(b^1, \dots, b^{2m}) \in \mathcal{B}_w(n)} \prod_{j=1}^{2m} K_{h_{j-1}}(a^{j-1}, b^j). \quad (44.8)$$

The number of pre-event coordinates for a fixed word is

$$D_w = \sum_{j=1}^{2m} h_{j-1} \leq m^2. \quad (44.9)$$

Proof. Every complete path determines exactly one event word, exactly one set of pre-event vectors, and exactly one chamber subpath in each gap. Conversely, any chamber subpaths counted on the right join through the forced boundary events to a legal full path. During a gap coordinates increase between endpoints in $[1, n-1]$, so no hidden birth or saturation is possible. The guards are exactly those needed at the junctions.

There is no extra choice of event times. The length of gap j is determined by its endpoints:

$$L_j = |b^j| - |a^{j-1}|.$$

Event j occurs at time $j + \sum_{r \leq j} L_r$. Since $a^0 = a^{2m} = ()$,

$$\sum_{j=1}^{2m} L_j = \sum_{j=1}^{2m} (|b^j| - |a^j|) = -m + m(n-1) = m(n-2).$$

Adding the $2m$ event increments gives exactly mn steps. No steps are possible before the first birth or after the last saturation.

Finally $h_j \leq \min(j, 2m-j)$, so

$$D_w \leq \sum_{j=0}^{2m-1} \min(j, 2m-j) = m^2.$$

Equality holds for $B^m S^m$. The dimension therefore depends on m alone. $\square$

For example, at $(m, n) = (3, 4)$ the five event patterns give

w	$BBBSSS$	$BBSBSS$	$BBSSBS$	$BSBBSS$	$BSBSBS$
count	4	16	4	4	1

and their sum is $g(3, 4) = 29$. At width two, only the alternating event word survives, giving $g(m, 2) = 1$. At width one the events coincide and we do not use the decomposition; directly $g(m, 1) = 1$.

44.5 Closure and the rational diagonal theorem

We use the algebra of multiple binomial sums over $\mathbb{Q}$ as defined in [10, Definition 1.1]. It contains the Kronecker delta, nonzero geometric sequences and (44.6); it is closed under addition, multiplication, integer-affine substitutions and directed indefinite summation. In particular

$$H(t) = \mathbf{1}_{\{t \geq 0\}}, \quad \delta(t) = \mathbf{1}_{\{t=0\}}$$

are binomial sums. Equation (6) of [10] gives closure under affine summation bounds; multiplying by H also gives ordinary sums with the empty-sum convention.

For fixed w , let every pre-event coordinate range independently from 0 to n . Impose (44.4) by products of

$$H(b_i^j - 1), \quad H(n - 1 - b_i^j), \quad H(b_i^j - b_{i+1}^j - 1).$$

The birth and saturation guards are respectively

$$H(b_{h_{j-1}}^j - 2) \quad (h_{j-1} > 0), \quad \delta(b_1^j - n + 1).$$

The optional monotonicity guards are $H(b_i^j - a_i^{j-1})$. Because each a^j is obtained from b^j by appending 1 or deleting its first entry, all these arguments are integer-affine in n and the summation coordinates. By (44.7), the same is true of every upper and lower binomial argument. The full guarded summand is therefore a binomial sum, and its D_w nested sums with bounds 0, n preserve that property. There are only finitely many w .

Write this everywhere-defined finite-sum expression as $E_m(n)$. The zero extension of our counting sequence from $\mathbb{N}$ to $\mathbb{Z}$ is

$$\delta(n) + \delta(n - 1) + H(n - 2)E_m(n). \quad (44.10)$$

Thus it too is a binomial sum. All factors are defined on all integer inputs, so this step does not conceal a negative-factorial singularity. Equivalently, one may use the bounded-fiber polyhedral summation closure in [10, Proposition 3.12]; the direct affine-bound route already suffices.

Theorem 3.5 of [10] now supplies a rational power series over $\mathbb{Q}$ whose diagonal is G_m . Its Corollary 3.6 supplies a polynomial-coefficient recurrence. The corresponding generating function is D-finite. This proves Theorem 44.1. The closure proof is constructive in principle; it does not assert that the resulting rational representation has few variables, or that a practically small recurrence follows without further simplification.

44.6 The leading equivalent used for transcendence

For completeness we give the needed asymptotic argument. It is the leading-order part of the earlier fixed-height report, Report 85 (Section 43 of this Part), reproduced here so that the transcendence conclusion does not rely on an unproved numerical constant. Put

$$d = \binom{m}{2}, \quad \nu = \frac{m^2 - 1}{2}, \quad C_m = 2^{-m(m-1)} \sqrt{m} (2\pi)^{-(m-1)/2} \prod_{j=0}^{m-1} j!.$$

Proposition 44.7. *For every fixed $m \geq 2$,*

$$g(m, n) \sim C_m m^{mn} n^{-\nu}. \quad (44.11)$$

Proof. Let f^λ be the number of ordinary standard tableaux of partition shape λ , with at most m rows, and put $\delta = (m-1, \dots, 0)$ and $\Delta(x) = \prod_{i < j} (x_i - x_j)$. The ordinary hook formula and the shifted hook formula are

$$f^\lambda = \frac{|\lambda|! \Delta(\lambda + \delta)}{\prod_i (\lambda_i + m - i)!}, \quad f^{(n^m)} = \frac{(mn)! \prod_{j=0}^{m-1} j!}{\prod_{j=0}^{m-1} (n+j)!}, \quad (44.12)$$

$$g^\lambda = \frac{|\lambda|!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (44.13)$$

The last formula applies to positive strict λ [11, equation (1)]. It counts our prefixes whenever $\lambda_1 < n$, since then no row has saturated. The ordinary formula also follows by shifting weakly ordered paths by δ and evaluating the reflection determinant.

Cut an ordinary rectangular-tableau path at time n . For $|\lambda| = n$, set

$$\bar{\lambda} = (n - \lambda_m, \dots, n - \lambda_1), \quad \mu_n(\lambda) = \frac{f^\lambda f^{\bar{\lambda}}}{f^{(n^m)}}.$$

These are probabilities on the weakly decreasing states. Complementation reverses the path and row order, and preserves (44.1). If $0 < \lambda_m < \dots < \lambda_1 < n$, the admissible through-count is exactly $g^\lambda g^{\bar{\lambda}}$. At any boundary state, the ratio of admissible to ordinary through-count lies in $[0, 1]$.

We record the concentration needed for those hook ratios. Let $\rho = 1/m$ and compare μ_n with the labelled multivariate hypergeometric law

$$\eta_n(\ell) = \frac{\prod_i \binom{n}{\ell_i}}{\binom{mn}{n}}, \quad \sum_i \ell_i = n.$$

Bounding ordinary prefixes and suffixes by unrestricted row words gives

$$\mu_n(\lambda) \leq R_n \eta_n(\lambda), \quad R_n = \frac{\prod_{j=0}^{m-1} (n+j)!}{(n!)^m \prod_{j=0}^{m-1} j!} = O_m(n^d). \quad (44.14)$$

Stirling bounds and strict concavity of binary entropy show that any fixed linear deviation from $\lambda_i = \rho n$ has exponentially small probability. More explicitly, the exponential part of η_n is $\exp\{n[\sum_i H(\lambda_i/n) - mH(\rho)]\}$, with a polynomial prefactor, where $H(x) = -x \log x - (1-x) \log(1-x)$. On the compact constraint $\sum_i \lambda_i/n = 1$ its exponent has a unique maximum at $(\rho, \dots, \rho)$. Equation (44.14) transfers the concentration to μ_n .

In a central region $\epsilon n \leq \lambda_i \leq (1-\epsilon)n$, put $y_i = (\lambda_i - \rho n)/\sqrt{n}$. The exact hook formulas give

$$\mu_n(\lambda) = R_n \eta_n(\lambda) \frac{\Delta(\lambda + \delta)^2}{\prod_i (\lambda_i + 1)_{m-i} (n - \lambda_i + 1)_{i-1}}, \quad (44.15)$$

where $(x)_j = x(x+1) \cdots (x+j-1)$. The denominator is bounded below by a positive constant times n^{2d} , and the numerator by $Cn^d(1 + \|y\|)^{2d}$. Uniform Stirling bounds on this region and the negative upper bound for H'' yield

$$\eta_n(\lambda) \leq Cn^{-(m-1)/2} e^{-c\|y\|^2}.$$

Hence

$$\mu_n(\lambda) \leq Cn^{-(m-1)/2} (1 + \|y\|)^{2d} e^{-c\|y\|^2}. \quad (44.16)$$

Gaussian lattice sums and the exponential outer-region bound imply tightness of y . In a fixed box $\|y\| \leq M$, requiring one gap $\lambda_i - \lambda_j \leq K$ leaves at most $O_{m,M}((K+1)n^{(m-2)/2})$ lattice points on $\sum \lambda_i = n$. Equation (44.16) bounds their probability by $O_{m,M}((K+1)/\sqrt{n})$. A finite union over pairs, then tightness, proves

$$\lambda_i/n \longrightarrow \rho, \quad \min_{i < j}(\lambda_i - \lambda_j) \longrightarrow \infty \quad \text{in probability under } \mu_n. \quad (44.17)$$

In particular the interior strict states have probability tending to one.

At such a state, division of (44.13) by (44.12) gives

$$\frac{g^\lambda}{f^\lambda} = \frac{\prod_i (\lambda_i + m - i)! / \lambda_i!}{\prod_{i < j} (\lambda_i + \lambda_j)} \frac{\Delta(\lambda)}{\Delta(\lambda + \delta)}.$$

The first factor tends to 2^{-d} by the coordinate concentration, and the second tends to 1 by the diverging gaps. The complementary ratio has the same limit. All exact through-count ratios lie in $[0, 1]$, so bounded convergence in probability gives

$$\frac{g(m, n)}{f(n^m)} \longrightarrow 2^{-2d}.$$

Finally Stirling's formula in the rectangular hook product gives

$$f(n^m) \sim \sqrt{m} (2\pi)^{-(m-1)/2} \left(\prod_{j=0}^{m-1} j! \right) m^{mn} n^{-\nu},$$

which proves (44.11). □

In particular the two principal OEIS rows have

$$g(4, n) \sim \frac{3}{1024\sqrt{2}\pi^{3/2}} \frac{256^n}{n^{15/2}}, \quad g(5, n) \sim \frac{9\sqrt{5}}{2^{17}\pi^2} \frac{3125^n}{n^{12}}.$$

The second equivalent is conjectured in A181199. The proof above does not use its conjectural recurrence. No all-orders estimate is needed for what follows.

44.7 Elementary algebraic obstructions

The following radial arguments avoid any assumption about the number or nature of other singularities on the circle of convergence.

Lemma 44.8. *If $b_n \sim C/n$ with $C > 0$, then*

$$\sum_{n \geq 1} b_n t^n \sim C \log \frac{1}{1-t} \quad (t \uparrow 1).$$

If $b_n \sim Cn^{-1/2}$, then

$$\sum_{n \geq 1} b_n t^n \sim C\sqrt{\pi} (1-t)^{-1/2} \quad (t \uparrow 1).$$

Proof. For the first claim compare the tail with $(C \pm \varepsilon)/n$ and use $\sum t^n/n = -\log(1-t)$; the finite head is negligible. For the second put $x = -\log t$ and $f(u) = u^{-1/2}e^{-u}$. Since f is positive and decreasing,

$$\int_x^\infty f(u) du \leq x \sum_{n \geq 1} f(nx) \leq \int_0^\infty f(u) du = \sqrt{\pi}.$$

Thus $\sqrt{x} \sum n^{-1/2} e^{-nx} \rightarrow \sqrt{\pi}$. Now $x \sim 1-t$, and tail comparison as before handles b_n . □

Lemma 44.9. *If $F \in \mathbb{Q}[[t]]$ is algebraic over $\mathbb{C}(t)$, it is algebraic over $\mathbb{Q}(t)$. Algebraic functions in characteristic zero remain algebraic under differentiation and multiplication by t .*

Proof. A complex polynomial relation is a dependence among finitely many series $t^i F^j$ having rational coefficients. Arrange their coefficients as the columns of a matrix with finitely many columns and possibly infinitely many rows. Its rational row space has finite dimension, so choose finitely many rows spanning it over $\mathbb{Q}$. A nonzero complex kernel implies that this finite rational matrix has rank less than its number of columns. It therefore has a nonzero rational kernel, which annihilates every row and gives a rational polynomial relation. For differentiation, implicit differentiation of a minimal polynomial gives $F' = -P_t(t, F)/P_y(t, F)$; the denominator is nonzero in characteristic zero. The remaining closure is immediate. $\square$

Lemma 44.10. *An algebraic function over $\mathbb{C}(t)$ cannot satisfy $F(1-s) \sim C \log(1/s)$ for $C > 0$ as $s \downarrow 0$. If a function algebraic over $\mathbb{Q}(t)$ satisfies $F(t) \sim L(1-t)^{-1/2}$ radially, then L is algebraic.*

Proof. For the first assertion, write a nonzero polynomial relation in the form

$$0 = P(1-s, F(1-s)) = \sum_{i \geq i_0} s^i p_i(F(1-s)), \quad p_{i_0} \neq 0.$$

Let $d = \deg p_{i_0}$ and divide by $s^{i_0} F(1-s)^d$. The first term tends to the nonzero leading coefficient of p_{i_0} . Every later term tends to zero, since any positive power of s dominates every fixed power of $\log(1/s)$. This is impossible.

For the second, take $P(t, y) \in \mathbb{Q}[t, y] \setminus \{0\}$ with $P(t, F(t)) = 0$, and let $d = \deg_y P$. The polynomial

$$Q_0(u, v) = u^d P(1-u^2, v/u)$$

is nonzero. Divide it by its largest common power of u , obtaining $Q(u, v) \in \mathbb{Q}[u, v]$ with $Q(0, v) \neq 0$. For $u > 0$ small, put $v(u) = uF(1-u^2)$. Then $Q(u, v(u)) = 0$ and $v(u) \rightarrow L$. Taking limits gives $Q(0, L) = 0$. $\square$

Proof of Theorem 44.2. Normalize the radius by

$$A_m(t) = G_m(m^{-m}t), \quad \vartheta = t \frac{d}{dt}.$$

Proposition 44.7 gives

$$[t^n] \vartheta^k A_m(t) \sim C_m n^{k-\nu}.$$

If $m \geq 3$ is odd, then ν is an integer. Choose $k = \nu - 1$. Lemma 44.8 gives

$$\vartheta^{\nu-1} A_m(t) \sim C_m \log \frac{1}{1-t},$$

contradicting Lemma 44.10 if G_m were algebraic.

If $m \geq 4$ is even, put $k = \lfloor \nu \rfloor = (m^2 - 2)/2$. Then

$$\vartheta^k A_m(t) \sim L_m (1-t)^{-1/2}, \quad L_m = C_m \sqrt{\pi} = \alpha_m \pi^{-(m-2)/2},$$

where

$$\alpha_m = 2^{-m(m-1)-(m-1)/2} \sqrt{m} \prod_{j=0}^{m-1} j! \in \overline{\mathbb{Q}} \setminus \{0\}.$$

The number L_m is transcendental: otherwise the positive integer power $\pi^{(m-2)/2}$ would be algebraic, contradicting the transcendence of π [17]. If G_m were algebraic over $\mathbb{C}(z)$, its rational coefficients, rational rescaling and Lemma 44.9 would make $\vartheta^k A_m$ algebraic over $\mathbb{Q}(t)$. Lemma 44.10 would then force L_m to be algebraic, a contradiction.

At height one $g(1, n) = 1$. At height two the row-choice path is a primitive Dyck path of semilength n : equality of its two coordinates is allowed only initially and finally. Removing the first up-step and last down-step leaves a Dyck path of semilength $n - 1$. Hence

$$g(2, n) = \text{Cat}_{n-1} = \frac{1}{n} \binom{2n-2}{n-1} \quad (n \geq 1),$$

and with the constant term 1 this gives the stated G_2 . □

Remark 44.11. The same obstruction is reusable whenever a rational-coefficient sequence has an equivalent $CR^{-n}n^{-\nu}$ with rational nonzero R . A positive integer ν produces the logarithmic contradiction. A half-integer $\nu > 0$ produces an obstruction when $C\sqrt{\pi}$ is transcendental. These are sufficient tests, not a classification of algebraic series from arbitrary coefficient asymptotics.

44.8 Exact verification and remaining questions

The accompanying standard-library Python checker uses exact integers and rational numbers. It compares independent implementations rather than a guessed recurrence:

1. A cell-poset algorithm counts linear extensions from predecessor masks for horizontal, vertical and both downward diagonal edges.
2. A row-state algorithm counts paths satisfying (44.1).
3. The full Dyck-event sum evaluates (44.8) with the signed binomial kernel (44.7).
4. Direct chamber walks are compared with both the factorial determinant and the binomial kernel, including reversed and zero-length endpoint pairs.

The recorded tests pass for 54 cell-poset/state instances with $1 \leq m \leq 6$ and $0 \leq n \leq 8$, including 44 event-sum instances. There are 2941 chamber comparisons with up to four coordinates, and 1554 binomial-product tests including negative lower arguments. All Dyck-word coordinate bounds are checked for $1 \leq m \leq 9$. Selected counts are

	$n = 0$	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
$m = 2$	1	1	1	2	5	14
$m = 3$	1	1	1	4	29	290
$m = 4$	1	1	1	8	169	6392
$m = 5$	1	1	1	16	985	141696

The mathematical proof covers every fixed height and all widths; these finite checks audit its indexing and boundary conventions.

The rational diagonal supplied by the closure theorem is not printed as one expanded rational function. Producing a compact representation and reducing it to a manageable differential equation are practical next steps. Proving the specific height-four and height-five conjectures, their minimal orders, and any finer subdominant exponential sectors remain separate problems. No assertion of priority beyond the cited sources and the existing project reports is intended.

[write] Added 5 October 2026, batch 101: the shipped layout of Report 91. Its `verify.py`, `code/check_exact.py` and `build_local.sh` are shipped as `code/06-strip-diagonals-verify.py`, `-check_exact.py` and `-build_local.sh`; `code/check_exact.json`, `CHECKS.json`, `SOURCES.json` and `verification.json` as `data/`

06-strip-diagonals-name. The delivered `verify.py` reads and rewrites `code/check_exact.json` and writes `verification.json` at its root, and `build_local.sh` changes to its own directory, so they run only on a copy with the delivered layout restored. At the write (5 October 2026, Python 3.14.4, on such a copy) `verify.py` passed in under two seconds, reported `matches_distributed_data: true`, and both refreshed records equal the shipped ones after removal of carriage returns.

45 [write] Chronology, correspondence and further questions of Part V

Added 5 October 2026, batch 101. This section is the write's.

45.1 What Reports 85 and 91 prove, against Parts I and II

Report 85 (1 October 2026) and Report 91 (2 October 2026) were written without access to this report, and the manuscripts of Part I (3 October) and Part II (4 October) without access to them. The table pairs their statements. “Same route” means the same chain of ideas, not the same text: the word 8-gram overlaps with Parts I and II are 0.4 % and 1.1 % for Report 85, and 0.0 % and 1.1 % for Report 91 (7.1 % of Report 91's 8-grams recur in Report 85).

Part V	Parts I–II	relation
Theorem 43.1, (43.2), (43.3)	Theorem 2.1, (2.7), Corollary 2.2	the same statements ($c_j(m) = b_{m,j}$, $C_m = K_m$); another route: an ordinary-tableau comparison and a cut at $k = n$, Poisson summation and cut independence, against Part I's random-threshold cut (Theorem 4.1)
(43.13)	Corollary 2.4, leading term	the same statement, proved directly by bounded convergence
Subsection 43.5, (43.19)	Section 5 and the integration by parts of Section 6	a second route to the coefficients: a loop-equation recursion for the trace-zero Gaussian–Vandermonde moments, and a traceless-GUE Wick check (classical tools)
(43.20)	(2.7)	a consequence (Remark 45.1)
Sun's height-three recurrence (Subsection 43.5.1)	—	not in Parts I–IV; see item (F12)
(43.21), (43.22)	Section 10.2	a second route to the W_{-1} inversion with two corrections, with the rounding qualification
Theorem 44.1	Theorem 14.1	the same theorem by the same route (Dyck event words, reflection determinants, everywhere-defined binomial products, Bostan–Lairez–Salvy); bound m^2 against $m(m-1)$ (note at the head of Section 44)
Theorem 44.2	Theorem 2.3	the same theorem and the same two obstructions (a logarithm at odd m , the amplitude $\alpha_m \pi^{-(m-2)/2}$ at even m , the constant of (8.4)); elementary radial Lemmas 44.8–44.10 replace the Puiseux argument
Proposition 44.7	Theorem 2.1, leading term	Report 85's Subsection 43.3, reproduced with more detail
the (3, 4) table of Subsection 44.4	Section 17.2	identical entries 4, 16, 4, 4, 1

So, within this report, the first proofs by date line of Theorems 2.1 and 2.3 and of Theorem 14.1 are those of Reports 85 and 91. This bears on item (F4), the priority of Theorem 14.1: that item asked for a search of the general literature, which is still missing; what is new is that an earlier manuscript in the same corpus proves the theorem by the same route. Nothing here is a priority claim outside ProveIt, and the two reports' own source checks were, in their words, bounded.

Remark 45.1 ([write] Equation (43.20) from Part I). Report 85’s identity $c_2 - c_1^2/2 = m^2(m^2 - 1)/96$ follows from Part I’s (2.7), since $c_j(m) = b_{m,j}$:

$$b_{m,2} - \frac{b_{m,1}^2}{2} = \frac{(m^2 - 1)((m^6 + 3m^2 - 1) - (m^2 - 1)^3)}{288m^2} = \frac{(m^2 - 1)3m^4}{288m^2} = \frac{m^2(m^2 - 1)}{96}.$$

45.2 Further questions and research recorded at the write

Under Vladimir Reshetnikov’s standing rule, the unproved and outline-level claims of the two manuscripts are recorded here. None of them is wrong as far as the intake could check; the intake confirmed the height-five and height-four displays, (43.20), the (3, 4) table and G_2 .

- (F11) *Outline steps of Report 85* (Subsection 43.4). Three steps are argued in outline: the growing-window Taylor expansion and its exponentiation (“Choosing ε small permits exponentiation of the remainder without losing an integrable Gaussian majorant”), the uniformity of Poisson summation over the affine lattice translations, and the passage from (43.16) to the rounding statement (43.22). The theorem itself is proved in full in Part I (Theorem 2.1, with the explicit Taylor bound (5.12)), so nothing depends on them. Missing: explicit window and remainder estimates for this route.
- (F12) *Sun’s height-three recurrence* (Subsection 43.5.1). The report states that “Sun’s proved recurrence simplifies to” $g(3, n+1) + g(3, n) = \frac{(3n)!}{(n!)^3} \frac{7n+1}{2(n+1)^2(4n^2-1)}$; its program `05-strip-asymptotics-ward_corrections.py` checks the expansion against this form, and the intake checked the displayed identity for $n = 1, \dots, 24$ against row-state counts. Missing: the derivation of this form from the recurrence as printed in [11], which neither the report nor the intake reproduced.
- (F13) *Growing height and smaller sectors* (Subsection 43.8). The scale m^3/n is “a proposed scale, not a growing-height theorem” (Research question 3); the exponentially smaller sectors and saturated boundary shapes are Research question 4 and items (F1)–(F2); effective inverse constants are Research question 6, now carried out at $m = 4$ and $m = 5$ in Parts III and IV.
- (F14) *A compact rational diagonal* (Subsection 44.8). “The rational diagonal supplied by the closure theorem is not printed as one expanded rational function”; a compact representation, its differential equation, the minimal orders and finer subdominant sectors are open. These are Research questions 9 and 11; at $m = 4$ and $m = 5$ Parts III and IV supply the operators, and Propositions 33.2 and 42.2 bound their orders.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the batch-101 write of Parts III–V, made by the intake after the write (ca1668212), reread every result the write added: Lemma 33.1, Proposition 33.2 with the remark after it and item (F5), Remark 42.1, Proposition 42.2 with item (F9), and Remark 45.1. It found all of them valid, with no counterexample and no gap in any proof chain. Two changes followed. Remark 42.1 called Part IV’s transfer, rank-two and scalar identities “literally the same statements” as Part III’s; they are the same after renaming Part III’s α, β to Part IV’s f, b , and Part IV’s $J_0 = I$ corresponds to the display (28.13), which the remark did not cite. The remark is reworded where it stands, with a dated note keeping the first wording. And the question left open after Proposition 33.2 and in item (F5), whether a recurrence of order three or more has degree below nine, has an answer: the check found and proved Propositions 33.3 and 42.3, now printed with their proofs. Items (F5) and (F9) are re-scoped with dated notes keeping their first wording.

The check used neither the delivered programs nor the write’s. Its own counter, in C, is a dynamic program over the order ideals of the literal $m \times n$ poset (row, column and antidiagonal relations), each ideal given by its row lengths; it runs modulo fourteen primes below 2^{62} and recombines by the Chinese remainder theorem, which is exact because $a_n \leq M_n$ and M_{64} is below the product of the primes. It gave $T_4(n)$ and $T_5(n)$ for $n \leq 64$. On these counts the operator of Theorem 30.1 has residual zero for $n = 1, \dots, 62$ and the

identity of Theorem 26.1 holds for $n \leq 63$; the operator of Theorem 34.2, with its coefficients typed from Appendix E, has residual zero for $n = 1, \dots, 61$; and the printed sums of Theorems 30.2 and 34.1 equal the counts for $2 \leq n \leq 64$ and $1 \leq n \leq 64$. Ingredients of the proofs were tested on the same counts: the affine formula (28.21) for $n \leq 64$ and the first difference (34.14) for $n \leq 63$, with Z_n from (29.1) and $Z_1 = 1/3$ (and $Z_1, Z_2, Z_3 = 1/3, 13/45, 461/1680$ by direct triangle integrals); $Y_1 = 0$ and $Y_n > 0$ for $2 \leq n \leq 12$, consistent with the proof of Proposition 42.2, which needs $Y_n \neq 0$ only for large n . In SymPy the check confirmed the hinge identities of both reports. For Report 231: the divergence certificate (29.4), the slanted-edge identity, the edge flux (29.8) (also against direct integration for $n \leq 6$), the cancellation of the Z_n coefficient with the rational return D_n , $q_n = J_{n+1}/J_n$ (30.3), and the three coefficient identities in the proof of Theorem 30.1. For Report 233: the γ -bridge (39.1) with $\gamma_1 = 1/3$, $[Z_n](F_{n+1} - \rho_n F_n) = 2\rho_n u(n)$ with the matching constant term, (E.12) and (E.4) coefficient by coefficient, and the factorization (39.17), through the exact right division of the new operators by L_{OEIS} with zero remainder. For the minimality statements it computed ranks of the matrices $(n^i a_{n+j})$ modulo $2^{61} - 1$ and $2^{62} - 57$. At height four: no solution of order one and degree at most 50, and at order two a solution space of dimension exactly $\max(0, d - 8)$ for every degree $d \leq 40$, which confirms the order-two statement of Proposition 33.2 in that range. At height five: no solution of order one or two and degree at most 60 on the windows used (for order two, $5 \leq n \leq 214$), and at order three nullity 0 up to degree 23 and 1 at degree 24. The same computations, with equations from $n = 30$, located the degrees of Propositions 33.3 and 42.3, and the right division by the proved operators turned them into proofs.

Before printing, the session that applied the check re-derived the four operators of those propositions by exact Ore multiplication and right division (SymPy 1.14.0) and found zero residual for $n = 1, \dots, 234$ on terms generated by Theorems 30.1 and 34.2; those terms agree with the check's counts for $n \leq 64$ and with this report's row-state program (code/tableaux.py) for $n \leq 18$ at height four and $n \leq 14$ at height five. It redid the rank computations of both proofs exactly over $\mathbb{Q}$ (python-flint 0.8.0), with equations from $n = 1$. The check was a careful reading with exact computations, not a formal verification or an external review.

A A direct proof of the shifted hook product

Let $\lambda_1 > \dots > \lambda_r > 0$ be a strict partition of k , and put

$$H(\lambda) = \frac{k!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}.$$

We prove that it satisfies the strict-partition path recursion

$$H(\lambda) = \sum_i H(\lambda - e_i), \quad (\text{A.1})$$

where only valid strict-partition removals are included and a trailing zero part is deleted. Together with $H(\emptyset) = 1$, this proves (3.5) by induction.

For a valid removal,

$$\frac{H(\lambda - e_i)}{H(\lambda)} = \frac{\lambda_i}{k} \prod_{j \neq i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1}. \quad (\text{A.2})$$

The same expression is zero for an invalid removal causing equal positive parts. If the last part is one, deleting its resulting zero gives the same ratio, since the extra zero factors cancel. Thus it suffices to prove

$$\sum_i \lambda_i \prod_{j \neq i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1} = k. \quad (\text{A.3})$$

Consider the rational function

$$R(z) = \prod_{j=1}^r \frac{z - \lambda_j - 1}{z - \lambda_j} \frac{z + \lambda_j}{z + \lambda_j - 1}, \quad f(z) = (z - \frac{1}{2})R(z).$$

Its possible finite poles are the disjoint sets λ_i and $1 - \lambda_i$. Direct substitution gives $R(1 - z) = R(z)$ and hence $f(1 - z) = -f(z)$. The residues at the paired poles are equal. At $z = \lambda_i$, the residue is the negative of the i th summand in (A.3).

At infinity, each pair of factors contributes $-2\lambda_j/z^2$ to the first nonconstant term, so

$$R(z) = 1 - \frac{2k}{z^2} + O(z^{-3}), \quad f(z) = z - \frac{1}{2} - \frac{2k}{z} + O(z^{-2}).$$

The sum of finite residues is the coefficient of z^{-1} , namely $-2k$. On the other hand it is minus twice the left side of (A.3). The identity follows, proving (A.1) and the hook product. All operations here are rational identities; the residue language can equivalently be replaced by partial fractions.

B An implementation-independent coefficient specification

For a fixed height m and requested order L , the following finite recipe fully specifies the coefficient calculation.

First compute $h_0, \dots, h_L$ from $\prod_{i < j} (1 - u(y_i + y_j)^2)^{-1}$ modulo u^{L+1} . Next compute the sign cumulants κ_r from

$$\log \cosh t = \sum_{r \geq 1} \kappa_r \frac{t^r}{r!}.$$

The even values begin $\kappa_2 = 1, \kappa_4 = -2, \kappa_6 = 16, \kappa_8 = -272$; the odd values vanish. The moment polynomials can be computed recursively as

$$\mu_0(z) = 1, \quad \mu_k(z) = \sum_{\substack{2 \leq r \leq k \\ r \text{ even}}} \binom{k-1}{r-1} \kappa_r z^{r/2-1} \mu_{k-r}(z) \quad (k > 0), \quad (\text{B.1})$$

with $\mu_k = 0$ for odd k . Apply (11.1) to each h_r , evaluate each monomial by (5.8), retain powers through z^{L-r} , and combine them as in (5.9). Finally apply the formal exponential (5.16). This recipe is sufficient to reproduce the printed coefficients independently of the shipped Python code.

For $m = 5$, the two outputs through order five are

$$(c_{5,j})_{j=0}^5 = \left(1, 10, \frac{225}{4}, \frac{1819}{8}, \frac{22045}{32}, 1353\right),$$

$$(b_{5,j})_{j=0}^5 = \left(1, \frac{48}{5}, \frac{5233}{100}, \frac{1028391}{5000}, \frac{60248377}{100000}, \frac{273939939}{250000}\right).$$

The first two nonconstant entries in each row also have the independent symbolic derivation in Section 6. Higher printed entries are exact evaluations of the proved finite formula, not inferred from observed numerical errors.

C A fully finite recipe for the exact sum

[write] Added 5 October 2026, batch 98. Appendices C and D belong to Part II: they are the manuscript's Appendices A and B.

For clarity, the diagonal theorem can be implemented without making any choices suggested by numerical data. The following recipe has fixed size once m is fixed.

First list all Dyck words of semilength m . For a chosen word, make one vector v_s of length r_s for each interval $0 \leq s < 2m$, and replace the top coordinate before each completion by $n - 1$. Sum every remaining coordinate from 0 to n . Construct $u_0 = ()$ and the later u_s using (16.1). Insert H factors for

$$v_{s,r_s} \geq 1, \quad v_{s,1} \leq n - 1, \quad v_{s,i} - v_{s,i+1} \geq 1,$$

where the conditions with coordinates are omitted when $r_s = 0$. For a birth with $r_s > 0$, also insert $H(v_{s,r_s} - 2)$. The already substituted completion coordinate enforces its event condition.

For each interval independently choose a permutation $\sigma_s \in \mathfrak{S}_{r_s}$, multiply its sign, and set

$$a_{s,i} = v_{s,i} - u_{s,\sigma_s(i)}.$$

Multiply the binomial product (16.5) for this displacement vector. Sum over the finitely many permutations and endpoint coordinates, and then over the words. This expression equals $T_m(n)$ for every $n \geq 2$ by the bijective proof of Theorem 16.1.

The endpoint-coordinate order in this recipe can be arbitrary; using nested partial sums and then affine substitution of their upper bounds realizes the rectangular sums in the binomial-sum algebra. Values outside the intended polytope are killed by the explicit H factors. In particular, one never applies a recurrence or divides by a factor that could vanish at a summation endpoint.

The delivered evaluator uses the same formula more efficiently by iterating over decreasing endpoint tuples directly and caching suffix computations. That optimization changes neither the mathematical formula nor its fixed-dimensional binomial nature. It is not claimed to beat the prefix dynamic program for every input.

D An implementation-independent coefficient algorithm

The symbolic routine can be reproduced using rational polynomial arithmetic alone. Here are sufficient details to fix every convention. For a nonnegative integer a , form the polynomial in p

$$C_a(n, p) = \sum_{b=0}^a \sum_{k=0}^b \binom{a}{b} 2^b (-n)^{a-b} \left\{ \begin{matrix} b \\ k \end{matrix} \right\} n^{\overline{k}} p^k. \quad (\text{D.1})$$

It is exactly $\mathbb{E}[(2X - n)^a \mid U = p]$. For a monomial $y_1^{a_1} \cdots y_r^{a_r}$, set $A = \sum_i a_i$ and multiply $\prod_i C_{a_i}(n, p) = \sum_{k=0}^A c_k(n) p^k$. Then

$$\mathbb{E} \prod_i Y_i^{a_i} = n^{-A/2} \sum_{k=0}^A c_k(n) \frac{(n)^{\overline{k}}}{(2n+1)^{\overline{k}}}. \quad (\text{D.2})$$

A convenient common denominator is $(2n+1)^{\overline{A}}$; its numerator is

$$\sum_{k=0}^A c_k(n) (n)^{\overline{k}} (2n+1+k)^{\overline{A-k}}. \quad (\text{D.3})$$

Reverse the numerator and denominator polynomials and perform ordinary power-series division at $1/n = 0$, retaining the extra factor $n^{-A/2}$. This is the exact moment functional $\mathcal{E}_\varepsilon$ used in (20.2). It avoids repeated simplification of many separate rational functions.

To expand the single-coordinate interaction in (19.8), write

$$\frac{1 - (2 + y^2)\varepsilon^2 - 2y\varepsilon^3}{1 - (6 + y^2)\varepsilon^2/9 + 2y\varepsilon^3/9} = \sum_{k \geq 0} c_k(y)\varepsilon^k. \quad (\text{D.4})$$

With $c_k = 0$ for $k < 0$, its coefficients satisfy

$$c_k = N_k + \frac{6 + y^2}{9}c_{k-2} - \frac{2y}{9}c_{k-3}, \quad N_0 = 1, \quad N_2 = -(2 + y^2), \quad N_3 = -2y, \quad (\text{D.5})$$

and all other N_k are zero. The pair interaction contributes the geometric series

$$\sum_{k \geq 0} \frac{(y_i + y_\ell)^{2k}}{4^k} \varepsilon^{2k}, \quad i < \ell.$$

Multiply these finitely truncated factors, then multiply by $\Delta(y)^2$. Apply (D.2) to each monomial, divide by $r!J_r$, and finally multiply by the truncated exponential (20.3). No asymptotic fitting or guessed differential equation enters this calculation. For the height-five coefficients in (21.1), a truncation at ε^6 suffices.

As intermediate checks, the engine gives

$$[z]Q_1 = -\frac{8}{3}, \quad [z]Q_2 = -\frac{119}{18}, \quad [z]Q_3 = -\frac{34}{3},$$

which also follow from (20.14). The package stores exact coefficient strings and reruns their derivation in `code/verify.py`.

E The full literal order three operator

[write] Added 5 October 2026, batch 101. Appendix E belongs to Part IV: it is the manuscript's Appendix A.

This appendix records all four source polynomials without requiring an external lookup. It also gives the shifted polynomial identities that connect the compact notation to the printed source. Write

$$B_{21}(n) = \sum_{j=0}^{21} b_j n^j, \quad C_{19}(n) = \sum_{j=0}^{19} d_j n^j,$$

with the integer coefficients in the table below. The distinct notation B_{21} avoids confusion with the integral operator $B = EA$. Define

$$Q_+(n) = 137855872n^{11} + 2377384288n^{10} + 18238806776n^9 \quad (\text{E.1})$$

$$+ 81945774178n^8 + 238738633847n^7 + 471293180347n^6 \quad (\text{E.2})$$

$$+ 638861237719n^5 + 588363536007n^4 + 354332674386n^3 \quad (\text{E.3})$$

$$+ 128320688700n^2 + 23068181160n + 1077753600. \quad (\text{E.4})$$

Then the literal coefficient list is

$$c_0(n) = 25(2n-1)^2(2n+1)(4n-1)(4n+1)Q_+(n) \quad (\text{E.5})$$

$$\cdot \prod_{j \in \{1,2,3,4,6,7,8,9\}} (5n+j), \quad (\text{E.6})$$

$$c_1(n) = -15(2n+1) \prod_{j=6}^9 (5n+j) C_{19}(n), \quad (\text{E.7})$$

$$c_2(n) = -(n+2)^2(2n+3)B_{21}(n), \quad (\text{E.8})$$

$$c_3(n) = 3(n+2)^2(2n+3)(2n+5)^2(3n+7)(3n+8) \quad (\text{E.9})$$

$$\cdot (4n+9)(4n+11)Q(n)(n+3)^4. \quad (\text{E.10})$$

Each has degree exactly 24. These expressions were transcribed from the A181199 recurrence before comparison with the factorization. Expanding the products and comparing all coefficients verifies (39.16) identically, with the four zero residuals recorded by the exact verifier.

The second degree-eight polynomial printed in the denominator of v is

$$P_+(n) = 25216n^8 + 211616n^7 + 760768n^6 + 1543976n^5 \quad (\text{E.11})$$

$$+ 1973632n^4 + 1683047n^3 + 971955n^2 + 353502n + 60480. \quad (\text{E.12})$$

The literal expansions satisfy $P_+(n) = P(n+1)$ and $Q_+(n) = Q(n+1)$. These equalities account for the compact forms used in (34.10) and (39.9); neither is an approximation.

Table 8: The complete integer coefficient arrays of B_{21} and C_{19}

j	b_j	d_j
0	-1104708217536000	257403484800
1	-22603977271411200	4014999151560
2	-180089041888657440	21777752002716
3	-685802053101717288	50327965592874
4	-850259283025327524	42448206362469
5	3499276436019940446	-54713722756179
6	19628106098287909191	-793918042456325
7	47097079083848116511	-4487751704725767
8	65692918122110754573	-12634296322883951
9	48833460779191449763	-17410620603191989
10	-1215620047630796049	-3351043407516635
11	-46634308960957698567	30740208891967721
12	-54791308745183340309	61042725728660150
13	-31771892184486994333	64302800289940820
14	-6080171043591548610	44061606220301336
15	5462710847171622988	20586994844739808
16	5556799141672247640	6541981632511232
17	2616700630350746208	1357324488921088
18	760770273998231808	166164038596608
19	139665528127686656	9113927409664
20	14962008250398720	0
21	717088749346816	0

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